SGA 5 — Exposé I

Dualizing Complexes

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Introduction

This exposé is devoted to the biduality theorem in étale cohomology. The theory developed here and in SGA A XVIII (SGA A = SGA 4), for the étale topology and constructible torsion sheaves on preschemes, is formally very analogous to the one developed in HARTSHORNE [H] for the Zariski topology and coherent sheaves: the latter served, to a very large extent, as its model. But, whereas in the case of "continuous" coefficients the existence of dualizing complexes presents no difficulties, in the case of "discrete" coefficients, on the other hand, it is much more delicate to prove, and it seems to require resolution of singularities in an essential way. See however Deligne's biduality theorem (SGA $4^{1/2}$, Finiteness Theorem 4.3.).

Paragraph 1 contains the definition of dualizing complexes and the formal properties that follow from it, notably the interpretation of dualizing functors as "exchangers of functors."

In paragraph 2, one shows that, on a connected locally noetherian prescheme, a dualizing complex is determined uniquely, up to a translation of degrees and tensoring by an invertible sheaf.

In paragraph 3, one comes to the central theorem of the exposé (3.4.1.), which asserts that, on a "good" regular prescheme X, for example on an excellent noetherian regular prescheme of characteristic zero[1], the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing.

Paragraph 4 gives the applications of the theory to the local duality theorem. The latter expresses that, on a "good" prescheme X endowed with a closed point x whose residue field is separably closed, and for a constructible sheaf F, one has a perfect pairing between $\underline{\mathrm{H}}^i_x(F')$ and $\underline{\mathrm{H}}^{-i}_x(F)$, where F' is the dual of F, i.e. $F' = \mathrm{R}\,\underline{\mathrm{Hom}}(F, K_X)$, where K_X is a dualizing complex.

Finally, in paragraph 5, one proves directly that on a regular prescheme of dimension 1 the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing (n prime to the residual characteristics).

1 Definition and formal properties of dualizing complexes

We fix a coefficient ring $A = \mathbb{Z}/n\mathbb{Z}$. If X is a prescheme, we denote by A_X the constant sheaf on $X_{\text{ét}}$ with value A, and by $\mathrm{D}(X)$ the derived category of the category of A_X-modules. We denote, on the other hand, by $\mathrm{D}_c(X)$ the full triangulated subcategory of $\mathrm{D}(X)$ formed by the complexes F such that $\mathrm{H}^i(F)$ is a constructible A_X-module for every i. We put $\mathrm{D}^*_c(X) = \mathrm{D}^*(X) \cap \mathrm{D}_c(X)$, where $* = \varnothing, +, -,$ or b.

[1] This last restriction will become unnecessary once one has resolution of singularities à la Hironaka for these preschemes, and the "purity theorem."

Definition 1.1

An A_X-module I is said to be *quasi-injective* if one has

$$\underline{\mathrm{Ext}}^i(F, I) = 0$$

for every *constructible* A_X-module F and every $i > 0$.

Remarks 1.2

a) Contrary to what happens in the case of "continuous" coefficients (cf. [H] II.7.17), a quasi-injective sheaf is not injective in general. For example, let $X = \mathrm{Spec}(\mathrm{R})$ and $A = \mathbb{Z}/2\mathbb{Z}$: the sheaf A_X is quasi-injective, but is not injective, nor even of finite injective dimension, since $\mathrm{H}^*(X; A_X) = \mathrm{H}^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}/2\mathbb{Z})$ is a polynomial algebra on one generator of dimension 1.

b) Let I be a quasi-injective A_X-module. It is false in general that the relation $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > 0$, valid for constructible F, is valid for every F. Indeed, take $A = \mathbb{F}_\ell$. Let k be a field, $\overline{k}$ a separable closure of k, and $f : \overline{x} = \mathrm{Spec}(\overline{k}) \to X = \mathrm{Spec}(k)$ the canonical morphism, and suppose that, for every connected étale covering $X' \to X$, there exists a connected étale covering $X'' \to X'$ such that $[X'' : X']$ is divisible by ℓ (take, for example, $k = \mathbb{Q}$). Consider the exact sequence

$$(*) \qquad 0 \longrightarrow A_X \xrightarrow{\varepsilon} f_* f^* A_X \longrightarrow F \longrightarrow 0$$

defined by the adjunction arrow ε. In view of the hypothesis, the section of $\mathrm{Ext}^1(F, A_X)$ defined by $(*)$ does not vanish above any U étale over X; hence $\underline{\mathrm{Ext}}^1(F, A_X)_{\overline{x}} \neq 0$. But one checks trivially that A_X is quasi-injective.

Proposition 1.3 (cf. [H] 1.7.6)

Let X be a prescheme and $N \in \mathbb{Z}$. The following conditions on $K \in \mathrm{ob}\, \mathrm{D}^+(X)$ are equivalent:

(i) K is isomorphic, in $\mathrm{D}(X)$, to a bounded complex I such that I^i is quasi-injective for every i and zero for $i > N$;

(ii) for every $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$ such that $\mathrm{H}^i(F) = 0$ for $i < 0$, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$;

(iii) for every constructible A_X-module F, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$.

Proof. *(i)* $\Rightarrow$ *(ii)*. Let $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$ be such that $\mathrm{H}^i(F) = 0$ for $i < 0$, and let $I \in \mathrm{ob}\, \mathrm{D}^+(X)$ be bounded and such that I^i is quasi-injective for every i and zero for $i > N$; we show that $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > N$. Arguing by induction on the number of indices i such that $\mathrm{H}^i \neq 0$, one is reduced, by dévissage, to the case where $N = 0$ and I is concentrated in degree 0. One then has a biregular spectral sequence

$$\mathrm{E}_2^{pq} = \underline{\mathrm{Ext}}^p(\mathrm{H}^{-q}(F), I) \Rightarrow \underline{\mathrm{Ext}}^*(F, I).$$

Since I is quasi-injective, $\mathrm{E}_2^{pq} = 0$ for $p > 0$ and every q, and hence one has $\mathrm{E}_2^{0i} = \underline{\mathrm{Hom}}(\mathrm{H}^{-i}(F), I) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(F, I)$, which gives the conclusion.

(ii) $\Rightarrow$ *(iii)* is trivial.

$(iii) \Rightarrow (i)$ We may suppose that K is bounded below and has injective components. The hypothesis, applied to $F = A_X$, implies that $\mathrm{H}^i(K) = 0$ for $i > N$. We may therefore replace K by an isomorphic complex K', bounded, and such that K'^i is injective for $i < N$ and zero for $i > N$. It is then observed that K'^N is quasi-injective, which completes the proof.

Remark 1.3.1

The writer does not know whether (1.3) is still true when, in condition (ii), the hypothesis "$F \in \mathrm{ob}\,\mathrm{D}^b_c(X)$" is replaced by "$F \in \mathrm{ob}\,\mathrm{D}^+_c(X)$".

Definition 1.4

The *quasi-injective dimension* of K, denoted $\dim . q. inj(K)$, is the lower bound (in $\overline{\mathrm{R}}$) of the integers N satisfying the equivalent conditions of (1.3).

If K is of finite quasi-injective dimension, the functor $\mathrm{R}\,\underline{\mathrm{Hom}}(\quad, K) : \mathrm{D}(X) \to \mathrm{D}(X)$ induces a functor $\mathrm{D}^b_c(X) \to \mathrm{D}^b(X)$.

We shall see below that if X is smooth of dimension d over a field k and n is prime to the characteristic of k, then $\dim . q. inj(A_X) = 2d$.

The example of (1.2 a)) shows that a complex of finite quasi-injective dimension is not necessarily of finite injective dimension. Nevertheless one has:

Proposition 1.5

Let X be a locally noetherian prescheme and $K \in \mathrm{ob}\,\mathrm{D}^+(X)$. Suppose that there exists an integer N such that $\mathrm{cd}_{\mathbf{n}}(X) \leqslant N$, where $\mathbf{n}$ is the set of prime divisors of n (notation of SGAA X 1). Then, for $N' \in \mathbb{Z}$, one has

$$\dim . q. inj(K) \leqslant N' \quad \Longrightarrow \quad \dim . inj(K) \leqslant N + N'.$$

We shall use, in the proof, the following

Lemma 1.5.1

Let C be an abelian category satisfying (AB5) and possessing a family of generators $(U_i)_{i \in I}$. Let $K \in \mathrm{ob}\,\mathrm{D}^+(C)$ and $N \in \mathbb{Z}$. The following conditions are equivalent:

(i) $\dim . inj(K) \leqslant N$;

(ii) for every $i \in I$ and every quotient F of U_i, one has

$$\mathrm{Ext}^n(F, K) = 0 \qquad \text{for } n > N.$$

Proof. It suffices to prove $(ii) \Rightarrow (i)$. By a well-known argument (cf. for example [H] 1.7.6), one is reduced to showing that if $K' \in \mathrm{ob}(C)$ is such that $\mathrm{Ext}^1(F, K') = 0$ every time F is a quotient of a U_i, then K' is injective. In other words, one has to prove that, if every morphism from a subobject

V of a U_i into K' extends to U_i, then K' is injective. But this follows immediately from the proof of Lemma 1, p. 136 of (Tohoku).

Proof of (1.5). Suppose $\dim.q.inj(K) \leqslant N'$ and prove $\dim.inj(K) \leqslant N + N'$. Since the category of A_X-modules admits as generators the $A_{U,X}$ where U is étale of finite type over X, it suffices, by (1.5.1), to prove that $\mathrm{Ext}^i(X; F, K) = 0$ for $i > N + N'$ when F is a quotient of such an $A_{U,X}$, hence constructible (SGAA IX 2.9). Consider the spectral sequence

$$\mathrm{E}_2^{pq} = \mathrm{H}^p(X; \underline{\mathrm{Ext}}^q(F, K)) \Rightarrow \mathrm{Ext}^*(X; F, K).$$

The hypothesis $\mathrm{cd}_{\mathbf{n}}(X) \leqslant N$ (resp. $\dim.q.inj(K) \leqslant N'$) implies $\mathrm{E}_2^{pq} = 0$ for $p > N$ (resp. $q > N'$). Hence $\mathrm{E}_2^{pq} = 0$ for $p + q > N + N'$, whence the conclusion.

Remark 1.5.2

The hypothesis of (1.5) is verified (with $N = 2d$) if X is a prescheme of finite type and dimension d over a separably closed field k, with n prime to $\mathrm{char}(k)$ (SGAA X 4).

Proposition 1.6

Let $f : X \to Y$ be a separated quasi-finite morphism of noetherian preschemes, and let $K \in \mathrm{ob}\,\mathrm{D}^+(Y)$. Then, for $N \in \mathbb{Z}$, one has

$$\dim.q.inj(K) \leqslant N \quad\Longrightarrow\quad \dim.q.inj(\mathrm{R}^! f(K)) \leqslant N.$$

Proof. According to the "Main theorem" (EGA IV 8.12.6), there exists a factorization of f as $f = uf'$, where f' is an open immersion and u is a finite morphism. We have $\mathrm{R}^! f = \mathrm{R}^! f' \,\mathrm{R}^! u$, which reduces us to examining separately the case of an open immersion and that of a finite morphism. If f is an open immersion, the assertion is trivial (since every constructible sheaf on X is then induced by a constructible sheaf on Y). Suppose therefore that f is a finite morphism, and let F be a constructible A_X-module. We then have the duality isomorphisms (SGAA XVIII 3.1.9.6)

$$f_* \underline{\mathrm{Ext}}^i(F, \mathrm{R}^! f(K)) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(f_* F, K).$$

Since $f_*(F)$ is constructible (SGAA IX 2.14), it follows that, if $\dim.q.inj(K) \leqslant N$, then $f_* \underline{\mathrm{Ext}}^i(F, \mathrm{R}^! f(K)) = 0$ for $i > N$, hence $\underline{\mathrm{Ext}}^i(F, \mathrm{R}^! f(K)) = 0$ for $i > N$ (SGAA VIII 5.5), as required.

Let now X be a prescheme, and let $K \in \mathrm{ob}\,\mathrm{D}^+(X)$. Denote by $\underline{D}_K$ the functor $\mathrm{R}\,\underline{\mathrm{Hom}}(\quad, K)$. We shall define, for $F \in \mathrm{ob}\,\mathrm{D}(X)$, a functorial homomorphism $F \to \underline{D}_K \underline{D}_K F$ (cf. [H] V.1.2). The construction which follows would be valid more generally on a ringed topos. We may first suppose that K is formed of injectives, which allows us to "remove the R". The identity morphism $\underline{\mathrm{Hom}}(F, K) \to \underline{\mathrm{Hom}}(F, K)$ defines, by the formula dear to Cartan, a homomorphism $F \otimes \underline{\mathrm{Hom}}(F, K) \to K$, whence, by a second application of the said formula, a homomorphism $F \to \underline{\mathrm{Hom}}(\underline{\mathrm{Hom}}(F, K), K)$, which is the announced homomorphism, and which we baptize the canonical homomorphism. We say that the pair (F, K) is *bidualizing*, or *satisfies biduality*, or again that F is *reflexive* relative to K, if the canonical homomorphism $F \to \underline{D}_K \underline{D}_K F$ is an isomorphism.

From now on, unless expressly mentioned otherwise, all preschemes considered will be assumed lo

Definition 1.7

Let X be a prescheme. A complex $K \in \operatorname{ob} \mathrm{D}^+(X)$ is said to be *dualizing* if the following conditions are satisfied:

(i) K is of finite quasi-injective dimension;

(ii) for every $F \in \operatorname{ob} \mathrm{D}_c^-(X)$, one has $\underline{D}_K(F) \in \operatorname{ob} \mathrm{D}_c^+(X)$;

(iii) every $F \in \operatorname{ob} \mathrm{D}_c^b(X)$ is reflexive relative to K.

Remark 1.8

a) By a standard dévissage (truncating F and using the fact that $\mathrm{D}_c(X)$ is triangulated), one sees that condition (ii) is equivalent to

(ii bis): for every constructible A_X-module F and every $i \in \mathbb{Z}$, $\underline{\operatorname{Ext}}^i(F, K)$ is constructible.

b) Condition (ii bis) implies $K \in \operatorname{ob} \mathrm{D}_c^+(X)$. Contrary to the example of coherent sheaves (EGA 0_{III} 12.3.3), the converse is perhaps not automatically verified. It is however verified (see no. 3) in the "good" cases, for example when one has on X resolution of singularities and purity (in particular if X is an excellent prescheme of characteristic zero).

c) By dévissage on F, one sees that condition (iii) is equivalent to (iii bis): every constructible A_X-module F is reflexive relative to K.

d) If $K \in \operatorname{ob} \mathrm{D}^+(X)$ satisfies (i) and (ii) (resp. is dualizing), the functor $\underline{D}_K$ induces a functor (resp. an equivalence of categories) $(\mathrm{D}_c^b(X))^\circ \to \mathrm{D}_c^b(X)$.

e) If $K \in \operatorname{ob} \mathrm{D}^+(X)$ is of finite *injective* dimension (cf. 1.5.2) and satisfies (ii), then, for every $F \in \operatorname{ob} \mathrm{D}_c(X)$, one has $\underline{D}_K(F) \in \mathrm{D}_c(X)$, and $\underline{D}_K$ induces a functor $(\mathrm{D}_c^+(X))^\circ \to \mathrm{D}_c^-(X)$.

If moreover K is dualizing, every $F \in \operatorname{ob} \mathrm{D}_c(X)$ is reflexive relative to K, and $\underline{D}_K$ induces equivalences of categories

$$(\mathrm{D}_c(X))^\circ \xrightarrow{\approx} \mathrm{D}_c(X), \quad (\mathrm{D}_c^-(X))^\circ \xrightarrow{\approx} \mathrm{D}_c^+(X), \quad (\mathrm{D}_c^+(X))^\circ \xrightarrow{\approx} \mathrm{D}_c^-(X).$$

It is not known whether this conclusion remains valid without the restriction that K be of finite injective dimension.

We now turn to the "exchange formulas." We shall not return to the notion of the tor-dimension of a complex, see SGA A, Exposé XVII, 4.1.9, and also SGA 6, I, 5. Retain only the following point: if X is a prescheme and $G \in \operatorname{ob} \mathrm{D}^-(X)$, the condition $\underline{\operatorname{Tor}}_i(F, G) = 0$ for $i > N$ and all A_X-modules F is equivalent to the same condition with F constructible, by SGA A, Exposé IX, 2.9 and the commutation of tensor product with filtered inductive limits. Thus there is no need to introduce a notion of "quasi-tor-dimension."

Lemma 1.9, cf. [H] II.4.3.

a) Let X be a prescheme and let $F, G \in \operatorname{ob} \mathrm{D}_c^-(X)$. One has $F \otimes^{\mathrm{L}} G \in \operatorname{ob} \mathrm{D}_c(X)$ if $F \in \operatorname{ob} \mathrm{D}_c^-(X)$, or if G has finite tor-dimension.

b) Let $f : X \to Y$ be a morphism of preschemes and let $G \in \operatorname{ob} \mathrm{D}_c^-(Y)$. Then $f^*(G) \in \operatorname{ob} \mathrm{D}_c(X)$.

Proof. Part (b) follows trivially from SGA A, Exposé IX, 2.4. We prove (a). Either hypothesis on (F, G) implies that the spectral sequence

$$E_2^{pq} = \sum_{q_1+q_2=q} \underline{\mathrm{Tor}}_{-p}(\mathrm{H}^{q_1}(F), \mathrm{H}^{q_2}(G)) \Rightarrow \mathrm{H}^*(F \otimes^{\mathrm{L}} G)$$

is biregular. Hence one is reduced to the case where F and G are concentrated in degree 0, that is, are constructible A_X-modules. The question is local, so one may suppose X noetherian. Then F admits, by SGA A, Exposé IX, 2.7, a left resolution F' in which the F'^i are of the form $A_{U,X}$ with U étale and of finite type over X. We have

$$F \otimes^{\mathrm{L}} G \xleftarrow{\sim} F' \otimes G,$$

and the result follows since the $A_{U,X} \otimes G$ are constructible. □

Notation 1.10. If X is a prescheme, we denote by $\mathrm{D}^b(X)_{\mathrm{torf}}$, respectively $\mathrm{D}^b(X)_{\mathrm{qinjf}}$, the full subcategory of $\mathrm{D}(X)$ consisting of objects of finite tor-dimension, respectively of finite quasi-injective dimension.

Proposition 1.11, cf. [H] V.2.6. Let X be a prescheme and $K \in \operatorname{ob} \mathrm{D}^+(X)$.

a) For $F, G \in \operatorname{ob} \mathrm{D}^-(X)$, respectively for $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}^b(X)_{\mathrm{torf}}$, there is a functorial isomorphism
$$\underline{D}_K(F \otimes^{\mathrm{L}} G) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, \underline{D}_K(G)).$$

b) If K satisfies conditions (i) and (ii) of Definition 1.7, then
$$G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}} \Longrightarrow \underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}}$$
and $\underline{D}_K(G)$ satisfies condition 1.7(ii).

c) If K is dualizing, then for $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^b(X)$, respectively for $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^b(X)$ with $\underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}}$, there is a functorial isomorphism
$$\underline{D}_K(F \otimes^{\mathrm{L}} \underline{D}_K(G)) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G).$$
Moreover,
$$\underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}} \Longrightarrow G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}},$$
and G satisfies condition 1.7(ii).

d) If K is dualizing and has finite injective dimension, then the implications in (b) and (c) are equivalences.

Proof. Part (a) is precisely Cartan's adjunction formula, SGA 6, I, 7.4.

For (b), let $G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}}$. If F is a variable constructible A_X-module, then the cohomology objects of $F \otimes^{\mathrm{L}} G$ are constructible, by Lemma 1.9, and vanish outside two fixed bounds independent of F. The same therefore holds for the cohomology objects of $\underline{D}_K(F \otimes^{\mathrm{L}} G)$, and (a) gives the assertion.

For (c), the first assertion follows from (a) applied to $\underline{D}_K(G)$ in place of G, together with the reflexivity of G. The second follows by a similar argument to (b).

For (d), prove for instance the converse implication of (b). Suppose that $\underline{D}_K(G) \in \operatorname{ob} \mathrm{D}^b_c(X)_{\mathrm{qinjf}}$ and satisfies condition 1.7(ii). Then, as F varies through constructible modules, the cohomology objects of $\underline{D}_K(F \otimes^{\mathrm{L}} G)$ are constructible and vanish outside two fixed bounds. The same is consequently true for the cohomology objects of $\underline{D}_K \underline{D}_K(F \otimes^{\mathrm{L}} G)$; but

$$F \otimes^{\mathrm{L}} G \xrightarrow{\sim} \underline{D}_K \underline{D}_K(F \otimes^{\mathrm{L}} G)$$

because K is dualizing and of finite injective dimension, cf. 1.8(d). Hence G has finite tor-dimension. The converse implication in (c) is proved analogously. □

Remarks 1.11.1.

a) The editor does not know whether assertion (d) of Proposition 1.11 is valid under the sole hypothesis that K be dualizing.

b) One expresses the preceding proposition pictorially by saying that the dualizing functor $\underline{D}_K$ exchanges the functors $\otimes^{\mathrm{L}}$ and $\mathrm{R}\,\underline{\mathrm{Hom}}$, as well as the notions of finite tor-dimension and finite quasi-injective dimension. The exchange is, however, fully satisfactory only when K has finite injective dimension.

Proposition 1.12. Let Y be a noetherian prescheme, let $f : X \to Y$ be a morphism of finite type, and let $K \in \operatorname{ob} \mathrm{D}^+(Y)$. Suppose that n is prime to the residual characteristics of Y, and put

$$K_X = \mathrm{R}^! f(K_Y), \qquad \underline{D}_X = \underline{D}_{K_X}, \qquad \underline{D}_Y = \underline{D}_{K_Y},$$

in the notation of SGA A, Exposé XVIII, 3.1.

a) For $F \in \operatorname{ob} \mathrm{D}^-(X)$ there is a functorial isomorphism

$$\mathrm{R} f_* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathrm{R} f_*(F). \tag{i}$$

If, moreover, K_X and K_Y are dualizing,[2] then for $F \in \operatorname{ob} \mathrm{D}^b_c(X)$ there is a functorial isomorphism

$$\mathrm{R} f_! \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathrm{R} f_*(F). \tag{ii}$$

b) For $F \in \mathrm{D}^-_c(Y)$ there is a functorial isomorphism[3]

$$\mathrm{R}^! f\, \underline{D}_Y(F) \xrightarrow{\sim} \underline{D}_X f^*(F). \tag{i}$$

If, moreover, K_X and K_Y are dualizing, then for $F \in \mathrm{D}^b_c(Y)$ there is a functorial isomorphism

$$f^* \underline{D}_Y F \xrightarrow{\sim} \underline{D}_X \mathrm{R}^! f\, F. \tag{ii}$$

[2] We shall see below that under fairly general conditions K_Y dualizing implies K_X dualizing.

[3] If the functor $\mathrm{R}^! f$ has finite cohomological dimension, as for example when Y is of finite type over a field (SGA A, Exposé X, 4.3 and Exposé XVIII, 3.1.7), the isomorphism (b)(i) is valid for $F \in \operatorname{ob} \mathrm{D}_c(Y)$.

Proof. The isomorphism (a)(i) is just the duality isomorphism of SGA A, Exposé XVIII, 3.1.9.6.

The isomorphism (a)(ii) is obtained by applying $\underline{D}_Y$ to both sides of (a)(i), written with argument $\underline{D}_X F$, and using the finiteness theorem SGA A, Exposé XVII, 5.3.6, which asserts that $\mathrm{R}f_*(\underline{D}_X(F)) \in \operatorname{ob} \mathrm{D}_c(Y)$.

The isomorphism (b)(i) is the induction formula of SGA A, Exposé XVIII, 3.1.12.2.

The isomorphism (b)(ii) is obtained by applying $\underline{D}_X$ to both sides of (b)(i), written with argument $\underline{D}_Y F$, and using 1.9(b). □

Scholium 1.12.1. In figurative language, the dualizing functors $\underline{D}_X$ and $\underline{D}_Y$ exchange the usual and unusual direct images, $\mathrm{R}f_*$ and $\mathrm{R}f_!$, and also the usual and unusual inverse images, f^* and $\mathrm{R}^!f$.

Corollary 1.13. Under the hypotheses of Proposition 1.12, suppose in addition that f is proper, and let $F \in \operatorname{ob} \mathrm{D}^-(X)$. If (F, K_X) is bidualizing, then $(\mathrm{R}f_*F, K_Y)$ is bidualizing. Conversely, if f is finite and $(\mathrm{R}f_*F, K_Y)$ is bidualizing, then (F, K_X) is bidualizing.

Proof. Suppose (F, K_X) is bidualizing. Applying $\mathrm{R}f_*$ to the canonical isomorphism $F \xrightarrow{\sim} \underline{D}_X^2 F$, one obtains

$$\begin{aligned} \mathrm{R}f_*F &\xrightarrow{\sim} \mathrm{R}f_*\underline{D}_X\underline{D}_X F \\ &\xrightarrow{\sim} \underline{D}_Y\mathrm{R}f_*\underline{D}_X F \qquad \text{by (a)(i)} \\ &\xrightarrow{\sim} \underline{D}_Y\underline{D}_Y\mathrm{R}f_*F \qquad \text{by (a)(i).} \end{aligned}$$

The courageous reader will convince himself that the isomorphism obtained is the canonical one; this proves the first assertion.

Now suppose f finite and (f_*F, K_Y) bidualizing, since $\mathrm{R}f_* = f_*$ for f finite. We prove that the canonical morphism $F \to \underline{D}_X^2 F$ is an isomorphism. By Lemma 1.14(a) below, it suffices to show that the induced morphism

$$f_*F \longrightarrow f_*\underline{D}_X^2 F$$

is an isomorphism. But by applying (a)(i) twice, one convinces oneself that this morphism is precisely the canonical isomorphism

$$f_*F \longrightarrow \underline{D}_Y^2 f_*F,$$

which completes the proof. □

Lemma 1.14. Let $f : X \to Y$ be a finite morphism of preschemes.

a) For a morphism $u : F \to G$ in $\mathrm{D}(X)$, u is an isomorphism if and only if f_*u is an isomorphism.

b) For $F \in \operatorname{ob} \mathrm{D}(X)$, one has $F \in \operatorname{ob} \mathrm{D}_c(X)$ if and only if $f_*F \in \operatorname{ob} \mathrm{D}_c(Y)$.

Proof. For (a), since f_* is exact, one may restrict to the case where F and G are concentrated in degree zero. The assertion then follows immediately from the computation of the fibres of f_*F and f_*G, SGA A, Exposé VIII, 5.5.

For (b), again one may restrict to the case where F is concentrated in degree zero. Necessity is known, SGA A, Exposé IX, 2.14. We prove sufficiency: f_*F constructible implies F constructible.

The question is local upstairs, hence a fortiori downstairs, so we may suppose Y and hence X affine. By EGA IV, 17.16.6, there exists a finite family (Y_i) of affine subpreschemes of Y, pairwise disjoint and with union Y, such that, if $X_i = X \times_Y Y_i$, the induced morphism $f_i : X_i \to Y_i$ factors as

$$X_i \xrightarrow{u_i} X_i' \xrightarrow{v_i} Y_i,$$

where u_i is finite radicial and surjective and v_i is finite étale. Put $F|_{X_i} = F_i$. By the base-change theorem for finite morphisms, SGA A, Exposé VIII, 5.5, one has

$$f_{i*}(F_i) \xrightarrow{\sim} (f_*F)|_{Y_i},$$

and by SGA A, Exposé IX, 2.8, one is reduced to proving the assertion for (f_i, F_i). Put $F_i' = u_{i*}(F_i)$, so that $f_{i*}(F_i) = v_{i*}(F_i')$. It is clear from the definition of constructibility, respectively follows from SGA A, Exposé VIII, 1.1, that F_i, respectively F_i', is constructible if and only if F_i', respectively $v_{i*}(F_i')$, is constructible. This proves the claim. □

Corollary 1.15. Let $f : X \to Y$ be a separated quasi-finite morphism of noetherian preschemes. Let $K_Y \in \operatorname{ob} \mathrm{D}^+(Y)$, and put, as in Proposition 1.12,

$$K_X = \mathrm{R}^! f(K_Y), \qquad \underline{D}_X = \underline{D}_{K_X}, \qquad \underline{D}_Y = \underline{D}_{K_Y}.$$

If K_Y is dualizing, then K_X is dualizing.

Proof. We already know, by Proposition 1.6, that if K_Y has finite quasi-injective dimension, then the same is true of K_X. We show that if K_Y satisfies condition (ii), respectively condition (iii), of Definition 1.7, then K_X satisfies the same condition.

By the Main Theorem, EGA IV, 8.12.6, f factors as $f = f'i$, where i is an open immersion and f' is finite. Since

$$\mathrm{R}^! f \simeq \mathrm{R}^! i \, \mathrm{R}^! f',$$

one must treat separately the case of an open immersion and the case of a finite morphism. If f is an open immersion, the assertion is trivial, since a constructible sheaf on X is induced by a constructible sheaf on Y, and formation of $\mathrm{R}\,\underline{\mathrm{Hom}}$ commutes with restriction to an open.

Suppose then that f is finite. It follows from Corollary 1.13 that if K_Y satisfies (iii), then K_X satisfies (iii), taking account of the finiteness theorem SGA A, Exposé IX, 2.14. It remains to prove that if K_Y satisfies (ii), then K_X satisfies (ii). Let $F \in \operatorname{ob} \mathrm{D}_c^-(X)$. We must show that $\underline{D}_X(F) \in \operatorname{ob} \mathrm{D}_c^+(X)$. By the finiteness theorem, $f_*F \in \operatorname{ob} \mathrm{D}_c^-(Y)$. Hence, by the hypothesis on K_Y,

$$\underline{D}_Y f_* F \in \operatorname{ob} \mathrm{D}_c^+(Y).$$

But

$$\underline{D}_Y f_* F \xrightarrow{\sim} f_* \underline{D}_X(F)$$

by Proposition 1.12(a)(i), and Lemma 1.14(b) completes the proof. □

2. Uniqueness of the dualizing complex

In this section we suppose that A is local, that is,

$$A = \mathbb{Z}/\ell^\nu \mathbb{Z},$$

where ℓ is prime.

Theorem 2.1, cf. [H] V.3.1. Let X be a connected locally noetherian prescheme, and let K be a dualizing complex on X in the sense of Definition 1.7. Let also $K' \in \operatorname{ob} \mathrm{D}_c^+(X)$. Then K' is dualizing if and only if there exist an invertible A_X-module L and an integer r, necessarily determined up to unique isomorphism, such that

$$K' \xrightarrow{\sim} K \otimes^{\mathrm{L}} L[r].$$

Proof. Let L be an invertible A_X-module and r an integer, and put $L^\vee = \underline{\mathrm{Hom}}(L, A_X)$. For $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}^+(X)$ there are canonical isomorphisms

$$\mathrm{R}\,\underline{\mathrm{Hom}}(F \otimes^{\mathrm{L}} L^\vee[-r], G) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G \otimes^{\mathrm{L}} L[r]) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \otimes^{\mathrm{L}} L[r]. \tag{*}$$

It follows immediately that, if $K' \simeq K \otimes^{\mathrm{L}} L[r]$, then K' is dualizing.

Moreover, if we write $\underline{D}_X = \underline{D}$ and $\underline{D}_{K'} = \underline{D}'$, then for $F \in \operatorname{ob} \mathrm{D}(X)$ the isomorphism $(*)$ gives

$$\underline{D}'F \xrightarrow{\sim} \underline{D}F \otimes^{\mathrm{L}} L^\vee[-r].$$

Since $\underline{D}K \simeq A_X$, one obtains

$$\underline{D}'K \xrightarrow{\sim} L^\vee[-r],$$

which shows that $L^\vee[r]$, hence L and r, are determined essentially uniquely by K'.

Conversely, suppose that K' is dualizing, and put

$$P = \underline{D}'K = \underline{D}'\underline{D}A_X = \mathrm{R}\,\underline{\mathrm{Hom}}(K, K').$$

We shall prove:

$$\text{(a)} \qquad K' \xrightarrow{\sim} K \otimes^{\mathrm{L}} P,$$

$$\text{(b)} \qquad P \xrightarrow{\sim} L[r] \quad \text{for an invertible } A_X\text{-module } L.$$

This will complete the proof of the theorem.

First observe that, by Proposition 1.9(b)(ii), for $F \in \operatorname{ob} \mathrm{D}_c(X)$ there is an isomorphism

$$F \otimes^{\mathrm{L}} P \xrightarrow{\sim} \underline{D}'\underline{D}F. \tag{**}$$

Applying $(**)$ to $F = K$ gives

$$K \otimes^{\mathrm{L}} P \xrightarrow{\sim} \underline{D}'\underline{D}K \xrightarrow{\sim} \underline{D}'A_X = K',$$

which proves (a).

Applying $(**)$ to

$$F = P' = \underline{D}\underline{D}'A_X$$

gives

$$P' \otimes^{\mathrm{L}} P = \underline{D}'\underline{D}\underline{D}\underline{D}'A_X \xrightarrow{\sim} \underline{D}'\underline{D}'A_X \xrightarrow{\sim} A_X.$$

Thus (b) follows from the following lemma. □

Lemma 2.2, cf. [H] V.3.3. Let X be a connected locally noetherian prescheme, and let $P, P' \in \operatorname{ob} \mathrm{D}_c^-(X)$ be such that

$$P \otimes^{\mathrm{L}} P' \xrightarrow{\sim} A_X.$$

Then there exist an invertible A_X-module L and an integer r such that

$$P \xrightarrow{\sim} L[r], \qquad P' \xrightarrow{\sim} L^\vee[-r].$$

Proof. The proof has two steps.

First suppose that $X = \operatorname{Spec}(k)$, with k separably closed. One merely repeats the easy proof given in [H]. The essential point is this: by EGA 0_{I}, 5.4.3, if M and M' are finite-type A-modules such that

$$M \otimes M' \xrightarrow{\sim} A,$$

then M and M' are invertible and $M' \simeq M^\vee$. This is where the hypothesis that A is local is used.

We now pass to the general case. By the special case just treated, for every $x \in X$ there exists an integer $r(x)$ such that

$$P_{\overline{x}} \simeq A_{\overline{x}}[r(x)], \qquad P'_{\overline{x}} \simeq A_{\overline{x}}[-r(x)].$$

Equivalently, $\mathrm{H}^i(P_{\overline{x}}) = 0$, respectively $\mathrm{H}^i(P'_{\overline{x}}) = 0$, except for $i = -r(x)$, respectively $i = r(x)$, and the corresponding nonzero cohomology modules are isomorphic to $A_{\overline{x}}$. It remains to show that the function $x \mapsto r(x)$ is locally constant.

Assume this for the moment. Since X is connected, r is constant, say equal to r_0. Then

$$\mathrm{H}^i(P) = 0 \ (i \neq -r_0), \qquad \mathrm{H}^i(P') = 0 \ (i \neq r_0),$$

and

$$\mathrm{H}^{-r_0}(P) \otimes \mathrm{H}^{r_0}(P') \xrightarrow{\sim} A.$$

Put $L = \mathrm{H}^{-r_0}(P)$ and $L' = \mathrm{H}^{r_0}(P')$. Let $u : \overline{x} \to \overline{y}$ be a specialization morphism, and let

$$u^* : L_{\overline{y}} \to L_{\overline{x}}, \qquad u'^* : L'_{\overline{y}} \to L'_{\overline{x}}$$

be the corresponding specialization morphisms. There is a commutative square

$$\begin{array}{ccc} L_{\overline{y}} \otimes L'_{\overline{y}} & \xrightarrow{\sim} & A_{\overline{y}} \\ \downarrow u^* \otimes u'^* & & \downarrow \\ L_{\overline{x}} \otimes L'_{\overline{x}} & \xrightarrow{\sim} & A_{\overline{x}}. \end{array} \qquad (*)$$

Choose a basis e, respectively f, of $L_{\overline{x}}$, respectively $L_{\overline{y}}$, and let e', respectively f', be the dual basis of $L'_{\overline{x}}$, respectively $L'_{\overline{y}}$, meaning that $e \otimes e'$ and $f \otimes f'$ map to 1. If $u^*(f) = \lambda e$ and $u'^*(f') = \lambda' e'$, the commutativity of $(*)$ gives $\lambda\lambda' = 1$. Hence u^* is an isomorphism, and u'^* is the contragredient isomorphism. By SGA A, Exposé IX, 2.13, it follows that L and L' are locally constant, hence of finite presentation as A_X-modules. Consequently, for every $x \in X$ the canonical map

$$\underline{\mathrm{Hom}}(L, F)_{\overline{x}} \longrightarrow \mathrm{Hom}(L_{\overline{x}}, F_{\overline{x}})$$

is an isomorphism for every A_X-module F. Since the geometric fibres of L are invertible, L itself is invertible. Similarly L' is invertible, and $L' \simeq L^\vee$. Hence

$$P \simeq L[-r_0], \qquad P' \simeq L^\vee[r_0].$$

It remains only to prove that r is locally constant. For $m \in \mathbb{Z}$, the condition $r(x) = m$ is equivalent to

$$\mathrm{H}^m(P')_{\overline{x}} \neq 0.$$

Thus, by SGA A, Exposé IX, 2.4, the function r is locally constructible. Therefore, to verify that it is locally constant, EGA IV, 1.10.1, allows one to suppose that X is local. Let x be its closed point and U the open complement.

We prove that r is constant. By induction on the dimension of X, we may already suppose that $r(y)$ is constant and equal to r_0 for $y \in U$; the point is to prove $r(x) = r_0$.

Assume the contrary, and derive a contradiction. By hypothesis

$$P|_U \xrightarrow{\sim} L[r_0]$$

with L an invertible A_U-module. After translating degrees we may suppose $r_0 = 0$. On the other hand $r(x) \neq 0$; write $r(x) = a > 0$. Let

$$i : U \to X, \qquad j : x \to X$$

be the canonical inclusions. Consider the exact sequence

$$0 \longrightarrow i_!(P|_U) \longrightarrow P \longrightarrow j_*j^*P \longrightarrow 0. \tag{**}$$

By hypothesis the only nonzero cohomology objects of P are $\mathrm{H}^0(P)$ and $\mathrm{H}^{-a}(P)$, and

$$\mathrm{H}^0(P) \xrightarrow{\sim} i_!L.$$

It follows that $(**)$ splits naturally, via the map $P \to \mathrm{H}^0(P)$, hence

$$P \xrightarrow{\sim} i_!(P|_U) \oplus j_*j^*P.$$

Therefore

$$\begin{aligned} P \otimes^{\mathrm{L}} P' &\xrightarrow{\sim} i_!(P|_U) \otimes^{\mathrm{L}} P' \oplus j_*j^*P \otimes^{\mathrm{L}} P' \\ &\xrightarrow{\sim} i_!\big(P|_U \otimes^{\mathrm{L}} P'|_U\big) \oplus j_*\big(j^*P \otimes^{\mathrm{L}} j^*P'\big) \\ &\xrightarrow{\sim} i_!A_U \oplus j_*A_x, \end{aligned}$$

which is impossible, since $i_!A_U \oplus j_*A_x$ is plainly not isomorphic to A_X; already the degree-zero cohomology sheaves are not isomorphic. This proves Lemma 2.2, and hence Theorem 2.1. ☐

3. Existence of dualizing complexes

3.1. Preliminaries

As was said in the introduction, the object of this section is to prove that, on a "good" regular prescheme X, the sheaf A_X is dualizing, where n is assumed prime to the residual characteristics.

In fact, one would like this to hold on excellent regular preschemes of equal characteristic; but there is no doubt that the proof given below will extend to them once resolution of singularities and the purity theorem are available in that case.

We first present the various ingredients which will be used.

3.1.1. Singular locus. Let X be a prescheme, locally noetherian. We shall say that X satisfies condition *(reg)* if the following equivalent properties hold (EGA IV, 6.12.3):

(i) for every sub-prescheme Y of X, the set $\mathrm{Reg}(Y)$ of regular points of Y is open in Y;

(ii) for every closed integral sub-prescheme Y of X, the set $\mathrm{Reg}(Y)$ contains a non-empty open subset of Y.

The condition (reg) is satisfied if X is locally of finite type over a complete noetherian local ring, for example over a field (EGA IV, 6.12.8), or over a Dedekind ring whose field of fractions has characteristic zero, for example over $\mathbb{Z}$ (EGA IV, 6.12.6); more generally, it is satisfied if X is excellent (EGA IV, 7.8.6).

3.1.2. Dévissage of constructible sheaves. Let X be a noetherian prescheme, and let C be a strictly full subcategory of the category $\mathrm{Cons}(X)$ of constructible A_X-modules. Suppose that C is stable under direct factors and extensions. The following conditions are equivalent:

(i) $C = \mathrm{Cons}(X)$;

(ii) C contains sheaves of the form $i_!(F)$, where $i : Y \to X$ is the immersion of an integral sub-prescheme and F is locally constant on Y;

(iii) C contains sheaves of the form $f_* i_!(A'_U)$, where $i : U \to Y$ is an open subset of an integral prescheme, $f : Y \to X$ is a finite morphism such that the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$, for ℓ a prime divisor of n.

If X satisfies (reg) (3.1.1), these conditions are again equivalent to the following:

(i bis) the analogue of (ii), with Y moreover regular;

(ii bis) the analogue of (iii), with U regular.

If, in addition, X is affine, then in (ii), respectively in (ii bis), one may suppose Y closed in an open subset of the form $D(f)$, with $f \in \Gamma(X, \mathcal{O}_X)$.

Proof. The equivalence of (i)–(iii) was proved in SGA A, Exposé IX, 2.5 and 5.8. The remaining assertions follow by noetherian induction. □

3.1.3. Cohomological dimension. Let X be a locally noetherian prescheme. We denote by $(\mathrm{cdloc})_n$ the following condition. For every sub-prescheme Y of X, every geometric point $\overline{y}$ of Y, and every

$$f \in \Gamma(\widetilde{Y}(\overline{y}), \mathcal{O}_{\widetilde{Y},\overline{y}}),$$

where $\widetilde{Y}(\overline{y})$ denotes the strict localization of Y at $\overline{y}$ (SGA A, Exposé VIII, 7.1), one has

$$\mathrm{cd}_{\mathbf{n}}(\widetilde{Y}(\overline{y}) - V(f)) \leq \dim \mathcal{O}_{Y,\overline{y}} \quad (= \dim \mathcal{O}_{Y,y} \text{ by EGA IV, 6.1.3}),$$

where $\mathbf{n}$ is the set of prime divisors of n.

The condition $(\mathrm{cdloc})_n$ is satisfied if X is locally of finite type over a field (SGA A, Exposé XIV, 3.5), or is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 6.2).[4]

3.1.4. Absolute cohomological purity. Let X be a regular prescheme, and let Y be a regular closed sub-prescheme of X, of codimension d at every point. One says that the pair (X, Y) satisfies the absolute cohomological purity theorem, relative to n, if

$$\underline{\mathrm{H}}^i_Y(A_X) = 0 \qquad (i \neq 2d),$$

and if the "local fundamental class" homomorphism (SGA $4^{1/2}$, Cycle 2.2)

$$A_Y \longrightarrow \underline{\mathrm{H}}^{2d}_Y((\mu_n)^{\otimes d}_X)$$

is an isomorphism.

One says that the absolute cohomological purity theorem is true on X if every pair (U, Y) as above satisfies it, where U is an open subset of X.

Absolute purity is true on X if X is smooth over a perfect field of characteristic prime to n (SGA A, Exposé XVI, 3.9), or if X is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 3.2), or if X is regular of dimension 1 (cf. 5.1).[5]

3.1.5. Hironaka-style resolution of singularities.

a) Let X be a regular prescheme, and let $D \geq 0$ be a divisor on X. One says that D has *strict normal crossings* if there exists a finite family $(f_i)_{i \in I}$ of elements of $\Gamma^*(X, \mathcal{O}_X)$ such that

$$D = \sum_{i \in I} \mathrm{Div}(f_i)$$

(EGA IV, 21), and such that, for every $x \in \mathrm{Supp}(D)$, the germs $(f_i)_x$ which lie in $\mathfrak{m}_x$ form part of a regular system of parameters. One says that D has *normal crossings* if, locally for the étale topology, D has strict normal crossings.

The standard example of a normal crossings divisor is furnished by a locally finite union of coordinate hyperplanes in affine space.

b) Let X be a locally noetherian prescheme. We shall say that X is *strongly desingularizable* if the following condition is satisfied: for every finite morphism $f : Y \to X$, where Y is integral and its generic point is separable over its image, and every regular open subset U of Y, there exists a proper birational morphism $h : Y' \to Y$, with Y' regular, such that h induces an isomorphism from $U' = h^{-1}(U)$ onto U, and $Y' - U'$ is a normal crossings divisor.

Here are cases where X is strongly desingularizable:

a) X is excellent of characteristic zero;

b) X is locally noetherian, regular, and of dimension ≤ 1;

[4]Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.

[5]Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.

c) X is of finite type over $\mathbb{Z}$ or over a perfect field, and of dimension ≤ 2.[6]

3.1.6. Cohomological finiteness. Let $f : X \to Y$ be a morphism of preschemes, and let $F \in \operatorname{ob} \mathrm{D}_c^+(X)$. One has $\mathrm{R}f_*(F) \in \operatorname{ob} \mathrm{D}_c^+(Y)$ in the following cases:

a) f is proper and of finite presentation (SGA A, Exposé XIV, 1.1);

b) "when resolution of singularities and the purity theorem are available", in particular:

  (i) if X and Y are locally of finite type over a field k of characteristic zero and f is a k-morphism of finite type (SGA A, Exposé XVI, 5.1);

  (ii) if Y is excellent of characteristic zero and f is of finite type (SGA A, Exposé XIX, 5.1);

  (iii) if X is locally of finite type over a field and $\dim(X) \leq 2$ (SGA A, Exposé XIX, 5.1).

3.2. The quasi-injective dimension of A_X

Proposition 3.2.1. Let X be a regular prescheme whose residual characteristics are prime to n. Suppose that X satisfies (reg) (3.1.1) and $(\mathrm{cdloc})_n$ (3.1.3), and that absolute purity is true on X (3.1.4). Then, for every constructible A_X-module F and every $x \in X$, one has

$$\underline{\mathrm{Ext}}^i(F, A_X)_x = 0 \qquad \text{for } i > 2\dim \mathcal{O}_{X,x}. \tag{*}$$

Proof. The question being local, we may suppose X affine. By dévissage (3.1.2), it is enough to verify $(*)$ for $F = i_!(G)$, where $i : Y \to X$ is the immersion of a regular integral sub-prescheme, closed in an open subset $D(f)$, $f \in A(X)$, and G is a locally constant constructible sheaf on Y. We must compute

$$\underline{\mathrm{Ext}}^q(F, A_X) = \mathrm{H}^q(\mathrm{R}\,\underline{\mathrm{Hom}}(F, A_X)).$$

The projection formula for i (SGA A, Exposé XVIII) gives

$$\mathrm{R}\,\underline{\mathrm{Hom}}(i_!G, A_X) \xrightarrow{\sim} \mathrm{R}i_*\mathrm{R}\,\underline{\mathrm{Hom}}(G, \mathrm{R}^!i(A_X)).$$

By purity,

$$\mathrm{R}^!i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes d}[-2d],$$

where $d = \operatorname{codim}(Y, X)$. Put $T = (\mu_n)_Y^{\otimes d}$. Since G is locally constant constructible, for every geometric point $\overline{y}$ of Y and every p one has

$$\underline{\mathrm{Ext}}^p(G, T)_{\overline{y}} \xrightarrow{\sim} \mathrm{Ext}^p(G_{\overline{y}}, T_{\overline{y}}).$$

But $T_{\overline{y}}$ is injective, being isomorphic to $A = \mathbb{Z}/n\mathbb{Z}$. Hence $\underline{\mathrm{Ext}}^p(G, T) = 0$ for $p \neq 0$, and

$$\mathrm{R}\,\underline{\mathrm{Hom}}(G, T) \xrightarrow{\sim} \underline{\mathrm{Hom}}(G, T).$$

[6]Added in 1977: more generally, if X is excellent noetherian of dimension ≤ 2, by H. Hironaka, *Desingularization of excellent surfaces*, notes by B. Bennett, Bowdoin College, 1967, to be supplemented if necessary by H. Hironaka, *Certain numerical characters of singularities*, J. Math. Kyoto Univ. 10 (1970), 151–187.

Thus

$$\mathrm{R}\,\underline{\mathrm{Hom}}(F, A_X) \xrightarrow{\sim} \mathrm{R}i_*(\mathcal{H})[-2d], \qquad \mathcal{H} = \underline{\mathrm{Hom}}(G, T).$$

Let Z be the schematic closure of Y in X (EGA I, 9.5.10). Then $Y = Z \cap D(f)$ (EGA 0_{I}, 2.1.6). Denote by $j : Y \to Z$ and $k : Z \to X$ the canonical immersions. We have $\mathrm{R}i_* = k_*\mathrm{R}j_*$, so we are reduced to computing $\mathrm{R}j_*(\mathcal{H})$. If $\overline{x}$ is a geometric point of Z, let $\widetilde{Z} = \mathrm{Spec}(\mathcal{O}_{Z,\overline{x}})$ be the strict localization, let $\widetilde{f}$ be the image of f in $\mathcal{O}_{Z,\overline{x}}$, let $\widetilde{Y} = \widetilde{Z} \times_Z Y = \widetilde{Z} - V(\widetilde{f})$, and let $\widetilde{\mathcal{H}}$ be the inverse image of $\mathcal{H}$ on $\widetilde{Y}$. By SGA A, Exposé VIII, 5.2,

$$\mathrm{R}^q j_*(\mathcal{H})_{\overline{x}} \xrightarrow{\sim} \mathrm{H}^q(\widetilde{Y}, \widetilde{\mathcal{H}}),$$

hence

$$\underline{\mathrm{Ext}}^q(F, A_X)_{\overline{x}} \xrightarrow{\sim} \mathrm{H}^{q-2d}(\widetilde{Y}, \widetilde{\mathcal{H}}).$$

The condition $(\mathrm{cdloc})_n$ implies that this group is zero for

$$q - 2d > \dim \mathcal{O}_{Z,\overline{x}}.$$

Using

$$d = \mathrm{codim}_{\overline{x}}(Z, X) \leq \dim \mathcal{O}_{X,\overline{x}}$$

(EGA IV, 5.1.3) and

$$\dim \mathcal{O}_{Z,\overline{x}} = \dim \mathcal{O}_{X,\overline{x}} - \mathrm{codim}_{\overline{x}}(Z, X)$$

(EGA 0_{IV}, 16.5.12 and EGA IV, 5.1.9), one obtains

$$\underline{\mathrm{Ext}}^q(F, A_X)_x = 0 \qquad (q > 2\dim \mathcal{O}_{X,x}),$$

as required. ☐

Remark 3.2.2. The bound obtained in 3.2.1 is the best possible. Indeed, let x be a point of X, and let Y be the closed integral sub-prescheme of X whose generic point is x. Since X satisfies (reg), there exists a regular open subset U of Y containing x. By purity,

$$\underline{\mathrm{Ext}}^{2d}(A_U, A_X)|U = (\mu_n)_U^{\otimes d},$$

where $d = \dim \mathcal{O}_{X,x} = \mathrm{codim}(Y, X)$.

Corollary 3.2.3. Let X be a regular prescheme of dimension d. If X is smooth over a field k of characteristic prime to n, or if X is excellent of characteristic zero, then X satisfies the conclusion of 3.2.1; consequently

$$\dim . q. inj(A_X) \leq 2d.$$

Proof. If X is excellent of characteristic zero, or if k is perfect of characteristic prime to n and X is smooth over k, then the hypotheses of 3.2.1 are satisfied by 3.1.1, 3.1.3, and 3.1.4. The conclusion is still valid when X is smooth over an arbitrary field k, as follows immediately after the base change $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$, where k' is a perfect closure of k (SGA A, Exposé VIII, 1.1). ☐

Lemma 3.2.4. Let X be a prescheme satisfying the conclusion of 3.2.1 and of dimension $\leq d$. If there exists an integer N such that X has cohomological dimension $\leq N$, then

$$\dim . inj(A_X) \leq 2d + N.$$

If, moreover, for every closed subset Y of X, the implication

$$\operatorname{codim}(Y, X) \geq p \quad \Rightarrow \quad \operatorname{cd}_{\mathbf{n}}(Y) \leq N - 2p$$

holds, then

$$\dim . \, inj(A_X) \leq N.$$

Proof. The first assertion follows immediately from 1.5. For the second, one argues as in 1.5. We are reduced to proving, for F constructible on X, the relation

$$\operatorname{Ext}^i(X; F, A_X) = 0 \qquad (i > N).$$

For this, inspect the spectral sequence

$$E_2^{pq} = \mathrm{H}^p(X; \underline{\operatorname{Ext}}^q(F, A_X)) \Rightarrow \operatorname{Ext}^*(X; F, A_X).$$

By 3.2.1, the sheaves $\underline{\operatorname{Ext}}^{2i}(F, A_X)$ and $\underline{\operatorname{Ext}}^{2i-1}(F, A_X)$ are concentrated in codimension $\geq i$. Therefore $E_2^{p,2i} = E_2^{p,2i-1} = 0$ for $p > N - 2i$, by the hypothesis. Hence $E_2^{pq} = 0$ for $p + q > N$, and the assertion follows. □

Corollary 3.2.5. If X is a smooth prescheme of dimension d over a separably closed field k of characteristic prime to n, then

$$\dim . \, inj(A_X) = 2d.$$

Proof. By 3.2.3 and SGA A, Exposé X, 4.2, we are in the situation of 3.2.4. The hypotheses of SGA A, Exposé X, 4.2 are satisfied by SGA A, Exposé X, 2.1 and 3.2; hence $\dim . \, inj(A_X) \leq 2d$. The equality follows from purity: if x is a closed point of codimension d in X, then

$$\operatorname{Ext}^{2d}(X, A_x, A_X) = \mathrm{H}_x^{2d}(A_X) \xrightarrow{\sim} (\mu_n)^{\otimes d}(k) \neq 0.$$

□

3.3. Constructibility of the sheaves $\underline{\operatorname{Ext}}^i(F, A_X)$

Proposition 3.3.1. Let X be a prescheme.

a) The following conditions are equivalent:

(i) for $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^+(X)$, one has

$$\mathrm{R}\, \underline{\operatorname{Hom}}(F, G) \in \operatorname{ob} \mathrm{D}_c^+(X);$$

(i bis) for every pair (F, G) of constructible A_X-modules, one has $\mathrm{R}\, \underline{\operatorname{Hom}}(F, G) \in \operatorname{ob} \mathrm{D}_c^+(X)$, i.e. the sheaves $\underline{\operatorname{Ext}}^i(F, G)$ are constructible for all i;

(ii) for every open immersion $i : U \to X$ and every constructible A_U-module F, the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p;

(ii bis) for every immersion $i : Y \to X$ and every constructible A_Y-module F, the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p;

(ii ter) for every immersion $i : Y \to X$ and every locally constant constructible A_Y-module F, the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p;

(iii) for every finite morphism, respectively every closed immersion, $f : Y \to X$, and every $F \in \mathrm{ob}\,\mathrm{D}_c^+(X)$, one has $\mathrm{R}^! f(F) \in \mathrm{ob}\,\mathrm{D}_c^+(Y)$.

b) These conditions are satisfied in each of the following two cases:

(i) $\dim(X) \leq 1$;

(ii) resolution of singularities and purity are available on X in the sense of 3.1.6(b).

Proof. The equivalence of (i) and (i bis) follows from the standard spectral sequence passing from sheaf-Ext to hyper-Ext.

(i bis) $\Rightarrow$ (ii). Let $\overline{F}$ be a constructible A_X-module extending F, for example $\overline{F} = i_! F$. By duality (SGA A, Exposé XVIII, 3.1.10),

$$\mathrm{R}i_* F = \mathrm{R}i_* \mathrm{R}\,\underline{\mathrm{Hom}}(A_U, i^! \overline{F}) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(i_! A_U, \overline{F}),$$

and this proves the claim.

(ii) $\Rightarrow$ (ii bis). Factor i as a closed immersion followed by an open immersion.

(ii bis) $\Rightarrow$ (i bis). The question is local, so we may suppose X affine. By the usual dévissage (SGA A, Exposé IX, 2.5), it is enough to verify (ii) for $F = i_! M$, where $i : Y \to X$ is the immersion of a sub-prescheme and M is a locally constant constructible A_Y-module. By the projection formula (SGA A, Exposé XVIII),

$$\mathrm{R}\,\underline{\mathrm{Hom}}(i_! M, G) \xrightarrow{\sim} \mathrm{R}i_* \mathrm{R}\,\underline{\mathrm{Hom}}(M, \mathrm{R}i^! G),$$

and by a spectral sequence we are reduced to proving that

$$\mathrm{R}\,\underline{\mathrm{Hom}}(M, \mathrm{R}i^! G) \in \mathrm{ob}\,\mathrm{D}_c^+(Y).$$

First prove $\mathrm{R}i^! G \in \mathrm{ob}\,\mathrm{D}_c^+(Y)$. After factoring i, we may suppose that Y is closed. Let $j : U = X - Y$ be the complementary open immersion. Applying $\mathrm{R}\,\underline{\mathrm{Hom}}(\ ,G)$ to

$$0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0$$

gives a distinguished triangle. Since $\mathrm{R}\,\underline{\mathrm{Hom}}(A_X, G) = G$, and

$$\mathrm{R}\,\underline{\mathrm{Hom}}(A_{U,X}, G) \xrightarrow{\sim} \mathrm{R}j_* j^! G$$

belongs to $\mathrm{D}_c^+(X)$ by hypothesis, it follows that

$$\mathrm{R}\,\underline{\mathrm{Hom}}(A_Y, G) \xrightarrow{\sim} i_* \mathrm{R}i^! G$$

belongs to $\mathrm{D}_c^+(X)$. Resolving M locally on Y, for the étale topology, by free A_Y-modules, another dévissage gives

$$\mathrm{R}\,\underline{\mathrm{Hom}}(M, \mathrm{R}i^! G) \in \mathrm{ob}\,\mathrm{D}_c^+(Y).$$

This proves the implication.

(ii bis) $\Rightarrow$ (ii ter) is trivial.

(ii ter) $\Rightarrow$ (ii). Let $i : U \to X$ be an open immersion and let F be a constructible A_U-module. We must prove $\mathrm{R}i_*F \in \mathrm{ob}\,\mathrm{D}_c^+(X)$. The question being local, suppose X noetherian and argue by noetherian induction on X. We may suppose $U \neq \varnothing$. Then there exists a non-empty open subset $j : V \to U$ such that j^*F is locally constant (SGA A, Exposé IX, 2.4). Let M be the mapping cylinder of

$$F \longrightarrow \mathrm{R}j_*j^*F.$$

Applying $\mathrm{R}i_*$ gives a distinguished triangle

$$\mathrm{R}i_*F \longrightarrow \mathrm{R}i_*\mathrm{R}j_*j^*F \longrightarrow \mathrm{R}i_*M \xrightarrow{[1]} .$$

Condition (ii ter) implies

$$\mathrm{R}i_*\mathrm{R}j_*j^*F \xrightarrow{\sim} \mathrm{R}(ij)_*j^*F \in \mathrm{ob}\,\mathrm{D}_c^+(X).$$

Consequently,

$$\mathrm{R}j_*j^*F \xrightarrow{\sim} i^*\mathrm{R}(ij)_*j^*F \in \mathrm{ob}\,\mathrm{D}_c^+(U).$$

Thus $M \in \mathrm{ob}\,\mathrm{D}_c^+(U)$, since $F \in \mathrm{ob}\,\mathrm{D}_c^+(U)$. Moreover, the cohomology of M is supported on a closed subset Z of U distinct from U. Applying the induction hypothesis to $U \cap \overline{Z} \to \overline{Z}$, we deduce $\mathrm{R}i_*M \in \mathrm{ob}\,\mathrm{D}_c^+(X)$, and hence $\mathrm{R}i_*F \in \mathrm{ob}\,\mathrm{D}_c^+(X)$.

The equivalence of (i)–(ii ter) with condition (iii) is left to the reader; use 1.14.

For b)(i), normalization reduces one to the case where X is regular, and one uses purity, which will be proved in 5.1. For b)(ii), see 3.1.6(b). ☐

3.4. The biduality theorem on regular preschemes

Theorem 3.4.1, cf. [H], V, 2.2. Let X be a regular locally noetherian prescheme of finite dimension. Suppose that X is strongly desingularizable (3.1.5), satisfies (reg) (3.1.1) and $(\mathrm{cdloc})_n$ (3.1.3), and that absolute purity (3.1.4) is true on every prescheme smooth over X. Then A_X is dualizing.

Proof. We must prove that A_X satisfies conditions (i)–(iii) of 1.6. Condition (i) follows from 3.2.1. Condition (ii) follows from 3.3.1(b). It remains to prove condition (iii). Thus, by 1.7, for every constructible A_X-module F we must prove that the canonical homomorphism

$$F \longrightarrow \underline{D}_X\underline{D}_XF, \qquad \underline{D}_X = \mathrm{R}\,\underline{\mathrm{Hom}}(\ ,A_X),$$

is an isomorphism. The question being local, suppose X noetherian. By dévissage (3.1.2, iii bis), it is enough to treat sheaves

$$F = f_*i_!(A'_U),$$

where $i : U \to Y$ is the inclusion of a regular open subset of an integral prescheme, $f : Y \to X$ is finite and the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$ for a prime divisor ℓ of n.

Because X is strongly desingularizable, there is a proper birational morphism $h : Y' \to Y$, with Y' regular, inducing an isomorphism from $U' = h^{-1}U$ to U, and such that $Y' - U'$ is a normal crossings divisor. By base change (SGA A, Exposé XVII),

$$i_!(A'_U) \xrightarrow{\sim} \mathrm{R}h_*i'_!(A'_{U'}),$$

where $i' : U' \to Y'$ is the inclusion. Hence, with $g = fh$,

$$f_* i_!(A'_U) \xrightarrow{\sim} f_* \mathrm{R} h_* i'_!(A'_{U'}) \xrightarrow{\sim} \mathrm{R} g_* i'_!(A'_{U'}).$$

By 1.11, it suffices to prove that the pair

$$(i'_!(A'_{U'}), \mathrm{R}^! g(A_X))$$

is bidualizing. Locally immersing Y' in a prescheme X' smooth over X, and applying purity on X', one sees that $\mathrm{R}^! g(A_X)$ is locally isomorphic to $A_{Y'}$, up to a translation of degrees. We are therefore reduced to the following special case.

Lemma 3.4.2. Let X be a locally noetherian regular prescheme on which absolute purity is true. Let $Y = Y_1 \cup \cdots \cup Y_p$ be a strict normal crossings divisor on X, and put $U = X - Y$. Then the pair $(A'_{U,X}, A_X)$ is bidualizing.

Proof. By the exact sequence

$$0 \longrightarrow A'_{U,X} \longrightarrow A'_X \longrightarrow A'_Y \longrightarrow 0,$$

it is equivalent to verify that (A'_Y, A_X) is bidualizing. We argue by induction on p.

If $p = 1$, let $i : Y \to X$ be the immersion. By 1.11, it is enough to verify that

$$(A'_Y, \mathrm{R}^! i(A_X))$$

is bidualizing. By purity,

$$\mathrm{R}^! i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes 1}[-2].$$

Thus the canonical homomorphism $A'_Y \to \underline{D}_Y \underline{D}_Y(A'_Y)$ is an isomorphism, as is checked trivially fibre by fibre, using Pontryagin duality for A-modules.

Suppose $p \geq 2$, and put $Y' = Y_1 \cup \cdots \cup Y_{p-1}$. We have an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_Y \longrightarrow A'_{Y'} \longrightarrow 0.$$

By the induction hypothesis, $(A'_{Y'}, A_X)$ is bidualizing. By the five lemma, we are reduced to biduality for $(A'_{(Y-Y'),X}, A_X)$. But again there is an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_{Y_p} \longrightarrow A'_{Y_p \cap Y'} \longrightarrow 0.$$

Since (A'_{Y_p}, A_X) is bidualizing, it remains to prove the same for $(A'_{Y_p \cap Y'}, A_X)$. Applying 1.11 again and using purity, we are finally reduced to biduality for this pair on Y_p, where the induction hypothesis applies. $\square$

This completes the proof of Theorem 3.4.1.

Corollary 3.4.3. Let S be an excellent noetherian regular prescheme of characteristic zero. Then A_S is dualizing. Moreover, if $f : X \to S$ is a morphism of finite type, then $\mathrm{R}^! f(A_S)$ is dualizing.

Proof. The first assertion follows because S satisfies the hypotheses of 3.4.1, as recalled in 3.1. For the second, one must first verify that $\mathrm{R}^! f(A_S)$ has finite quasi-injective dimension. Cover S by finitely many affine open subsets S_i such that $X_i = f^{-1}(S_i)$ is a finite union of affine open subsets

X_{ij} immersed as closed subschemes in preschemes X'_{ij} smooth of finite dimension over S_i. From 3.2.3 and 1.6 it follows that

$$\dim . q. inj(\mathrm{R}^! f(A_S)) \leq 2 \sup \dim X'_{ij}.$$

It remains to verify conditions (ii) and (iii) of 1.7. Since these are local on X for the étale topology, we may suppose that f factors as $f = f'i$, where $f' : X' \to S$ is smooth of relative dimension d, and $i : X \to X'$ is a closed immersion. Then

$$\mathrm{R}^! f(A_S) \simeq \mathrm{R}^! i\, \mathrm{R}^! f'(A_S).$$

But $\mathrm{R}^! f'(A_S)$ is locally isomorphic to $A_{X'}$, up to the translation by $2d$, hence is dualizing by the first assertion. By 1.15, $\mathrm{R}^! f(A_S)$ is dualizing, and in particular satisfies conditions (ii) and (iii) of 1.7. ☐

Corollary 3.4.4. Let k be a field, and let $f : X \to S = \mathrm{Spec}(k)$ be a morphism locally of finite type, with $\dim(X) \leq 2$. Then $\mathrm{R}^! f(A_S)$ is dualizing.

Proof. Replacing k by its perfect closure, which is harmless for the étale topology (SGA A, Exposé VIII, 1.1), we may suppose k perfect. Cover X by affine open subsets X_i of finite type over k. Since $\dim(X_i) \leq 2$, the normalization lemma makes X_i finite over a scheme X'_i smooth over S and of dimension ≤ 2. By 1.15 this reduces us to the case where f is smooth, hence locally

$$f^!(A_S) \simeq A_X[2d].$$

Thus it remains to show that A_X is dualizing in this case; this is exactly 3.4.1, since the hypotheses are satisfied (see 3.1). ☐

Remark 3.4.5. We shall see below that if X is regular of dimension ≤ 1, then A_X is dualizing.

4. Local duality

The object of this section is to present the notion of "biduality" developed in the preceding sections in a form closer to geometric intuition.

4.1

Let X be a prescheme, let $i : x \to X$ be a closed point whose residue field is separably closed, and let K_X be a dualizing complex on X (1.7). We know (1.15) that $K_{\{x\}} = \mathrm{R}^! i(K_X)$ is a dualizing complex on x, hence, by (2.1) and (3.4.4), isomorphic to $A[r]$ for some $r \in \mathbb{Z}$ (if A is local). We shall say that K_X is *normalized at* x if $r = 0$ and if an isomorphism $K_{\{x\}} \xrightarrow{\sim} A$ has been given.

For example, if X is smooth of relative dimension d over the spectrum of a separably closed field, the complex $(\mu_n)_X^{\otimes d}[2d]$ is, by the purity theorem (SGA XVI 3, SGA $4^{1/2}$ Cycle 2.3), canonically normalized at every closed point of X.

4.2

Let X be a prescheme, $i : Y \to X$ a closed sub-prescheme, and K_X a dualizing complex on X. Then (1.15) $K_Y = \mathrm{R}^! i(K_X)$ is a dualizing complex on Y, and one has the *complementary induction formula* (1.12 b) (ii)), for $F \in \operatorname{ob} \mathrm{D}_c^b(X)$,

$$(4.2.1) \qquad i^* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathrm{R}^! i(F),$$

where $\underline{D}_X = \mathrm{R}\,\underline{\mathrm{Hom}}(\ , K_X)$ and $\underline{D}_Y = \mathrm{R}\,\underline{\mathrm{Hom}}(\ , K_Y)$.

Apply (4.2.1) in particular in the case where $Y = \{x\}$ is a closed point of X with separably closed residue field, K_X being normalized at x (4.1). Taking into account that A_x is injective, (4.2.1) is then written

$$(4.2.2) \qquad \underline{D}_X(F)_x \xrightarrow{\sim} \mathrm{Hom}^*(\mathrm{R}^! i(F), A).$$

Noting that $\mathrm{R}^! i(F)$ is also written, by definition, $\mathrm{R}\Gamma_x(F)$, one therefore has a perfect pairing in $\mathrm{D}(A)$:

$$(4.2.2)' \qquad \underline{D}_X(F)_x \times \mathrm{R}\Gamma_x(F) \longrightarrow A,$$

giving rise to perfect pairings between finite groups:

$$(4.2.2)'' \qquad \underline{\mathrm{H}}^i(\underline{D}_X(F))_x \times \underline{\mathrm{H}}_x^{-i}(F) \longrightarrow A.$$

For example, let X be a prescheme smooth of dimension d over a separably closed field, and let $x \in X$ be a closed point. Then, for $F \in \operatorname{ob} \mathrm{D}_c^b(X)$, one has a canonical perfect duality between finite groups:

$$(4.2.2)''' \qquad \mathrm{Ext}^{2d-i}(F, \mu_n^{\otimes d})_x \times \underline{\mathrm{H}}_x^i(F) \longrightarrow A.$$

Exercise 4.2.3

Let X be a strictly local scheme (SGA VIII 4.2), with closed point x, and let $i : x \to X$ be the canonical morphism. Let K be a dualizing complex on X, normalized at x. Denote by

$$\overset{\mathbf{L}}{\otimes} i_? : \mathrm{D}^b(A) \longrightarrow \mathrm{D}^-(X)$$

the functor defined by

$$\overset{\mathbf{L}}{\otimes} i_?(M) = M \overset{\mathbf{L}}{\otimes}_A K.$$

(If K has finite tor-dimension over A, $\overset{\mathbf{L}}{\otimes} i_?$ extends to a functor from $\mathrm{D}^+(A)$ to $\mathrm{D}(X)$.)

Show that there exists, for $L \in \mathrm{D}^-(X)$ and $M \in \mathrm{D}^b(A)$, a canonical functorial isomorphism

$$(*) \qquad \mathrm{R}\,\mathrm{Hom}(L, \overset{\mathbf{L}}{\otimes} i_?(M)) \xrightarrow{\sim} \mathrm{R}\,\mathrm{Hom}(\mathrm{R}^! i(L), M),$$

(assuming of course that $\mathrm{R}^! i : \mathrm{D}^-(X) \to \mathrm{D}^-(A)$ is defined $\ldots$).

Hint: First define a "trace" morphism $\mathrm{R}^! i \overset{\mathbf{L}}{\otimes} i_?(M) \to M$; then, to check that $(*)$ is an isomorphism, reduce by dévissage to the case where $L \in \operatorname{ob} \mathrm{D}_c^b(X)$ and $M = A$, and apply (4.2.2). Note that, for $M = A$, $(*)$ reduces to the perfect pairing ("local duality", cf. [H] V 6.2)

$$\mathrm{R}\Gamma_x(L) \times \mathrm{R}\,\mathrm{Hom}(L, K) \longrightarrow A,$$

or

$$\mathrm{H}_x^i(L) \times \mathrm{Ext}^{-i}(L, K) \longrightarrow A.$$

4.3

We shall now generalize (4.2.2) to the case of a geometric point $\overline{x}$ of X localized at a point x which is not necessarily closed. For this we need a few preliminaries.

Lemma 4.3. Let X be a prescheme, X' the strict localization of X at a geometric point $\overline{x}$, and $f : X' \to X$ the canonical map. Then, for $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}^+(X)$, the canonical functorial homomorphism (SGA VI I 3.7)

$$f^*\mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R}\,\underline{\mathrm{Hom}}(f^*F, f^*G)$$

is an isomorphism.

Proof. As usual, one reduces to F and G concentrated in degree 0. Since the question is local in a neighborhood of x, one may suppose X affine. Then F admits (SGA IX 2.7) a left resolution by sheaves of the form $i_!(A_U)$, where $i : U \to X$ is étale of finite presentation; hence, by dévissage, one is reduced to $F = i_!(A_U)$. Form the cartesian square

$$\begin{tikzcd} U & U' \arrow[l, "g"'] \\ X \arrow[u, "i"] & X' \arrow[l, "f"] \arrow[u, "i'"'] \end{tikzcd}$$

One has canonical isomorphisms, trivial special cases of duality (SGA XVIII 3.1.10):

$$\mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \xleftarrow{\sim} \mathrm{R}i_*i^*G, \qquad \mathrm{R}\,\underline{\mathrm{Hom}}(f^*F, f^*G) \xleftarrow{\sim} \mathrm{R}i'_*g^*i^*G.$$

Under these isomorphisms, the canonical arrow

$$f^*\mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R}\,\underline{\mathrm{Hom}}(f^*F, f^*G)$$

is identified with the base-change arrow (SGA XVII)

$$f^*\mathrm{R}i_*i^*(G) \longrightarrow \mathrm{R}i'_*g^*i^*(G).$$

But this is an isomorphism, as one sees at once by passing to the limit (SGA VII 5.11).

Lemma 4.4. Let X be a prescheme, and let $K \in \operatorname{ob} \mathrm{D}_c^+(X)$ be a complex satisfying conditions (i) and (ii) of (1.7). In order that K be dualizing, it is necessary and sufficient that, for every strict localization $f : \overline{X}(\overline{x}) \to X$, f^*K be dualizing.

Proof. Use the fact that every constructible sheaf on a strict localization $\overline{X}(\overline{x})$ is induced by a constructible sheaf on an étale $\overline{x}$-pointed scheme over X (SGA IX 2.7.4), and apply (4.3).

4.5

Let X be a prescheme, $\overline{x}$ a geometric point of X, and $f : X' = \overline{X}(\overline{x}) \to X$ the corresponding strict localization.

Let Y be the integral closed sub-prescheme which is the closure of x in X; denote by $g : \overline{x} \to Y$ and $i : Y \to X$ the canonical maps, and form the cartesian square

$$\begin{tikzcd} Y \arrow[r, "i"] & X \\ \overline{x} \arrow[u, "g"] \arrow[r, "i'"'] & X' \arrow[u, "f"'] \end{tikzcd}$$

($\overline{x}$ is both the closed point of X' and the strict localization of Y at $\overline{x}$).

We shall denote by

$$\mathrm{R}\Gamma_{\overline{x}} : \mathrm{D}^+(X) \longrightarrow \mathrm{D}^+(\overline{x})$$

the functor defined by

$$(4.5.1) \qquad \mathrm{R}\Gamma_{\overline{x}} = g^*\mathrm{R}^!i.$$

Using the well-known relation between the functors $\mathrm{R}^!i$ and $\mathrm{R}j_*j^*$, where $j : U \to X$ is the open complement of Y, and applying passage to the limit (SGA VII 5.11), one obtains a canonical isomorphism

$$(4.5.2) \qquad \mathrm{R}\Gamma_{\overline{x}} \xrightarrow{\sim} \mathrm{R}^!i'f^*.$$

When x is a closed point with separably closed residue field, the functor $\mathrm{R}\Gamma_{\overline{x}}$ coincides with the usual functor $\mathrm{R}\Gamma_x$ (SGA V 4.3), which justifies the notation.

Let $K_X \in \operatorname{ob}\mathrm{D}^+(X)$, and put $K_Y = \mathrm{R}^!j(K_X)$, $K_{X'} = f^*K_X$, and $K_{\overline{x}} = \mathrm{R}\Gamma_{\overline{x}}(K_X)$ (warning: $K_{\overline{x}}$ is not the fiber of K_X at $\overline{x}$!). Suppose that K_X is dualizing. Then, by (1.15) and (4.4), the same is true of $K_{X'}$, K_Y, and $K_{\overline{x}}$. Hence (2.1) one has $K_{\overline{x}} \simeq A[d]$ for some $d \in \mathbb{Z}$.

We shall say that K_X is *normalized at* $\overline{x}$ if $d = 0$ and if an isomorphism $K_{\overline{x}} \simeq A$ has been chosen. In the case where x is closed with separably closed residue field, this recovers the notion introduced in (4.1).

Let K_X be a dualizing complex, and put $\underline{D}_X = \mathrm{R}\underline{\mathrm{Hom}}(\ ,K_X)$, $\underline{D}_Y = \mathrm{R}\underline{\mathrm{Hom}}(\ ,K_Y)$, etc. Let $F \in \operatorname{ob}\mathrm{D}^b_c(X)$; let us compute $(\underline{D}_X F)_{\overline{x}}$. One has:

$$\begin{aligned} (\underline{D}_X F)_{\overline{x}} &\xrightarrow{\sim} g^*i^*\underline{D}_X F \\ &\xrightarrow{\sim} g^*\underline{D}_Y \mathrm{R}^!iF \qquad \text{(by the complementary induction formula (4.2.1))} \\ &\xrightarrow{\sim} \underline{D}_{\overline{x}} g^*\mathrm{R}^!i(F) \qquad \text{(by (4.3)), that is,} \end{aligned}$$

$$(4.5.3) \qquad (\underline{D}_X F)_{\overline{x}} \xrightarrow{\sim} \underline{D}_{\overline{x}}\mathrm{R}\Gamma_{\overline{x}}(F).$$

Of course, one could have made an analogous computation by passing through X' instead of Y; one would then have obtained a canonical isomorphism

$$(\underline{D}_X F)_{\overline{x}} \xrightarrow{\sim} \underline{D}_{\overline{x}}\mathrm{R}^!i'f^*(F),$$

compatible with (4.5.3) by means of the identification (4.5.2).

If moreover K_X is normalized at $\overline{x}$, one obtains perfect pairings generalizing (4.2.2)' and (4.2.2)":

$$(4.5.3)' \qquad \mathrm{H}^i_Y(F)_{\overline{x}} \times \mathrm{R}^{-i}\Gamma_{\overline{x}}(F) \longrightarrow A,$$

$$(4.5.3)'' \qquad \mathrm{H}^i(\underline{D}_X(F))_{\overline{x}} \times \mathrm{H}^{-i}_{\overline{x}}(F) \longrightarrow A,$$

where $\mathrm{H}^i_{\overline{x}} = \mathrm{H}^i\mathrm{R}\Gamma_{\overline{x}}$.

Conversely, let $K_X \in \operatorname{ob} \mathrm{D}^+(X)$ satisfy conditions (i) and (ii) of (1.7), and suppose that $K_{\overline{x}} = \mathrm{R}\Gamma_{\overline{x}}(K_X)$ is dualizing for every geometric point $\overline{x}$ of X. Then, for every $\overline{x}$, one can define (cf. 1.12 b) (ii)) a canonical arrow

$$(4.5.3)^{\mathrm{bis}} \qquad \underline{D}_X(F)_{\overline{x}} \longrightarrow \underline{D}_{\overline{x}}\mathrm{R}\Gamma_{\overline{x}}(F) \qquad (F \in \operatorname{ob} \mathrm{D}^b_c(X)).$$

If this is an isomorphism for every $\overline{x}$ and every F, then K_X is dualizing. Indeed, it remains to verify that the canonical arrow

$$F \longrightarrow \underline{D}_X\underline{D}_X(F)$$

is an isomorphism for every $F \in \operatorname{ob} \mathrm{D}^b_c(X)$. At an arbitrary geometric point $\overline{x}$, one has

$$\begin{aligned} F_{\overline{x}} &\xrightarrow{\sim} \underline{D}_{\overline{x}}\underline{D}_{\overline{x}}(F_{\overline{x}}) && \text{(since } K_{\overline{x}} \text{ is dualizing)} \\ &\xrightarrow{\sim} \underline{D}_{\overline{x}}\mathrm{R}\Gamma_{\overline{x}}\underline{D}_X(F) && \text{(by (4.3) and the induction formula 1.12 b) (i))} \\ &\xrightarrow{\sim} \underline{D}_{\overline{x}}\underline{D}_{\overline{x}}(F)_{\overline{x}} && \text{(by hypothesis),} \end{aligned}$$

and therefore, modulo a compatibility check, (F, K_X) is bidualizing, which proves the assertion. In summary:

Proposition 4.5.4. Let X be a prescheme, and let $K_X \in \operatorname{ob} \mathrm{D}^+(X)$ be a complex satisfying conditions (i) and (ii) of (1.7). In order that K_X be dualizing, it is necessary and sufficient that, for every geometric point $\overline{x}$ of X, the complex $K_{\overline{x}}$ be dualizing and the canonical arrow $(4.5.3)^{\mathrm{bis}}$ be an isomorphism for every $F \in \operatorname{ob} \mathrm{D}^b_c(X)$.

Remark 4.5.5. The existence of dualizing complexes should be regarded, in a sense, as an important complement to the global duality theorem. Indeed, the latter expresses that, for a separated morphism of finite type $f : X \to Y$ (Y quasi-compact), one has a canonical isomorphism

$$\underline{D}_Y \mathrm{R}_! f(F) \xrightarrow{\sim} \mathrm{R}f_*\underline{D}_X(F)$$

(with the notation of 1.12). Its use for computing $\mathrm{R}_! f(F)$ (when $F \in \operatorname{ob} \mathrm{D}^b_c(X)$) requires information about $\underline{D}_X(F)$. When K_X is dualizing, this information is supplied by the isomorphism (4.5.3), which makes possible, at least in theory, a computation of the fibers of $\underline{D}_X(F)$ as invariants of "purely spatial" cohomology.

Exercise 4.5.6. Let X be a prescheme endowed with a dualizing complex (1.7). The family of functors (cf. (4.5.3)")

$$\mathrm{H}^i_{\overline{x}} : \mathrm{D}^b_c(X) \longrightarrow (A\text{-modules})$$

(i running through $\mathbb{Z}$ and $\overline{x}$ through the set of geometric points of X) is "conservative", i.e. a morphism $F \to G$ in $\mathrm{D}^b_c(X)$ such that the maps $\mathrm{H}^i_{\overline{x}}(F) \to \mathrm{H}^i_{\overline{x}}(G)$ are isomorphisms, is an isomorphism.

4.6

Let X be a prescheme, $j : Y \to X$ a closed sub-prescheme, and $i : \overline{x} \to Y$ a geometric point of Y. Let, on the other hand, K_X be a dualizing complex on X, normalized at $\overline{x}$ (cf. (4.5)). We saw in (4.5) that, if Y is the closure of x, one has, for $F \in \operatorname{ob} \mathrm{D}^b_c(X)$, a perfect pairing

$$\mathrm{R}^! j(F)_{\overline{x}} \times \underline{D}_X(F)_{\overline{x}} \longrightarrow A.$$

It is not difficult to define an analogous pairing in the general case. Introduce the factorization of i as

$$\overline{x} \xrightarrow{i'} \{\overline{x}\} \xrightarrow{i''} Y.$$

For $F \in \operatorname{ob} \mathrm{D}^b_c(X)$, one has canonical isomorphisms:

$$\begin{aligned}
\mathrm{R}\Gamma_{\overline{x}} j^* \underline{D}_X(F) &\xrightarrow{\sim} \mathrm{R}\Gamma_{\overline{x}} \underline{D}_Y \mathrm{R}^! j(F) \qquad \text{(complementary induction formula (4.2.1))}\\
&\xrightarrow{\sim} i'^* \mathrm{R}^! i'' \underline{D}_Y \mathrm{R}^! j(F) \qquad \text{(definition of } \mathrm{R}\Gamma_{\overline{x}} \text{ (4.5.1))}\\
&\xrightarrow{\sim} i'^* \underline{D}_{\overline{x}} i''^* \mathrm{R}^! j(F) \qquad \text{(ordinary induction (1.12 b) (i))}\\
&\xrightarrow{\sim} \underline{D}_{\overline{x}} i^* \mathrm{R}^! j(F) \qquad \text{(by 4.3)), i.e. finally a}
\end{aligned}$$

canonical isomorphism

$$\text{(4.6.1)} \qquad \mathrm{R}^! j(F)_{\overline{x}} \xrightarrow{\sim} \underline{D}_{\overline{x}} \mathrm{R}\Gamma_{\overline{x}} j^* \underline{D}_X(F),$$

and hence perfect pairings

$$\text{(4.6.1)}' \qquad \mathrm{R}^! j(F)_{\overline{x}} \times \mathrm{R}\Gamma_{\overline{x}} j^* \underline{D}_X(F) \longrightarrow A,$$

$$\text{(4.6.1)}'' \qquad \mathrm{H}^i_Y(F)_{\overline{x}} \times \mathrm{H}^{-i}_{\overline{x}}(\underline{D}_X(F)|Y) \longrightarrow A.$$

Example 4.6.2. Let X be a regular prescheme, let $\overline{x}$ be a geometric point of X, and put $d = \dim(\mathcal{O}_{X,\overline{x}})$. Set $K_X = (\mu_n)^{\otimes d}_X[2d]$. If X satisfies condition (reg) (3.1.1) and the purity theorem (3.1.4), then K_X is canonically normalized at $\overline{x}$. Thus, if K_X is dualizing, one has perfect pairings

$$\mathrm{H}^i_Y(F)_{\overline{x}} \times \mathrm{H}^{2d-i}_{\overline{x}}(F'|Y) \longrightarrow A,$$

for every closed subset Y containing x and every $F \in \operatorname{ob} \mathrm{D}^b_c(X)$, with $F' = \mathrm{R}\operatorname{Hom}(F, (\mu_n)^{\otimes d}_X)$.

To finish, let us give a strictly local variant of (4.6.1).

4.7

Let X be a noetherian strictly local scheme (SGA VIII 4.2), with closed point x. Let Y be a non-empty closed subset of X, and consider the canonical immersions

$$U = X - Y \xrightarrow{i} V = X - \{x\} \xrightarrow{j} X.$$

Finally, let K_X be a dualizing complex normalized at x (4.1), giving dualizing complexes "corresponding" on U, V, Y, x (and by hypothesis $\underline{D}_x = \mathrm{R}\operatorname{Hom}(\ , A)$):

$$K_V = j^* K_X, \qquad K_U = i^* K_V, \qquad K_Y = \mathrm{R}\Gamma_Y(K_X).$$

Let $G \in \operatorname{ob} \mathrm{D}^b_c(U)$, and put $G' = \underline{D}_U(G)$. We propose to define a natural pairing

$$\text{(4.7.1)} \qquad \mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(V, i_! G')[-1] \longrightarrow A,$$

i.e. a functorial homomorphism

$$\text{(4.7.1)}' \qquad \mathrm{R}\Gamma(U, G) \overset{\mathbf{L}}{\otimes}_A \mathrm{R}\Gamma(V, i_! G')[-1] \longrightarrow A.$$

(Since the ring $A = \mathbb{Z}/\ell\mathbb{Z}$ has infinite cohomological dimension for $\ell \geqslant 2$, one knows how to define only the tensor products

$$\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}^-(A) \longrightarrow \mathrm{D}^-(A)$$

and

$$\overset{\mathbf{L}}{\otimes} : \mathrm{D}(A) \times \mathrm{D}^b(A)_{\mathrm{torf}} \longrightarrow \mathrm{D}(A) \quad (*).$$

We therefore leave it to the reader to make suitable hypotheses ensuring that the two factors of $\overset{\mathbf{L}}{\otimes}$ appearing in (4.7.1)' are in $\mathrm{D}^b(A)$, or that one of them has finite tor-dimension.)

First note that, by the complementary duality formula (1.12 a) (ii), one has a canonical isomorphism

$$(4.7.2) \qquad i_!(G') \xrightarrow{\sim} \underline{D}_V(\mathrm{R}i_*G).$$

On the other hand, one has the ordinary duality isomorphism

$$(4.7.3) \qquad \mathrm{R}\,\mathrm{Hom}(A_U, G) \xrightarrow{\sim} \mathrm{R}\,\mathrm{Hom}(A_V, \mathrm{R}i_*G).$$

By the definition of $\underline{D}_V$ (and independently of the fact that K_V is dualizing), one has a canonical arrow

$$(4.7.4) \qquad \mathrm{R}i_*G \overset{\mathbf{L}}{\otimes} \underline{D}_V(\mathrm{R}i_*G) \longrightarrow K_V.$$

From (4.7.2), (4.7.3), and (4.7.4) one obtains a (canonical) arrow

$$(4.7.5) \qquad \mathrm{R}\Gamma(U, G) \overset{\mathbf{L}}{\otimes}_A \mathrm{R}\Gamma(V, i_!G') \longrightarrow \mathrm{R}\,\mathrm{Hom}(A_V, K_V).$$

But $K_V = j^*K_X$, and one has the duality isomorphism (SGA 3.1.10):

$$(4.7.6) \qquad \mathrm{R}\,\mathrm{Hom}(A_V, K_V) \xrightarrow{\sim} \mathrm{R}\,\mathrm{Hom}(A_{V,X}, K_X).$$

Finally, the exact sequence

$$(4.7.7) \qquad 0 \longrightarrow A_{V,X} \longrightarrow A_X \longrightarrow A_x \longrightarrow 0$$

gives a homomorphism of degree $+1$:

$$(4.7.8) \qquad \mathrm{R}\,\mathrm{Hom}(A_{V,X}, K_X) \longrightarrow \mathrm{R}\,\mathrm{Hom}(A_x, K_X) \xrightarrow{\sim} A.$$

Composing (4.7.5), (4.7.6), and (4.7.8), one obtains the announced arrow (4.7.1)'.

(*) (added in 1977) Inexact: in fact it is easy to extend $\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}^-(A) \to \mathrm{D}^-(A)$ *to* $\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}(A) \to \mathrm{D}(A)$.

We shall now see that the pairing (4.7.1) is "perfect", i.e. that the arrow (4.7.1)' identifies $\mathrm{R}\Gamma(V, i_!G')[-1]$ with $\underline{D}_x\mathrm{R}\Gamma(U, G)$. Indeed suppose, as is allowed, that

$$G = (ji)^*F,$$

with $F \in \mathrm{ob}\,\mathrm{D}^b_c(X)$. Put $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_!G'$. One has the exact sequence

$$(4.7.9) \qquad 0 \longrightarrow F'_{U,X} \longrightarrow F' \longrightarrow F'|Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{tikzcd} & \mathrm{R}\Gamma_x(F'_{U,X}) \arrow[dl] & \\ \mathrm{R}\Gamma_x(F'|Y) \arrow[rr] & & \mathrm{R}\Gamma_x(F') \arrow[ul, "+1"'] \end{tikzcd} \qquad (b)$$

By definition of $\mathrm{R}\Gamma_x$, one has

$$\mathrm{R}\Gamma_x(F'_{U,X}) = \mathrm{R}\operatorname{Hom}(A_x, F'_{U,X});$$

hence, using (4.7.7) and the fact that $(F'_{U,X})_x = \mathrm{R}\Gamma(X, F'_{U,X}) = 0$, one has a canonical isomorphism

$$\mathrm{R}\operatorname{Hom}(A_{V,X}, F'_{U,X})[1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

But one has

$$\begin{aligned} \mathrm{R}\operatorname{Hom}(A_{V,X}, F'_{U,X}) &= \mathrm{R}\operatorname{Hom}(j_! A_V, j_! i_! G') \\ &\xrightarrow{\sim} \mathrm{R}\operatorname{Hom}(A_V, i_! G') \qquad \text{by duality (SGA 3.1.10),} \end{aligned}$$

and hence finally a canonical isomorphism

$$\text{(4.7.10)} \qquad \mathrm{R}\Gamma(V, i_! G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

On the other hand one has the exact sequence

$$\text{(4.7.11)} \qquad 0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{tikzcd} & \mathrm{R}\Gamma(U, F|U) \arrow[dl] & \\ \mathrm{R}\Gamma_Y(F) \arrow[rr] & & \mathrm{R}\Gamma_X(F) \arrow[ul, "+1"'] \end{tikzcd} \qquad (a)$$

By virtue of (4.7.1) and (4.7.10), one has a natural pairing

$$\text{(4.7.12)} \qquad \mathrm{R}\Gamma(U, F|U) \times \mathrm{R}\Gamma_x(F'_{U,X}) \longrightarrow A.$$

By virtue of (4.6.1), one has a perfect pairing

$$\text{(4.7.13)} \qquad \mathrm{R}\Gamma_Y(F) \times \mathrm{R}\Gamma_x(F'|Y) \longrightarrow A.$$

Finally, by the ordinary induction formula (1.12 b) (i)), one has a perfect pairing

$$\text{(4.7.14)} \qquad \mathrm{R}\Gamma(X, F) \times \mathrm{R}\Gamma_x(F') \longrightarrow A.$$

The reader who has the patience to orient himself in this maze of "canonical arrows" will check that the pairings (4.7.12), (4.7.13), and (4.7.14) are "compatible" with the arrows of the triangles (a) and (b), and consequently that the pairings (4.7.12) and (4.7.1) are perfect.

The pairing of the triangles (a) and (b) gives rise to two exact sequences "transpose to one another":

$$\text{(4.7.15)} \qquad \cdots \longrightarrow \mathrm{H}^i_Y(X, F) \longrightarrow \mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(U, F) \longrightarrow \cdots$$

$$\cdots \longleftarrow \mathrm{H}_x^{-i}(Y, F'|Y) \longleftarrow \mathrm{H}_x^{-i}(X, F') \longleftarrow \mathrm{H}_x^{-i-1}(V, F'_{U,V}) \longleftarrow \cdots .$$

In summary, we may state:

Proposition 4.7.16. Let X be a noetherian strictly local scheme, with closed point x. Let Y be a non-empty closed subset of X, let $U = X - Y$, $V = X - x$, and let $i : U \to V$ and $j : V \to X$ be the canonical inclusions. Let K_X be a dualizing complex on X normalized at x (hence giving associated dualizing complexes on V, U, Y). For $G \in \operatorname{ob} \mathrm{D}_c^b(U)$, one has a perfect pairing:

$$(*) \qquad \mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(V, i_! G')[-1] \longrightarrow A$$

(where $G' = \underline{D}_U(G)$). If $G = F|U$, where $F \in \operatorname{ob} \mathrm{D}_c^b(X)$, one has a canonical isomorphism

$$(**) \qquad \mathrm{R}\Gamma(V, i_! G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X})$$

(where $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_!(F'|U) = (ji)_!(G')$), and a perfect pairing between the distinguished triangles

$$\mathrm{R}\Gamma_Y(F) \longrightarrow \mathrm{R}\Gamma(X, F) \longrightarrow \mathrm{R}\Gamma(U, F|U) \xrightarrow{+1},$$
$$\mathrm{R}\Gamma_x(F'_{U,X}) \longrightarrow \mathrm{R}\Gamma_x(F') \longrightarrow \mathrm{R}\Gamma_x(F'|Y) \xrightarrow{+1} .$$

compatible with the pairing $(*)$ and the isomorphism $(**)$, and giving rise to two exact sequences transpose to one another (4.7.15).

Remark 4.7.17. In the particular case where $Y = \{x\}$, the pairing (4.7.1) is defined under the sole hypothesis that K_X is normalized at x (and not necessarily dualizing), i.e. such that $\mathrm{R}\Gamma_x(K_X) \simeq A$ (the isomorphism being given): the isomorphism (4.7.2) (which used biduality) then reduces to the identity. Since the pairing (4.7.14) is in any case perfect, one sees therefore that local duality on X (in the form (4.2.2)') is equivalent to the "global" duality on $U = X - x$, i.e. to the assertion that the pairing (4.7.1)

$$\mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(U, G')[-1] \longrightarrow A$$

is perfect, or equivalently that the pairings

$$\mathrm{H}^i(U, G) \times \mathrm{H}^{-i-1}(U, G') \longrightarrow A$$

are perfect.

Suppose that one has on X a dualizing complex K_X normalized at x. If X is excellent and U is regular of dimension d (x may therefore be an isolated singularity), one should be able to check that $K_U = K_X|U$ is canonically isomorphic to $(\mu_n)_U^{\otimes d}[2d]$ (this is easily checked, at any rate, if X is the strict localization of a prescheme locally of finite type over a field of characteristic 0). Then local duality on X implies that, for every locally constant constructible A_U-module G, one has perfect pairings:

$$\mathrm{H}^i(U, G) \times \mathrm{H}^{2d-1-i}(U, G^\circ) \longrightarrow A,$$

where $G^\circ = \mathrm{Hom}(G, (\mu_n)_U^{\otimes d})$, corresponding formally to Poincaré duality on a scheme of dimension $2d - 1$.

5. Local duality on curves

Theorem 5.1. Let X be a noetherian regular prescheme of dimension 1. Suppose that n is prime to the residual characteristics of X. Then the purity theorem is true on X (3.1.4), and the complex A_X is dualizing (1.7).

Proof. Suppose purity has been proved. Since X plainly satisfies condition (reg) (3.1.1), and since condition $(\text{cdloc})_n$ is verified by (SGA X 2.2), it follows from (3.2.1) that $\dim .q.inj(A_X) = 2$. On the other hand, condition (ii) of (1.7) is verified by (3.3.1 b) (i)) (which was proved assuming purity). Hence, by (4.4), we are reduced to proving that A_X is dualizing when X is strictly local. But, to prove purity, which is a local question in a neighborhood of a closed point, one may equally well suppose X strictly local. Finally, we are reduced to proving the theorem when X is strictly local.

Thus suppose that X is strictly local. Let x be its closed point, let $(\pi = \operatorname{car}(k(x)))$, let η be its generic point, and let $U = X - x = \operatorname{Spec}(k(\eta))$. Let $G = \operatorname{Gal}(k(\overline{\eta})/k(\eta))$ be the fundamental group of U. It is known [CL, Chap. IV § 2 Exer. 2] that one has a canonical exact sequence

$$(5.1.1) \qquad 1 \longrightarrow P \longrightarrow G \longrightarrow H \longrightarrow 1,$$

where $H = \prod_{\ell \neq \pi} \mathbb{Z}_\ell(1)$, $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^\nu}$, and P is a pro-π-group. (Of course, H is isomorphic, but not canonically, to $\prod_{\ell \neq \pi} \mathbb{Z}_\ell$.)

It is also known (SGA VIII 2) that the étale cohomology of U is interpreted as the Galois cohomology of G, sheaves (respectively constructible sheaves) of A_U-modules corresponding to A-modules (respectively finite type A-modules) on which G acts continuously.

This being so, the $\mathrm{H}^i_x(A_X)$ are determined by the exact sequence

$$(5.1.2) \qquad 0 \longrightarrow \mathrm{H}^0_x(A_X) \longrightarrow \mathrm{H}^0(X, A_X) \longrightarrow \mathrm{H}^0(U, A_U) \longrightarrow \mathrm{H}^1_x(A_X) \longrightarrow 0,$$

and the isomorphisms

$$(5.1.3) \qquad \mathrm{H}^i_x(A_X) \xrightarrow{\sim} \mathrm{H}^{i-1}(U, A_U) \qquad \text{for } i \geqslant 2.$$

But trivially $\mathrm{H}^0(X, A_X) = \mathrm{H}^0(U, A_U) = A$, so $\mathrm{H}^0_x(A_X) = \mathrm{H}^1_x(A_X) = 0$.

On the other hand, for a Galois G-A-module M, one has the Hochschild–Serre spectral sequence

$$E_2^{pq} = \mathrm{H}^p(H, \mathrm{H}^q(P, M)) \Longrightarrow \mathrm{H}^{p+q}(G, M).$$

Since n is prime to π and P is a pro-π-group, one has

$$\mathrm{H}^q(P, M) = 0 \quad \text{for } q \neq 0,$$

hence $E_2^{pq} = 0$ for $q \neq 0$, giving a canonical isomorphism

$$(5.1.4) \qquad \mathrm{H}^p(G, M) \xrightarrow{\sim} \mathrm{H}^p(H, M^P).$$

By an analogous argument, one sees moreover that $\mathrm{H}^p(H, N)$ is identified with $\mathrm{H}^p(\widehat{\mathbb{Z}}(1), N)$ when the factor $\widehat{\mathbb{Z}}(1)$ is made to act trivially on N. Thus, finally, with the preceding convention, one has a canonical isomorphism

$$(5.1.5) \qquad \mathrm{H}^p(G, M) \xrightarrow{\sim} \mathrm{H}^p(\widehat{\mathbb{Z}}(1), M^P).$$

Now the Galois cohomology of $\widehat{\mathbb{Z}}$ is well known (CG chap. I p. 31): for a G-A-module N, one has

$$(5.1.6) \qquad \mathrm{H}^i(\widehat{\mathbb{Z}}, N) = 0 \quad \text{for } i \geqslant 2, \qquad \mathrm{H}^0(\widehat{\mathbb{Z}}, N) = N^{\widehat{\mathbb{Z}}}, \qquad \mathrm{H}^1(\widehat{\mathbb{Z}}, N) = N_{\widehat{\mathbb{Z}}}.$$

From (5.1.3), (5.1.5), and (5.1.6) one immediately deduces that

$$\mathrm{H}^i(U, A_U) = 0 \quad \text{for } i \geqslant 2,$$

and

$$\mathrm{H}^1(U, A_U) \xrightarrow{\sim} A.$$

This last isomorphism is not canonical, since it depends on the choice of an identification of $\widehat{\mathbb{Z}}(1)$ with $\widehat{\mathbb{Z}}$. But, using the Kummer exact sequence, one finds (SGA XIX 1.3) a canonical isomorphism (given by the "local fundamental class")

$$(5.1.7) \qquad \mathrm{H}^1(U, \mu_n) \xrightarrow{\sim} A.$$

Purity is therefore proved.

Let us now prove that A_X is dualizing. By (4.7.17), it is a matter of showing that, for every locally constant constructible A_U-module M, the pairing

$$(5.1.8) \qquad \mathrm{H}^i(U, M) \times \mathrm{H}^{1-i}(U, \mathrm{Hom}(M, A_U)) \longrightarrow \mathrm{H}^1(U, A_U) \xrightarrow{\sim} (\mu_n)_x^{\otimes -1}$$

is a perfect duality. We shall interpret this pairing in terms of Galois cohomology. We shall use the following elementary lemma.

Lemma 5.1.9. Let $D = \mathrm{Hom}(\ , A)$ denote the Pontrjagin dualizing functor on the category of finite type A-modules. Let M be a finite type A-module on which a group G acts A-linearly.

(i) One has a canonical isomorphism

$$D(M^G) \xrightarrow{\sim} (DM)_G.$$

(ii) If G is a profinite group acting continuously on M, and if G has pro-order prime to n, then the canonical arrow

$$M^G \longrightarrow M_G$$

is an isomorphism.

Proof. By definition one has

$$M^G = \varprojlim_{s \in G} \mathrm{Ker}(M \xrightarrow{s-1} M),$$

hence

$$D(M^G) \xrightarrow{\sim} \varinjlim_{s \in G} \mathrm{Coker}(DM \xrightarrow{s-1} DM) \xrightarrow{\sim} (DM)_G,$$

which proves (i).

Let us prove (ii). Let M' be the A-module defined by the exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M_G \longrightarrow 0.$$

One has $\mathrm{H}^1(G, M') = 0$ because G has pro-order prime to n. The exact cohomology sequence therefore implies that the arrow $M^G \to M_G$ is an epimorphism. But, applying the functor D and taking (i) into account, one deduces that it is a monomorphism, which completes the proof.

Applying (5.1.9) to M and P, one finds that $\mathrm{Hom}(M, A)^P$ is canonically identified with $\mathrm{Hom}(M^P, A)$. If one chooses an isomorphism from $\widehat{\mathbb{Z}}(1)$ to $\widehat{\mathbb{Z}}$, the pairing (5.1.8) is written, taking (5.1.5) into account, as

$$(5.1.8)' \qquad \mathrm{H}^i(\widehat{\mathbb{Z}}, M^P) \times \mathrm{H}^{1-i}(\widehat{\mathbb{Z}}, D(M^P)) \longrightarrow \mathrm{H}^1(\widehat{\mathbb{Z}}, A) \xrightarrow{\sim} A.$$

It follows immediately from (5.1.6) and (5.1.9(i)) that (5.1.8)' is a perfect pairing. This completes the proof of (5.1).

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Appendix

by L. Illusie

The preceding theory extends easily to base Rings more general than the constant Ring $\mathbb{Z}/n\mathbb{Z}$. We shall briefly indicate how the theory is written for locally constant base Rings, killed by $n > 0$, and whose geometric fibres are noetherian rings. This is, moreover, only a provisional framework[7]: sooner or later one will have to globalize, by taking as "coefficients" on a prescheme X relative preschemes over the site $(X_{\text{ét}}, \mathbb{Z}/n\mathbb{Z})$.

1. Constructible sheaves

1.1. Let X be a prescheme, and let A_X be a sheaf of rings on X (for the étale topology), locally constant, whose geometric fibres are left noetherian rings. We denote by A^X the category of left

[7] According to Grothendieck!

A_X-Modules, and by $\mathrm{D}_A(X)$ (or $\mathrm{D}(X)$ if there is no risk of confusion) the corresponding derived category. The notion of a constructible left A_X-Module is defined as in (SGAA IX 2.3), by replacing the expression "locally constant with value of finite presentation" by "of finite presentation as an A_U-Module".

We denote by $\mathrm{D}_c(X)$ the full subcategory of $\mathrm{D}(X)$ consisting of complexes E such that $\mathrm{H}^i(E)$ is a constructible A_X-Module for every i. It is a triangulated subcategory of $\mathrm{D}(X)$.

1.2. Let $E \in \operatorname{ob} \mathrm{D}(X)$. We say (cf. Exp. II 3.11) that E is pseudo-coherent if, for every i, $\mathrm{H}^i(E)$ is an A_X-Module of finite presentation (or locally constant constructible, if one prefers). We denote by $\mathrm{D}(X)_{\mathrm{coh}}$ (or $\mathrm{D}_{\mathrm{coh}}(X)$) the full subcategory of $\mathrm{D}(X)$ consisting of pseudo-coherent objects. It follows from (SGAA IX 2.1) that this is a triangulated subcategory of $\mathrm{D}_c(X)$.

Proposition 1.3 (dévissage lemma). Let X be a quasi-compact and quasi-separated prescheme (respectively a noetherian prescheme), and let C be a strictly full subcategory of the category $\mathrm{Cons}(X)$ of constructible left A_X-Modules. Suppose that C is stable under extensions. The following conditions are equivalent:

(i) $C = \mathrm{Cons}(X)$;

(ii) C contains the sheaves of the form $i_!(F)$, where $i : Y \to X$ is a sub-prescheme of finite presentation (respectively an integral sub-prescheme), and F is locally constant constructible on Y.

If A_X is constant with value A, a left noetherian ring killed by an integer $n > 0$, the preceding conditions are equivalent to:

(iii) C is stable under direct factors and contains the sheaves of the form $f_* i_!(M_U)$, where $i : U \to Y$ is an open immersion (respectively an open subset of an integral prescheme), $f : Y \to X$ is a finite morphism (respectively finite and such that the generic point of Y is separable over its image), and M_U is a left A_U-Module constant with value a monogenic A-module killed by a prime divisor of n.

Finally, if X is noetherian and satisfies condition (reg) (I 3.1.1), condition (ii) (respectively (iii)) is equivalent to (i) if one further assumes Y (respectively U) regular.

Proof. The equivalence of conditions (i) and (ii) is proved as in the case where A_X is constant; see (SGAA IX 2.4 and 2.5).

Suppose that A_X is constant with value A and killed by n, and prove that (iii) implies (ii). Let $E \in \operatorname{ob} \mathrm{Cons}(X)$; we show that $E \in \operatorname{ob} C$. By the equivalence (i)$\Leftrightarrow$(ii), we may suppose that $E = j_! F$, where $j : Z \to X$ is the immersion of a sub-prescheme of finite presentation (respectively an integral one), and F is a constructible locally constant A_Z-Module, which we may even suppose, after a little more dévissage, to be trivialized by an étale covering $g : Z' \to Z$ with group G. The sheaf F is therefore defined by a finite type A-module M on which G acts. If ℓ is a prime divisor of n, the subgroup of ℓ-torsion in M is a sub-G-A-module M_ℓ of M, and

$$M = \bigoplus_\ell M_\ell,$$

where ℓ runs over the set of prime divisors of n. Since C is stable under direct factors, we are reduced to the case where M is ℓ-torsion. Let H be an ℓ-Sylow subgroup of G, and let $h : U = Z'/H \to Z$

be the corresponding intermediate covering. Since the degree of h is prime to ℓ, the "trace method" (SGAA IX 5.1) shows that the adjunction map

$$F \longrightarrow h_* h^*(F)$$

is an isomorphism onto a direct factor; it is therefore enough to prove that $j_! h_* h^*(F) \in \operatorname{ob} C$. By the "Main theorem" (EGA IV 8.12.6), there exists a commutative diagram

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[d, "h"'] & Y \arrow[d, "f"] \\ Z \arrow[r, "j"'] & X, \end{tikzcd}$$

where i is an open immersion and f is finite. We have

$$j_! h_* h^*(F) \simeq f_* i_! h^*(F).$$

But $h^*(F)$ is defined by M considered as an H-A-module. Thus (iii)⇒(ii) follows from the next lemma.

Lemma 1.3.1. Let A be a left noetherian ring, ℓ a prime number, H a finite ℓ-group, and M a left H-A-module of finite type killed by ℓ^ν ($\nu \geq 1$). There exists a finite filtration of M by sub-H-A-modules such that the successive quotients are monogenic A-modules killed by ℓ, and on which H acts trivially.

Proof. The exact sequence of H-A-modules

$$0 \longrightarrow \ell M \longrightarrow M \longrightarrow M/\ell M \longrightarrow 0$$

reduces, by induction on ν, to the case where M is killed by ℓ. We show that then M has a finite filtration by sub-H-A-modules on whose successive quotients H acts trivially. Arguing by noetherian induction on the set of sub-H-A-modules of M having the stated property, we are reduced to proving that $M \neq 0$ implies $M^H \neq 0$. But the kernel of the augmentation $\mathbb{F}_\ell[H] \to \mathbb{F}_\ell$ is plainly nilpotent, and Nakayama's lemma implies that every nonzero H-$\mathbb{F}_\ell$-module has a nonzero invariant vector. This proves the assertion. Thus M has a finite filtration whose successive quotients are finite type A-modules killed by ℓ and on which H acts trivially. By noetherian induction one may even arrange for the successive quotients to be monogenic A-modules, completing the proof of (1.3.1).

The last assertion of (1.3) is immediate and left to the reader.

2. Quasi-injective dimension

With the notation of (1.1), one defines, as in (I 1.1 and 1.4), the notions of a quasi-injective A_X-Module and of quasi-injective dimension of an object of $\mathrm{D}(X)$. Propositions (I 1.3, 1.5 and 1.6) extend without change.

It is sometimes convenient to introduce a notion of pointwise quasi-injective dimension. Let $x \in X$ and let $K \in \operatorname{ob} \mathrm{D}^+(X)$. We say that K has quasi-injective dimension $\leq N$ at $\bar{x}$ if, for every constructible A_X-Module F, one has

$$\underline{\mathrm{Ext}}^i(F, K)_{\bar{x}} = 0 \qquad \text{for } i > N.$$

The quasi-injective dimension of K at $\bar{x}$, denoted $\dim .\mathrm{q.inj}_{\bar{x}}(K)$, is the greatest lower bound of the N satisfying the preceding condition. Of course,

$$\dim .\mathrm{q.inj}(K) = \sup_{\bar{x}} \dim .\mathrm{q.inj}_{\bar{x}}(K),$$

where $\bar{x}$ runs through the set of geometric points of X.

It is interesting to compare this with the notion of strict topological dimension introduced by Verdier in [1]. We say that $K \in \operatorname{ob} \mathrm{D}^+(X)$ has strict topological dimension $\leq N$ at a geometric point $\bar{x}$ of X if, for every sub-prescheme Y of X,

$$\underline{\mathrm{Ext}}^i(A_Y, K)_{\bar{x}} = \underline{\mathrm{H}}^i_Y(K)_{\bar{x}} = 0 \qquad \text{for } i > N.$$

The strict topological dimension of K at $\bar{x}$, denoted $\dim .\mathrm{top}_{\bar{x}}(K)$, is the greatest lower bound of the N satisfying the preceding condition. Whereas quasi-injective dimension depends a priori on the base Ring, strict topological dimension depends only on the underlying complex of abelian sheaves.

One plainly has

$$\dim .\mathrm{top}_{\bar{x}}(K) \leq \dim .\mathrm{q.inj}_{\bar{x}}(K).$$

We shall see below examples where equality holds.

3. Dualizing complexes: definition and formal properties

Keep the preceding notation, but suppose A_X is commutative. The notions of bidualizing pair and dualizing complex are defined as in (I 1.7). The statements (I 1.9, 1.11, 1.14) remain valid without modification. The statements (I 1.12, 1.13, 1.15) are also valid, but on condition that A be killed by an integer $n > 0$ prime to the residual characteristics of X. Finally, the uniqueness theorem (I 2.1) is valid provided one assumes $\operatorname{Spec}(A_{\bar{x}})$ connected for every geometric point $\bar{x}$ of X.

4. Absolute purity

Return to the situation of absolute cohomological purity (I 3.1.4). Let X be a locally noetherian regular prescheme, let $n > 0$ be an integer prime to the residual characteristics of X, and let $i : Y \to X$ be a regular sub-prescheme of codimension $d > 0$ at every point. To say that the pair (Y, X) satisfies absolute cohomological purity means, recall, that the "local fundamental class" homomorphism (SGA $4^{1/2}$ Cycle 2.2)

$$(\mu_n^{\otimes d})_Y[-2d] \longrightarrow \mathrm{R}i^!(\mathbb{Z}/n\mathbb{Z})_X \tag{4.1}$$

is an isomorphism.

Now put on X a sheaf of rings A_X killed by n. For the moment set $G = (\mathbb{Z}/n\mathbb{Z})_X$. Then, for every $F \in \operatorname{ob} \mathrm{D}^+_A(X)$, one can define a canonical arrow in $\mathrm{D}^+_A(X)$:

$$i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} \mathrm{R}i^!(G) \longrightarrow \mathrm{R}i^!(F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} G). \tag{4.2}$$

Indeed, by duality (SGAA XVIII 3.1.6), this is equivalent to defining an arrow

$$i_!(i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} \mathrm{R}i^!(G)) \longrightarrow F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} G;$$

we take this to be the composite of the Künneth arrow (SGAA XVII 5.4.2.2)

$$i_!(i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} \mathrm{R}i^!(G)) \longrightarrow F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} i_!\mathrm{R}i^!(G),$$

and the arrow deduced from the trace homomorphism (SGAA XVIII 3.1.6)

$$F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} i_!\mathrm{R}i^!(G) \longrightarrow F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} G.$$

Notice that in the definition of (4.2) only the fact that G and $\mathrm{R}i^!(G)$ have finite Tor-dimension over $\mathbb{Z}/n\mathbb{Z}$ was used. Suppose on the other hand that the functor $i^!$ has finite cohomological dimension on the category of $(\mathbb{Z}/n\mathbb{Z})_X$-Modules: this is the case, for example, if X is of finite type over a field (SGAA XIV 3.1), or more generally if X is excellent of equal characteristic (SGAA XIX 6.1) when resolution of singularities is available, or again if X is of finite type over $\mathrm{Spec}(\mathbb{Z})$ (SGAA X 6). Then the arrow (4.2) is defined for all $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$.

Composing (4.1) and (4.2), one obtains, for $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$, a canonical arrow

$$i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} (\mu_n^{\otimes d})_Y[-2d] \longrightarrow \mathrm{R}i^!(F). \tag{4.3}$$

Assume that A_X is locally constant, and let $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$ be such that every $\mathrm{H}^i(F)$ is a locally constant A_X-Module. Then, if $i^!$ has finite cohomological dimension, the arrow (4.2), and hence the arrow (4.3) if (Y, X) satisfies purity, is an isomorphism.

Indeed, by dévissage, one first reduces to the case where F is concentrated in degree zero. Since the question is local, one may suppose X affine and A_X and F constant. Resolving F on the left by free A_X-Modules and applying the hypothesis that $i^!$ has finite cohomological dimension, one reduces to the case where F is free. By an immediate passage to the limit, one reduces to F free of finite type, and finally to $F = A_X$. It is therefore enough to prove the assertion when $A_X = (\mathbb{Z}/n\mathbb{Z})_X$. But the preceding dévissage reduces to $F = (\mathbb{Z}/n\mathbb{Z})_X$, and in that case (4.2) is the identity.

Definition 4.4. Given the Ring A_X (locally constant and killed by n), we shall say that the pair (Y, X) satisfies cohomological purity in the strong sense if, for every $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$ such that the $\mathrm{H}^i(F)$ are locally constant, the canonical arrow (4.3) is an isomorphism.

We shall say that the purity theorem is true in the strong sense on X if, locally for the étale topology, every pair (Y, X) as above satisfies purity in the strong sense. From what has been shown, for (Y, X) to satisfy purity in the strong sense it is enough that (Y, X) satisfy ordinary purity (I 3.1.) and that the functor $i^!$ have finite cohomological dimension.

5. Bound for quasi-injective dimension

Proposition 5.1. Let X be a locally noetherian regular prescheme, equipped with a locally constant sheaf of rings A_X, killed by an integer $n > 0$ prime to the residual characteristics of X, whose geometric fibres are left noetherian rings. Suppose that X satisfies conditions (reg) (I 3.1.1) and $(\mathrm{cdloc})_n$ (I 3.1.3), and that the purity theorem in the strong sense is true on X (4.4) (conditions all satisfied if X is smooth over a perfect field, or if X is excellent of characteristic zero).

Let $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$ be such that the $\mathrm{H}^i(F)$ are locally constant, and let x be a point of X such that $F_{\bar{x}} \neq 0$ and $\dim.\mathrm{inj}_{A_{\bar{x}}}(F_{\bar{x}}) < +\infty$. Then

$$\dim.\mathrm{q.inj}_{\bar{x}}(F) = 2\dim(\mathcal{O}_{X,x}) + \dim.\mathrm{inj}(F_{\bar{x}}),$$

with the notation of no. 2.

Proof. One merely repeats step by step the proof of (I 3.2.1). First show the inequality

$$\dim.\mathrm{q.inj}_{\bar{x}}(F) \leq 2\dim(\mathcal{O}_{X,x}) + \dim.\mathrm{inj}(F_{\bar{x}}) = N,$$

that is, the relation

$$\underline{\mathrm{Ext}}^i(E,F)_{\bar{x}} = 0 \quad \text{for } i > N \text{ and every constructible } A_X\text{-Module } E. \tag{*}$$

Since the question is local, by dévissage (1.3) one may restrict to checking $(*)$ for $E = i_!(M)$, where $i : Y \to X$ is an integral regular sub-prescheme closed in an open $D(f)$ (with $f \in \Gamma(X, \mathcal{O}_X)$), of codimension d at every point of $D(f)$, and M is a locally constant constructible A_Y-Module ($A_Y = i^* A_X$).

Now there is the duality isomorphism (SGAA XVIII)

$$\mathrm{R}\underline{\mathrm{Hom}}(i_!(M), F) \simeq \mathrm{R}i_*\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F).$$

By the purity theorem, there is also the isomorphism (4.3)

$$i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} (\mu_n^{\otimes d})_Y[-2d] \xrightarrow{\sim} \mathrm{R}i^!(F).$$

Let $\bar{y}$ be a geometric point of Y. Since M is locally constant constructible, one has canonically

$$\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F)_{\bar{y}} \simeq \mathrm{R}\,\mathrm{Hom}(M_{\bar{y}}, (\mathrm{R}i^! F)_{\bar{y}}),$$

hence, by (4.3),

$$\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F)_{\bar{y}} \simeq \mathrm{R}\,\mathrm{Hom}(M_{\bar{y}}, F_{\bar{y}})[-2d].$$

After replacing X by a connected neighbourhood of x, one has $F_{\bar{x}} \simeq F_{\bar{y}}$ and $A_{\bar{x}} \simeq A_{\bar{y}}$, since A_X and the $\mathrm{H}^i(F)$ are locally constant. Thus

$$\mathrm{Ext}^i(M_{\bar{y}}, F_{\bar{y}}) = 0 \quad \text{for } i > \dim.\mathrm{inj}(F_{\bar{x}}),$$

and hence

$$\underline{\mathrm{Ext}}^q(M, \mathrm{R}i^! F) = 0 \quad \text{for } q > 2d + \dim.\mathrm{inj}(F_{\bar{x}}).$$

Set $G = \mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F)$ and compute $\mathrm{R}i_*(G)_{\bar{x}}$. Introduce, as in (I 3.2.1), the schematic closure Z of Y in X, giving a cartesian diagram of immersions

$$\begin{tikzcd} Y \arrow[r] \arrow[d, "j"'] & D(f) \arrow[d] \\ Z \arrow[r, "k"'] & X. \end{tikzcd}$$

Then $\mathrm{R}i_*(G) \simeq k_*\mathrm{R}j_*(G)$. Assume of course that $x \in Z$, since otherwise $\mathrm{R}i_*(G)_{\bar{x}} = 0$. Let $\bar{Z} = \mathrm{Spec}(\mathcal{O}_{Z,\bar{x}})$ be the strict localization of Z at $\bar{x}$, let $\bar{f}$ be the image of f in $\mathcal{O}_{Z,\bar{x}}$, let $\bar{Y} = \bar{Z} \times_Z Y = \bar{Z} \cap D(f)$, and let $\bar{G}$ be the inverse image of G on $\bar{Y}$. By (SGAA VIII 5.2),

$$\mathrm{R}i_*(G)_{\bar{x}} = \mathrm{R}j_*(G)_{\bar{x}} \simeq \mathrm{R}\Gamma(\bar{Y}, \bar{G}).$$

Condition $(\mathrm{cdloc})_n$ therefore implies, by a spectral sequence argument, that

$$R^p i_*(G)_{\bar{x}} = 0 \quad \text{for } p > 2d + \dim.\mathrm{inj}(F_{\bar{x}}) + \dim(\mathcal{O}_{Z,x}).$$

But (cf. the proof of I 3.2.1)

$$\dim(\mathcal{O}_{Z,x}) = \dim(\mathcal{O}_{X,x}) - d \geq 0,$$

so

$$R^p i_*(G)_{\bar{x}} = 0 \quad \text{for } p > d + \dim(\mathcal{O}_{X,x}) + \dim.\mathrm{inj}(F_{\bar{x}}),$$

and a fortiori for $p > 2\dim(\mathcal{O}_{X,x}) + \dim.\mathrm{inj}(F_{\bar{x}})$. This proves $(*)$.

It remains to prove that the bound $(*)$ is the best possible. Put $f = \dim.\mathrm{inj}(F_{\bar{x}})$, and choose a finite type $A_{\bar{x}}$-Module M' such that

$$\mathrm{Ext}^f(M', F_{\bar{x}}) \neq 0.$$

Let Z be the integral closed sub-prescheme of X which is the closure of x, and let $Y = U \cap Z$ be a regular open subset of Z containing x and of codimension $d = \dim(\mathcal{O}_{X,x})$ at every point. After possibly shrinking Y, there exists a locally constant constructible A_Y-Module M such that $M_{\bar{x}} = M'$. Applying the purity theorem, one finds

$$\underline{\mathrm{Ext}}^{2d+f}(M, F|U)_{\bar{x}} \simeq \mathrm{Ext}^f(M'_{\bar{x}}, F_{\bar{x}}) \neq 0,$$

which completes the proof.

If F is an A_X-Module such that $F_{\bar{x}}$ is injective, then in the last part of the proof one may take $M = A_Y$; hence:

Corollary 5.1.1. Under the hypotheses of (5.1), suppose that F is a locally constant A_X-Module such that $F_{\bar{x}}$ is injective. Then

$$\dim.\mathrm{q.inj}_{\bar{x}}(F) = \dim.\mathrm{top}_{\bar{x}}(F) = 2\dim(\mathcal{O}_{X,x}),$$

with the notation of no. 2.

Remarks 5.1.2. a) With the notation of (5.1), if $F_{\bar{x}} = 0$, then $F = 0$ in a neighbourhood of x, hence $\mathrm{R}\underline{\mathrm{Hom}}(E, F)_{\bar{x}} = 0$ for all $E \in \mathrm{ob}\,\mathrm{D}^-(X)$.

b) If $\dim.\mathrm{inj}(F_{\bar{x}}) = +\infty$, then $\dim.\mathrm{q.inj}_{\bar{x}}(F) = +\infty$; indeed, for every p there exists a finite type $A_{\bar{x}}$-module M' such that $\mathrm{Ext}^p(M', F_{\bar{x}}) \neq 0$, hence a constructible A_X-Module M, locally constant in a neighbourhood of x, such that $\underline{\mathrm{Ext}}^p(M, F)_{\bar{x}} \neq 0$.

6. Constructibility of $\mathrm{R}\underline{\mathrm{Hom}}$

The equivalence of conditions a) of (I 3.5.1) remains valid when the Ring $(\mathbb{Z}/n\mathbb{Z})_X$ is replaced by a commutative Ring, locally constant, whose geometric fibres are noetherian rings: nothing has to be changed in the proof.

The preceding conditions are verified when one has resolution of singularities and purity in the strong sense (4.4), in particular if X has dimension ≤ 1, or if X is excellent of characteristic zero, or locally of finite type over a field and of dimension ≤ 2.

7. Existence of dualizing complexes

In this number, X denotes a prescheme equipped with a commutative sheaf of rings A_X, locally constant, whose geometric fibres are noetherian rings.

Lemma 7.1. If $F \in \operatorname{ob} \mathrm{D}^-_{\mathrm{coh}}(X)$ and $G \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$, then $\mathrm{R}\underline{\mathrm{Hom}}(F, G) \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$ (notation of (1.2)).

Proof. This is standard. One reduces, by a spectral sequence, to F and G concentrated in degree zero, and then argues as in (EGA 0_{III} 12.3.3).

Lemma 7.2. Let $g : X' \to X$ be a morphism of preschemes, and endow X' with the sheaf of rings $A_{X'} = g^*(A_X)$. Then, for $F \in \operatorname{ob} \mathrm{D}^-_{\mathrm{coh}}(X)$ and $G \in \operatorname{ob} \mathrm{D}^+(X)$, the canonical map

$$g^* \mathrm{R}\underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(g^*F, g^*G)$$

is an isomorphism.

Proof. This too is standard: one reduces by dévissage to $F = A_X$ (cf. EGA 0_{III} 12.3.5).

Remark 7.2.1. The preceding lemmas are special cases of more general statements appearing in (SGA 6 I).

Lemma 7.3. Suppose X is quasi-compact, and let $K \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$ be such that, for every geometric point $\bar{x}$ of X, $K_{\bar{x}}$ is a dualizing complex in the sense of [H] V 2.1. Then, for every $F \in \operatorname{ob} \mathrm{D}^b_{\mathrm{coh}}(X)$, one has $\mathrm{R}\underline{\mathrm{Hom}}(F, K) \in \operatorname{ob} \mathrm{D}^b_{\mathrm{coh}}(X)$, and F is reflexive relative to K.

Proof. It suffices, of course, to check this for F concentrated in degree zero. By (7.1), $\mathrm{R}\underline{\mathrm{Hom}}(F, K) \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$. To prove boundedness, let $\bar{x}$ be a geometric point of X. By (7.2),

$$\mathrm{R}\underline{\mathrm{Hom}}(F, K)_{\bar{x}} \simeq \mathrm{R}\,\mathrm{Hom}(F_{\bar{x}}, K_{\bar{x}}).$$

Putting $\dim.\mathrm{inj}(K_{\bar{x}}) = N(\bar{x})$, one obtains

$$\mathrm{Ext}^i(F, K)_{\bar{x}} = 0 \quad \text{for } i > N(\bar{x}).$$

Since the sheaves $\underline{\mathrm{Ext}}^i(F, K)$ are locally constant, there is an open neighbourhood U of x such that $\underline{\mathrm{Ext}}^i(F, K)|U = 0$ for $i > N(\bar{x})$, and quasi-compactness of X gives the assertion. It remains to check that the canonical arrow

$$F \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(\mathrm{R}\underline{\mathrm{Hom}}(F, K), K)$$

is an isomorphism. It is enough to evaluate at a geometric point $\bar{x}$ of X. In the commutative diagram

$$\begin{tikzcd}
F_{\bar{x}} \arrow[r] \arrow[d, "\mathrm{Id}"'] & \mathrm{R}\underline{\mathrm{Hom}}(\mathrm{R}\underline{\mathrm{Hom}}(F, K), K)_{\bar{x}} \arrow[d] \\
F_{\bar{x}} \arrow[r] & \mathrm{R}\,\mathrm{Hom}(\mathrm{R}\,\mathrm{Hom}(F_{\bar{x}}, K_{\bar{x}}), K_{\bar{x}}),
\end{tikzcd}$$

the right vertical arrow is an isomorphism by (7.2), and the bottom horizontal arrow is an isomorphism because $K_{\bar{x}}$ is dualizing; hence the top horizontal arrow is an isomorphism, q.e.d.

Lemma 7.4. Suppose X is a noetherian regular prescheme, A_X is killed by an integer $n > 0$ prime to the residual characteristics of X, and the purity theorem in the strong sense (4.4) is true on X.

Let D be a divisor with normal crossings on X (I 3.1.5 a)), and denote by $j : Y = V(D) \to X$ and $i : X - Y = U \to X$ the canonical inclusions. Let $K \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$ be such that, for every geometric point $\bar{x}$ of X, $K_{\bar{x}}$ is a dualizing complex for the ring $A_{\bar{x}}$ (in the sense of [H] V 2.1). Then, for every $E \in \operatorname{ob} \mathrm{D}^b_{\mathrm{coh}}(X)$, one has $\mathrm{R}\underline{\mathrm{Hom}}(i_! i^*(E), K) \in \operatorname{ob} \mathrm{D}^b_c(X)$, and the pair $(i_! i^*(E), K)$ is bidualizing.

Proof. We may suppose E is concentrated in degree zero, i.e. a locally constant constructible A_X-Module. Since the question is local on X for the étale topology, we may suppose that (a) A_X is constant with value A, and E is constant with value E; and (b) D has strict normal crossings, say $D = \sum_{i\in I} \operatorname{div}(s_i)$ with $s_i \in \Gamma(X, \mathcal{O}_X)$.

Argue by induction on $p = \operatorname{card}(I)$. For $p = 0$, the assertions follow from (7.3). Suppose $I = \{1, \dots, p\}$, put $Y_i = V(s_i)$, $Y' = V(\sum_{1\le i\le p-1} \operatorname{div}(s_i))$, and $U' = X - Y'$. We get a cartesian diagram of immersions

$$\begin{tikzcd} \varnothing \arrow[r] \arrow[d] & U' \cap Y_p \arrow[r, "i'_p"] \arrow[d, "j'_p"'] & Y_p \arrow[d, "j_p"] \\ U \arrow[r] & U' \arrow[r, "i'"'] & X. \end{tikzcd}$$

Consider the exact sequence

$$0 \longrightarrow E_{U,X} \longrightarrow E_{U',X} \longrightarrow E_{U'\cap Y_p,X} \longrightarrow 0.$$

By the induction hypothesis the theorem is true for $(E_{U',X}, K)$. For fixed K, the F such that $\mathrm{R}\underline{\mathrm{Hom}}(F, K) \in \operatorname{ob} \mathrm{D}^b_c(X)$ and (F, K) is bidualizing form a triangulated subcategory of $\mathrm{D}_c(X)$; hence we are reduced to the case $F = E_{U'\cap Y_p,X}$. But

$$E_{U'\cap Y_p,X} = j_{p*} i'_{p!}(E_{U'\cap Y_p}),$$

and, by (I 1.13), it is equivalent to prove the theorem for $(i'_{p!}(E_{U'\cap Y_p}), \mathrm{R}j^!_p(K))$ on Y_p. By purity, however,

$$\mathrm{R}j^!_p(K) \simeq j^*_p(K) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} (\mu_n)^{\otimes -1}_{Y_p}[-2].$$

Thus $\mathrm{R}j^!_p(K) \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(Y_p)$ and, for every geometric point $\bar{y}$ of Y_p, $(\mathrm{R}j^!_p(K))_{\bar{y}}$ is dualizing. The induction hypothesis then applies.

Theorem 7.5. Assume the following: X is a noetherian regular prescheme; A_X is killed by an integer $n > 0$ prime to the residual characteristics of X; X is strongly desingularizable (I 3.1.5), satisfies conditions (reg) (I 3.1.1) and $(\mathrm{cdloc})_n$ (I 3.1.3); and the purity theorem in the strong sense (4.4) is true on every prescheme smooth over X (conditions satisfied if X is excellent of characteristic zero, or of dimension ≤ 2 and smooth over a perfect field).

Then, if $K \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$ is such that $K_{\bar{x}}$ is dualizing (in the sense of [H] V 2.1) for every geometric point $\bar{x}$ of X, the complex K is dualizing.

Proof. We must show that K satisfies conditions (i) to (iii) of (I 1.7). For every geometric point $\bar{x}$ of X, $K_{\bar{x}}$ has finite injective dimension, and since $K \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$, the function $x \mapsto \dim .\mathrm{inj}(K_{\bar{x}})$ is locally constant; it is therefore bounded because X is noetherian. It follows from (5.1) that K has finite quasi-injective dimension.

By what was seen in no. 6, condition (ii) of loc. cit. is also satisfied.

It remains to prove condition (iii), i.e. that every $F \in \operatorname{ob} \mathrm{D}^b_c(X)$ is reflexive relative to K. By dévissage we may suppose F is concentrated in degree zero. Since the question is local on X for the étale topology, we may suppose A_X constant with value A. Dévissage (1.3) then reduces us to $F = f_* i_!(E_U)$, where $f : Y \to X$ is finite, Y is integral with generic point separable over its image, $i : U \to Y$ is a nonempty regular open, and E_U is an A_U-Module constant with value a finite type A-module E. Since X is strongly desingularizable, there exists (I 3.1.5) a projective birational morphism $h : Y' \to Y$ such that Y' is regular, h induces an isomorphism $h^{-1}(U) \to U$, and $h^{-1}(U)$

is the complement of a normal crossings divisor in Y'. Identifying $h^{-1}(U)$ with U via h, we have a commutative diagram

$$\begin{tikzcd} & Y' \arrow[dl, "i'"'] \arrow[d, "h"] \arrow[dr, "g"] & \\ U \arrow[r, "i"'] & Y \arrow[r, "f"'] & X. \end{tikzcd}$$

Consequently,

$$f_* i_!(E_U) \simeq f_* \mathrm{R}h_* i'_!(E_U) \simeq \mathrm{R}g_* i'_!(E_U) \simeq \mathrm{R}g_* i'_! i'^*(E_{Y'}).$$

By (I 1.13), it is enough to prove biduality for the pair $(i'_! i'^*(E_{Y'}), \mathrm{R}g^!(K))$. But, locally embedding Y' into a prescheme X' smooth over X and applying strong purity on X' (4.4), one finds that $\mathrm{R}g^!(K)$ is locally isomorphic, up to degree shift, to $g^*(K)$. Hence $\mathrm{R}g^!(K) \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(Y')$ and $\mathrm{R}g^!(K)_{\bar{x}}$ is dualizing for every geometric point $\bar{x}$ of Y'. It remains only to apply (7.4).

The preceding proof also shows that $\mathrm{R}\underline{\mathrm{Hom}}(F, K) \in \operatorname{ob} \mathrm{D}^b_c(X)$ (using the finiteness theorem for a proper morphism (SGAA XIV)), which will reassure the reader who did not trust the spiel of no. 6.

This completes the proof.

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Bibliography

[1] Verdier, J.-L., *Dimension des espaces localement compacts*, Note aux C.R. Ac. Sc. Paris, 261, pp. 5293–5296 (1965). See also: Verdier, J.-L., *Dualité dans la cohomologie des espaces localement compacts*, Séminaire Bourbaki, no. 300, 1965–66.

SGA A = SGA 4.

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SGA 5 — Exposé III
LEFSCHETZ FORMULA

by A. Grothendieck, written up by L. Illusie

In this exposé we prove the Lefschetz formula in étale cohomology, for correspondences between complexes of sheaves, in Verdier's formulation (cf. [18]). In an earlier version, distributed by the IHES about ten years ago, the formula was stated, under resolution hypotheses, but the required compatibilities were not proved. Deligne's finiteness theorems (SGA $4^{1/2}$, Finiteness) made it possible to dispense with these hypotheses, provided one works with schemes separated and of finite type over a field. As for the compatibilities, the editor has verified them (painfully).

Let us say at once that the Lefschetz formula in question here, just like the trace formula for constant coefficients (SGA $4^{1/2}$, Cycle), which it generalizes, is essentially trivial: it only translates certain compatibilities in the formalism of duality. Roughly speaking, it says that, under certain conditions, the trace of an endomorphism of a hypercohomology complex $R\Gamma(X, L)$, where L is a complex of sheaves on X, can be calculated as an "integral" of certain cohomology classes on $X \times X$. In practice, the endomorphisms one considers are associated with correspondences $C \subset X \times X$, and the classes recovered on the product have support in $C \cap \Delta$, where Δ is the diagonal; these classes are "local in nature" in a neighbourhood of the fixed points, which in principle facilitates their calculation. See 4.2 and 4.7 for precise statements. In fact, the calculation of these "local" classes in terms of more comprehensible invariants, without which the present formula would have no interest, has so far been undertaken only in very special situations (see 4.10, 4.11, and the following exposé, where in particular the case of the Frobenius correspondence on a curve is treated).

For lack of a good derived category of ℓ-adic sheaves (Jouanolou's thesis unfortunately not having been published), we work systematically with torsion coefficients (prime to the residual characteristics). Despite minor technical difficulties in the definition of traces, this point of view in fact leads to finer formulae, also more manageable insofar as, for the calculation of local terms, one wants to trivialize the sheaves involved by coverings (see notably (SGA $4^{1/2}$, Report), III B, and XII).

In no. 1, we fix the notation and recall various Künneth formulae from (SGA $4^{1/2}$, Finiteness), which play an essential role in what follows. We also introduce a certain number of fibred and cofibred categories, whose consideration later facilitates certain verifications of compatibilities. No. 2, which is very technical, may be skipped on first reading. Its main result (2.5) expresses a compatibility between external tensor product of RHom, direct image, and base change. In no. 3, we define cohomological correspondences between complexes, and show how they act on hypercohomology; the result quoted above makes it possible to describe this action in several equivalent ways. The core of the exposé is no. 4, where Verdier's pairing between cohomological correspondences is defined and the Lefschetz formula is proved, in a relative situation appreciably more general than the one usually considered, with a view to later applications to computations of local terms. We give the most evident corollaries for correspondences with constant coefficients. In no. 5, we indicate rapidly how to generalize to correspondences between complexes certain usual constructions on correspondences with constant coefficients (cf. [12]), such as passage to the dual, composition, and external tensor product. Most of the verifications, analogous to those in the preceding numbers, have been omitted. Finally, in the appendix, we paraphrase the preceding theory in the context of coherent sheaves, using the formalism of duality of [8]. Here one may work over any noetherian base S, provided that, when necessary, one imposes hypotheses of properness, finite tor-dimension, and tor-independence on the S-schemes considered. We show that the very general Lefschetz formula thereby obtained contains as a special case the Woods Hole formula [1], and we point out some applications.

Contents

2 Notation and recalls of Künneth formulae

1.1

We denote by S the spectrum of a field. Unless otherwise stated, the S-schemes considered in what follows will be S-schemes separated and of finite type.

We denote by A a commutative noetherian ring, annihilated by an integer invertible on S. If X is an S-scheme, we denote by $D(X) = D(X, A)$ the derived category of complexes of A-modules on X (for the étale topology), and by $D_{\mathrm{ctf}}(X)$ the full subcategory consisting of complexes of finite tor-dimension, with constructible cohomology. The category $D_{\mathrm{ctf}}(X)$ is a triangulated subcategory of $D(X)$. If $E, F \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, it follows readily from the definitions that $E \otimes^L F \in \operatorname{ob} D_{\mathrm{ctf}}(X)$. According to (SGA $4^{1/2}$, Finiteness Theorem 1.6, 1.7), one also has $\mathrm{RHom}(E, F) \in \operatorname{ob} D_{\mathrm{ctf}}(X)$.

1.2

Let $f : X \to Y$ be an S-morphism. One has trivially

$$f^* D_{\mathrm{ctf}}(Y) \subset D_{\mathrm{ctf}}(X),$$

and, by (loc. cit.),

$$Rf_* D_{\mathrm{ctf}}(X) \subset D_{\mathrm{ctf}}(Y), \qquad Rf_! D_{\mathrm{ctf}}(X) \subset D_{\mathrm{ctf}}(Y), \qquad Rf^! D_{\mathrm{ctf}}(Y) \subset D_{\mathrm{ctf}}(X),$$

(the inclusion relative to $Rf_!$ uses only the finiteness theorem for proper morphisms (SGA 4 XIV) and the sorites (SGA 4 XVII 5.2.11), and hence holds more generally for every compactifiable morphism whose fibres have bounded dimension). In what follows, we shall sometimes write f_*, $f_!$, $f^!$ instead of Rf_*, $Rf_!$, $Rf^!$; the hypotheses of 1.1 imply that these functors are defined on the whole derived category.

1.3

For $f : X \to S$, we shall put

$$K_X = f^! A,$$

and shall denote by D_X (or D) the functor $\mathrm{RHom}(-, K_X)$. According to (SGA $4^{1/2}$, Finiteness Theorem 4.3), the complex K_X is dualizing: for $E \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, one has canonically

$$E \xrightarrow{\sim} DDE.$$

1.4

Let $f : X \to Y$ be an S-morphism, $L \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D(Y)$. We put

$$\operatorname{Hom}_f^{(1)}(L, M) = \operatorname{Hom}(f^*M, L) \quad (\simeq \operatorname{Hom}(M, f_*L)),$$

$$\operatorname{Hom}_f^{(2)}(L, M) = \operatorname{Hom}(L, f^*M),$$

$$\operatorname{Hom}_f^{(3)}(L, M) = \operatorname{Hom}(f_!L, M) \quad (\simeq \operatorname{Hom}(L, f^!M)),$$

$$\operatorname{Hom}_f^{(4)}(L, M) = \operatorname{Hom}(f^!M, L).$$

The elements of $\operatorname{Hom}_f^{(i)}(L, M)$ will be called *arrows of type i from L to M over f* (or f-arrows of type i from L to M). The transitivity isomorphisms for the functors f^*, f_*, $f_!$, $f^!$ allow one to compose arrows of type i for fixed i, and one thus obtains certain fibred or cofibred categories over the category of S-schemes: for example, for $i = 1$ (resp. $i = 3$) one obtains the fibred and cofibred category considered in (SGA 4 XVII 4.1.3) (resp. (SGA 4 XVIII 3.1.13)), whose fibre over X is the category opposite to $D(X)$ (resp. $D(X)$).

1.5

Let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, and $L_i \in \operatorname{ob} D^-(X_i)$. We put

$$L_1 \otimes_S^L L_2 = \operatorname{pr}_1^* L_1 \otimes^L \operatorname{pr}_2^* L_2,$$

where $\operatorname{pr}_i : X \to X_i$ is the canonical projection.

1.6

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism,

$$f = f_1 \times_S f_2 : X = X_1 \times_S X_2 \to Y = Y_1 \times_S Y_2,$$

$L_i \in \operatorname{ob} D^-(X_i)$, $M_i \in \operatorname{ob} D^-(Y_i)$. The evident canonical isomorphism

$$f_1^*M_1 \otimes_S^L f_2^*M_2 \xrightarrow{\sim} f^*(M_1 \otimes_S^L M_2) \tag{1.6.1}$$

allows one to define, for $u_i \in \operatorname{Hom}_{f_i}^{(1)}(L_i, M_i)$ (resp. $\operatorname{Hom}_{f_i}^{(2)}(L_i, M_i)$),

$$u_1 \otimes_S^L u_2 \in \operatorname{Hom}_f^{(1)}(L, M) \quad (\text{resp. } \operatorname{Hom}_f^{(2)}(L, M)), \tag{1.6.2}$$

where $L = L_1 \otimes_S^L L_2$, $M = M_1 \otimes_S^L M_2$. If u_i, v_i are composable, one has

$$(v_1u_1) \otimes_S^L (v_2u_2) = (v_1 \otimes_S^L v_2)(u_1 \otimes_S^L u_2). \tag{1.6.3}$$

Taking $M_i = f_{i*}L_i$, and u_i of type 1, given by the identity of $f_{i*}L_i$, 1.6.2 gives a *Künneth arrow* (cf. SGA 4 XVII 5.4.1.4)

$$f_{1*}L_1 \otimes_S^L f_{2*}L_2 \longrightarrow f_*(L_1 \otimes_S^L L_2). \tag{1.6.4}$$

According to (SGA $4^{1/2}$, Finiteness Theorem 1.9), 1.6.4 is an isomorphism: by 1.6.3 one is indeed reduced to the case where $X_2 = Y_2$, $f_2 = 1$, and one applies (loc. cit. App.). When f_1 and f_2 are proper, it is not necessary to invoke (loc. cit.): 1.6.4 is a special case of the Künneth isomorphism (SGA 4 XVII 5.4.3).

1.7

Choose a compactification $f_i = p_i j_i$, where j_i is an open immersion and p_i is a proper morphism. Composing the evident canonical isomorphism (where $j = j_1 \times_S j_2$)

$$j_{1!}L_1 \otimes^L_S j_{2!}L_2 \xrightarrow{\sim} j_!(L_1 \otimes^L_S L_2)$$

with the isomorphism 1.6.4 relative to $(p_i, j_{i!}L_i)$, one obtains a Künneth isomorphism (SGA 4 XVII 5.4.3)

$$f_{1!}L_1 \otimes^L_S f_{2!}L_2 \xrightarrow{\sim} f_!(L_1 \otimes^L_S L_2). \tag{1.7.1}$$

One checks easily (loc. cit.) that 1.7.1 does not depend on the compactifications chosen, and is compatible with the transitivity isomorphisms for composites $g_i f_i$ (cf. (loc. cit. 5.2.4)). Thanks to 1.7.1, one can define, for $u_i \in \mathrm{Hom}^{(3)}_{f_i}(L_i, M_i)$,

$$u_1 \otimes^L_S u_2 \in \mathrm{Hom}^{(3)}_f(L, M) \tag{1.7.2}$$

with L and M as in 1.6.2, and one has a transitivity formula analogous to 1.6.3.

For $L_i = f_i^! M_i$, and u_i given by the identity of L_i, 1.7.2 gives a Künneth arrow

$$f_1^! M_1 \otimes^L_S f_2^! M_2 \longrightarrow f^!(M_1 \otimes^L_S M_2). \tag{1.7.3}$$

Proposition 1.7.4

The arrow 1.7.3 is an isomorphism.

By the transitivity formula, it suffices to treat the case where $X_2 = Y_2$, $f_2 = 1$. By dévissage, one reduces to $M_2 = g_* j_! A_V$, where $j : V \to U$ is an open immersion and $g : U \to Y_2$ a finite morphism, and then, by (SGA 4 XVIII 3.1.12.3), to the case where $M_2 = A_{Y_2}$. By localization on X_2, one may suppose moreover that $f_1 = g_1 j_1$, where j_1 is a closed immersion and g_1 a smooth morphism purely of relative dimension n_1. By transitivity, it suffices to treat the cases

$$\text{a) } f_1 = g_1, \qquad \text{b) } f_1 = j_1.$$

In case a), the trace morphism supplies by adjunction (SGA 4 XVIII 3.2.5) an isomorphism

$$f_1^* M_1(n_1)[2n_1] \xrightarrow{\sim} f_1^! M_1;$$

this is compatible with the base change $Y_2 \to S$, because the trace morphism is so (SGA 4 XVIII 2.9), whence 1.7.4 in this case. In case b), if i_1 denotes the inclusion of the complementary open, one has an exact sequence

$$0 \longrightarrow j_{1*} R j_1^! M_1 \longrightarrow M_1 \longrightarrow R i_{1*} i_1^* M_1 \longrightarrow 0,$$

and the compatibility of Ri_{1*} with base change, a special case of 1.6.4, gives the conclusion.

Thanks to 1.7.3, one can define, for $u_i \in \mathrm{Hom}^{(4)}_{f_i}(L_i, M_i)$,

$$u_1 \otimes^L_S u_2 \in \mathrm{Hom}^{(4)}_f(L, M) \tag{1.7.5}$$

(L and M as in 1.6.2), with a transitivity formula analogous to 1.6.3.

On the other hand, with the notation of 1.3, 1.7.3 gives an isomorphism

$$K_{X_1} \otimes^L_S K_{X_2} \xrightarrow{\sim} K_X. \tag{1.7.6}$$

1.8

Let $f : X \to Y$ be an S-morphism, $L \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D(Y)$, S' an S-scheme, $P \in \operatorname{ob} D(S')$, and $f' : X' \to Y'$ the morphism deduced from f by the base change $S' \to S$. The identity arrow of P may be regarded indifferently as an arrow of type i $(1 \le i \le 4)$ over the identity arrow of S'; for $u \in \operatorname{Hom}_f^{(i)}(L, M)$ $(1 \le i \le 4)$, one may therefore, by 1.6.2, 1.7.2, 1.7.5, consider

$$u \otimes_S^L \operatorname{Id}_P \in \operatorname{Hom}_{f'}^{(i)}(L \otimes_S^L P, M \otimes_S^L P). \tag{1.8.1}$$

We shall sometimes write $u \otimes_S^L P$ instead of $u \otimes_S^L \operatorname{Id}_P$.

3 Functoriality of RHom and external tensor products

2.1

Let $f : X \to Y$ be an S-morphism, $P \in \operatorname{ob} D(Y)$, $Q \in \operatorname{ob} D^+(Y)$. The evident canonical arrow at the level of complexes

$$f^*\mathcal{H}om^\bullet(P, Q) \longrightarrow \mathcal{H}om^\bullet(f^*P, f^*Q)$$

gives by derivation a canonical arrow

$$f^*\mathrm{RHom}(P, Q) \longrightarrow \mathrm{RHom}(f^*P, f^*Q). \tag{2.1.1}$$

Let $L \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D^+(X)$. Put

$$\operatorname{Hom}_f^{(i,j)}((L, M), (P, Q)) = \operatorname{Hom}_f^{(i)}(L, P) \times \operatorname{Hom}_f^{(j)}(M, Q), \tag{2.1.2}$$

with the notation of 1.4. For $(u, v) \in \operatorname{Hom}_f^{(2,1)}((L, M), (P, Q))$ one defines

$$\mathrm{RHom}(u, v) \in \operatorname{Hom}_f^{(1)}(\mathrm{RHom}(L, M), \mathrm{RHom}(P, Q)) \tag{2.1.3}$$

by composing 2.1.1 with the arrow $\mathrm{RHom}(f^*P, f^*Q) \to \mathrm{RHom}(L, M)$ defined by u and v. For $L = f^*P$, $Q = f_*M$, u (resp. v) given by the identity of L (resp. Q), $\mathrm{RHom}(u, v)$ is the "trivial duality" isomorphism

$$\mathrm{RHom}(P, f_*M) \xrightarrow{\sim} f_*\mathrm{RHom}(f^*P, M). \tag{2.1.4}$$

For $(u, v) \in \operatorname{Hom}_f^{(3,4)}((L, M), (P, Q))$, one defines

$$\mathrm{RHom}(u, v) \in \operatorname{Hom}_f^{(1)}(\mathrm{RHom}(L, M), \mathrm{RHom}(P, Q)) \tag{2.1.5}$$

as the composite of the arrows

$$\mathrm{RHom}(P, Q) \xrightarrow{a} \mathrm{RHom}(f_!L, Q) \xrightarrow{b} f_*\mathrm{RHom}(L, f^!Q) \xrightarrow{c} f_*\mathrm{RHom}(L, M),$$

where a (resp. c) is defined by u (resp. v), and b is the inverse of the duality isomorphism (SGA 4 XVIII 3.1.9.6).

It follows easily from the definitions that the formation of $\mathrm{RHom}(u, v)$ is compatible with composition: for composable (u, v), (u', v') of type $(2, 1)$ (resp. $(3, 4)$), one has

$$\mathrm{RHom}(u'u, v'v) = \mathrm{RHom}(u', v')\,\mathrm{RHom}(u, v). \tag{2.1.6}$$

2.2

Let X be an S-scheme. Recall that for $L \in \operatorname{ob} D^-(X)$, $M \in \operatorname{ob} D(X)$, $N \in \operatorname{ob} D^+(X)$, one has the adjunction isomorphism ("dear to Cartan," as Grothendieck says)

$$\operatorname{Hom}(L \otimes^L M, N) = \operatorname{Hom}(L, \operatorname{RHom}(M, N)). \tag{2.2.1}$$

For $E \in \operatorname{ob} D^-(X)$, $F \in \operatorname{ob} D^+(X)$, the identity arrow of $\operatorname{RHom}(E, F)$ defines via 2.2.1 an arrow

$$E \otimes^L \operatorname{RHom}(E, F) \longrightarrow F, \tag{2.2.2}$$

called the *evaluation arrow*: it is derived from the arrow at the level of complexes

$$\mathcal{H}om^\bullet(E, F) \otimes E \longrightarrow F, \qquad f \otimes x \longmapsto f(x).$$

Let, for $i = 1, 2$, $E_i, F_i \in \operatorname{ob} D_{\mathrm{ctf}}(X)$. By adjunction (2.2.1), the arrow

$$E_1 \otimes^L E_2 \otimes \operatorname{RHom}(E_1, F_1) \otimes \operatorname{RHom}(E_2, F_2) \longrightarrow F_1 \otimes^L F_2,$$

composed of a symmetry isomorphism and of the tensor product of the evaluation arrows relative to (E_1, F_1) and (E_2, F_2), supplies an arrow

$$\operatorname{RHom}(E_1, F_1) \otimes \operatorname{RHom}(E_2, F_2) \longrightarrow \operatorname{RHom}(E_1 \otimes^L E_2, F_1 \otimes^L F_2). \tag{2.2.3}$$

Let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, $p_i : X \to X_i$ the canonical projection, $E_i, F_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$, $E = E_1 \otimes_S^L E_2$, $F = F_1 \otimes_S^L F_2$. One has a canonical arrow

$$\operatorname{RHom}(E_1, F_1) \otimes_S^L \operatorname{RHom}(E_2, F_2) \longrightarrow \operatorname{RHom}(E, F), \tag{2.2.4}$$

obtained by composing the tensor product of the arrows $p_i^* \operatorname{RHom}(E_i, F_i) \to \operatorname{RHom}(p_i^* E_i, p_i^* F_i)$ and 2.2.3 for $(p_i^* E_i, p_i^* F_i)$.

Proposition 2.3

The arrow 2.2.4 is an isomorphism.

By localization and dévissage, one reduces to $E_i = f_{i!} A_{U_i}$ for $f_i : U_i \to X_i$ étale. One has canonical isomorphisms (duality)

$$\operatorname{RHom}(f_{i!} A_{U_i}, F_i) \xrightarrow{\sim} f_{i*} f_i^* F_i, \qquad \operatorname{RHom}(E, F) \xrightarrow{\sim} f_* f^* F \quad (f = f_1 \times_S f_2). \tag{1}$$

Taking into account the Künneth formula 1.6.4, it suffices, to conclude, to verify that these isomorphisms make the square

$$\begin{tikzcd}
\operatorname{RHom}(f_{1!} A_{U_1}, F_1) \otimes_S^L \operatorname{RHom}(f_{2!} A_{U_2}, F_2) \arrow[r, "2.2.3"] \arrow[d, "\sim"'] & \operatorname{RHom}(f_! A_U, F) \arrow[d, "\sim"] \\
f_{1*} f_1^* F_1 \otimes_S^L f_{2*} f_2^* F_2 \arrow[r, "1.6.4"'] & f_* f^* F
\end{tikzcd}$$

commute, where $U = U_1 \times_S U_2$. Since f_i is étale, the isomorphism (1) relative to f_i (resp. f) is adjoint to the composite arrow

$$f_i^* \operatorname{RHom}(f_{i!} A_{U_i}, F_i) \xrightarrow{\sim} \operatorname{RHom}(f_i^* f_{i!} A_{U_i}, f_i^* F_i) \longrightarrow f_i^* F_i, \tag{3}$$

where the second arrow is deduced from the adjunction $A_{U_i} \to f_i^* f_{i!} A_{U_i}$ (resp. the analogous composite arrow with f). The commutativity of (2) is equivalent to that of the square

$$\begin{tikzcd}
f_1^*\mathrm{RHom}(f_{1!}A_{U_1}, F_1) \otimes_S^L f_2^*\mathrm{RHom}(f_{2!}A_{U_2}, F_2) \arrow[r] \arrow[d] & f^*\mathrm{RHom}(f_!A_U, F) \arrow[d] \\
f_1^*F_1 \otimes_S^L f_2^*F_2 \arrow[r, "\sim"'] & f^*F,
\end{tikzcd}$$

where the vertical arrows are given by (3). But the commutativity of this square follows from the fact that 2.2.3 is functorial in E_i and reduces to the identity for $E_i = A$, whence the proposition.

2.4

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism,

$$f = f_1 \times_S f_2 : X \to Y,$$

$E_i, F_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$, $P_i, Q_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y_i)$,

$$E = E_1 \otimes_S^L E_2, \quad F = F_1 \otimes_S^L F_2, \quad P = P_1 \otimes_S^L P_2, \quad Q = Q_1 \otimes_S^L Q_2.$$

Consider the diagram

$$\begin{tikzcd}
(E_1, F_1) \arrow[d, "f_1"'] & X_1 & X_1 \times_S Y_2 \arrow[l] \arrow[d] & X \arrow[l] \arrow[d] \arrow[ddl, "f" description] & (E, F) \\
(P_1, Q_1) & Y_1 & Y \arrow[l] \arrow[d] & Y_1 \times_S X_2 \arrow[l] \arrow[d] & \\
S & & Y_2 \arrow[ll] & X_2 \arrow[l, "f_2"] & (E_2, F_2).
\end{tikzcd} \tag{2.4.0}$$

Give ourselves $(u_1, v_1) \in \mathrm{Hom}_{f_1}^{(3,4)}((E_1, F_1), (P_1, Q_1))$, $(u_2, v_2) \in \mathrm{Hom}_{f_2}^{(2,1)}((E_2, F_2), (P_2, Q_2))$ (notation of 2.1.2 and 1.4). By 1.8, one deduces from (u_i, v_i) a square

$$\begin{tikzcd}
(E, F) \arrow[d, "(u_1 \otimes_S^L E_2,\, v_1 \otimes_S^L F_2)"] & (E_1 \otimes_S^L P_2, F_1 \otimes_S^L Q_2) \arrow[l, "(E_1 \otimes_S^L u_2,\, F_1 \otimes_S^L v_2)"'] \arrow[d, "(u_1 \otimes_S^L P_2,\, v_1 \otimes_S^L Q_2)"] \\
(P_1 \otimes_S^L E_2, Q_1 \otimes_S^L F_2) \arrow[r, "(P_1 \otimes_S^L u_2,\, Q_1 \otimes_S^L v_2)"'] & (P, Q),
\end{tikzcd} \tag{2.4.1}$$

where the vertical arrows (resp. horizontal arrows) are of type $(3, 4)$ (resp. $(2, 1)$).

We may now state the main compatibility of this number, which will play an essential role in the verification of the Lefschetz–Verdier formula.

Proposition 2.5

The diagram 2.5.1 below, where the square on the right is deduced from 2.4.1 by applying RHom in the sense of 2.1.3 and 2.1.5, is commutative:

$$\begin{tikzcd}
\mathrm{RHom}(E_1, F_1) \otimes^L_S \mathrm{RHom}(E_2, F_2) \arrow[r, "2.2.4"] \arrow[d] & \mathrm{RHom}(E, F) \arrow[dr] & \\
\mathrm{RHom}(u_1, v_1) \otimes^L_S \mathrm{RHom}(u_2, v_2) \arrow[d] & & \mathrm{RHom}(E_1 \otimes^L_S P_2, F_1 \otimes^L_S Q_2) \arrow[d] \\
\mathrm{RHom}(P_1, Q_1) \otimes^L_S \mathrm{RHom}(P_2, Q_2) \arrow[r, "2.2.4"'] & \mathrm{RHom}(P, Q) & \mathrm{RHom}(P_1 \otimes^L_S E_2, Q_1 \otimes^L_S F_2) \arrow[l]
\end{tikzcd} \tag{2.5.1}$$

2.6

The proof of 2.5 will occupy the rest of no. 2. It follows from 1.6.3 that, if $w_i : L_i \to M_i$ is an arrow of type 1 over f_i, the arrow $w = w_1 \otimes^L_S w_2$ makes the diagram

$$\begin{tikzcd}
& L_1 \otimes^L_S L_2 \arrow[dl, "L_1 \otimes^L_S w_2"'] \arrow[dr, "w_1 \otimes^L_S L_2"] & \\
L_1 \otimes^L_S M_2 \arrow[dr, "w_1 \otimes^L_S M_2"'] & & M_1 \otimes^L_S L_2 \arrow[dl, "M_1 \otimes^L_S w_2"] \\
& M_1 \otimes^L_S M_2 &
\end{tikzcd}$$

commute. Applying this remark to $w_i = \mathrm{RHom}(u_i, v_i)$, one sees that it suffices to prove the following special cases of 2.5.

Corollary 2.6.1

Let $f : X \to Y$ be an S-morphism, S' an S-scheme, $f' : X' \to Y'$ the morphism deduced from f by the base change $S' \to S$, $E, F \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, $P, Q \in \operatorname{ob} D_{\mathrm{ctf}}(Y)$, $P', Q' \in \operatorname{ob} D_{\mathrm{ctf}}(S')$, and $(u, v) \in \mathrm{Hom}^{(2,1)}_f((E, F), (P, Q))$ (resp. $\mathrm{Hom}^{(3,4)}_f((E, F), (P, Q))$). Then one has a commutative square, whose vertical arrows are of type 1 over f':

$$\begin{tikzcd}
\mathrm{RHom}(E, F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "2.2.4"] \arrow[d, "\mathrm{RHom}(u,v) \otimes^L_S \mathrm{RHom}(P', Q')"'] & \mathrm{RHom}(E \otimes^L_S P', F \otimes^L_S Q') \arrow[d, "\mathrm{RHom}(u \otimes^L_S P', v \otimes^L_S Q')"] \\
\mathrm{RHom}(P, Q) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "2.2.4"'] & \mathrm{RHom}(P \otimes^L_S P', Q \otimes^L_S Q').
\end{tikzcd} \tag{2.6.1.1}$$

First examine the case where (u, v) is of type $(2, 1)$. By functoriality, one may suppose that $E = f^*P$, $F = f^*Q$, (u, v) being given by the identity of (f^*P, f^*Q). The commutativity of 2.6.1.1 in the case

$P' = Q' = A_{S'}$ is evident; to establish it in the general case, it therefore suffices to verify that, for $L, M \in \operatorname{ob} D_{\mathrm{ctf}}(Y)$, the square

$$\begin{tikzcd}
f^*\mathrm{RHom}(E,F) \otimes^L f^*\mathrm{RHom}(L,M) \arrow[r, "f^*(2.2.3)"] \arrow[d, "2.1.1\otimes 2.1.1"'] & f^*\mathrm{RHom}(E\otimes^L L, F\otimes^L M) \arrow[d, "2.1.1"] \\
\mathrm{RHom}(f^*E, f^*F) \otimes^L \mathrm{RHom}(f^*L, f^*M) \arrow[r, "2.2.3"'] & \mathrm{RHom}(f^*(E\otimes^L L), f^*(F\otimes^L M))
\end{tikzcd}$$

is commutative. But one sees immediately that the two composites of this square are adjoint to the tensor product of the evaluation arrows

$$f^*E \otimes^L f^*\mathrm{RHom}(E,F) \longrightarrow f^*F \quad \text{and} \quad f^*L \otimes^L f^*\mathrm{RHom}(L,M) \longrightarrow f^*M,$$

which proves the assertion.

Now pass to the case where (u,v) is of type $(3,4)$. By functoriality, one may suppose that $P = f_!E$, $F = f^!Q$, (u,v) being defined by $(\mathrm{Id}_P, \mathrm{Id}_F)$. One must prove the commutativity of the square

$$\begin{tikzcd}
f_*\mathrm{RHom}(E, f^!Q) \otimes^L_S \mathrm{RHom}(P',Q') \arrow[r, "b"] \arrow[d, "a\otimes\mathrm{RHom}(P',Q')"'] & f'_*\mathrm{RHom}(E\otimes^L_S P', f^!Q\otimes^L_S Q') \arrow[d, "a'"] \\
\mathrm{RHom}(f_!E, Q) \otimes^L_S \mathrm{RHom}(P',Q') \arrow[r, "2.2.4"'] & \mathrm{RHom}(f_!E\otimes^L_S P', Q\otimes^L_S Q'),
\end{tikzcd}$$

where b is the composite of $f'_*(2.2.4)$ and a base-change isomorphism, a is the duality isomorphism (SGA 4 XVIII 3.1.9.6), and a' is the composite of the duality isomorphism (relative to $E \otimes^L_S P'$ and $Q \otimes^L_S Q'$) and the base-change isomorphisms

$$f'_!(E\otimes^L_S P') \xrightarrow{\sim} f_!E \otimes^L_S P', \qquad f^!Q\otimes^L_S Q' \xrightarrow{\sim} f'^!(Q\otimes^L_S Q').$$

Recall that a is the composite of the canonical arrow (SGA 4 XVIII 3.1.9.5)

$$f_*\mathrm{RHom}(E, f^!Q) \longrightarrow \mathrm{RHom}(f_!E, f_!f^!Q)$$

and of the arrow defined by the adjunction arrow $f_!f^!Q \to Q$. There is an analogous description of a', as the composite of the quoted arrow

$$f'_*\mathrm{RHom}(E\otimes^L_S P', f'^!(Q\otimes^L_S Q')) \longrightarrow \mathrm{RHom}(f'_!(E\otimes^L_S P'), f'_!f'^!(Q\otimes^L_S Q')),$$

of the arrows defined by c and d, and of the arrow defined by the adjunction arrow

$$f'_!f'^!(Q\otimes^L_S Q') \longrightarrow Q\otimes^L_S Q'.$$

By definition, d is adjoint to the arrow

$$f'_!(f^!Q\otimes^L_S Q') \longrightarrow Q\otimes^L_S Q'$$

defined by the base-change isomorphism $f'_!(f^!Q\otimes^L_S Q') \simeq f_!f^!Q\otimes^L_S Q'$ and the adjunction arrow $f_!f^!Q \to Q$; in other words, the square

$$\begin{tikzcd}
f'_!(f^!Q\otimes^L_S Q') \arrow[r, "f'_!d"] \arrow[d, "\text{base ch.}\ \sim"'] & f'_!f'^!(Q\otimes^L_S Q') \arrow[d, "\text{adj}"] \\
f_!f^!Q\otimes^L_S Q' \arrow[r, "\text{adj}\otimes 1"'] & Q\otimes^L_S Q'
\end{tikzcd}$$

is commutative. It follows that one may also describe a' as the composite of the arrow above, of the arrow defined by c, and of those defined by the base-change isomorphism

$$f'_!(f^!Q \otimes^L_S Q') \xrightarrow{\sim} f_!f^!Q \otimes^L_S Q'$$

and the adjunction arrow $f_!f^!Q \to Q$. This decomposition of a' and that of a given above, together with the commutativity of the square

$$\begin{tikzcd}
\mathrm{RHom}(f_!E, f_!f^!Q) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r] \arrow[d] & \mathrm{RHom}(f_!E \otimes^L_S P', f_!f^!Q \otimes^L_S Q') \arrow[d] \\
\mathrm{RHom}(f_!E, Q) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r] & \mathrm{RHom}(f_!E \otimes^L_S P', Q \otimes^L_S Q'),
\end{tikzcd}$$

defined by 2.2.4 and the adjunction arrow $f_!f^!Q \to Q$, show that, to establish the commutativity of 2.6.2, it suffices to prove the following compatibility.

Lemma 2.6.3

Let f, f', E, F, P', Q' be as in 2.6.1. Then one has a commutative square

$$\begin{tikzcd}
f_*\mathrm{RHom}(E, F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "(1)"] \arrow[d, "(2)"'] & f'_*\mathrm{RHom}(E \otimes^L_S P', F \otimes^L_S Q') \arrow[d, "(4)"] \\
\mathrm{RHom}(f_!E, f_!F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "(3)"'] & \mathrm{RHom}(f_!E \otimes^L_S P', f_!F \otimes^L_S Q'),
\end{tikzcd}$$

where (1) is the composite of the base-change isomorphism

$$f_*\mathrm{RHom}(E, F) \otimes^L_S - \xrightarrow{\sim} f'_*(\mathrm{RHom}(E, F) \otimes^L_S -)$$

and f'_*(2.2.4), (2) is deduced from the canonical arrow

$$f_*\mathrm{RHom}(E, F) \longrightarrow \mathrm{RHom}(f_!E, f_!F)$$

(SGA 4 XVIII 3.1.9.5) by applying $\otimes^L_S \mathrm{RHom}(P', Q')$, (3) is given by 2.2.4, and finally (4) is the composite of the canonical arrow

$$f'_*\mathrm{RHom}(E \otimes^L_S P', F \otimes^L_S Q') \longrightarrow \mathrm{RHom}(f'_!(E \otimes^L_S P'), f'_!(F \otimes^L_S Q'))$$

(loc. cit.) and the arrows defined by the base-change isomorphisms

$$f'_!(E \otimes^L_S P') \xrightarrow{\sim} f_!E \otimes^L_S P', \qquad f'_!(F \otimes^L_S Q') \xrightarrow{\sim} f_!F \otimes^L_S Q'.$$

Returning to the definition of the arrow (SGA 4 XVIII 3.1.9.5), choose a compactification of f[8]; one is then reduced to proving the lemma separately in the case where f is an open immersion and in the case where f is a proper morphism.

[8]The arrow quoted is defined with the aid of such a compactification, but it does not depend on it: this point, tacitly admitted in (loc. cit.), is easily verified.

a) Case where f is an open immersion. By adjunction (2.2.1), it suffices to see that, if one tensors the square 2.6.3 by $f_!E \otimes^L_S P'$ and composes with the evaluation arrow 2.2.2

$$(f_!E \otimes^L_S P') \otimes^L \mathrm{RHom}(f_!E \otimes^L_S P', f_!F \otimes^L_S Q') \longrightarrow f_!F \otimes^L_S Q',$$

the square obtained is commutative. But the sums of this square are zero outside X', so that it suffices to verify that the square induced on X' commutes, which is trivial.

b) Case where f is a proper morphism. Then $f_! = f_*$, $f'_! = f'_*$. One sees easily that the composite of

$$f_*\mathrm{RHom}(E, F) \longrightarrow \mathrm{RHom}(f_*E, f_*F)$$

with the trivial-duality isomorphism

$$\mathrm{RHom}(f_*E, f_*F) \xrightarrow{\sim} f_*\mathrm{RHom}(f^*f_*E, F)$$

is none other than the arrow deduced from the adjunction arrow $f^*f_*E \to E$ by applying $f_*\mathrm{RHom}(-, F)$. Similarly, the composite of (4) with the trivial-duality isomorphism

$$\mathrm{RHom}(f_*E \otimes^L_S P', f_*F \otimes^L_S Q') \xrightarrow{\sim} f'_*\mathrm{RHom}(f^*f_*E \otimes^L_S P', F \otimes^L_S Q')$$

($f_*F \otimes^L_S Q'$ being identified with $f'_*(F \otimes^L_S Q')$ by the canonical isomorphism) is none other than the arrow deduced from the adjunction arrow $f^*f_*E \to E$ by applying $f'_*\mathrm{RHom}(- \otimes^L_S P', F \otimes^L_S Q')$. Thus one is reduced to verifying the commutativity of the square

$$\begin{tikzcd}
\mathrm{RHom}(f_*E, f_*F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "(3)"] \arrow[d] & \mathrm{RHom}(f_*E \otimes^L_S P', f_*F \otimes^L_S Q') \arrow[d] \\
f_*\mathrm{RHom}(f^*f_*E, F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "(1')"'] & f'_*\mathrm{RHom}(f^*f_*E \otimes^L_S P', F \otimes^L_S Q'),
\end{tikzcd}$$

where the vertical arrows are the trivial-duality isomorphisms and $(1')$ is (1) with E replaced by f^*f_*E. But this is a special case of 2.6.1.1 in the case, already treated, where (u, v) is of type $(2, 1)$ (take $u : f^*f_*E \to f_*E$ given by the identity of f^*f_*E, and $v : F \to f_*F$ given by the identity of f_*F). We have therefore proved 2.6.3. As we have seen, the commutativity of 2.6.2 follows, which completes the proof of 2.6.1 and, finally, of 2.5.

4 Cohomological Correspondences

3.1

Let, for $i = 1, 2$, X_i be an S-scheme and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$, and let

$$X = X_1 \times_S X_2, \qquad p_i : X \longrightarrow X_i, \qquad q_i : X_i \longrightarrow S$$

be the canonical projections:

$$\begin{tikzcd} & X \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X_1 \arrow[dr, "q_1"'] & & X_2 \arrow[dl, "q_2"] \\ & S & \end{tikzcd}$$

With the notation of 1.3, one has a canonical isomorphism (2.2.4)

$$DL_1 \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} \mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, q_1^! A_S \overset{\mathbf{L}}{\otimes}_S L_2).$$

Composing with the Künneth isomorphism

$$q_1^! A_S \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} p_2^! L_2 \qquad (1.7.3),$$

one obtains an isomorphism

$$DL_1 \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} \mathrm{RHom}(p_1^* L_1, p_2^! L_2). \tag{3.1.1}$$

This isomorphism will play a fundamental role throughout. For $X_i = S$, 3.1.1 reduces to

$$L_1^\vee \overset{\mathbf{L}}{\otimes} L_2 \xrightarrow{\sim} \mathrm{RHom}(L_1, L_2),$$

a special case of an isomorphism valid for perfect complexes on an arbitrary ringed topos (SGA 6 I 7.7). We set

$$\mathrm{RHom}(p_1^* L_1, p_2^! L_2) = \mathrm{RHom}_S(L_1, L_2), \qquad \mathrm{Hom}(p_1^* L_1, p_2^! L_2) = \mathrm{Hom}_S(L_1, L_2). \tag{3.1.2}$$

The elements of $\mathrm{Hom}_S(L_1, L_2)$ will be called *cohomological correspondences from L_1 to L_2*. Instead of "cohomological correspondence from A_{X_1} to A_{X_2}", we shall simply say *cohomological correspondence from X_1 to X_2* (understood: with coefficients in A).

If X_1 is smooth, purely of relative dimension r_1, then the same is true of p_2, and by the duality theorem (SGA 4 XVIII 3.2.5) one has

$$p_2^! L_2 \xrightarrow{\sim} p_2^* L_2(r_1)[2r_1],$$

therefore

$$\mathrm{RHom}_S(L_1, L_2) = \mathrm{RHom}(p_1^* L_1, p_2^* L_2)(r_1)[2r_1], \qquad \mathrm{Hom}_S(L_1, L_2) = \mathrm{H}^{2r_1}(X, \mathrm{RHom}(p_1^* L_1, p_2^* L_2)(r_1)). \tag{3.1.3}$$

In particular, the group of cohomological correspondences from X_1 to X_2 is identified in this case with $\mathrm{H}^{2r_1}(X, A(r_1))$.

3.2

The cohomological correspondences met in practice are associated with S-morphisms $X_2 \to X_1$, or more generally with cycles on X, or with schemes over X. Let

$$c : C \longrightarrow X$$

be a scheme over X (one sometimes says that c is a correspondence between X_1 and X_2), and put $c_i = p_i c : C \to X_i$ for the projections. We shall call *cohomological correspondences from L_1 to L_2 with support in c* (or in C, if there is no risk of confusion) the elements of the group

$$\mathrm{H}^0(C, \mathrm{RHom}(c_1^* L_1, c_2^! L_2)) \quad (= \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \text{ if } c \text{ is proper}).$$

By the induction formula (SGA 4 XVIII 3.1.12.2), one has a canonical isomorphism

$$c^! \mathrm{RHom}(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \mathrm{RHom}(c_1^* L_1, c_2^! L_2), \tag{3.2.1}$$

whence, applying $c_!$ and composing with the adjunction arrow $c_! c^! \to \mathrm{Id}$, canonical arrows

$$c_! \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \longrightarrow \mathrm{RHom}(p_1^* L_1, p_2^! L_2), \tag{3.2.2}$$

$$\mathrm{H}^0(C, \mathrm{RHom}(c_1^* L_1, c_2^! L_2)) \longrightarrow \mathrm{Hom}_S(L_1, L_2). \tag{3.2.3}$$

Thus every correspondence from L_1 to L_2 with support in C canonically defines, by 3.2.3, a correspondence (without support) from L_1 to L_2.

Examples. a) Suppose that c is proper and that c_2 has finite tor-dimension and relative dimension $\leq N$. The trace morphism (SGA $4^{1/2}$ Cycle 2.3.3)

$$\mathrm{Tr}_{c_2} : c_{2!} A_C \longrightarrow A_{X_2}(-N)[-2N]$$

defines by adjunction a cohomological correspondence from A_{X_1} to $A_{X_2}(-N)[2N]$ with support in C, which will be called the *class of* (c, N), and will be denoted

$$\mathrm{cl}(c, N) \in \mathrm{H}^{-2N}(C, c_2^! A_{X_2}(-N)). \tag{3.2.4}$$

The most important case is that in which $N = 0$, i.e. c_2 is finite (and of finite tor-dimension); in that case $\mathrm{cl}(c)$ is a cohomological correspondence from X_1 to X_2 with support in C. If X_1 is smooth, purely of relative dimension r_1, then, by 3.1.3 and 3.2.1,

$$\mathrm{H}^{-2N}(C, c_2^! A_{X_2}(-N)) = \mathrm{H}^{2(r_1 - N)}(C, c^! A_X(r_1 - N)). \tag{3.2.5}$$

If moreover c is a closed immersion, the second member rewrites as

$$\mathrm{H}_C^{2(r_1 - N)}(X, A_X(r_1 - N)),$$

and the class of (c, N), considered as an element of this group, is precisely the cohomology class associated with c in the sense of (SGA $4^{1/2}$ Cycle 2.3.1), as follows immediately from the definitions.

b) Let $F : X_2 \to X_1$ be an S-morphism, and denote by $c(F) : C(F) \to X$ its graph, i.e. the image of the section of p_2 defined by F ("the set of points (Fx, x)"): c_2 is an isomorphism and $c_1 c_2^{-1} = F$. One therefore has canonical isomorphisms

$$\mathrm{RHom}(c_1(F)^* L_1, c_2(F)^! L_2) = \mathrm{RHom}(F^* L_1, L_2), \qquad \mathrm{Hom}(c_1(F)^* L_1, c_2(F)^! L_2) = \mathrm{Hom}(F^* L_1, L_2). \tag{3.2.6}$$

In other words, the cohomological correspondences from L_1 to L_2 with support in the graph of F are the "lifts" of F to a homomorphism $F^*L_1 \to L_2$. Here are two important examples of such correspondences:

(i) $X_1 = X_2$, $L_1 = L_2$, $F = 1$ is the identity of X_1, whose graph Δ is the diagonal. The correspondence with support in Δ defined by the identity of L_1 is called the *diagonal correspondence*. It is the element

$$1 \in \mathrm{Hom}(L_1, L_1) = \mathrm{H}^0(X, \Delta_X^! \mathrm{RHom}_S(L_1, L_1))$$

(a group that is identified with $\mathrm{H}^{2r}_{\Delta}(X, \mathrm{RHom}_S(L_1, L_1)(r))$ when X_1 is smooth, purely of relative dimension r).

(ii) Suppose that S is the spectrum of a finite field $\mathbb{F}_q$, that $X_1 = X_2$, $L_1 = L_2$, and take for $F : X_1 \to X_1$ the relative Frobenius endomorphism over $\mathbb{F}_q$, the identity on the underlying space and raising to the q-th power on $\mathcal{O}_{X_1}$ (cf. XIV), and take as lift of F to L_1 the canonical isomorphism

$$F^*L_1 \xrightarrow{\sim} L_1$$

(loc. cit. §2, Prop. 1) (more precisely, the a-th iterate of the isomorphism of (loc. cit.) if $q = p^a$). The cohomological correspondence from L_1 to L_1 with support in the graph of F so defined is called the *Frobenius correspondence* (relative to $\mathbb{F}_q$).

c) Certain cohomological correspondences between sheaves on curves, related to Hecke operators, have been studied by Langlands in ([13] §7).

d) Cohomological correspondences between constant sheaves (the most frequent ones in nature) furnish in a natural way, by "integration", cohomological correspondences between complexes, as we shall now see.

3.3

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism, and put

$$f = f_1 \times_S f_2 : X \to Y, \qquad f'_1 = f_1 \times_S Y_2, \qquad f'_2 = Y_1 \times_S f_2,$$

$$p'_1 = X_1 \times_S f_2, \qquad p'_2 = f_1 \times_S X_2.$$

$$\begin{tikzcd}
X_1 \times_S Y_2 & & X \arrow[ll, "p'_1"'] \arrow[dll, "f"'] \arrow[d, "p'_2"] \\
Y \arrow[d, equal] & & Y_1 \times_S X_2 \arrow[ll, "f'_2"] \\
Y_1 \times_S Y_2 & &
\end{tikzcd} \tag{3.3.0}$$

Let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. There exists a unique isomorphism

$$f_*\mathrm{RHom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) \tag{3.3.1}$$

whose inverse makes the square

$$\begin{tikzcd}
D(f'_{1!}L_1) \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2 \arrow[r, "3.1.1"] & \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) \arrow[d] \\
f_{1!}DL_1 \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2 \arrow[r, "(2)"'] & f_*\mathrm{RHom}_S(L_1, L_2),
\end{tikzcd}$$

commutative, where (1) is deduced from the inverse of the duality isomorphism (SGA 4 XVIII 3.1.9.6)

$$Df'_{1!}L_1 \xrightarrow{\sim} f_{1*}DL_1$$

by applying $- \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2$, and (2) is the composite of the Künneth isomorphism

$$f_{1*}DL_1 \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2 \xrightarrow{\sim} f_*(DL_1 \overset{\mathbf{L}}{\otimes}_S L_2)$$

and $f_*(3.1.1)$. Applying $R\Gamma(Y, -)$ to 3.3.1, one obtains an isomorphism

$$f_* : \mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_S(f'_{1!}L_1, f'_{2*}L_2). \tag{3.3.2}$$

For $u \in \mathrm{Hom}_S(L_1, L_2)$, the correspondence f_*u will be called the *direct image* of u by f. When $Y_1 = Y_2 = S$, we shall write u_* rather than f_*u. If $Y_1 = Y_2 = S$ is the spectrum of a separably closed field, 3.3.2 reads

$$\mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}(R\Gamma_c(X_1, L_1), R\Gamma(X_2, L_2)). \tag{3.3.3}$$

Thus cohomological correspondences "operate" on cohomology. The following statement will allow us to interpret this operation in several ways, and in particular to relate it to the usual definitions in situations such as a) or b) above.

Proposition 3.4. With the notation of 3.3.0, one has a commutative diagram of isomorphisms in $D_{\mathrm{ctf}}(Y)$:

$$\begin{tikzcd}
 & \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) \arrow[dl, "(1)"'] \arrow[d] \arrow[dr, "(2)"] & \\
f'_{1*}\mathrm{RHom}_S(L_1, f'_{2*}L_2) \arrow[dr, "f'_{1*}(3)"'] & f_*\mathrm{RHom}_S(L_1, L_2) & f'_{2*}\mathrm{RHom}_S(f_{1!}L_1, L_2) \arrow[dl, "f'_{2*}(4)"] \\
 & f_*\mathrm{RHom}_S(L_1, L_2) &
\end{tikzcd} \tag{3.4.1}$$

where (5) is the inverse of 3.3.1 and the arrows (1) to (4) are defined by the commutative diagrams below; in these diagrams "base ch." denotes an arrow deduced from a base-change or Künneth isomorphism of one of the types considered in no. 1, "dual." an arrow deduced from a duality isomorphism of type 2.1.4 or (SGA 4 XVIII 3.1.9.6), and "triv." a trivial isomorphism:

$$\begin{tikzcd}
\mathrm{RHom}(f'_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2) \arrow[r, "\text{base ch.}"] \arrow[d, "\text{base ch.}"'] & \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) \\
\mathrm{RHom}(f'_{1!}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}), K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2) \arrow[d, "\text{dual.}"'] & \\
f'_{1*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, f'^{!}_1(K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2)) \arrow[d, "\text{base ch.}"'] & \\
f'_{1*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2) \arrow[r, "\text{base ch.}"'] & f'_{1*}\mathrm{RHom}_S(L_1, f'_{2*}L_2),
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[r, "\text{base ch.}"] \arrow[d, "\text{base ch.}"'] & \mathrm{RHom}_S(f_{1!}L_1, L_2) \\
\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, f'_{2*}(K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2)) \arrow[d, "\text{dual.}"'] & \\
f'_{2*}\mathrm{RHom}(f'^{*}_2(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}), K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[d, "\text{triv.}"'] & \\
f'_{2*}\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[r, "\text{base ch.}"'] & f'_{2*}\mathrm{RHom}_S(f_{1!}L_1, L_2),
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2) \arrow[r, "\text{base ch.}"] \arrow[d, "\text{base ch.}"'] & \mathrm{RHom}_S(L_1, f_{2*}L_2) \\
\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, p'_{1*}(K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2)) \arrow[d, "\text{dual.}"'] & \\
p'_{1*}\mathrm{RHom}(p'^{*}_1(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}), K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[d, "\text{triv.}"'] & \\
p'_{1*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[r, "\text{base ch.}"'] & p'_{1*}\mathrm{RHom}_S(L_1, L_2),
\end{tikzcd}$$

$$\begin{array}{ccc}
\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) & \xrightarrow{\text{base ch.}} & \mathrm{RHom}_S(f_{1!}L_1, L_2) \\
\big\downarrow{\scriptstyle \text{base ch.}} & & \\
\mathrm{RHom}(p'_{2!}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}), K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) & & \\
\big\downarrow{\scriptstyle \text{dual.}} & & \\
p'_{2*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, p'^{!}_2(K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2)) & & \\
\big\downarrow{\scriptstyle \text{base ch.}} & & \\
p'_{2*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2) & \xrightarrow[\text{base ch.}]{} & p'_{2*}\mathrm{RHom}_S(L_1, L_2).
\end{array}$$

This is only an explication of 2.5 in the particular case where $(u_1, v_1) : (L_1, K_{X_1}) \to (f_{1!}L_1, K_{Y_1})$ is given by the identity of $f_{1!}L_1$ and the canonical isomorphism $f_1^! K_{Y_1} \xrightarrow{\sim} K_{X_1}$, and where $(u_2, v_2) : (A_{X_2}, L_2) \to (A_{Y_2}, f_{2*}L_2)$ is given by the canonical isomorphism $A_{X_2} \xrightarrow{\sim} f_2^! A_{Y_2}$ and the identity of $f_{2*}L_2$.

Corollary 3.5. Assume that $Y_1 = Y_2 = S$. Let $u \in \mathrm{Hom}_S(L_1, L_2)$. The direct image $f_*u \in \mathrm{Hom}(f_{1!}L_1, f_{2*}L_2)$ (3.3.2) can be obtained in the following two equivalent ways:

a) Identify u, by trivial duality, with an arrow

$$L_1 \longrightarrow p_{1*}p_2^! L_2;$$

compose with the base-change isomorphism

$$p_{1*}p_2^! L_2 \xrightarrow{\sim} f_1^! f_{2*} L_2;$$

f_*u is the adjoint of this composite arrow.

b) Identify u, by adjunction, with an arrow

$$p_{2!}p_1^* L_1 \longrightarrow L_2;$$

compose with the base-change isomorphism

$$f_2^* f_{1!} L_1 \xrightarrow{\sim} p_{2!}p_1^* L_1;$$

f_*u is the adjoint of this composite arrow.

Description a) (respectively b)) indeed corresponds to the path $(f_{1*}(3) \circ (1))^{-1}$ (respectively $f_{2*}(4) \circ (2)$) in 3.4.1.

3.6

In practice, it is mainly description 3.5 b) that is used. Let us spell it out at the level of cohomology classes, assuming, in order to lighten the notation, that S is the spectrum of a separably closed field. Let

$$u \in \mathrm{Hom}_S(L_1, L_2) = \mathrm{Hom}(p_1^* L_1, p_2^! L_2).$$

For $x \in \mathrm{H}^i_c(X_1, L_1)$, $u_*(x) \in \mathrm{H}^i(X_2, L_2)$ is defined as follows: consider the element $p_1^* x \in \mathrm{H}^i(X_2, p_{2!}p_1^* L_1)$ deduced from x by inverse image, apply u to it, i.e. form the product

$$u \cdot p_1^* x \in \mathrm{H}^i(X_2, p_{2!}p_2^! L_2),$$

then apply the arrow

$$a : \mathrm{H}^i(X_2, p_{2!}p_2^! L_2) \longrightarrow \mathrm{H}^i(X_2, L_2)$$

defined by the adjunction arrow $p_{2!}p_2^! \to \mathrm{Id}$; in short:

$$u_*(x) = a(u \cdot p_1^* x). \tag{3.6.1}$$

Let $c : C \to X$ be a proper morphism, and let $u \in \mathrm{Hom}(c_1^* L_1, c_2^! L_2)$. Denote by

$$u_* \in \mathrm{Hom}\big(R\Gamma_c(X_1, L_1), R\Gamma(X_2, L_2)\big)$$

the image of u under the composite of 3.2.3 and 3.3.3. It follows from 3.6.1 that, for $x \in \mathrm{H}^i_c(X_1, L_1)$, one has

$$u_*(x) = a(u \cdot c_1^* x), \tag{3.6.2}$$

where $c_1^* x$ is the inverse image of x in $\mathrm{H}^i(X_2, c_{2!}c_1^* L_1)$ and

$$a : \mathrm{H}^i(X_2, c_{2!}c_2^! L_2) \longrightarrow \mathrm{H}^i(X_2, L_2)$$

is the arrow deduced from the adjunction $c_{2!}c_2^! L_2 \to \mathrm{Id}$. In the situation of Example 3.2 a), one thus finds

$$\mathrm{cl}(c, N)_*(x) = \mathrm{Tr}_{c_2}(c_1^* x), \tag{3.6.3}$$

where

$$\mathrm{Tr}_{c_2} : \mathrm{H}^i(X_2, Rc_{2!}A_C) \longrightarrow \mathrm{H}^{i-2N}(X_2, A(-N))$$

is the arrow induced by the trace morphism relative to c_2. For $N = 0$, this formula agrees with (SGA $4^{1/2}$ Cycle 3.6): $\mathrm{cl}(c)_*$ is the homomorphism denoted γ^*_{X,X_1} there. If X_1 is smooth, purely of relative dimension r_1, and c is a closed immersion, it follows from 3.6.1 and from the remark made at the end of 3.2 a) that one also has

$$\mathrm{cl}(c, N)_*(x) = \mathrm{Tr}_{p_2}(\mathrm{cl}(c)\, p_1^* x), \tag{3.6.4}$$

where $\mathrm{cl}(c) \in \mathrm{H}^{2(r_1-N)}_C(X, A(r_1 - N))$ is the cohomology class of C and

$$\mathrm{Tr}_{p_2} : \mathrm{H}^*(X, Rp_{2!}A_X) \longrightarrow \mathrm{H}^{*-2r_1}(X_2, A(-r_1))$$

is the arrow defined by the trace morphism relative to p_2. This recovers the usual definition of the homomorphism induced on cohomology by a correspondence ([12] 1.3), (SGA $4^{1/2}$ Cycle 3.2).

3.6.5

In the situation of 3.2 b), it follows from 3.6.2 that, for $u \in \mathrm{Hom}(F^* L_1, L_2)$,

$$u_* : \mathrm{H}^i_c(X_1, L_1) \longrightarrow \mathrm{H}^i(X_2, L_2)$$

is the composite of the "inverse image" homomorphism

$$\mathrm{H}^i_c(X_1, L_1) \longrightarrow \mathrm{H}^i(X_2, F^* L_1)$$

and the homomorphism induced by u (more generally, $f_* u : Rf_{1!}L_1 \to Rf_{2*}L_2$ is the composite of the canonical homomorphism

$$Rf_{1!}L_1 \longrightarrow Rf_{2*}F^*L_1$$

and of $Rf_{2*}u$). In particular, the diagonal correspondence induces the canonical arrow

$$Rf_{1!}L_1 \longrightarrow Rf_{1*}L_1$$

(the identity if X_1 is proper over S).

3.7

Return to the situation of the beginning of 3.3, assuming in addition that a commutative square

$$\begin{tikzcd} X & C \arrow[l, "c"'] \arrow[d, "g"] \\ Y \arrow[u, "f"] & D \arrow[l, "d"] \end{tikzcd} \tag{3.7.1}$$

is given, with c proper. Put $c_i = p_i c$, $d_i = q_i d$, where $p_i : X \to X_i$, $q_i : Y \to Y_i$ are the projections. From 3.7.1 one deduces a commutative diagram whose inner square is cartesian:

$$\begin{tikzcd} & C \arrow[dl, "c"'] \arrow[d, "i"] \arrow[dr, "g"] & \\ X & C' \arrow[l, "c'"'] \arrow[d, "g'"] & D \arrow[l] \\ Y & D \arrow[l, "d"] & \end{tikzcd} \tag{3.7.2}$$

For $K \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, there is a canonical arrow

$$f_* c_! c^! K \longrightarrow d_* d^! f_* K, \tag{3.7.3}$$

which is the composite of the arrow

$$f_* c_! c^! K = f_* c'_* (i_* i^!) c'^! K \longrightarrow f_* c'_! c'^! K = d_* g'_! c'^! K$$

defined by the adjunction arrow $i_* i^! \to \mathrm{Id}$, and the isomorphism

$$d_* g'_! c'^! K \xrightarrow{\sim} d_* d^! f_* K$$

defined by the canonical isomorphism $g'_! d'^! \xrightarrow{\sim} d^! f_*$ (SGA 4 XVIII 3.1.12.3). Taking $K = \mathrm{RHom}_S(L_1, L_2)$, and composing 3.7.3 with the isomorphism deduced from 3.3.1, one obtains a canonical arrow

$$f_* c_! c^! \mathrm{RHom}_S(L_1, L_2) \longrightarrow d_* d^! \mathrm{RHom}_S(f_{1!}L_1, f_{2*}L_2), \tag{3.7.4}$$

which, by 3.2.1, may be rewritten as

$$f_* c_! \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \longrightarrow d_* \mathrm{RHom}(d_1^*(f_{1!}L_1), d_2^!(f_{2*}L_2)). \tag{3.7.5}$$

One deduces

$$f_* : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \longrightarrow \mathrm{Hom}(d_1^*(f_{1!}L_1), d_2^!(f_{2*}L_2)). \tag{3.7.6}$$

The arrow 3.7.5 (respectively 3.7.6) generalizes 3.3.1 (respectively 3.3.2), with which it coincides when c and d are the identity. Also note that, for c proper, 3.2.2 (respectively 3.2.3) is nothing other than 3.7.5 (respectively 3.7.6) when f_1, f_2, and d are the identity arrows. When $Y_1 = Y_2 = S = D$, we shall abbreviate $f_* u$ as u_*, as above.

5 Pairings of Correspondences. Lefschetz Formula

4.1

Let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, $p_i : X \to X_i$ the projections, $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$, and K_{X_i} the dualizing complex (1.3). The canonical arrows

$$p_i^* \mathrm{RHom}(L_i, K_{X_i}) \overset{\mathbf{L}}{\otimes} p_i^* L_i \longrightarrow p_i^* K_{X_i}, \tag{4.1.1}$$

composed from 2.1.1 and the evaluation arrow 2.2.2, give, by tensor product on X and composition with the Künneth isomorphism 1.7.6, a pairing

$$(DL_1 \overset{\mathbf{L}}{\otimes}_S L_2) \overset{\mathbf{L}}{\otimes} (L_1 \overset{\mathbf{L}}{\otimes}_S DL_2) \longrightarrow K_X, \tag{4.1.2}$$

which may be rewritten, by 3.1.1 and 3.1.2, as

$$\mathrm{RHom}_S(L_1, L_2) \overset{\mathbf{L}}{\otimes} \mathrm{RHom}_S(L_2, L_1) \longrightarrow K_X. \tag{4.1.3}$$

One deduces a pairing, called cup product,

$$\langle\ ,\ \rangle : \mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_2, L_1) \longrightarrow \mathrm{H}^0(X, K_X). \tag{4.1.4}$$

One may also regard 4.1.2 as the external tensor product of the evaluation arrows

$$DL_i \overset{\mathbf{L}}{\otimes} L_i \longrightarrow K_{X_i}.$$

In view of the Künneth isomorphism 2.2.4, one thus sees that 4.1.2 is a "perfect duality", i.e. identifies by the adjunction isomorphism 2.2.1 each of the complexes $DL_1 \overset{\mathbf{L}}{\otimes}_S L_2$ and $L_1 \overset{\mathbf{L}}{\otimes}_S DL_2$ with the dual of the other (with values in K_X).

When $X_1 = X_2 = S$, 4.1.3 is a special case of a pairing valid for perfect complexes on an arbitrary ringed topos (SGA 6 I 7.8), and, with the notation of (SGA 6 I 8.3), one has

$$\langle u, v\rangle = \mathrm{Tr}(vu) = \mathrm{Tr}(uv). \tag{4.1.5}$$

4.2

Let $c : C \to X$, $d : D \to X$, and $e = c \times_X d : E = C \times_X D \to X$. For $P, Q \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, one has a canonical arrow (which is not in general an isomorphism)

$$c^! P \overset{\mathbf{L}}{\otimes}_X d^! Q \longrightarrow e^!(P \overset{\mathbf{L}}{\otimes} Q), \tag{4.2.1}$$

defined, in the same way as 1.7.3, as the adjoint of the composite arrow

$$e_!(c^! P \overset{\mathbf{L}}{\otimes}_X d^! Q) \underset{(1)}{\xrightarrow{\sim}} c_! c^! P \overset{\mathbf{L}}{\otimes} d_! d^! Q \xrightarrow{(2)} P \overset{\mathbf{L}}{\otimes} Q,$$

where (1) is the Künneth isomorphism (SGA 4 XVII 5.4.2.2) and (2) is the tensor product of the adjunction arrows. Applying $e_!$ to 4.2.1 and composing with the Künneth isomorphism just cited, one obtains a canonical arrow

$$c_! c^! P \overset{\mathbf{L}}{\otimes} d_! d^! Q \longrightarrow e_! e^!(P \overset{\mathbf{L}}{\otimes} Q). \tag{4.2.2}$$

Similarly, composing the Künneth arrow

$$c_*c^!P \overset{\mathbf{L}}{\otimes} d_*d^!Q \longrightarrow e_*(c^!P \overset{\mathbf{L}}{\otimes}_X d^!Q)$$

with the arrow deduced from 4.2.1 by applying e_*, one obtains a canonical arrow

$$c_*c^!P \overset{\mathbf{L}}{\otimes} d_*d^!Q \longrightarrow e_*e^!(P \overset{\mathbf{L}}{\otimes} Q). \tag{4.2.3}$$

For $P = DL_1 \overset{\mathbf{L}}{\otimes}_S L_2$, $Q = L_1 \overset{\mathbf{L}}{\otimes}_S DL_2$, one deduces from 4.2.2 and 4.2.3, by composition with 4.1.2 and the isomorphism $e^!K_X = K_E$, and with the identifications 3.1.1 and 3.2.1, pairings

$$c_!\mathrm{RHom}(c_1^*L_1, c_2^!L_2) \overset{\mathbf{L}}{\otimes} d_!\mathrm{RHom}(d_2^*L_2, d_1^!L_1) \longrightarrow e_!K_E, \tag{4.2.4}$$

$$c_*\mathrm{RHom}(c_1^*L_1, c_2^!L_2) \overset{\mathbf{L}}{\otimes} d_*\mathrm{RHom}(d_2^*L_2, d_1^!L_1) \longrightarrow e_*K_E, \tag{4.2.4'}$$

$$\langle\ ,\ \rangle : \mathrm{Hom}(c_1^*L_1, c_2^!L_2) \otimes \mathrm{Hom}(d_2^*L_2, d_1^!L_1) \longrightarrow \mathrm{H}^0(E, K_E). \tag{4.2.5}$$

These pairings are compatible with étale localization: given a commutative square (not necessarily cartesian)

$$\begin{tikzcd} & E' \arrow[dl] \arrow[dr] & \\ C' \arrow[dr] & & D' \arrow[dl] \\ & X' & \end{tikzcd} \qquad \text{étale over} \qquad \begin{tikzcd} & E \arrow[dl] \arrow[dr] & \\ C \arrow[dr] & & D \arrow[dl] \\ & X & \end{tikzcd}$$

if c', d', e' denote the composites $C' \to C \to X$, $D' \to D \to X$, $E' \to E \to X$, one has a commutative square.

$$\begin{tikzcd} \mathrm{Hom}(c_1^*L_1, c_2^!L_2) \otimes \mathrm{Hom}(d_2^*L_2, d_1^!L_1) \arrow[r, "4.2.5"] \arrow[d] & \mathrm{H}^0(E, K_E) \arrow[d] \\ \mathrm{Hom}(c_1'^*L_1, c_2'^!L_2) \otimes \mathrm{Hom}(d_2'^*L_2, d_1'^!L_1) \arrow[r] & \mathrm{H}^0(E', K_{E'}) \end{tikzcd} \tag{4.2.6}$$

where the vertical arrows are the restrictions, and the lower horizontal arrow is the composite of the arrow analogous to 4.2.5 and the restriction

$$\mathrm{H}^0(E'', K_{E''}) \longrightarrow \mathrm{H}^0(E', K_{E'}),$$

with $E'' = C' \times_{X'} D'$ (there are analogous commutative squares with arrows of type 4.2.4, 4.2.4', which we leave to the reader to write down). If Z is étale over E, we shall denote, for

$$u \in \mathrm{Hom}(c_1^*L_1, c_2^!L_2), \qquad v \in \mathrm{Hom}(d_2^*L_2, d_1^!L_1),$$

$$\langle u, v\rangle_Z \in \mathrm{H}^0(Z, K_Z) \tag{4.2.7}$$

the image of $\langle u, v\rangle$ in $\mathrm{H}^0(Z, K_Z)$ by restriction (in practice, Z will be a connected component of E). If z is an isolated closed point of E, and $\bar{z}$ a geometric point above z, it follows from 4.2.6 that

$$\langle u, v\rangle_z \in \mathrm{H}^0(z, K_z) = A$$

depends only on the situation deduced by strict localization at the images of $\bar{z}$ in C, D, X.

Example 4.3. Assume that X_i is smooth, purely of relative dimension r_i, and that c and d are closed immersions, with c_2 (respectively d_1) finite. For $L_1 = A_{X_1}$, $L_2 = A_{X_2}$, we have, by 3.1.3,

$$DL_1 \otimes_S^{\mathbf{L}} L_2 = A_X(r_1)[2r_1], \qquad L_1 \otimes_S^{\mathbf{L}} DL_2 = A_X(r_2)[2r_2],$$

and 4.1.2 is identified with the product

$$A_X(r_1)[2r_1] \otimes_X^{\mathbf{L}} A_X(r_2)[2r_2] \longrightarrow A_X(r)[2r],$$

where $r = r_1 + r_2$, and hence, taking 3.2.1 into account, 4.2.5 is identified with the usual cup product

$$\mathrm{H}_C^{2r_1}(X, A(r_1)) \otimes \mathrm{H}_D^{2r_2}(X, A(r_2)) \longrightarrow \mathrm{H}_{C\cap D}^{2r}(X, A(r)).$$

Take as cohomological correspondence from X_1 to X_2 (respectively from X_2 to X_1) the class of c (respectively d) (3.2 a)). It follows from (SGA $4^{1/2}$ Cycle 2.3.8) that, at an isolated point z of $C \cap D$, $\langle \mathrm{cl}(c), \mathrm{cl}(d)\rangle_z$ is nothing but the image in A of the intersection multiplicity of C and D at z.

We now give the main result of the exposé.

Theorem 4.4. Let, for $i = 1, 2$, $f_i : X_i \longrightarrow Y_i$ be a proper S-morphism,

$$f = f_1 \times_S f_2 : X \longrightarrow Y, \qquad p_i : X \longrightarrow X_i, \qquad q_i : Y \longrightarrow Y_i$$

the projections. Let moreover

$$\begin{tikzcd}
& C' \arrow[rr] \arrow[dl, "c'"'] \arrow[dd, "f'" near start] & & C \arrow[dl, "c"] \arrow[dd, "g"] \\
X \arrow[dd, "f"'] & & C'' \arrow[ll, "c''" near start] \arrow[dd, "f''" near end] & \\
& D' \arrow[rr] \arrow[dl, "d'"'] & & D \arrow[dl, "d"] \\
Y & & D'' \arrow[ll, "d''"] &
\end{tikzcd} \tag{4.4.0}$$

be a commutative cube with cartesian horizontal faces, with c', c'', d', d'' proper. Put

$$c_i' = p_i c', \qquad c_i'' = p_i c'', \qquad d_i' = q_i d', \qquad d_i'' = q_i d''.$$

Finally, let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. Then there is a commutative square

$$\begin{tikzcd}
f_* c'_* \mathrm{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes^{\mathbf{L}} f_* c''_* \mathrm{RHom}(c_2''^* L_2, c_1''^! L_1) \arrow[r, "(1)"] \arrow[d, "(2)"'] & f_* c_* K_C \arrow[d, "(4)"] \\
d'_* \mathrm{RHom}(d_1'^! M_1, d_2'^! M_2) \otimes^{\mathbf{L}} d''_* \mathrm{RHom}(d_2''^! M_2, d_1''^! M_1) \arrow[r, "(3)"'] & d_* K_D,
\end{tikzcd} \tag{4.4.1}$$

where $M_i = f_{i*} L_i$, (3) is given by 4.2.4, (1) is deduced from 4.2.4 by applying f_* and composing with the canonical arrow

$$f_* c'_*(-) \otimes^{\mathbf{L}} f_* c''_*(-) \longrightarrow f_*(c'_*(-) \otimes^{\mathbf{L}} c''_*(-)),$$

(2) is the tensor product of the arrows 3.7.5 relative to the faces containing f, and (4) is defined by the adjunction arrow $g_*K_C = g_*g^!K_D \longrightarrow K_D$.

Proof. Put

$$DL_1 \otimes_S^{\mathbf{L}} L_2 = P, \qquad L_1 \otimes_S^{\mathbf{L}} DL_2 = Q, \qquad DM_1 \otimes_S^{\mathbf{L}} M_2 = E, \qquad M_1 \otimes_S^{\mathbf{L}} DM_2 = F.$$

Consider the diagram

$$\begin{tikzcd}
f_*c'_*c'^!P \otimes^{\mathbf{L}} f_*c''_*c''^!Q \arrow[rr, "(1)"] \arrow[d, "(8)"'] & & f_*c'_*(P \otimes^{\mathbf{L}} Q) \arrow[r, "(2)"] \arrow[d, "(9)"'] & f_*c'_*K_X \arrow[d, "(10)"] \\
d'_*d'^!f_*P \otimes^{\mathbf{L}} d''_*d''^!f_*Q \arrow[r, "(3)"'] \arrow[d, "(11)"'] & d_*d'^!(f_*P \otimes^{\mathbf{L}} f_*Q) \arrow[r, "(4)"'] \arrow[d, "(12)"'] & d_*d'^!(P \otimes^{\mathbf{L}} Q) \arrow[r, "(5)"'] & d_*d'^!K_X \arrow[d, "(13)"] \\
d'_*d'^!E \otimes^{\mathbf{L}} d''_*d''^!F \arrow[r, "(6)"'] & d_*d'^!(E \otimes^{\mathbf{L}} F) \arrow[rr, "(7)"'] & & d_*d^!K_Y.
\end{tikzcd} \tag{4.4.2}$$

The arrows are as follows: (1) is deduced from 4.2.2 by applying f_*; (2) is defined by 4.1.2; (3) is given by 4.2.2; (4) is given by the canonical arrow $f_*P \otimes^{\mathbf{L}} f_*Q \longrightarrow f_*(P \otimes^{\mathbf{L}} Q)$; (5) by 4.1.2; (6) by 4.2.2; (7) by 4.1.2; (8) is the tensor product of arrows 3.7.3; (9) and (10) are defined by 3.7.3; (11) and (12) by the isomorphisms $f_*P \longrightarrow E$, $f_*Q \longrightarrow F$ (3.3.1); and (13) by the adjunction arrow $f_*K_X \longrightarrow K_Y$. By definition, 4.4.1 is identified, via 3.1.1 and 3.2.1, with the outer boundary of 4.4.2. It remains only to prove that the squares (A), (B), (C), (D) are commutative. The commutativity of (B) and (C) is clear by functoriality. It remains to prove that of (A) and (D).

a) Commutativity of (D). It is enough to prove that the square

$$\begin{tikzcd}
f_*P \otimes^{\mathbf{L}} f_*Q \arrow[r] \arrow[d] & f_*K_X \arrow[d] \\
E \otimes^{\mathbf{L}} F \arrow[r] & K_Y
\end{tikzcd} \tag{4.4.3}$$

(corresponding to (D) for $d = \mathrm{Id}$) commutes. Recall that the pairing

$$P \otimes^{\mathbf{L}} Q \longrightarrow K_X \quad (\text{respectively } E \otimes^{\mathbf{L}} F \longrightarrow K_Y)$$

is the external tensor product of the evaluation arrows

$$DL_i \otimes^{\mathbf{L}} L_i \longrightarrow K_{X_i} \quad (\text{respectively } DM_i \otimes^{\mathbf{L}} M_i \longrightarrow K_{Y_i}).$$

By the Künneth isomorphisms

$$f_{1*}DL_1 \otimes_S^{\mathbf{L}} f_{2*}L_2 \xrightarrow{\sim} f_*P, \qquad f_{1*}L_1 \otimes_S^{\mathbf{L}} f_{2*}DL_2 \xrightarrow{\sim} f_*Q,$$

4.4.3 is therefore identified with the external tensor product of the squares

$$\begin{tikzcd}
f_{i*}DL_i \otimes^{\mathbf{L}} f_{i*}L_i \arrow[r, "(1)"] \arrow[d, "(3)"'] & f_{i*}K_{X_i} \arrow[d, "(4)"] \\
DM_i \otimes^{\mathbf{L}} M_i \arrow[r, "(2)"'] & K_{Y_i},
\end{tikzcd} \tag{4.4.4}$$

where (2) is the evaluation arrow, (1) the composite of the canonical arrow

$$f_{i*}DL_i \otimes^{\mathbf{L}} f_{i*}L_i \longrightarrow f_{i*}(DL_i \otimes^{\mathbf{L}} L_i)$$

and the arrow defined by the evaluation arrow, (3) is the tensor product of the duality isomorphism $f_{i*}DL_i \longrightarrow DM_i$ and the identity arrow of M_i, and (4) is the adjunction arrow. It is therefore enough to verify the commutativity of the squares 4.4.4. Since the above duality isomorphism is the composite of the canonical arrow (SGA 4 XVIII 3.1.9.5)

$$f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \longrightarrow \mathrm{RHom}(f_{i*}L_i, f_{i*}K_{X_i})$$

and the arrow deduced from the adjunction arrow

$$f_{i*}K_{X_i} \longrightarrow K_{Y_i},$$

it is enough to see that the following square commutes:

$$\begin{tikzcd}
f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \otimes^{\mathbf{L}} f_{i*}L_i \arrow[r, "(1)"] \arrow[d, "(3)"'] & f_{i*}K_{X_i} \arrow[d, "\mathrm{Id}"] \\
\mathrm{RHom}(f_{i*}L_i, f_{i*}K_{X_i}) \otimes^{\mathbf{L}} f_{i*}L_i \arrow[r, "(2)"'] & f_{i*}K_{X_i},
\end{tikzcd} \tag{4.4.5}$$

where (1) is the arrow (1) of 4.4.4, (2) the evaluation arrow, and (3) the tensor product by $f_{i*}L_i$ of the canonical arrow just mentioned. More generally, for any arrow $N_i \longrightarrow f_{i*}L_i$ in $D_{\mathrm{ctf}}(X_i)$, i.e. any arrow $L_i \xrightarrow{u} N_i$ of type 1 over f_i, the triangle

$$\begin{tikzcd}
f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \otimes^{\mathbf{L}} N_i \arrow[d] \arrow[dr] & \\
\mathrm{RHom}(N_i, f_{i*}K_{X_i}) \otimes^{\mathbf{L}} N_i \arrow[r] & f_{i*}K_{X_i}
\end{tikzcd}$$

deduced from 4.4.5 commutes: indeed, by functoriality, one may assume that $L_i = f_i^* N_i$, with u given by the identity of L_i; the arrow

$$f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \longrightarrow \mathrm{RHom}(N_i, f_{i*}K_{X_i})$$

is then the inverse of the trivial duality isomorphism, and the commutativity of the triangle is verified immediately at the level of complexes, resolving N_i flatly and K_{X_i} injectively.

b) Commutativity of (A). Referring to the definition of 3.7.3, one sees that it is enough to verify this commutativity in each of the following cases:

(i) $X_i = Y_i$, $\quad f_i =$ the identity arrow of X_i; $\qquad$ (ii) the lateral faces of 4.4.0 are cartesian.

Place ourselves in case (i). The square (A) is the outer boundary of the diagram

$$\begin{tikzcd}
c'_*(c'^! P \otimes^{\mathbf{L}}_X c''^! Q) \arrow[r] \arrow[d] & c_*(c'^! P \otimes^{\mathbf{L}}_X c''^! Q) \arrow[r] \arrow[d] & c_* c'^!(P \otimes Q) \arrow[d] \\
d'_* d'^! P \otimes^{\mathbf{L}}_X d''_* d''^! Q \arrow[r] & d_*(d'^! P \otimes^{\mathbf{L}}_X d''^! Q) \arrow[r] & d_* d'^!(P \otimes Q),
\end{tikzcd}$$

in which the horizontal isomorphisms of A_1 are Künneth isomorphisms, the horizontal arrows of A_2 are given by 4.2.1, and the middle vertical arrow is deduced by applying d_* to the arrow

$$g_*(c'^!P \otimes^{\mathbf{L}}_X c''^!Q) \longrightarrow d'^!P \otimes^{\mathbf{L}}_X d''^!Q$$

defined, via Künneth, by the external tensor product of the adjunction arrows

$$f'_* c'^!P \longrightarrow d'^!P, \qquad f''_* c''^!Q \longrightarrow d''^!Q.$$

These adjunction arrows and the identity arrows of $d'^!d'^!P$, $d''^!d''^!Q$ define commutative triangles of arrows of type 3 (1.4)

$$\begin{tikzcd} & c'^!P \arrow[dl] \arrow[d] \\ d'^!d'^!P & d'^!P \arrow[l] \end{tikzcd} \qquad \begin{tikzcd} & c''^!Q \arrow[dl] \arrow[d] \\ d''^!d''^!Q & d''^!Q \arrow[l] \end{tikzcd}$$

over

$$\begin{tikzcd} & C' \arrow[dl, "c'"'] \arrow[d, "f'"] \\ X & D' \arrow[l, "d'"] \end{tikzcd} \qquad \begin{tikzcd} & C'' \arrow[dl, "c''"'] \arrow[d, "f''"] \\ X & D'' \arrow[l, "d''"] \end{tikzcd} .$$

By external tensor product (over X), these triangles give a commutative triangle

$$\begin{tikzcd} & c'^!P \otimes^{\mathbf{L}}_X c''^!Q \arrow[dl] \arrow[d] \\ d'^!d'^!P \otimes^{\mathbf{L}} d''^!d''^!Q & d'^!P \otimes^{\mathbf{L}}_X d''^!Q \arrow[l] \end{tikzcd} \quad \text{over} \quad \begin{tikzcd} & C \arrow[dl, "c"'] \arrow[d, "g"] \\ X & D \arrow[l, "d"] \end{tikzcd} .$$

The commutativity of this triangle is equivalent to that of the square deduced from A_1 by inverting the horizontal arrows. On the other hand, there are commutative triangles of arrows of type 3

$$\begin{tikzcd} & c'^!P \arrow[dl] \arrow[d] \\ P & d'^!P \arrow[l] \end{tikzcd} \qquad \begin{tikzcd} & c''^!Q \arrow[dl] \arrow[d] \\ Q & d''^!Q \arrow[l] \end{tikzcd}$$

defined by the identity arrows of $d'^!P$, $c'^!P$, $d''^!Q$, $c''^!Q$. By external tensor product, one obtains a commutative triangle

$$\begin{tikzcd} & c'^!P \otimes^{\mathbf{L}}_X c''^!Q \arrow[dl] \arrow[d] \\ P \otimes Q & d'^!P \otimes^{\mathbf{L}}_X d''^!Q \arrow[l] \end{tikzcd}$$

which can be interpreted as a morphism of arrows of type 3 over $g : C \longrightarrow D$:

$$\begin{tikzcd} c'^! P \otimes^{\mathbf{L}}_X c''^! Q \arrow[r] \arrow[d] & c'^!(P \otimes^{\mathbf{L}} Q) \arrow[d] \\ d'^! P \otimes^{\mathbf{L}}_X d''^! Q \arrow[r] & d'^!(P \otimes Q), \end{tikzcd}$$

or again as a commutative square over D:

$$\begin{tikzcd} g_*(c'^! P \otimes^{\mathbf{L}}_X c''^! Q) \arrow[r] \arrow[d] & g_* c'^!(P \otimes^{\mathbf{L}} Q) \arrow[d] \\ d'^! P \otimes^{\mathbf{L}}_X d''^! Q \arrow[r] & d^!(P \otimes Q). \end{tikzcd}$$

But the square deduced from the latter by applying d_* is none other than A_2, whence the commutativity of (A) in case (i).

It remains to treat case (ii). Since the faces of 4.4.0 containing f are cartesian, the arrows

$$f'_* c'^! P \longrightarrow d'^! f_* P, \qquad f''_* c''^! Q \longrightarrow d''^! f_* Q$$

(3.7.3) are isomorphisms. Their inverses define arrows of type 1

$$c'^! P \longrightarrow d'^! f_* P, \qquad c''^! Q \longrightarrow d''^! f_* Q$$

over f', f'', whence, by direct image, commutative squares of arrows of type 1

$$\begin{tikzcd} c'_* c'^! P & c'^! P \arrow[l] \arrow[d] \\ d'_* d'^! P & d'^! f_* P \arrow[l] \end{tikzcd} \qquad \begin{tikzcd} c''_* c''^! Q & c''^! Q \arrow[l] \arrow[d] \\ d''_* d''^! Q & d''^! f_* Q \arrow[l] \end{tikzcd}$$

over the faces of 4.4.0 containing f. Taking the tensor product, one obtains a commutative square in $D_{\mathrm{ctf}}(Y)$

$$\begin{tikzcd} d'_* d'^! f_* P \otimes^{\mathbf{L}}_Y d''_* d''^! f_* Q \arrow[r] \arrow[d] & d_*(d'^! f_* P \otimes^{\mathbf{L}}_Y d''^! f_* Q) \arrow[d] \\ f_* c'_* c'^! P \otimes^{\mathbf{L}} f_* c''_* c''^! Q \arrow[r] & d_* g_*(c'^! P \otimes^{\mathbf{L}}_X c''^! Q), \end{tikzcd} \tag{4.4.6}$$

where the left vertical arrow is the inverse of (8), the horizontal arrows are given by Künneth, and the right vertical arrow by the external tensor product over Y of the arrows

$$f'_* c'^! P \longrightarrow d'^! f_* P, \qquad f''_* c''^! Q \longrightarrow d''^! f_* Q.$$

The same tensor product defines the left vertical arrow of the square

$$\begin{tikzcd} d'^! f_* P \otimes^{\mathbf{L}}_Y d''^! f_* Q \arrow[r] \arrow[d] & d'^! f_*(P \otimes^{\mathbf{L}} Q) \arrow[d] \\ g_*(c'^! P \otimes^{\mathbf{L}}_X c''^! Q) \arrow[r] & g_* c'^!(P \otimes Q), \end{tikzcd} \tag{4.4.7}$$

where the right vertical arrow is 3.7.3, and the horizontal arrows are defined by 4.2.1. The outer boundary of the diagram composed of 4.4.6 and $d_*(4.4.7)$ is the square deduced from (A) by inverting the vertical arrows. It is therefore enough to prove that 4.4.7 commutes, in other words that the arrows 4.2.1

$$c'^! P \otimes^{\mathbf{L}}_X c''^! Q \longrightarrow c'^!(P \otimes^{\mathbf{L}} Q), \qquad d'^! f_* P \otimes^{\mathbf{L}}_Y d''^! f_* Q \longrightarrow d'^! f_*(P \otimes^{\mathbf{L}} Q)$$

define a morphism from

$$c'^! P \otimes^{\mathbf{L}}_X c''^! Q \longrightarrow d'^! f_* P \otimes^{\mathbf{L}}_Y d''^! f_* Q$$

to

$$c'^!(P \otimes Q) \longrightarrow d'^! f_*(P \otimes Q),$$

viewed as arrows of type 1 over g. By adjunction, we are reduced to verifying that the tensor products of adjunction arrows

$$c'_* c'^! P \otimes^{\mathbf{L}}_X c''_* c''^! Q \longrightarrow P \otimes^{\mathbf{L}} Q, \qquad d'_* d'^! f_* P \otimes^{\mathbf{L}}_Y d''_* d''^! f_* Q \longrightarrow f_* P \otimes^{\mathbf{L}} f_* Q$$

define a morphism, between arrows of type 1 over f, from the tensor product of the arrows

$$c'_* c'^! P \longrightarrow d'_* d'^! f_* P, \qquad c''_* c''^! Q \longrightarrow d''_* d''^! f_* Q$$

to the tensor product of the arrows

$$P \longrightarrow f_* P, \qquad Q \longrightarrow f_* Q.$$

For this it is enough to verify that the adjunction arrows

$$c'_* c'^! P \longrightarrow P, \qquad d'_* d'^! f_* P \longrightarrow f_* P$$

define a morphism from

$$c'_* c'^! P \longrightarrow d'_* d'^! f_* P$$

to

$$P \longrightarrow f_* P,$$

and likewise for the pair (c'', d''). It is enough to consider the case of (c', d'), the statement being the same in the two cases. We have to see that the triangle

$$\begin{tikzcd}
 & f_* c'_* c'^! P \arrow[dl] \arrow[d, "\sim"] \\
f_* P & d'_* d'^! f_* P \arrow[l]
\end{tikzcd}$$

where the horizontal and oblique arrows are given by adjunction and the vertical arrow by the isomorphism 3.7.3, is commutative. Taking into account the definition of 3.7.3 (SGA 4 XVIII 3.1.12.3) as the transpose of a base-change isomorphism, we are reduced, by applying the functor $\mathrm{Hom}(R, -)$ $(R \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y))$, to verifying the commutativity of the square

$$\begin{tikzcd}
c'_* c'^* f^* R \arrow[r, "\mathrm{Id}"] & c'_* f'^* d'^* R \\
f^* R \arrow[u] \arrow[r] & f^* d'_* d'^* R \arrow[u, "\sim"']
\end{tikzcd}$$

where the left vertical arrow (respectively the lower horizontal arrow) is defined by the adjunction arrow $1 \longrightarrow c'_* c'^*$ (respectively $1 \longrightarrow d'_* d'^*$), and the right vertical arrow by the base-change isomorphism

$$f^* d'_* \xrightarrow{\sim} c'_* f'^*.$$

But this square commutes by the very definition of the base-change arrow. The commutativity of (A) is therefore established, and this completes the proof of the theorem.

Corollary 4.5. Under the hypotheses of 4.4, there is a commutative square

$$\begin{tikzcd}
\mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1) \arrow[r, "4.2.5"] \arrow[d, "3.7.6 \otimes 3.7.6"'] & \mathrm{H}^0(C, K_C) \arrow[d, "g_*"] \\
\mathrm{Hom}(d_1'^* f_{1*} L_1, d_2'^! f_{2*} L_2) \otimes \mathrm{Hom}(d_2''^* f_{2*} L_2, d_1''^! f_{1*} L_1) \arrow[r, "4.2.5"'] & \mathrm{H}^0(D, K_D),
\end{tikzcd}$$

where g_* is defined by the adjunction arrow

$$g_* K_C = g_* g^! K_D \longrightarrow K_D.$$

In other words, for

$$u \in \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2), \qquad v \in \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1),$$

one has

$$\langle f_* u, f_* v \rangle = g_* \langle u, v \rangle. \tag{4.5.1}$$

For every proper S-scheme $t : T \longrightarrow S$, denote by

$$\int_T : \mathrm{H}^0(T, K_T) \longrightarrow \mathrm{H}^0(S, K_S) = \mathrm{H}^0(S, A) \tag{4.6}$$

the arrow defined by the adjunction arrow

$$t_! K_T \longrightarrow A.$$

Corollary 4.7 – Lefschetz–Verdier formula. Under the hypotheses of 4.4, suppose that $Y_1 = Y_2 = S = D' = D''$. For

$$u \in \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2), \qquad v \in \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1),$$

one has

$$\langle u_*, v_* \rangle = \int_C \langle u, v \rangle, \tag{4.7.1}$$

where

$$u_* \in \mathrm{Hom}(f_{1*} L_1, f_{2*} L_2), \qquad v_* \in \mathrm{Hom}(f_{2*} L_2, f_{1*} L_1)$$

are the homomorphisms deduced from u, v by direct image over S (3.7.6), and

$$\langle u_*, v_* \rangle = \mathrm{Tr}(u_* v_*) = \mathrm{Tr}(v_* u_*)$$

is the cup product, in the sense of (SGA 6 I 8.3), of the homomorphisms of perfect complexes u_*, v_* (4.1.5).

Taking 3.6.5 into account, in particular one has:

Corollary 4.8. Let $t : T \longrightarrow S$ be a proper S-scheme, $\Delta \subset T \times_S T$ the diagonal, $F : T \longrightarrow T$ an S-endomorphism of T, $T^F = (\text{graph of } F) \times_{(T \times_S T)} \Delta$ the scheme of fixed points of F, $L \in \operatorname{ob} D_{\mathrm{ctf}}(T)$, $u \in \operatorname{Hom}(F^*L, L)$ a correspondence from L to L with support in the graph of F, and $1 \in \operatorname{Hom}(L, L)$ the identity correspondence, with support in Δ (3.2 b)). Then

$$\operatorname{Tr}(u_*) = \int_{T^F} \langle u, 1 \rangle,$$

where u_* is the endomorphism of t_*L obtained by composing $t_*L \longrightarrow t_*F^*L$ (inverse image) with $t_*u : t_*F^*L \longrightarrow t_*L$.

From the remarks of 3.2 a) and 3.6 one also deduces:

Corollary 4.9. Under the hypotheses of 4.7, assume that X_i is smooth over S, purely of relative dimension r_i, and that C' (respectively C'') is a closed subscheme of X finite and of finite tor-dimension over X_2 (respectively X_1). Let

$$\operatorname{cl}(C') \in \mathrm{H}^{2r_1}_{C'}(X, A(r_1)), \qquad \operatorname{cl}(C'') \in \mathrm{H}^{2r_2}_{C''}(X, A(r_2))$$

be the cohomology classes of C' and C'', and let

$$\operatorname{cl}(C')_* \in \operatorname{Hom}(\mathrm{R}f_{1*}A, \mathrm{R}f_{2*}A), \qquad \operatorname{cl}(C'')_* \in \operatorname{Hom}(\mathrm{R}f_{2*}A, \mathrm{R}f_{1*}A)$$

be the corresponding homomorphisms (3.2 a), 3.7.6). Then

$$\langle \operatorname{cl}(C')_*, \operatorname{cl}(C'')_* \rangle = \int_{C' \cap C''} \operatorname{cl}(C) \operatorname{cl}(D),$$

where

$$\operatorname{cl}(C) \operatorname{cl}(D) \in \mathrm{H}^{2(r_1+r_2)}_{C' \cap C''}(X, A(r_1 + r_2)) = \mathrm{H}^0(C' \cap C'', K_{C' \cap C''})$$

is the usual cup product.

Taking 4.3 into account, in particular one has:

Corollary 4.10. Under the hypotheses of 4.9, suppose that $C' \cap C''$ is finite. For $z \in C' \cap C''$, denote by $(C' \cdot C'')_z$ the image in A of the intersection multiplicity of C' and C'' at z. Then

$$\langle \operatorname{cl}(C')_*, \operatorname{cl}(C'')_* \rangle = \sum_{z \in C' \cap C''} (C' \cdot C'')_z.$$

4.11. Under the hypotheses of 4.9, one has in general

$$\langle \operatorname{cl}(C')_*, \operatorname{cl}(C'')_* \rangle = \sum_{Z \in \pi_0(C' \cap C'')} (\operatorname{cl}(C') \operatorname{cl}(C''))_Z, \tag{4.11.1}$$

where $\pi_0(C' \cap C'')$ denotes the set of connected components of $C' \cap C''$, and $(\operatorname{cl}(C') \operatorname{cl}(C''))_Z$ the restriction of $\operatorname{cl}(C') \operatorname{cl}(C'')$ to $\mathrm{H}^0(Z, K_Z)$. When Z is a component of dimension ≥ 1, the calculation of $(\operatorname{cl}(C') \operatorname{cl}(C''))_Z$ presents problems, because one is in an "excess" situation. If Z is smooth, purely of dimension d, if C' and C'' are smooth in a neighbourhood of Z, and if X is projective, one can show that

$$\int_Z (\mathrm{cl}(C')\,\mathrm{cl}(C''))_Z = \int_Z c_d(N_Z), \tag{4.11.2}$$

where N_Z is the locally free sheaf of rank d defined by the exact sequence of normal bundles

$$0 \longrightarrow N_{Z/C'} \oplus N_{Z/C''} \longrightarrow N_{Z/X} \longrightarrow N_Z \longrightarrow 0,$$

and $c_d(N_Z) \in \mathrm{H}^{2d}(Z, A(d))$ denotes its d-th Chern class (SGA 5 VII) (use (SGA 6 XIV 6.3) to reduce to a calculation of intersection number in K-theory, and apply (SGA 6 VII 2.5), cf. ([10] 4.1); the projectivity hypothesis is no doubt unnecessary).

Let T be a smooth projective S-scheme; take as correspondence C' (respectively C'') the graph of an S-endomorphism F (respectively the diagonal), and suppose that the scheme of fixed points $T^F = C' \cap C''$ is smooth. The sheaf $N_{(T^F)}$ above is nothing other than the tangent sheaf to T^F, so that 4.11.2 implies, by the Gauss–Bonnet formula (SGA 5 VII 4.9), the analogue of a formula well known to topologists:

$$\mathrm{Tr}(F, \mathrm{R}\Gamma(T, A)) = \mathrm{EP}(T_s^F), \tag{4.11.3}$$

where $\mathrm{EP}(T_s^F)$ denotes the Euler–Poincaré characteristic of T_s^F, for s a geometric point above S (loc. cit.).

The calculation of $(\mathrm{cl}(C')\,\mathrm{cl}(C''))_Z$ for a non-smooth component Z of dimension ≥ 1 has not been treated in general, at least to the knowledge of the editor.

4.12. In the situation of 4.7, let z be an isolated point of C, with projection z_i on X_i, a' (respectively a'') on C' (respectively C''), and let $\bar{z}$ be a geometric point above z, with image $\bar{z}_i$ above z_i. Assume that c_2' (respectively c_1'') is quasi-finite and flat at a' (respectively a''), that L_i is perfect in a neighbourhood of $\bar{z}_i$, and that u (respectively v) is given in a neighbourhood of a' (respectively a'') by

$$\mathrm{Tr}_{c_2'} : c'_{2!}c_2'^{*}L_2 \longrightarrow L_2 \quad \text{and} \quad s \in \mathrm{Hom}(c_1'^{*}L_1, c_2'^{*}L_2)$$

(respectively

$$\mathrm{Tr}_{c_1''} : c''_{1!}c_1''^{*}L_1 \longrightarrow L_1 \quad \text{and} \quad t \in \mathrm{Hom}(c_2''^{*}L_2, c_1''^{*}L_1)).$$

Let $s_{\bar{z}} \in \mathrm{Hom}(L_{1\bar{z}_1}, L_{2\bar{z}_2})$ (respectively $t_{\bar{z}} \in \mathrm{Hom}(L_{2\bar{z}_2}, L_{1\bar{z}_1})$) be the homomorphism deduced from s (respectively t) by passage to the fibres. Let again, as in 3.2 a), $\mathrm{cl}(c')$ (respectively $\mathrm{cl}(c'')$) denote the correspondence from X_1 to X_2 (respectively from X_2 to X_1) defined, in a neighbourhood of a' (respectively a''), by c_2' (respectively c_1''), and let $(C' \cdot C'')_z$ be the corresponding local term $\langle \mathrm{cl}(c'), \mathrm{cl}(c'')\rangle_z$ (equal to the image in A of the intersection multiplicity of C' and C'' at z when C' and C'' are closed subschemes of X, with X_1 and X_2 smooth). Then:

$$\langle u, v\rangle_z = (C' \cdot C'')_z \langle s_{\bar{z}}, t_{\bar{z}}\rangle. \tag{4.12.1}$$

This follows easily from 4.2.6.

4.13. If T is a scheme over S, denote by $DF(T)$ the filtered derived category of complexes of A_T-modules with finite filtration ([9] V 1), and by $DF_{\mathrm{ctf}}(T)$ the full subcategory formed by the L such that $\mathrm{gr}(L) \in \mathrm{ob}\, D_{\mathrm{ctf}}(T)$. The kind of compatibility of filtered derived functors with passage to associated graded objects implies that DF_{ctf} is stable under the six operations f^*, f_*, $f_!$, $f^!$, $\otimes^{\mathbf{L}}$, RHom. The definitions and results of the preceding numbers are valid, mutatis mutandis, in DF_{ctf}. The pairing 4.1.2 is compatible with passage to associated graded objects (K_X being regarded as endowed with the trivial filtration), and it follows that, with the notation of 4.2.5, for $L_i \in \mathrm{ob}\, DF_{\mathrm{ctf}}(X_i)$,

$$u \in \mathrm{Hom}_{DF}(c_1^* L_1, c_2^! L_2), \qquad v \in \mathrm{Hom}_{DF}(d_2^* L_2, d_1^! L_1),$$

one has the following equality in $\mathrm{H}^0(E, K_E)$:

$$\langle u, v \rangle = \sum_i \langle \mathrm{gr}^i\, u, \mathrm{gr}^i\, v \rangle. \tag{4.13.1}$$

As Deligne observes, this remark applies in particular to the situation considered by Langlands in ([13] §7). The correspondences that he considers in (loc. cit. 7.11) indeed preserve, near the fixed points, finite filtrations on the sheaves with associated graded objects inducing, on the strict localization at the image of a fixed point, either a constant sheaf or a constant sheaf on the complement of the closed point extended by zero. The calculation of the local terms is easily reduced, by 4.13, to 4.12.1 and to the case of sheaves concentrated on the closed points that are the images of the fixed points, a case treated directly. Applying 4.7, one obtains the Lefschetz formula required in ([13] 7.11). We point out that the general statement (loc. cit. 7.12) is true; the proof, more delicate, will be given in III B.

5. Complements

5.1. Dual of a correspondence

Let, for $i = 1, 2$, X_i be an S-scheme, let $X = X_1 \times_S X_2$, and let $p_i : X \to X_i$ be the projections. Let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. Every cohomological correspondence u from L_1 to L_2 gives, by applying D_X and identifying $Dp_2^!$ with $p_2^* D$ and Dp_1^* with $p_1^! D$, a correspondence Du from DL_2 to DL_1, called the *dual of u*. In other words, D defines a homomorphism

$$D : \mathrm{Hom}(p_1^* L_1, p_2^! L_2) \longrightarrow \mathrm{Hom}(p_2^* DL_2, p_1^! DL_1). \tag{5.1.1}$$

One checks that 5.1.1 is obtained, by applying $H^0(X, -)$, from the isomorphism

$$D : \mathrm{RHom}(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \mathrm{RHom}(p_2^* DL_2, p_1^! DL_1) \tag{5.1.2}$$

which makes the square

$$\begin{tikzcd} p_1^* DL_1 \otimes^{\mathbf{L}} p_2^* L_2 \arrow[r, "3.1.1"] \arrow[d] & \mathrm{RHom}(p_1^* L_1, p_2^! L_2) \arrow[d] \\ p_2^* DDL_2 \otimes^{\mathbf{L}} p_1^* DL_1 \arrow[r, "3.1.1"] & \mathrm{RHom}(p_2^* DL_2, p_1^! DL_1), \end{tikzcd} \tag{5.1.3}$$

commutative. Here the left vertical arrow is the composite of the symmetry isomorphism and the one defined by the biduality isomorphism $L_2 \to DDL_2$. It follows easily from the definitions that the following square is commutative:

$$\begin{tikzcd} \mathrm{RHom}_S(L_1, L_2) \otimes^{\mathbf{L}} \mathrm{RHom}_S(L_2, L_1) \arrow[r, "4.1.3"] \arrow[d, "D\otimes D"'] & K_X \arrow[d, "\mathrm{Id}"] \\ \mathrm{RHom}_S(DL_2, DL_1) \otimes^{\mathbf{L}} \mathrm{RHom}_S(DL_1, DL_2) \arrow[r, "4.1.3"] & K_X. \end{tikzcd} \tag{5.1.4}$$

Let $c : C \to X$ be an S-morphism and put $c_i = p_i c$. Applying $c_* c^!$ to 5.1.2, and using (3.2.1), one deduces an isomorphism

$$D : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \xrightarrow{\sim} \mathrm{Hom}(c_2^* DL_2, c_1^! DL_1), \tag{5.1.5}$$

which is checked to identify with the one obtained by applying the functor D_X and the identifications of Dc_1^* with $c_1^! D$ and $Dc_2^!$ with $c_2^* D$. If $c' : C' \to X$ and $c'' : C'' \to X$ are S-morphisms, it follows from 5.1.4 that, for

$$u \in \mathrm{Hom}({c'_1}^* L_1, {c'_2}^! L_2), \qquad v \in \mathrm{Hom}({c''_2}^* L_2, {c''_1}^! L_1),$$

one has

$$\langle u, v \rangle = \langle Du, Dv \rangle, \tag{5.1.6}$$

with the notation of 4.2.5.

On the other hand, one checks that, in the situation of 3.3, the following square is commutative:

$$\begin{array}{ccc} f_* \mathrm{RHom}_S(L_1, L_2) & \xrightarrow{3.3.1} & \mathrm{RHom}_S(f_1^! L_1, f_2^* L_2) \\ {\scriptstyle f_* D}\downarrow & & \downarrow{\scriptstyle D} \\ f_* \mathrm{RHom}_S(DL_2, DL_1) & \xrightarrow{3.3.1} & \mathrm{RHom}_S(f_2^! DL_2, f_1^* DL_1), \end{array} \tag{5.1.7}$$

and consequently, in the situation of 3.7.6, one has

$$f_* Du = D f_* u. \tag{5.1.8}$$

5.2. Composition of correspondences

Let, for $i = 1, 2, 3$, X_i be an S-scheme, let $X_{ij} = X_i \times_S X_j$, $X = X_1 \times_S X_2 \times_S X_3$, and let $p_i : X \to X_i$, $p_{ij} : X \to X_{ij}$ be the projections. Let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. One has $X = X_{12} \times_{X_2} X_{23}$, and

$$(DL_1 \otimes_S^{\mathbf{L}} L_2) \otimes_{X_2}^{\mathbf{L}} (DL_2 \otimes_S^{\mathbf{L}} L_3) = p_1^* DL_1 \otimes^{\mathbf{L}} p_2^* L_2 \otimes^{\mathbf{L}} p_2^* DL_2 \otimes^{\mathbf{L}} p_3^* L_3.$$

The evaluation arrow $L_2 \otimes^{\mathbf{L}} DL_2 \to K_{X_2}$ therefore defines a pairing

$$(DL_1 \otimes_S^{\mathbf{L}} L_2) \otimes_{X_2}^{\mathbf{L}} (DL_2 \otimes_S^{\mathbf{L}} L_3) \longrightarrow p_{13}^! (DL_1 \otimes_S^{\mathbf{L}} L_3), \tag{5.2.1}$$

the second member being identified with $(DL_1 \otimes_S^{\mathbf{L}} L_3) \otimes_S^{\mathbf{L}} K_{X_2}$ by Künneth (1.7.3). Taking 3.1.1 and 3.1.2 into account, one may rewrite 5.2.1 in the form

$$\mathrm{RHom}_S(L_1, L_2) \otimes_{X_2}^{\mathbf{L}} \mathrm{RHom}_S(L_2, L_3) \longrightarrow p_{13}^! \mathrm{RHom}_S(L_1, L_3) = \mathrm{RHom}(p_1^* L_1, p_3^! L_3), \tag{5.2.2}$$

the identification with the second member being given by the induction isomorphism (SGA 4 XVIII 3.1.12.2). Hence one obtains a pairing, called *composition of correspondences*,

$$\mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_2, L_3) \longrightarrow \mathrm{Hom}(p_1^* L_1, p_3^! L_3), \tag{5.2.3}$$

which will be denoted $(u, v) \mapsto vu$. One checks that, if $p_{ij,i} : X_{ij} \to X_i$ and $p_{ij,j} : X_{ij} \to X_j$ denote the projections, 5.2.2 can also be obtained by composing the arrow

$$\mathrm{RHom}_S(L_1, L_2) \otimes_{X_2}^{\mathbf{L}} \mathrm{RHom}_S(L_2, L_3) \longrightarrow \mathrm{RHom}(p_{12}^* p_{12,1}^* L_1, p_{12}^! p_{12,2}^* L_2) \otimes \mathrm{RHom}(p_{23}^! p_{23,2}^* L_2, p_{23}^! p_{23,3}^! L_3)$$

(the tensor product of the canonical arrows $p_{12}^*\mathrm{RHom}_S(L_1, L_2) \to \mathrm{RHom}(p_{12}^* L_1, p_{12}^! L_2)$ and $p_{23}^*\mathrm{RHom}_S(L_2, L_3) \to \mathrm{RHom}(p_{23}^! L_2, p_{23}^! L_3)$), the arrow defined by the base-change arrow $p_{12}^! p_{12,2}^* L_2 \to p_{23}^* p_{23,2}^* L_2$, and a usual composition arrow on the RHom's.

Let $c' : C' \to X_{12}$ and $c'' : C'' \to X_{23}$ be correspondences, and let $c : C = C' \times_{X_2} C'' \to X$ be the "composite" correspondence. From 5.2.2 one deduces, or else defines directly by a composition analogous to the one just described, a pairing

$$\mathrm{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes^{\mathbf{L}}_{X_2} \mathrm{RHom}(c_2''^* L_2, c_3''^! L_3) \longrightarrow \mathrm{RHom}(c_1^* L_1, c_3^! L_3), \tag{5.2.4}$$

hence a "composition" pairing $(u, v) \mapsto vu$:

$$\mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(c_2''^* L_2, c_3''^! L_3) \longrightarrow \mathrm{Hom}(c_1^* L_1, c_3^! L_3). \tag{5.2.5}$$

One checks that composition of correspondences is compatible with direct images in the following sense. Let, for $1 \le i \le 3$, $f_i : X_i \to Y_i$ be an S-morphism, with f_2 proper, and let c', c'', c be morphisms as above, with c' and c'' proper. Let $d' : D' \to Y_{12}$ and $d'' : D'' \to Y_{23}$ be proper morphisms, $d : D = D' \times_{Y_2} D'' \to Y$, and suppose given the commutative squares

$$\begin{tikzcd} X_{12} & C' \arrow[l] \arrow[d] \\ Y_{12} \arrow[u, "f_{12}"] & D' \arrow[l] \end{tikzcd} \qquad \begin{tikzcd} X_{23} & C'' \arrow[l] \arrow[d] \\ Y_{23} \arrow[u, "f_{23}"] & D'' \arrow[l] \end{tikzcd}$$

Then one has a commutative square

$$\begin{tikzcd} \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(c_2''^* L_2, c_3''^! L_3) \arrow[r, "5.2.5"] \arrow[d, "f_{12*} \otimes f_{23*}"'] & \mathrm{Hom}(c_1^* L_1, c_3^! L_3) \arrow[d, "f_{13*}"] \\ \mathrm{Hom}(d_1'^* M_1, d_2'^! M_2) \otimes \mathrm{Hom}(d_2''^* M_2, d_3''^! M_3) \arrow[r, "5.2.5"] & \mathrm{Hom}(d_1^* M_1, d_3^! M_3), \end{tikzcd} \tag{5.2.6}$$

where $M_i = f_{i!} L_i$.

Finally, suppose that $X_1 = X_3$ and $L_1 = L_3$. The fundamental pairing 4.1.2 may be interpreted as the composite of the pairing

$$q_1^* DL_1 \otimes^{\mathbf{L}} q_2^* L_2 \otimes^{\mathbf{L}} q_2^* DL_2 \otimes^{\mathbf{L}} q_1^* L_1 \longrightarrow q_1^* DL_1 \otimes^{\mathbf{L}} q_2^* K_{X_2} \otimes^{\mathbf{L}} q_1^* L_1 \tag{5.2.7}$$

defined by $L_2 \otimes^{\mathbf{L}} DL_2 \to K_{X_2}$, or, equivalently, induced by 5.2.1 on X_{12} via the diagonal $X_{12} \to X_1 \times_S X_2 \times_S X_1$ ($q_i : X_{12} \to X_i$ denoting the projection), and the pairing

$$q_1^* DL_1 \otimes^{\mathbf{L}} q_2^* K_{X_2} \otimes^{\mathbf{L}} q_1^* L_1 \longrightarrow K_{X_{12}} \tag{5.2.8}$$

defined by $DL_1 \otimes^{\mathbf{L}} L_1 \to K_{X_1}$. It follows that, for $u \in \mathrm{Hom}_S(L_1, L_2)$ and $v \in \mathrm{Hom}_S(L_2, L_1)$, one has, in $H^0(X_1 \times_S X_2, K_{X_1 \times_S X_2})$,

$$\langle u, v \rangle = \langle vu, 1 \rangle, \tag{5.2.9}$$

where 1 denotes the identity correspondence from L_1 to L_1 on $X_1 \times_S X_1$, supported on the diagonal (3.2 b)), and the cup product in the second member is taken in the sense of 4.2.5, taking account of the fact that the fibred product $X \times_{(X_1 \times_S X_1)} X_1$ (via $p_{13} : X \to X_{13} = X_1 \times_S X_1$ and the diagonal $X_1 \to X_1 \times_S X_1$) identifies with $X_1 \times_S X_2$. Likewise, more generally, if $c' : C' \to X_{12}$ and $c'' : C'' \to X_{21}$ are correspondences, if $c : C = C' \times_{X_{12}} C'' \to X_1 \times_S X_2 \times_S X_1$ is the composite correspondence, and if $d : C' \times_{X_{12}} C'' = D \to X_{12}$ is the intersection correspondence, then, for $u \in \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2)$ and $v \in \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1)$, one has the following equality in $H^0(D, K_D)$:

$$\langle u, v \rangle = \langle vu, 1 \rangle, \tag{5.2.10}$$

where 1 denotes, as above, the diagonal correspondence, and D is identified with the fibred product of $C \to X_1 \times_S X_1$ and the diagonal $X_1 \to X_1 \times_S X_1$. Formulas 5.2.9 and 5.2.10 generalize a well-known formula in the case of constant coefficients ([12] 1.3.6 (ii) c)).

5.3. External tensor product of correspondences

Let, for $1 \leq i \leq 4$, X_i be an S-scheme, let $X = X_1 \times_S X_2 \times_S X_3 \times_S X_4$, $X_{ij} = X_i \times_S X_j$, and let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. The Künneth isomorphism (1.7.6, 2.2.4)

$$DL_1 \otimes_S^{\mathbf{L}} DL_3 \xrightarrow{\sim} D(L_1 \otimes_S^{\mathbf{L}} L_3)$$

defines an isomorphism on X

$$(DL_1 \otimes_S^{\mathbf{L}} L_2) \otimes_S^{\mathbf{L}} (DL_3 \otimes_S^{\mathbf{L}} L_4) \xrightarrow{\sim} D(L_1 \otimes_S^{\mathbf{L}} L_3) \otimes_S^{\mathbf{L}} (L_2 \otimes_S^{\mathbf{L}} L_4), \tag{5.3.1}$$

which, by 3.1.1, may be rewritten in the form

$$\mathrm{RHom}_S(L_1, L_2) \otimes_S^{\mathbf{L}} \mathrm{RHom}_S(L_3, L_4) \xrightarrow{\sim} \mathrm{RHom}_S(L_1 \otimes_S^{\mathbf{L}} L_3, L_2 \otimes_S^{\mathbf{L}} L_4). \tag{5.3.2}$$

Hence a pairing, called the *external tensor product*,

$$\mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_3, L_4) \longrightarrow \mathrm{Hom}_S(L_1 \otimes_S^{\mathbf{L}} L_3, L_2 \otimes_S^{\mathbf{L}} L_4), \tag{5.3.3}$$

which will be denoted $(u, v) \mapsto u \otimes_S^{\mathbf{L}} v$. The pairing 5.3.2 may also be interpreted as the composite of the external tensor product

$$\mathrm{RHom}(E_1, E_2) \otimes_S^{\mathbf{L}} \mathrm{RHom}(E_3, E_4) \xrightarrow{\sim} \mathrm{RHom}(E_1 \otimes_S^{\mathbf{L}} E_3, E_2 \otimes_S^{\mathbf{L}} E_4)$$

(where $E_1 = p_{12}^* p_{12,1}^* L_1$, etc., with the evident notation) and of the isomorphism defined by Künneth

$$E_2 \otimes_S^{\mathbf{L}} E_4 \xrightarrow{\sim} p_{24}^! (L_2 \otimes_S^{\mathbf{L}} L_4) \quad (1.7.3).$$

Let $c' : C' \to X_{12}$ and $c'' : C'' \to X_{34}$ be correspondences, and let $c : C = C' \times_S C'' \to X = X_{13} \times_S X_{24}$ be the product correspondence. From 5.3.2 one deduces, or else defines analogously, a pairing

$$\mathrm{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes_S^{\mathbf{L}} \mathrm{RHom}(c_3''^* L_3, c_4''^! L_4) \longrightarrow \mathrm{RHom}(c_{13}^* (L_1 \otimes_S^{\mathbf{L}} L_3), c_{24}^! (L_2 \otimes_S^{\mathbf{L}} L_4)), \tag{5.3.4}$$

hence an external tensor product pairing, $(u, v) \mapsto u \otimes_S^{\mathbf{L}} v$,

$$\mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(c_3''^* L_3, c_4''^! L_4) \longrightarrow \mathrm{Hom}(c_{13}^* (L_1 \otimes_S^{\mathbf{L}} L_3), c_{24}^! (L_2 \otimes_S^{\mathbf{L}} L_4)). \tag{5.3.5}$$

One checks that these pairings are compatible with direct images, in the following sense. Let, for $1 \leq i \leq 4$, $f_i : X_i \to Y_i$ be an S-morphism, let c', c'', c be as above, with c' and c'' proper, let

$d' : D' \to Y_{12}$ and $d'' : D'' \to Y_{34}$ be proper morphisms, let $d : D = D' \times_S D'' \to Y = Y_{12} \times_S Y_{34}$, and suppose given the commutative squares

$$\begin{tikzcd} X_{12} & C' \arrow[l] \arrow[d] \\ Y_{12} \arrow[u, "f_{12}"] & D' \arrow[l] \end{tikzcd} \qquad \begin{tikzcd} X_{34} & C'' \arrow[l] \arrow[d] \\ Y_{34} \arrow[u, "f_{34}"] & D'' \arrow[l] \end{tikzcd}$$

Then one has a commutative square

$$\begin{tikzcd} \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(c_3''^* L_3, c_4''^! L_4) \arrow[r, "5.3.5"] \arrow[d, "f_{12*} \otimes f_{34*}"'] & \mathrm{Hom}(c_{13}^* L_{13}, c_{24}^! L_{24}) \arrow[d, "f_{13,24*}"] \\ \mathrm{Hom}(d_1'^* M_1, d_2'^! M_2) \otimes \mathrm{Hom}(d_3''^* M_3, d_4''^! M_4) \arrow[r, "5.3.5"] & \mathrm{Hom}(d_{13}^* M_{13}, d_{24}^! M_{24}), \end{tikzcd} \tag{5.3.6}$$

where $M_i = f_{i!} L_i$ if $i = 1, 3$ and $f_{i*} L_i$ otherwise, $L_{ij} = L_i \otimes_S^{\mathbf{L}} L_j$, and $M_{ij} = M_i \otimes_S^{\mathbf{L}} M_j$ (there is also an analogous square with the RHom's).

Suppose that $X_3 = X_2$, $X_4 = X_1$, $L_3 = L_2$, and $L_4 = L_1$. A reasoning analogous to that of the last paragraph of 5.2 shows that, for $u \in \mathrm{Hom}_S(L_1, L_2)$ and $v \in \mathrm{Hom}_S(L_2, L_1)$, one has

$$\langle u, v \rangle = \langle u \otimes_S^{\mathbf{L}} v, 1 \rangle, \tag{5.3.7}$$

where 1 denotes the diagonal correspondence from L_{12} to L_{12} on X_{12}, and the cup product in the second member is taken in the sense of 4.2.5. More generally, if $c' : C' \to X_{12}$ and $c'' : C'' \to X_{21}$ are correspondences, their "intersection" $C = C' \times_{X_{12}} C''$ identifies with the fibred product of $C' \times_S C'' \to X = X_{12} \times_S X_{21}$ and the diagonal $X_{12} \to X$, and, for $u \in \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2)$ and $v \in \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1)$, one has the following equality in $H^0(C, K_C)$:

$$\langle u, v \rangle = \langle u \otimes_S^{\mathbf{L}} v, 1 \rangle, \tag{5.3.8}$$

where 1 denotes, as above, the diagonal correspondence. Thus there are two "diagonal" procedures (5.2.10, 5.3.8) for the calculation of the "Verdier local term" $\langle u, v \rangle$.

6. Appendix. Lefschetz formula for coherent sheaves

In this number, S denotes a noetherian scheme. The S-schemes considered will be assumed separated and of finite type. Recall that, by Nagata, every S-morphism between such schemes is compactifiable, i.e. factors as an open immersion followed by a proper morphism. For every S-scheme X, we shall write $D(X)$ for the derived category of the category of $\mathcal{O}_X$-modules.

6.1

Recall ([8] App.) that for every S-morphism $f : X \to Y$ one has a functor

$$f^! : D^+(Y) \longrightarrow D^+(X)$$

(and transitivity isomorphisms $(gf)^! = f^! g^!$ for a composite gf, with a cocycle condition). If f is proper, $f^!$ is right "adjoint" to the functor $f_* : D^+(X)_{\mathrm{coh}} \to D^+(Y)_{\mathrm{coh}}$ (where the subscript "coh" denotes the full subcategory formed by complexes with coherent cohomology), this adjunction being made precise by an isomorphism

$$f_* \mathrm{RHom}(E, f^! F) \xrightarrow{\sim} \mathrm{RHom}(f_* E, F) \tag{6.1.1}$$

for $E \in \mathrm{ob}\, D(X)_{\mathrm{coh}}$, $F \in \mathrm{ob}\, D^+(Y)$, defined analogously to the isomorphism (SGA 4 XVIII 3.1.9.6). The considerations of ([8] App.5), together with the duality theorem for projective space, show that, if f is smooth, purely of dimension d, there is a canonical isomorphism

$$f^! L \xleftarrow{\sim} f^* L \otimes \Omega^d_{X/Y}[d]; \tag{6.1.2}$$

and, if moreover f is proper, the arrow

$$\mathrm{Tr}_f : f_*(f^* L \otimes \Omega^d_{X/Y}[d]) \longrightarrow L \tag{6.1.3}$$

deduced, by 6.1.2, from the adjunction arrow $f_* f^! \to I$, coincides with the trace morphism ([8] VI).

6.2

Contrary to what happens in the case of discrete coefficients, there is no reasonable subcategory of D here that is stable under the "six operations" $(f^*, f_*, f^!, f_!, \otimes^{\mathbf{L}}, \mathrm{RHom})$. The most natural finiteness notion seems to be relative perfection (SGA 6 III 4). If X is an S-scheme and $L \in \mathrm{ob}\, D(X)$, recall that L is said to be perfect relative to S if L has finite tor-dimension relative to S and coherent cohomology. We shall denote by $D(X)_{\mathrm{parf}/S}$ the full subcategory of $D(X)$ formed by the complexes perfect relative to S. It is not stable in general either under $\otimes^{\mathbf{L}}$ or under RHom. However, if E is perfect in the absolute sense and F is perfect relative to S, then $\mathrm{RHom}(E, F)$ is perfect relative to S (SGA 6 III 4.9). On the other hand, let, for $i = 1, 2$, X_i be an S-scheme and $L_i \in \mathrm{ob}\, D^-(X_i)$. We shall put, as in 1.5,

$$L_1 \otimes^{\mathbf{L}}_S L_2 = Lp_1^* L_1 \otimes^{\mathbf{L}} Lp_2^* L_2,$$

where $p_i : X_1 \times_S X_2 \to X_i$ is the projection. Considering this object is reasonable only if X_1 and X_2 are tor-independent over S, i.e. (SGA 6 III 1.5) if $\mathrm{Tor}_i^S(\mathcal{O}_X, \mathcal{O}_Y) = 0$ for $i \neq 0$. Suppose this is the case: then, if L_1 and L_2 are perfect relative to S, so is $L_1 \otimes^{\mathbf{L}}_S L_2$ (SGA 6 III 4.7).

Let $f : X \to Y$ be an S-morphism. If f is proper, then $f_* D(X)_{\mathrm{parf}/S} \subset D(Y)_{\mathrm{parf}/S}$ (SGA 6 III 4.8). If f has finite tor-dimension, then $f^* D(Y)_{\mathrm{parf}/S} \subset D(X)_{\mathrm{parf}/S}$ (SGA 6 III 4.5), and $f^! D(Y)_{\mathrm{parf}/S} \subset D(X)_{\mathrm{parf}/S}$ (SGA 6 III 4.9).

For every S-scheme $a : X \to S$, we put

$$K_X = a^! \mathcal{O}_S, \tag{6.2.1}$$

and write D_X (or simply D) for the functor $\mathrm{RHom}(-, K_X)$. If X has finite tor-dimension over S, then K_X is perfect relative to S and, for every $L \in \mathrm{ob}\, D(X)_{\mathrm{parf}/S}$, one has $DL \in \mathrm{ob}\, D(X)_{\mathrm{parf}/S}$, and the canonical arrow $L \to DDL$ is an isomorphism (SGA 6 III 4.9).

Let $f : X \to Y$ be an S-morphism, $L \in \mathrm{ob}\, D(X)$, and $M \in \mathrm{ob}\, D(Y)$. If $f^* M$ is defined (for example if $M \in \mathrm{ob}\, D^-(Y)$, or if f has finite tor-dimension), put

$$\mathrm{Hom}_f^{(1)}(L, M) = \mathrm{Hom}(f^* M, L), \qquad \mathrm{Hom}_f^{(2)}(L, M) = \mathrm{Hom}(L, f^* M).$$

If $f^! M$ is defined (for example if $M \in \operatorname{ob} D^+(Y)$, or if f is smooth, or finite of finite tor-dimension, or a composite of such morphisms, since in these cases $f^!$ extends, by 6.1.1 and 6.1.2, to a functor defined on the whole category $D(Y)$), put

$$\operatorname{Hom}_f^{(3)}(L, M) = \operatorname{Hom}(L, f^! M), \qquad \operatorname{Hom}_f^{(4)}(L, M) = \operatorname{Hom}(f^! M, L).$$

The elements of $\operatorname{Hom}_f^{(i)}(L, M)$ will be called, as in 1.4, *arrows of type i from L to M over f* (or f-arrows of type i from L to M). We leave to the reader the task of making explicit the composition of arrows of type i, for fixed i, and the corresponding fibred and cofibred categories. We recall only the partial adjunction formulas

$$\operatorname{Hom}_f^{(1)}(L, M) = \operatorname{Hom}(M, f_* L) \quad (M \in \operatorname{ob} D^-(Y),\ L \in \operatorname{ob} D^+(X)),$$

$$\operatorname{Hom}_f^{(3)}(L, M) = \operatorname{Hom}(f_* L, M) \quad (f \text{ proper},\ L \in \operatorname{ob} D(X)_{\mathrm{coh}},\ M \in \operatorname{ob} D^+(Y)).$$

6.3

In the situation of 1.6, suppose that X_1 and X_2 are tor-independent over S, and likewise Y_1 and Y_2. Then, for $L_i \in \operatorname{ob} D(X_i)_{\mathrm{parf}/S}$, the Künneth arrow 1.6.4 is defined (use the fact that finite relative tor-dimension is preserved by direct image (SGA 6 III 3.7.1)) and is an isomorphism (to verify this, one may assume S and the Y_i affine, and, for a Čech calculation, reduce to also assuming X_1 and X_2 affine; the conclusion is then a trivial consequence of the tor-independence hypotheses). On the other hand, for $M_i \in \operatorname{ob} D(Y_i)_{\mathrm{parf}/S}$, with $f_1^! M_1$ or $f_2^! M_2$ in D^b, the Künneth arrow 1.7.3 is defined[9], and is an isomorphism when f_1 and f_2 have finite tor-dimension, as is easily checked with the help of the Künneth isomorphism of type 1.6.4 just indicated, conjugated with 6.1.2, and with 6.1.1 in the case where f_i is a closed immersion. In particular, if X_1 and X_2 have finite tor-dimension over S and are tor-independent, 1.7.6 is an isomorphism.

One defines $\operatorname{RHom}(u, v)$ as in 2.1 for (u, v) of type $(2, 1)$ (respectively $(3, 4)$, with f proper in order to avoid the pro-objects caused by an $f_!$). Under suitable degree hypotheses (see for example ([9] V 2.3)), the evaluation arrows (2.2.2) and tensor-product arrows (2.2.3, 2.2.4) are defined, and the arrows 2.2.3 and 2.2.4 (provided they are defined) are isomorphisms as soon as E_1 and E_2 are perfect in the absolute sense.

In the situation of 2.4, suppose that the pairs (X_1, X_2), (X_1, Y_2), (Y_1, X_2), (Y_1, Y_2) are tor-independent over S, that f_1 is proper, that E_i, F_i, P_i, Q_i, $\operatorname{RHom}(E_i, F_i)$, $\operatorname{RHom}(P_i, Q_i)$ are perfect relative to S, and that the Künneth arrows

$$f_1^! Q_1 \otimes_S^{\mathbf{L}} Q_2 \to (f_1 \times_S Y_2)^! (Q_1 \otimes_S^{\mathbf{L}} Q_2), \qquad f_1^! Q_1 \otimes_S^{\mathbf{L}} F_2 \to (f_1 \times_S Y_2)^! (Q_1 \otimes_S^{\mathbf{L}} F_2)$$

are isomorphisms. Then 2.4.1 and the diagram of 2.5 are defined, and the latter is commutative (however, the horizontal arrows are not necessarily isomorphisms). The verification is the same as for 2.5.

6.4

Let, for $i = 1, 2$, X_i be an S-scheme, let $X = X_1 \times_S X_2$, and let $p_i : X \to X_i$, $q_i : X_i \to S$ be the canonical projections. Suppose that X_1 and X_2 are tor-independent and that q_1 has finite

[9] In the proper or lissifiable case, the definition of the arrow is immediate; the general case can be reduced to the lissifiable case by cohomological descent.

tor-dimension. Let $L_i \in \operatorname{ob} D(X_i)_{\mathrm{parf}/S}$. By the analogue of 2.2.4, one has a canonical arrow

$$DL_1 \otimes_S^{\mathbf{L}} L_2 \longrightarrow \mathrm{RHom}(L_1 \otimes^{\mathbf{L}} \mathcal{O}_{X_2}, q_1^! \mathcal{O}_S \otimes_S^{\mathbf{L}} L_2), \tag{6.4.1}$$

and hence, by composing with the Künneth isomorphism (6.3) $q_1^! \mathcal{O}_S \otimes_S^{\mathbf{L}} L_2 \xrightarrow{\sim} p_2^! L_2$, an arrow

$$DL_1 \otimes_S^{\mathbf{L}} L_2 \longrightarrow \mathrm{RHom}(p_1^* L_1, p_2^! L_2). \tag{6.4.2}$$

The arrow 6.4.1, and therefore also 6.4.2, is an isomorphism, as one sees by reducing to the case where q_1 is a closed immersion.

As in 3.1.2, we shall put

$$\mathrm{RHom}(p_1^* L_1, p_2^! L_2) = \mathrm{RHom}_S(L_1, L_2), \qquad \mathrm{Hom}(p_1^* L_1, p_2^! L_2) = \mathrm{Hom}_S(L_1, L_2), \tag{6.4.3}$$

and call the elements of $\mathrm{Hom}_S(L_1, L_2)$ *cohomological correspondences from L_1 to L_2*. If $c : C \to X$ is a correspondence (proper in practice), then by the induction formula ([8] III 8.8)[10]

$$c^! \mathrm{RHom}(p_1^* L_1, p_2^! L_2) = \mathrm{RHom}(c_1^* L_1, c_2^! L_2), \tag{6.4.4}$$

and the elements of

$$\mathrm{Hom}(c_1^* L_1, c_2^! L_2) = H^0(X, c_* c^! \mathrm{RHom}(p_1^* L_1, p_2^! L_2))$$

will be called *cohomological correspondences from L_1 to L_2 with support in C*. We shall see below examples analogous to those of 3.2.

6.5

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be a proper S-morphism, and let $f = f_1 \times_S f_2 : X \to Y$. Suppose that the X_i and Y_i have finite tor-dimension over S, and that the squares of 2.4.0 are tor-independent. Let $L_i \in \operatorname{ob} D(X_i)_{\mathrm{parf}/S}$. As in 3.3.1, one defines an isomorphism

$$f_* \mathrm{RHom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{RHom}_S(f_{1*} L_1, f_{2*} L_2), \tag{6.5.1}$$

hence, by applying $R\Gamma(Y, -)$, an isomorphism

$$f_* : \mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_S(f_{1*} L_1, f_{2*} L_2), \tag{6.5.2}$$

which we shall simply denote by $u \mapsto u_*$ when $Y_1 = Y_2 = S$. There is a commutative diagram analogous to 3.4.1, as follows from the analogue of 2.5 indicated in 6.3. One deduces from it a corollary analogous to 6.5 and, in the case where $Y_1 = Y_2 = S$ is the spectrum of a field, a description, analogous to that of 3.6, of the operations u_* on cohomology.

Given a commutative square 3.7.1, with c proper, one has, for $K \in \operatorname{ob} D^+(C)$, a canonical arrow

$$f_* c_* c^! K \longrightarrow d_* d^! f_* K, \tag{6.5.3}$$

defined in the same way as 3.7.3 (the isomorphism $g'_* c^! \xrightarrow{\sim} d^! f_*$ is also valid in the present context: definition of the arrow by transposition of a base-change arrow, and verification that it is an isomorphism by reduction to the cases where d is smooth and where d is a closed immersion). Taking $K = \mathrm{RHom}_S(L_1, L_2)$, one deduces from 6.5.3 and 6.5.1 canonical arrows

$$f_* c_* \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \longrightarrow d_* \mathrm{RHom}(d_1^*(f_{1*} L_1), d_2^!(f_{2*} L_2)), \tag{6.5.4}$$

$$f_* : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \longrightarrow \mathrm{Hom}(d_1^*(f_{1*} L_1), d_2^!(f_{2*} L_2)), \tag{6.5.5}$$

analogous to 3.7.5 and 3.7.6.

[10] The "lissifiability" hypothesis of (loc. cit.) is unnecessary.

6.6

Let, for $i = 1, 2$, X_i be S-schemes of finite tor-dimension and tor-independent over S, and let $L_i \in \operatorname{ob} D(X_i)_{\mathrm{parf}/S}$. The external tensor product of the evaluation arrows $DL_i \otimes^{\mathbf{L}} L_i \to K_{X_i}$ defines, as in 4.1, "cup-product" pairings analogous to 4.1.2, 4.1.3, and 4.1.4.

Let $c : C \to X$ and $d : D \to X$ be proper morphisms, let $e = c \times_X d : E \to X$, and let $P, Q \in \operatorname{ob} D(X)_{\mathrm{parf}/S}$. By 6.1.1, one has

$$c_* c^! P = \mathrm{RHom}(Rc_* \mathcal{O}_C, P)$$

(and an analogous isomorphism with d). Assuming c and d of finite tor-dimension at the points above $e(E)$, one deduces, by tensor product, a canonical arrow

$$c_* c^! P \otimes^{\mathbf{L}} d_* d^! Q \longrightarrow \mathrm{RHom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, P \otimes^{\mathbf{L}} Q). \tag{6.6.1}$$

When c and d are tor-independent, the second member is rewritten, by Künneth and 6.1.1, as $e_* e^! (P \otimes^{\mathbf{L}} Q)$, and 6.6.1 takes the form 4.2.3. For $P = DL_1 \otimes^{\mathbf{L}}_S L_2$, $Q = L_1 \otimes^{\mathbf{L}}_S DL_2$, one deduces from 6.6.1 and the pairings indicated above pairings analogous to 4.2.4 and 4.2.5:

$$c_* \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \otimes^{\mathbf{L}} d_* \mathrm{RHom}(d_2^* L_2, d_1^! L_1) \longrightarrow \mathrm{RHom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, K_X), \tag{6.6.2}$$

$$\langle\ ,\ \rangle : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}(d_2^* L_2, d_1^! L_1) \longrightarrow \mathrm{Hom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, K_X). \tag{6.6.3}$$

Just like the pairings of 4.2, these are compatible with étale localization. When c and d are tor-independent, the second member of 6.6.2 (respectively 6.6.3) identifies canonically with $e_* K_E$ (respectively $H^0(E, K_E)$).

Theorem 4.4 has the following analogue:

Theorem 6.7. Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be a proper S-morphism, let $f = f_1 \times_S f_2 : X \to Y$, and let $p_i : X \to X_i$, $q_i : Y \to Y_i$ be the projections. Suppose given, on the other hand, a cube 4.4.0 with cartesian horizontal faces, with c', c'', d', d'' proper, and with c', c'' (respectively d', d'') of finite tor-dimension at the points above $c(C)$ (respectively $d(D)$). Put $c'_i = p_i c'$, $c''_i = p_i c''$, $d'_i = q_i d'$, and $d''_i = q_i d''$. Finally, let $L_i \in \operatorname{ob} D(X_i)_{\mathrm{parf}/S}$. Assume that the pairs (X_1, X_2), (X_1, Y_2), (Y_1, X_2), (Y_1, Y_2) are tor-independent over S, and that X_i and Y_i have finite tor-dimension over S. Then one has a commutative square

$$\begin{tikzcd}
f_* c'_* \mathrm{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes f_* c''_* \mathrm{RHom}(c_2''^* L_2, c_1''^! L_1) \arrow[r, "(1)"] \arrow[d, "(2)"] & f_* \mathrm{RHom}(Rc'_* \mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_* \mathcal{O}_{C''}, K_X) \arrow[d, "(4)"] \\
d'_* \mathrm{RHom}(d_1'^* M_1, d_2'^! M_2) \otimes d''_* \mathrm{RHom}(d_2''^* M_2, d_1''^! M_1) \arrow[r, "(3)"] & \mathrm{RHom}(Rd'_* \mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_* \mathcal{O}_{D''}, K_Y),
\end{tikzcd} \tag{6.7.1}$$

where $M_i = f_{i!} L_i$, (3) is given by 6.6.2, (1) is deduced from 6.6.2 by application of f_* and composition with the canonical arrow $f_* c'_* \otimes f_* c''_* \to f_*(c'_* \otimes c''_*)$, (2) is the tensor product of the arrows 6.5.4 relative to the faces containing f, and (4) is the composite of the duality isomorphism 6.1.1 and of the arrow defined by the canonical arrow

$$Rd'_* \mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_* \mathcal{O}_{D''} \longrightarrow f_*(Rc'_* \mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_* \mathcal{O}_{C''})$$

which is adjoint to the tensor product of the base-change arrows $f^* Rd'_* \mathcal{O}_{D'} \to Rc'_* \mathcal{O}_{C'}$, $f^* Rd''_* \mathcal{O}_{D''} \to Rc''_* \mathcal{O}_{C''}$.

The proof is essentially the same as that of 4.4; we leave to the reader the small modifications to be made.

When c' and c'' are tor-independent, and likewise d' and d'', the arrow (4) of 6.7.1 is rewritten in the form $f_*c_*K_C \to d_*K_D$; it is simply defined by the adjunction arrow $g_*K_C = g_*g^!K_D \to K_D$ (as is arrow (4) of 4.4.1). However, it would not be natural to consider only this case, since that would exclude correspondences with excess intersection.

From 6.7 one deduces formulas analogous to 4.5.1 and 4.7.1: under the hypotheses of 6.7, one has, for $u \in \mathrm{Hom}(c_1'^*L_1, c_2'^!L_2)$ and $v \in \mathrm{Hom}(c_2''^*L_2, c_1''^!L_1)$,

$$\langle f_*u, f_*v\rangle = f_*\langle u, v\rangle, \tag{6.7.2}$$

where f_*u (respectively f_*v) denotes the direct image correspondence of u (respectively v) in the sense of 6.5.5, and f_* in the second member is the homomorphism

$$\mathrm{Hom}(Rc'_*\mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_*\mathcal{O}_{C''}, K_X) \longrightarrow \mathrm{Hom}(Rd'_*\mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_*\mathcal{O}_{D''}, K_Y) \tag{6.7.3}$$

which is the composite of the duality isomorphism 6.1.1 and the arrow induced by the canonical arrow

$$Rd'_*\mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_*\mathcal{O}_{D''} \longrightarrow f_*(Rc'_*\mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_*\mathcal{O}_{C''})$$

cited in 6.7. When $Y_1 = Y_2 = S = D' = D''$, 6.7.2 is the *Lefschetz-Verdier formula.*

As an illustration, we shall show that this formula implies the Woods Hole formula [1].

6.8

For this purpose we shall need a few reminders on the "Hodge" cohomology class associated with a regularly embedded subscheme of a smooth scheme and on Grothendieck residues ([8] III 9). Let $f : X \to S$ be a smooth morphism purely of relative dimension N, and let $i : Y \hookrightarrow X$ be a regular immersion of codimension d, i.e. ((SGA IV 16.9.2), (SGA 6 VII 1)) a closed subscheme defined locally by a regular sequence of d equations, or, in equivalent terminology ((8 III 1), (SGA 6 VIII 1)), a closed immersion, locally a complete intersection, of codimension d. Put $g = fi$. Then one associates with Y an element

$$\mathrm{cl}^d_{X/S}(Y) \in \mathrm{Ext}^d_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S}), \tag{6.8.1}$$

called the *cohomology class associated with* Y, defined as follows. By the "purity" theorem ([8] III 7.2), one has

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S}) = \begin{cases} 0, & i \neq d, \\ \mathrm{Hom}(\bigwedge^d N_{Y/X}, \Omega^d_{X/S} \otimes \mathcal{O}_Y), & i = d, \end{cases} \tag{6.8.2}$$

where $N_{Y/X} = I/I^2$ is the conormal sheaf ($I = \mathrm{Ker}\ \mathcal{O}_X \to \mathcal{O}_Y$). Therefore

$$\mathrm{Ext}^d_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S}) = H^0(Y, \underline{\mathrm{Hom}}(\bigwedge^d N_{Y/X}, \Omega^d_{X/S} \otimes \mathcal{O}_Y)). \tag{6.8.3}$$

The differential $d_{X/S} : \mathcal{O}_X \to \Omega^1_{X/S}$ induces an $\mathcal{O}_Y$-linear map

$$d_{X/S} \otimes 1 : N_{Y/X} \longrightarrow \Omega^1_{X/S} \otimes \mathcal{O}_Y.$$

The d-th exterior power of this map is a global section, on Y, of $\Lambda^d N^\vee_{Y/X} \otimes \Omega^d_{X/S}$; its image in $\mathrm{Ext}^d_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S})$ under the isomorphism 6.8.3 is, by definition, the class 6.8.1.

Assume g finite and flat (hence $d = N$). Let $\omega \in H^0(Y, \Lambda^N N^\vee_{Y/X} \otimes \Omega^N_{X/S})$. By 6.8.2 and 6.1, one may view ω as a homomorphism

$$\mathcal{O}_Y \longrightarrow g^!\mathcal{O}_S (= \Lambda^N N^\vee_{Y/X} \otimes \Omega^N_{X/S}),$$

and therefore, by adjunction, as a homomorphism $g_*\mathcal{O}_Y \to \mathcal{O}_S$. The residue (relative to g) of ω is defined as the section of $\mathcal{O}_S$ which is the image of 1 under this homomorphism, and is denoted $\mathrm{Res}_{Y/S}(\omega)$ (cf. ([8] III 9)). Let, on the other hand, $i_*\omega \in H^N(X, \Omega^N_{X/S})$ denote the image of ω by the composite of the isomorphism 6.8.3 and the canonical arrow

$$\mathrm{Ext}^N_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^N_{X/S}) \longrightarrow H^N(X, \Omega^N_{X/S}).$$

When f is proper, it follows from transitivity for trace morphisms (or adjunction arrows) $i_*i^! \to 1$, $f_*f^! \to 1$, that

$$\mathrm{Res}_{Y/S}(\omega) = \mathrm{Tr}_f(i_*\omega), \tag{6.8.4}$$

where $\mathrm{Tr}_f : H^N(X, \Omega^N_{X/S}) \to H^0(S, \mathcal{O}_S)$ is defined by the adjunction arrow $f_*f^!\mathcal{O}_S = Rf_*\Omega^N_{X/S}[N] \to \mathcal{O}_S$ (6.1). On the other hand, without any properness hypothesis, one has the important compatibility ([8] III 9 R6)

$$\mathrm{Res}_{Y/S}(y \,\mathrm{cl}^N_{X/S}(Y)) = \mathrm{Tr}_{Y/S}(y), \tag{6.8.5}$$

for $y \in H^0(Y, \mathcal{O}_Y)$, where $\mathrm{Tr}_{Y/S} : g_*\mathcal{O}_Y \to \mathcal{O}_S$ denotes the trace in the classical sense, i.e. $\mathrm{Tr}_{Y/S}(y)$ is the trace of the endomorphism of $g_*\mathcal{O}_Y$ defined by multiplication by y (this compatibility is not proved in loc. cit.; the case $N = 1$ is treated in [Raynaud, Anneaux locaux henséliens, VII 1, Lecture Notes in Math. n° 169, Springer Verlag, 1970]). In other words, $\mathrm{Tr}_{Y/S}$ corresponds, by adjunction, to the homomorphism $\mathcal{O}_Y \to g^!\mathcal{O}_S$ defined by multiplication by $\mathrm{cl}^N_{X/S}(Y)$. It follows that, for $F \in \mathrm{ob}\, D(S)_{\mathrm{coh}}$, the arrow

$$g^*F \longrightarrow g^!F = g^*F \otimes^{\mathbf{L}} g^!\mathcal{O}_S \tag{6.8.6}$$

given by multiplication by $\mathrm{cl}^N_{X/S}(Y)$ corresponds by adjunction to the arrow

$$g_*g^*F = F \otimes^{\mathbf{L}} g_*\mathcal{O}_Y \longrightarrow F \tag{6.8.7}$$

defined by $\mathrm{Tr}_{Y/S} : g_*\mathcal{O}_Y \to \mathcal{O}_S$.

6.9

Let X (respectively Y) be a proper smooth S-scheme of relative dimension m (respectively n), $Z = X \times_S Y$, and let $a : A \hookrightarrow Z$, $b : B \hookrightarrow Z$ be regular immersions such that $a_2 : A \to Y$, $b_1 : B \to X$ are finite and flat (write $p_1 : Z \to X$, $p_2 : Z \to Y$ for the projections, and $a_i = p_i a$, $b_i = p_i b$). Let $c : C = A \times_Z B \to Z$, $L \in \mathrm{ob}\, D(X)_{\mathrm{parf}}$, $M \in \mathrm{ob}\, D(Y)_{\mathrm{parf}}$ (the subscript parf means "perfect in the absolute sense", which is also equivalent to being perfect rel. to S, since X and Y are smooth). Assume that C is finite and flat over S. The hypotheses made therefore imply that c is a regular immersion and that a and b are tor-independent.

One has $p_1^*\Omega^1_{X/S} = \Omega^1_{Z/Y}$, $p_2^*\Omega^1_{Y/S} = \Omega^1_{Z/X}$, and a decomposition

$$\Omega^1_{Z/S} = \Omega^1_{Z/Y} \oplus \Omega^1_{Z/X},$$

whence, for every integer r, a decomposition

$$\Omega^r_{Z/S} = \bigoplus_{i+j=r} \Omega^i_{Z/Y} \otimes \Omega^j_{Z/X}. \tag{6.9.0}$$

On the other hand, by 6.1.2, one has $p_2^! M = p_2^* M \otimes \Omega^m_{Z/Y}[m]$, hence

$$H^0(Z, a_* a^! \mathrm{RHom}(p_1^* L, p_2^! M)) = \mathrm{Ext}^m_{\mathcal{O}_Z}(a_1^* L, a_2^* M \otimes \Omega^m_{Z/Y}),$$

whence, by 6.4.4 and 6.8.2, an isomorphism

$$\mathrm{Hom}(a_1^* L, a_2^! M) = H^0(A, \mathrm{RHom}(a_1^* L, a_2^* M) \otimes \Lambda^m N^\vee_{A/Z} \otimes \Omega^m_{Z/Y}). \tag{6.9.1}$$

Similarly one has an isomorphism

$$\mathrm{Hom}(b_2^* M, b_1^! L) = H^0(B, \mathrm{RHom}(b_2^* M, b_1^* L) \otimes \Lambda^n N^\vee_{B/Z} \otimes \Omega^n_{Z/X}). \tag{6.9.2}$$

Let $u \in \mathrm{Hom}(a_1^* L, a_2^* M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^* L)$, and write

$$\langle u, v \rangle \in H^0(C, \mathcal{O}_C) \tag{6.9.3}$$

for the section $\mathrm{Tr}(u_C v_C) = \mathrm{Tr}(v_C u_C)$, where $u_C \in \mathrm{Hom}(c_1^* L, c_2^* M)$ and $v_C \in \mathrm{Hom}(c_2^* M, c_1^* L)$ are induced by u and v. Consider the cohomological correspondence $u^! \in \mathrm{Hom}(a_1^* L, a_2^! M)$ (respectively $v^! \in \mathrm{Hom}(b_2^* M, b_1^! L)$) deduced from u (respectively v) by composition with the trace homomorphism 6.8.7 $a_{2*} a_2^* M \to M$ (respectively $b_{1*} b_1^* L \to L$), or equivalently (6.8.6) by product with $\mathrm{cl}_{Z/Y}(A)$ (respectively $\mathrm{cl}_{Z/X}(B)$). Note in passing that $\mathrm{cl}_{Z/Y}(A)$ (respectively $\mathrm{cl}_{Z/X}(B)$) is the component in $H^0(A, \Lambda^m N^\vee_{A/Z} \otimes \Omega^m_{Z/Y})$ (respectively $H^0(B, \Lambda^n N^\vee_{B/Z} \otimes \Omega^n_{Z/X})$) of $\mathrm{cl}_{Z/S}(A)$ (respectively $\mathrm{cl}_{Z/S}(B)$) defined by the projection $\Omega^m_{Z/S} \to \Omega^m_{Z/Y}$ (respectively $\Omega^n_{Z/S} \to \Omega^n_{Z/X}$) of 6.9.0. One checks easily that the cup-product

$$\langle u^!, v^! \rangle \in \mathrm{Ext}^{m+n}_{\mathcal{O}_Z}(\mathcal{O}_C, \Omega^{m+n}_{Z/S}) = H^0(C, \Lambda^{m+n} N^\vee_{C/Z} \otimes \Omega^{m+n}_{Z/S})$$

defined in 6.6.3 is given by

$$\langle u^!, v^! \rangle = \langle u, v \rangle \, \mathrm{cl}_{Z/Y}(A) \, \mathrm{cl}_{Z/X}(B). \tag{6.9.4}$$

On the other hand one checks that the homomorphism 6.7.3

$$\mathrm{Ext}^{m+n}_{\mathcal{O}_Z}(\mathcal{O}_C, \Omega^{m+n}_{Z/S}) \longrightarrow H^0(S, \mathcal{O}_S)$$

is the composite of the canonical arrow $\mathrm{Ext}^{m+n}_{\mathcal{O}_Z}(\mathcal{O}_C, \Omega^{m+n}_{Z/S}) \to H^{m+n}(Z, \Omega^{m+n}_{Z/S})$ and the trace morphism $\mathrm{Tr}_{Z/S} : H^{m+n}(Z, \Omega^{m+n}_{Z/S}) \to H^0(S, \mathcal{O}_S)$. The Lefschetz-Verdier formula 6.7.2 therefore gives the relation

$$\langle (u^!)_*, (v^!)_* \rangle = \mathrm{Tr}_{Z/S}(\langle u, v \rangle \, \mathrm{cl}_{Z/Y}(A) \, \mathrm{cl}_{Z/X}(B)). \tag{6.9.5}$$

Thanks to 6.8.4 one may rewrite the second member as a residue; thus, in summary, one has obtained:

Theorem 6.10. Under the hypotheses of the first paragraph of 6.9, let $u \in \mathrm{Hom}(a_1^* L, a_2^* M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^* L)$, and consider the cohomological correspondence $u^!$ (respectively $v^!$) from L to M (respectively from M to L), with support in A (respectively B), deduced from u (respectively v) by composition with the trace $a_{2*} a_2^* M \to M$ (respectively $b_{1*} b_1^* L \to L$) (6.8.7), and the homomorphism $(u^!)_* : f_* L \to g_* M$ (respectively $(v^!)_* : g_* M \to f_* L$) deduced from $u^!$ (respectively $v^!$) by direct image (6.5.5) ($f : X \to S$, $g : Y \to S$ denoting the projections). Then, with the notation of 6.8, 6.9.3, one has

$$\langle (u^!)_*, (v^!)_* \rangle = \mathrm{Res}_{C/S}(\langle u, v\rangle \, \mathrm{cl}_{Z/Y}(A) \, \mathrm{cl}_{Z/X}(B)). \tag{6.10.1}$$

Remarks 6.11. a) As in 3.6, using the description 3.5 b) of the direct image of a correspondence, one checks that, if $h : Z \to S$ is the projection, $(u^!)_*$ (respectively $(v^!)_*$) is the composite of the arrows

$$f_*L \xrightarrow{(1)} (ha)_* a_1^* L \xrightarrow{(ha)_* u} (ha)_* a_2^* M = g_* a_{2*} a_2^* M \xrightarrow{(2)} g_* M,$$

$$(\text{respectively } g_*M \xrightarrow{(1)} (hb)_* b_2^* M \xrightarrow{(hb)_* v} (hb)_* b_1^* L = f_* b_{1*} b_1^* L \xrightarrow{(2)} f_* L),$$

where (1) is the usual functoriality arrow (defined by the adjunction arrow $1 \to a_{1*}a_1^*$ (respectively $1 \to b_{2*}b_2^*$)) and (2) is given by the trace $a_{2*}a_2^*M \to M$ (respectively $b_{1*}b_1^*L \to L$) (6.8.7).

b) Suppose that S is the spectrum of a field. The intersection $C = A \times_{(X \times_S Y)} B$ is then formed of isolated points. At each point z of C, choose, near the images of z, regular sequences $(s_1, \dots, s_m)$, $(t_1, \dots, t_n)$ of equations of A, B. Then 6.10.1 is written, with the notation of the residue symbols of ([8] III 9), as

$$\langle (u^!)_*, (v^!)_* \rangle = \sum_{z \in C} \mathrm{Res}_z \left(\langle u, v \rangle \prod_{1 \le i \le m} \frac{d_{Z/Y} s_i}{s_i} \prod_{1 \le j \le n} \frac{d_{Z/X} t_j}{t_j} \right), \tag{6.11.1}$$

where $d_{Z/Y}s_i$ (respectively $d_{Z/X}t_j$) denotes the component of type $(1,0)$ (respectively $(0,1)$) of $d_{Z/S}s_i$ (respectively $d_{Z/S}t_j$) in the decomposition 6.9.0.

Corollary 6.12. (Woods-Hole formula [1]). Assume that S is the spectrum of a field. Let X be a proper smooth S-scheme, F an S-endomorphism of X, L an object of $D(X)_{\mathrm{parf}}$ (i.e. a perfect complex on X), and $u \in \mathrm{Hom}(F^*L, L)$. Assume that the scheme of fixed points X^F is finite and consists of transverse points (i.e. points where the graph of F meets the diagonal transversely). Then

$$\sum_i (-1)^i \,\mathrm{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \frac{\mathrm{Tr}(u_x)}{\det(1 - dF_x)},$$

where (F, u) is the endomorphism of $H^i(X, L)$ which is the composite of $H^i(X, L) \to H^i(X, F^*L)$ (inverse image) and $H^i(X, u)$, $\mathrm{Tr}(u_x)$ denotes the value at x of the endomorphism induced by u on the fibre of L at x, and $\det(1 - dF_x)$ is the value at x of the determinant of the map $(1 - \text{map induced by } F \text{ on } \Omega^1_{X,x})$.

Proof. Apply 6.10, taking account of 6.11 a), b). Put $Z = X \times_S X$. Let $x \in X^F$. Near x, choose a system of local coordinates $(x_1, \dots, x_n)$ (defining an étale morphism into $\mathbb{A}^n_S$), put $F_i = x_i F$, and write $(x_1, \dots, x_n, y_1, \dots, y_n)$ for the corresponding local coordinates on Z near $z = (x, x)$ ($x_i = x_i \otimes 1$, $y_i = 1 \otimes x_i$). Then, near z, $(x_1 - y_1, \dots, x_n - y_n)$ (respectively $(y_1 - F_1, \dots, y_n - F_n)$) is a regular system of equations for the diagonal A (respectively for the graph B of F). The contribution at z of the second member of 6.11.1 is

$$\langle u, 1 \rangle_z = \mathrm{Res}_z \left(\mathrm{Tr}(u) \frac{dx_1 \cdots dx_n \, dy_1 \cdots dy_n}{(x_1 - y_1) \cdots (x_n - y_n)(y_1 - F_1) \cdots (y_n - F_n)} \right).$$

Observe that in the numerator one may replace dy_i by

$$dy_i - dF_i = dy_i - \sum_{1 \le j \le n} (dF_i/dx_j) dx_j.$$

Applying then formula ([8] III 9 R3) to the situation $(B \hookrightarrow Z \to S)$, $(s_1, \dots, s_p) = (y_1 - F_1, \dots, y_n - F_n)$, $(t_1, \dots, t_n) = (x_1 - y_1, \dots, x_n - y_n)$, $\omega = (\mathrm{Tr}(u) dx_1 \cdots dx_n)$, one obtains

$$\langle u, 1\rangle_z = \mathrm{Res}_x \left(\mathrm{Tr}(u) \frac{dx_1 \cdots dx_n}{(x_1 - F_1) \cdots (x_n - F_n)} \right). \tag{*}$$

Since x is transverse, the $x_i - F_i$ form a regular system of parameters of $\mathcal{O}_{X,x}$, and hence one may write, near x,

$$x_i = \sum_j (x_j - F_j) c_{ij},$$

where (c_{ij}) is a matrix of sections of $\mathcal{O}_X$ invertible at x, whose determinant has value at x equal to $\det(1 - dF_x)^{-1}$. Thanks to ([8] III 9 R1), one therefore deduces from (*) that

$$\langle u, 1\rangle_z = \mathrm{Res}_x \left(\mathrm{Tr}(u) \det(c_{ij}) \frac{dx_1 \cdots dx_n}{x_1 \cdots x_n} \right),$$

whence, by ([8] III 9 R6),

$$\langle u, 1\rangle_z = \mathrm{Tr}(u(x)) \det(c_{ij}(x)) = \mathrm{Tr}(u(x)) \det(1 - dF_x)^{-1}.$$

The corollary therefore follows from 6.11.1.

Remark 6.12.1. If X^F is assumed only finite, the preceding calculation shows that one has

$$\sum_i (-1)^i \mathrm{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \mathrm{Res}_x \left(\mathrm{Tr}(u) \frac{dx_1 \cdots dx_n}{(x_1 - F_1) \cdots (x_n - F_n)} \right),$$

where (x_i) is a system of local coordinates at x, and the F_i are the coordinates of F (near x) in this system.

The reader will convince himself, moreover, that the preceding formula is still valid if X is proper and is assumed only to be smooth at the points of X^F.

6.13

Assume that S is the spectrum of an algebraic closure of the finite field $\mathbb{F}_q$ and that X/S is obtained by extension of scalars from a proper smooth scheme $X_0/\mathbb{F}_q$. Take for F the Frobenius endomorphism of X/S "defined by raising coordinates to the q-th power". One has $X^F = X_0(\mathbb{F}_q)$, and X^F consists of transverse points since $dF = 0$. If L is a locally free sheaf of finite type on X and $u \in \mathrm{Hom}(F^*L, L)$, the Woods-Hole formula is therefore written as

$$\sum_i (-1)^i \mathrm{Tr}((F, u), H^i(X, L)) = \sum_{x \in X_0(\mathbb{F}_q)} \mathrm{Tr}(u_x, L_x). \tag{6.13.1}$$

In particular, for $L = \mathcal{O}_X$, and u the canonical isomorphism $F^*\mathcal{O}_X \xrightarrow{\sim} \mathcal{O}_X$, one finds:

$$\sum_i (-1)^i \mathrm{Tr}(F, H^i(X, \mathcal{O}_X)) = \mathrm{Card}(X_0(\mathbb{F}_q)) \pmod{p}. \tag{6.13.2}$$

Artin-Schreier theory makes it possible to rewrite the first member in the form $\sum_i (-1)^i \mathrm{Tr}(F, H^i(X, \mathbb{F}_p))$, where $H^*(X, \mathbb{F}_p)$ denotes the étale cohomology of X with values in the constant sheaf $\mathbb{F}_p$. By a refinement of this argument, Deligne, in (SGA $4^{1/2}$ Fonctions L), deduced from 6.13.1 the following generalization of 6.13.2:

Theorem 6.13.3 (Deligne). Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, let X be the scheme deduced from X_0 by extension of scalars to an algebraic closure k of $\mathbb{F}_q$, and let F be the Frobenius endomorphism of X/k. Then, for every $L_0 \in \operatorname{ob} D^b_{\mathrm{ctf}}(X_0, \mathbb{F}_p)$, if L denotes the inverse image of L_0 on X, one has

$$\sum_i (-1)^i \operatorname{Tr}(F, H^i_c(X, L)) = \sum_{x \in X^F} \operatorname{Tr}(F, L_x).$$

Let us point out that the particular case of 6.13.3 corresponding to $L_0 = \mathbb{F}_p$ was established by Katz in (SGA 7 XXII) by a completely different method, based on Dwork's calculation of the zeta function of a hypersurface. Moreover, 6.13.2 in the proper smooth case is also an immediate consequence of the Lefschetz formula in crystalline cohomology ([5] VII 3.1.9). In fact, crystalline cohomology gives the more precise information that each factor $\det(1 - Ft, H^i(X_0/W)) \in W[t]$ of the zeta function is congruent mod p to $\det(1 - Ft, H^i(X_0, \mathcal{O}_{X_0}))$ (at least if $H^*(X_0/W)$ is torsion-free). Katz showed [11] that, using estimates of B. Mazur [14], one could obtain higher congruences (involving the $H^i(X_0, \Omega^j_{X_0})$ and the Cartier operator) for smooth hypersurfaces (or complete intersections) for which the Hodge numbers (in the interesting dimension) are assumed to vanish up to a certain level.

On the other hand, if X_0 is a proper scheme over $\mathbb{F}_q$, geometrically integral, and such that $H^i(X_0, \mathcal{O}_{X_0}) = 0$ for $i > 0$, the Lefschetz-Verdier formula 6.7.2, applied to $X = X_0 \otimes k$ and the Frobenius endomorphism of X, immediately implies that X_0 has at least one rational point over $\mathbb{F}_q$ (since $\operatorname{Tr}(F, R\Gamma(X, \mathcal{O}_X)) = 1$). Serre asks whether this conclusion is still true when one replaces $\mathbb{F}_q$ by a field satisfying (C_1) ([17] II 3), for example $\mathbb{C}(t)$ (Tsen), or $\mathbb{C}((t))$ (Lang) (see loc. cit. for other examples); indeed, he observes that if X_0 is a complete intersection of multidegree $(m_1, \ldots, m_r)$ in $\mathbb{P}^n$, the condition of triviality of cohomology is equivalent ([16] no. 78) to the condition $m_1 + \cdots + m_r < n + 1$, which is also the condition that occurs when one writes condition (C_1).

6.14

The Woods Hole formula has many other applications, for which we refer to [2] and to Beauville's exposé [4].

6.15

The editor has not examined the case of components of fixed points of dimension ≥ 1, but it is not impossible that one could deduce from 6.7.2 (probably conjugated with Riemann-Roch) Lefschetz-Riemann-Roch formulae in the style of those of Donovan [7], Iversen [10], Nielsen [15] (at least with values in the base field). Nor is it impossible, on the other hand, that the formula 6.7.2, applied over the dual numbers, might provide residue formulae for vector fields analogous to those of Bott [6], [3].

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Exposé III B
Calculations of Local Terms

by Luc Illusie

Introduction

This exposé, written in January 1977, does not correspond to any oral exposé of the seminar. It comprises two independent parts. In the first, we compute the Lefschetz–Verdier local terms for cohomological correspondences between smooth curves over an algebraically closed field, supported on correspondences satisfying certain general-position hypotheses. The formula that we obtain gives as corollaries Verdier's formula [6] for transversal endomorphisms (which applies in particular to the case of the Frobenius endomorphism), and the formula stated by Langlands in ([4] 7.12). The method of proof is parallel to that followed by Artin–Verdier in [6]. One reduces to proving the vanishing of the local term when the given complexes of sheaves have zero fibre at the images of the fixed point under consideration, and in fact one proves a stronger property, namely that a certain restriction map is zero (2.5). To establish this property, one first reduces to the tamely ramified case by covers of the same degree of the two given curves (so as not to destroy the general-position hypotheses), and then treats this case directly by means of Abhyankar's lemma and the relative purity theorem. The second part of this exposé, of a much more technical nature, is inspired by the method used by Grothendieck to establish the Lefschetz formula for certain cohomological correspondences on curves (see XII and (SGA $4^{1/2}$ Rapport)). In no. 5, for complexes of modules on a topos, we develop a theory of non-commutative traces which generalizes the commutative theory of (SGA 6 I) and, in the case of a punctual topos, recovers the non-commutative theory developed by Grothendieck in the oral seminar (this was written up in Exposé XI, unfortunately lost). We apply the techniques of no. 5 to define, in no. 6, Lefschetz–Verdier local terms for cohomological correspondences between complexes of modules over rings that are not necessarily commutative. These local terms behave like non-commutative traces and make it possible, in particular when one takes as base ring the algebra of a group, to divide canonically the ordinary local terms attached to equivariant correspondences. As an application, we prove Grothendieck's divisibility conjecture of (XII 4.5), in a more general form. We also show that the local terms defined by Grothendieck in the Lefschetz formula of XII are indeed the Lefschetz–Verdier local terms, and we generalize the formula of (loc. cit.).

I thank P. Deligne for the useful suggestions he made to me during many discussions; I am particularly grateful to him for having pointed out, and helped me correct, an error in an early proof of Theorem 1.2.

Summary

I. Correspondences in General Position between Curves

In this part, S denotes the spectrum of an algebraically closed field k of characteristic p, and Λ a noetherian commutative ring killed by an integer prime to p. For every scheme X over S we denote by $D(X)$ the derived category of sheaves of Λ-modules on X, and by $D_{\mathrm{ctf}}(X)$ the full subcategory of $D(X)$ formed by complexes of finite tor-dimension and constructible cohomology.

1. Statement of the theorem and corollaries

1.1

Let X (resp. Y) be the strict localization, at a closed point, of a smooth separated S-curve, let A (resp. B) be a strictly local S-scheme, and let

$$a : A \longrightarrow X \times_S Y, \qquad b : B \longrightarrow X \times_S Y$$

be S-morphisms. We denote by $p_1 : X \times_S Y \to X$, $p_2 : X \times_S Y \to Y$ the projections, put $a_i = p_i a$, $b_i = p_i b$, and denote by x (resp. y) the closed point of X (resp. Y). We assume that a_2 and b_1 are finite and surjective, and that $A \times_{(X \times_S Y)} B$ is reduced to the closed point z whose image in $X \times_S Y$ is (x, y).

Let $L \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, $M \in \operatorname{ob} D_{\mathrm{ctf}}(Y)$, $u \in \operatorname{Hom}(a_1^* L, a_2^! M)$, and $v \in \operatorname{Hom}(b_2^* M, b_1^! L)$. Denote by

$$u_z \in \operatorname{Hom}(L_x, M_y) \qquad (\text{resp. } v_z \in \operatorname{Hom}(M_y, L_x))$$

the composite homomorphism

$$L_x = R\Gamma(X, L) \xrightarrow{a_1^*} R\Gamma(A, a_1^* L) \xrightarrow{R\Gamma(A,u)} R\Gamma(A, a_2^! M) \xrightarrow{a_{2*}} R\Gamma(Y, M) = M_y,$$

where a_{2*} is defined by the adjunction map $a_{2*} a_2^! M \to M$ (resp. the analogous composite homomorphism defined by means of (b, v)). Since L_x and M_y are perfect complexes of Λ-modules, one can form the cup-product (SGA 6 I 8.3)

$$\langle u_z, v_z \rangle = \operatorname{Tr}(u_z v_z) = \operatorname{Tr}(v_z u_z) \in \Lambda. \tag{1.1.1}$$

Put $\Lambda_X(1)[2] = K_X$, $\Lambda_Y(1)[2] = K_Y$. By (I 5) (or (SGA $4^{1/2}$ Duality 1, or Finitude 4.1)) the functor $D_X = \operatorname{RHom}(-, K_X)$ (resp. $D_Y = \operatorname{RHom}(-, K_Y)$) on $D_{\mathrm{ctf}}(X)$ (resp. $D_{\mathrm{ctf}}(Y)$) is dualizing, i.e. for every $E \in \operatorname{ob} D_{\text{ctf}}(X)$ (resp. $E \in \operatorname{ob} D_{\text{ctf}}(Y)$), one has $E \xrightarrow{\sim} D_X D_X E$ (resp. $E \xrightarrow{\sim} D_Y D_Y E$). We shall sometimes write D instead of D_X, D_Y.

On the other hand, denote by $p_i^!$ the functor $p_i^*(-)(1)[2]$. We have trivially

$$K_X \otimes_S^{\mathbf{L}} M \xrightarrow{\sim} p_2^! M, \qquad L \otimes_S^{\mathbf{L}} K_Y \xrightarrow{\sim} p_1^! L,$$

and it follows from (III 2.3), by passage to the limit, that the Künneth maps analogous to (III 3.1.1)

$$DL \otimes_S^{\mathbf{L}} M \longrightarrow \mathrm{RHom}(p_1^* L, p_2^! M), \qquad L \otimes_S^{\mathbf{L}} DM \longrightarrow \mathrm{RHom}(p_2^* M, p_1^! L)$$

are isomorphisms. Since a_2 is finite, a is finite, and hence the functor $a^!$ is defined; one verifies easily that one has the transitivity isomorphism $a^! p_2^! = a_2^!$. By the induction formula for a (SGA 4 XVIII 3.1.12.2), one therefore has a canonical isomorphism analogous to (III 3.2.1)

$$a^! \mathrm{RHom}(p_1^* L, p_2^! M) \xrightarrow{\sim} \mathrm{RHom}(a_1^* L, a_2^! M),$$

whence an isomorphism

$$\mathrm{Hom}(a_1^* L, a_2^! M) \xrightarrow{\sim} H^0(X \times_S Y, a_* a^! (DL \otimes_S^{\mathbf{L}} M)). \tag{*}$$

Likewise one has an isomorphism

$$\mathrm{Hom}(b_2^* M, b_1^! L) \xrightarrow{\sim} H^0(X \times_S Y, b_* b^! (L \otimes_S^{\mathbf{L}} DM)). \tag{**}$$

As in (III 4.1.2), one defines a natural pairing

$$(DL \otimes_S^{\mathbf{L}} M) \otimes^{\mathbf{L}} (L \otimes_S^{\mathbf{L}} DM) \longrightarrow K_X \otimes_S^{\mathbf{L}} K_Y = \Lambda_{X \times_S Y}(2)[4] \overset{\text{dfn}}{=} K_{X \times_S Y}.$$

Composing with the pairing analogous to (III 4.2.3)

$$a_* a^! P \otimes^{\mathbf{L}} b_* b^! Q \longrightarrow c_* c^! (P \otimes^{\mathbf{L}} Q),$$

where $P = DL \otimes_S^{\mathbf{L}} M$, $Q = L \otimes_S^{\mathbf{L}} DM$, and $c : Z = A \times_{(X \times_S Y)} B \to X \times_S Y$ is the canonical projection, and taking into account the isomorphisms (*), (**), one deduces a pairing analogous to (III 4.2.5)

$$\langle\ ,\ \rangle : \mathrm{Hom}(a_1^* L, a_2^! M) \otimes \mathrm{Hom}(b_2^* M, b_1^! L) \longrightarrow H^0(X \times_S Y, c_* c^! K_{X \times_S Y}).$$

One verifies easily, by passage to the limit, that $c^! K_{X \times_S Y} = \Lambda_z$, so that $H^0(X \times_S Y, c_* c^! K_{X \times_S Y}) = \Lambda$. Thus one has a "cup-product"

$$\langle u, v \rangle \in \Lambda. \tag{1.1.2}$$

1.1.3

Let X' (resp. Y') be a smooth S-curve, A' (resp. B') a separated S-curve,

$$a' : A' \longrightarrow X' \times_S Y', \qquad b' : B' \longrightarrow X' \times_S Y'$$

finite S-morphisms, x' (resp. y') a closed point of X' (resp. Y'), z' an isolated closed point of $A' \times_{(X' \times_S Y')} B'$, with image s' (resp. t') in A' (resp. B'), and such that $a'_2 (= p_2 a')$ (resp. $b'_1 (= p_1 b')$) is quasi-finite at s' (resp. t'),

$$L' \in \operatorname{ob} D_{\text{ctf}}(X'), \quad M' \in \operatorname{ob} D_{\text{ctf}}(Y'), \quad u' \in \mathrm{Hom}(a_1'^* L', a_2'^! M'), \quad v' \in \mathrm{Hom}(b_2'^* M', b_1'^! L').$$

One can then consider the local term (III 4.2.7) $\langle u', v'\rangle_{z'} \in \Lambda$. On the other hand, if X, Y, A, B are the strict localizations of X', Y', A', B' at x', y', s', t', if $a : A \to X \times_S Y$, $b : B \to X \times_S Y$ are the morphisms deduced from a', b', if $L = L'|X$, $M = M'|Y$, and if $u \in \operatorname{Hom}(a_1^* L, a_2^! M)$, $v \in \operatorname{Hom}(b_2^* M, b_1^! L)$ are the maps deduced from u', v', then the data (X, Y, a, b, u, v) satisfy the hypotheses at the beginning of 1.1, and hence a cup-product 1.1.2 $\langle u, v\rangle \in \Lambda$ is defined. It follows from the compatibility of the formation of local terms (III 4.2.7) with localization (III 4.2.6) that $\langle u', v'\rangle_{z'} = \langle u, v\rangle$. It is also easy to see that any data (X, Y, a, b, u, v) satisfying the hypotheses at the beginning of 1.1 are induced by data $(X', Y', a', b', x', y', z', s', t', u', v')$ as above, where one may also assume, if desired, that $A' \times_{(X' \times_S Y')} B'$ is reduced to z'.

In general the cup-products 1.1.1 and 1.1.2 are unequal. Nevertheless one has the following theorem, which is the main result of this first part:

Theorem 1.2. Under the hypotheses at the beginning of 1.1, assume moreover that A and B are in general position, i.e. that the intersection of the tangent cones at (x, y) to the images of A and B in $X \times_S Y$ is reduced to (x, y), and that likewise A and $X \times \{y\}$ are in general position, as are $\{x\} \times Y$ and B. Then

$$\langle u_z, v_z\rangle = \langle u, v\rangle. \tag{1.2.1}$$

By the Lefschetz–Verdier formula (III 4.7), taking 1.1.3 into account, one deduces:

Corollary 1.3. Let F be an endomorphism of a proper smooth curve X/S such that the fixed-point scheme X^F is finite and formed of points where the graph of F is transverse to the diagonal. Let $L \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, and $u \in \operatorname{Hom}(F^* L, L)$. Then

$$\operatorname{Tr}((F, u), R\Gamma(X, L)) = \sum_{x \in X^F} \operatorname{Tr}(u_x, L_x),$$

where the trace on both sides is taken in the sense of (SGA 6 I 8).

For L concentrated in degree 0, 1.3 is the result of Artin–Verdier ([6] 4.1). By a passage to the limit explained in XV, one deduces from 1.3 an analogous result in ℓ-adic cohomology ([6] 1.1), (XV §2 no. 3 prop. 2):

Corollary 1.3.1. With X and F as in 1.3, let L be a constructible $\mathbb{Q}_\ell$-sheaf on X (ℓ prime to p), and let $u \in \operatorname{Hom}(F^* L, L)$. Then:

$$\sum_i (-1)^i \operatorname{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \operatorname{Tr}(u_x, L_x).$$

The hypotheses of 1.3 and 1.3.1 apply in particular to the case where (F, u) is a Frobenius correspondence (XV 2), whence Grothendieck's trace formula ([2] 5.1), (SGA $4^{1/2}$ Rapport 3.2).

1.4

With the notation of 1.1, assume that b_2 and a_1 are finite and surjective (instead of a_2 and b_1), and consider the dual correspondences (III 5.1)

$$Du \in \operatorname{Hom}(a_2^* DM, a_1^! DL), \qquad Dv \in \operatorname{Hom}(b_1^* DL, b_2^! DM),$$

and the homomorphisms $(Du)_z \in \mathrm{Hom}((DM)_y, (DM)_x)$, $(Dv)_z \in \mathrm{Hom}((DL)_x, (DM)_y)$. Assume that $(A, \{x\} \times Y)$ is in general position, as are $(X \times \{y\}, B)$ and (A, B). Applying 1.2 to Du, Dv, one obtains

$$\langle Du, Dv \rangle = \langle (Du)_z, (Dv)_z \rangle,$$

whence, by (III 5.1.6) and (III 4.2.6),

$$\langle u, v \rangle = \langle (Du)_z, (Dv)_z \rangle. \tag{1.4.1}$$

From this remark and from 1.2 one deduces, by the Lefschetz–Verdier formula (III 4.7):

Corollary 1.5. Let X, Y be proper smooth curves over S,

$$a : A \longrightarrow X \times_S Y, \qquad b : B \longrightarrow X \times_S Y$$

correspondences where A and B are proper curves over S, let $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, $M \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$, $u \in \mathrm{Hom}(a_1^* L, a_2^! M)$, and $v \in \mathrm{Hom}(b_2^* M, b_1^! L)$. Assume that $C = A \times_{(X \times_S Y)} B$ is finite over S and of the form $C = C_1 \amalg C_2$, with the following properties: for $z \in C_1$, the morphisms deduced from a_2, b_1 by strict localization at the images of z are finite, and, if (x, y) denotes the image of z in $X \times_S Y$, each of the pairs (A, B), $(A, X \times \{y\})$, $(\{x\} \times Y, B)$ is in general position at (x, y); for $z \in C_2$, the morphisms deduced from a_1, b_2 by strict localization at the images of z are finite, and each of the pairs (A, B), $(A, \{x\} \times Y)$, $(B, X \times \{y\})$ is in general position at (x, y) ("general position" meaning that the intersection of the tangent cones at (x, y) to the images of the curves of the pair under consideration is reduced to (x, y)). Then, if $u_* \in \mathrm{Hom}(R\Gamma(X, L), R\Gamma(Y, M))$, $v_* \in \mathrm{Hom}(R\Gamma(Y, M), R\Gamma(X, L))$ are the morphisms defined by u, v (III 3.7.6), one has, with the notation of 1.1.1, (III 5.1.5),

$$\langle u_*, v_* \rangle = \sum_{z \in C_1} \langle u_z, v_z \rangle + \sum_{z \in C_2} \langle (Du)_z, (Dv)_z \rangle. \tag{1.5.1}$$

1.6

Assuming the formalism of ℓ-adic cohomology, one deduces from 1.5, by the passage-to-the-limit technique of XIV, that, under the hypotheses of 1.5 on X, Y, a, b, for $L \in \mathrm{ob}\, D^b(X, \mathbb{Q}_\ell)$ (ℓ prime $\neq p$), $M \in \mathrm{ob}\, D^b(Y, \mathbb{Q}_\ell)$, $u \in \mathrm{Hom}(a_1^* L, a_2^! M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^! L)$, one has the analogue of 1.5.1 in $\mathbb{Q}_\ell$. This formula contains, as a special case, Proposition 7.12 of Langlands [4]: apply the formula with $X = Y$, M the sheaf F considered by Langlands; $L = \alpha M$ ($\alpha \in \mathbb{Q}_\ell^*$), A the diagonal of $X \times X$, b the correspondence Φ of (loc. cit.), u the correspondence supported on A given by $\alpha^{-1} : L \to M$, v the correspondence Φ of (loc. cit.); with the notation of (loc. cit.), C_1 (resp. C_2) is the set of fixed points where $a < d$ (resp. $a > d$).

1.7

Under the hypotheses of 1.2, take for u (resp. v) the correspondence $\mathrm{cl}(A)$ (resp. $\mathrm{cl}(B)$) defined by $\mathrm{Tr}_{a_2} : a_{2*}\Lambda \to \Lambda$ (resp. $\mathrm{Tr}_{b_1} : b_{1*}\Lambda \to \Lambda$). Formula 1.2.1 becomes

$$\deg(a_2) \deg(b_1) = \langle \mathrm{cl}(A), \mathrm{cl}(B) \rangle, \tag{1.7.1}$$

where $\deg(a_2)$ (resp. $\deg(b_1)$) denotes the generic degree of a_2 (resp. b_1). Taking into account the hypotheses of general position, the left hand side of 1.7.1 is none other than the intersection multiplicity at (x, y) of the cycles a_*A, b_*B, the direct images of A and B. It is not difficult to deduce 1.7.1 directly from the compatibility of the formation of the cohomology class associated with a cycle with intersections (SGA $4^{1/2}$ Cycle 2.3.8).

Exposé III B: Calculations of Local Terms

2. Reduction to a vanishing theorem

2.1

Under the hypotheses of 1.1, let s (resp. t) denote the image of z in A (resp. B), and let $i : U = X - \{x\} \hookrightarrow X$ and $j : V = Y - \{y\} \hookrightarrow Y$ be the canonical inclusions. Since a_2 and b_1 are finite, one has $a_2^{-1}(y) = s$, $b_1^{-1}(x) = t$, hence

$$a_1^{-1}(U) \subset a_2^{-1}(V), \qquad b_2^{-1}(V) \subset b_1^{-1}(U),$$

and there are commutative diagrams

$$\begin{tikzcd} U & a_1^{-1}(U) \arrow[l, "a_1'"'] \arrow[r, "a_2'"] \arrow[d] & V \arrow[d, "j"] \\ X & A \arrow[l, "a_1"] \arrow[r, "a_2"'] & Y \end{tikzcd} \qquad \begin{tikzcd} U & b_2^{-1}(V) \arrow[l, "b_1'"'] \arrow[r, "b_2'"] \arrow[d] & V \arrow[d, "j"] \\ X & B \arrow[l, "b_1"] \arrow[r, "b_2"'] & Y, \end{tikzcd}$$

where the squares corresponding to $U \leftarrow a_1^{-1}(U) \to A \to X$ and to $b_2^{-1}(V) \to V \to Y$ are cartesian.

Lemma 2.2. To prove 1.2, it is enough to consider the following two cases:

$$\text{a) } L_x = 0, \quad M_y = 0; \qquad \text{b) } L|U = 0, \quad M|V = 0.$$

Consider the decreasing filtrations on L and M defined by the subcomplexes

$$F^n L = \begin{cases} L, & n \le 0, \\ i_! i^* L, & n = 1, \\ 0, & n > 1, \end{cases} \qquad F^n M = \begin{cases} M, & n \le 0, \\ j_! j^* M, & n = 1, \\ 0, & n > 1. \end{cases}$$

The complex $a_{2*}a_1^*L$ (resp. $b_{1*}b_2^*M$) is then filtered by

$$F^n a_{2*}a_1^*L = a_{2*}a_1^*F^nL.$$

We have $F^n a_{2*}a_1^*L = a_{2*}a_1^*L$ for $n \le 0$, and 0 for $n > 1$; moreover, since a_2 is finite and the square is cartesian,

$$F^1 a_{2*}a_1^*L = j_!(a_2')_!(a_1')^*i^*L.$$

Similarly,

$$F^n b_{1*}b_2^*M = \begin{cases} b_{1*}b_2^*M, & n \le 0, \\ i_!(b_1')_!(b_2')^*j^*M, & n = 1, \\ 0, & n > 1. \end{cases}$$

We may suppose that u (resp. v) is defined by a homomorphism of complexes

$$u : a_{2*}a_1^*L \longrightarrow M \quad (\text{resp. } v : b_{1*}b_2^*M \longrightarrow L).$$

The preceding formulas show that the composite

$$F^1 a_{2*}a_1^*L \longrightarrow a_{2*}a_1^*L \xrightarrow{u} M$$

(resp. $F^1 b_{1*}b_2^*M \to b_{1*}b_2^*M \xrightarrow{v} L$) factors uniquely through F^1M (resp. F^1L); in other words, u (resp. v) is a homomorphism of filtered complexes. Thus u (resp. v) underlies a filtered correspondence, in the sense of (III 4.13), which we again denote by

$$u \in \mathrm{Hom}_{DF}(a_1^*L, a_2^!M), \qquad v \in \mathrm{Hom}_{DF}(b_2^*M, b_1^!L).$$

By additivity of cup-products ([3] V 3.8.7), (III 4.13.1), one has

$$\langle u_z, v_z\rangle = \langle \mathrm{gr}^0 u_z, \mathrm{gr}^0 v_z\rangle + \langle \mathrm{gr}^1 u_z, \mathrm{gr}^1 v_z\rangle,$$

and

$$\langle u, v\rangle = \langle \mathrm{gr}^0 u, \mathrm{gr}^0 v\rangle + \langle \mathrm{gr}^1 u, \mathrm{gr}^1 v\rangle.$$

The lemma follows, since $\mathrm{gr}^1 L = i_! i^* L$ (resp. $\mathrm{gr}^1 M = j_! j^* M$) has zero fibre at x (resp. y), whereas $\mathrm{gr}^0 L$ (resp. $\mathrm{gr}^0 M$) is concentrated at x (resp. y).

Lemma 2.3. Under the hypotheses of 1.1, suppose that $L|U = 0$ and $M|V = 0$. Then

$$\langle u_z, v_z\rangle = \langle u, v\rangle.$$

The idea of the proof is that then u and v are direct images of correspondences between complexes concentrated at the closed points, and that the formula to be proved merely expresses the Lefschetz formula for the projections $x \to S$, $y \to S$. In view of 1.1.3, the data of 1.1 may be replaced by the following data: X (resp. Y) is a smooth S-curve endowed with a closed point x (resp. y), A (resp. B) is a separated S-curve, $a : A \to X \times_S Y$, $b : B \to X \times_S Y$ are finite S-morphisms such that $A \times_{X\times_S Y} B$ is reduced to a closed point z above (x, y),

$$u \in \mathrm{Hom}(a_1^*L, a_2^!M), \qquad v \in \mathrm{Hom}(b_2^*M, b_1^!L),$$

with

$$L = h_{1*}L', \quad M = h_{2*}M', \quad L' \in \mathrm{ob}\, D_{\mathrm{ctf}}(\{x\}), \quad M' \in \mathrm{ob}\, D_{\mathrm{ctf}}(\{y\}),$$

where $h_1 : \{x\} \to X$ and $h_2 : \{y\} \to Y$ are the canonical inclusions. We must show that

$$\langle u_z, v_z\rangle = \langle u, v\rangle_z,$$

where

$$u_z \in \mathrm{Hom}(\mathrm{R}\Gamma(X, L), \mathrm{R}\Gamma(Y, M)), \qquad v_z \in \mathrm{Hom}(\mathrm{R}\Gamma(Y, M), \mathrm{R}\Gamma(X, L))$$

are defined as in 1.1, and $\langle u, v\rangle_z$ is the value at z of the cup-product (III 4.2.7). We have $\mathrm{R}\Gamma(X, L) = \mathrm{R}\Gamma_c(X, L) = L_x$, $\mathrm{R}\Gamma(Y, M) = \mathrm{R}\Gamma_c(Y, M) = M_y$, and u_z (resp. v_z) is nothing other than the direct image u_* (resp. v_*) of u (resp. v) on S in the sense of (III 3.7.6), as is shown by the calculation procedure (III 3.5 b)).

On the other hand, if $h = h_1 \times_S h_2 : (x, y) \to X \times_S Y$ is the inclusion, and if $a' : A' \to (x, y)$, $b' : B' \to (x, y)$ denote the fibres of a, b at (x, y), each reduced to a point, then there are canonical isomorphisms

$$h_* a'_* a'^! (DL_x \overset{\mathrm{L}}{\otimes}_\Lambda M_y) \xrightarrow{\sim} a_* a^! (DL \overset{\mathrm{L}}{\otimes}_\Lambda M),$$

$$h_* b'_* b'^! (L_x \overset{\mathrm{L}}{\otimes}_\Lambda DM_y) \xrightarrow{\sim} b_* b^! (L \overset{\mathrm{L}}{\otimes}_\Lambda DM),$$

and the arrows

$$h_* : \mathrm{Hom}(a_1'^* L_x, a_2'^! M_y) \to \mathrm{Hom}(a_1^* L, a_2^! M),$$

$$h_* : \mathrm{Hom}(b_2'^* M_y, b_1'^! L_x) \to \mathrm{Hom}(b_2^* M, b_1^! L)$$

(III 3.7.6) are isomorphisms. The correspondences u and v are therefore written uniquely as

$$u = h_* u', \qquad v = h_* v',$$

with

$$u' \in \mathrm{Hom}(a_1'^* L_x, a_2'^! M_y), \qquad v' \in \mathrm{Hom}(b_2'^* M_y, b_1'^! L_x).$$

By the Lefschetz formula (III 4.5.1) for $(h_1, h_2, A' \to A, B' \to B)$, one has

$$\langle u, v \rangle_z = \langle u', v' \rangle_z.$$

On the other hand, by the Lefschetz formula for $\{x\} \to S$, $\{y\} \to S$, one has

$$\langle u_*, v_* \rangle = \langle u', v' \rangle_z.$$

In view of the preceding remark, the conclusion follows by comparing these two equalities.

Lemma 2.4. To prove 1.2, one may suppose that a and b are closed immersions, and it is enough to prove that, if $L_x = 0$ and $M_y = 0$, the restriction map

$$H^0_A(X \times_S Y, DL \overset{\mathrm{L}}{\otimes}_\Lambda M) \longrightarrow H^0_z(B, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)) \tag{2.4.1}$$

is zero.

Put $X \times_S Y = Z$. The closed subschemes $a' : A' \to Z$, $b' : B' \to Z$, images of a and b, satisfy the same hypotheses as a and b. Let u' (resp. v') be the correspondence supported on A' (resp. B') which is the direct image of u (resp. v) by the arrow deduced from the adjunction arrow $f_* f^! \to 1$ (resp. $g_* g^! \to 1$), where $f : A \to A'$ (resp. $g : B \to B'$) is the projection. It is immediate that $u_z = u'_z$ and $v_z = v'_z$. On the other hand, it follows easily from the definition of the cup-product 1.1.2 that

$$\langle u, v \rangle = \langle u', v' \rangle,$$

hence the first assertion.

By 2.2 and 2.3, one may suppose that $L_x = 0$ and $M_y = 0$, and the conclusion of 1.2 then becomes $\langle u, v \rangle = 0$. Let $P, Q \in \mathrm{ob}\, D^b_{\mathrm{ctf}}(Z)$. The cup-product

$$H^0_A(Z, P) \otimes H^0_B(Z, Q) \longrightarrow H^0_z(Z, P \overset{\mathrm{L}}{\otimes}_\Lambda Q)$$

decomposes as

$$H^0_A(Z, P) \otimes H^0_B(Z, Q) \xrightarrow{b^* \otimes 1} H^0_z(B, b^* P) \otimes H^0_B(Z, Q) \longrightarrow H^0_z(Z, P \overset{\mathrm{L}}{\otimes}_\Lambda Q),$$

where b^* is the restriction and the second arrow, a variant of the product considered in (SGA $4^{1/2}$ Cycle I 2.1), is derived from the evident product at the level of sections of sheaves. This assertion is easily checked, for example by means of the flasque resolutions of (SGA 4 XVII 4.2). If $a' : \{z\} \to B$, $b' : \{z\} \to A$, $c : \{z\} \to Z$ are the inclusions, one may also invoke the standard decomposition of the Künneth arrow (III 4.2.1)

$$a'^! P \overset{\mathrm{L}}{\otimes}_\Lambda b'^! Q \longrightarrow c^! (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)$$

into a base-change arrow

$$a'^! P \overset{\mathrm{L}}{\otimes}_\Lambda b'^! Q \longrightarrow a'^! b^* P \overset{\mathrm{L}}{\otimes}_\Lambda a^* b'^! Q,$$

followed by projection arrows

$$a'^! b^* P \overset{\mathrm{L}}{\otimes}_\Lambda a^* b'^! Q \longrightarrow a'^! (b^* P \overset{\mathrm{L}}{\otimes}_\Lambda b'^! Q) \longrightarrow a'^! b'^! (P \overset{\mathrm{L}}{\otimes}_\Lambda Q).$$

Taking $P = DL \overset{\mathrm{L}}{\otimes}_\Lambda M'$, $Q = L \overset{\mathrm{L}}{\otimes}_\Lambda DM$, and returning to the definition of $\langle u, v \rangle$ (III 4.2.5), one sees that it is enough to prove that $b^* u = 0$, whence the conclusion.

In view of 2.4, 1.2 will therefore follow from the following more precise result.

Proposition 2.5. Under the hypotheses of 1.2, suppose that a and b are closed immersions and that $L_x = 0$, $M_y = 0$. Let T be the strict localization of $X \times_S Y$ at $z = (x, y)$, and identify A, B with closed subschemes of T. Then the restriction arrows

$$b^* : \mathrm{R}\Gamma_A(T, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T) \longrightarrow \mathrm{R}\Gamma_z(B, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)),$$

$$a^* : \mathrm{R}\Gamma_B(T, (L \overset{\mathrm{L}}{\otimes}_\Lambda DM)|T) \longrightarrow \mathrm{R}\Gamma_z(A, a^*(L \overset{\mathrm{L}}{\otimes}_\Lambda DM))$$

are zero.

2.6

The proof of 2.5 will occupy the rest of number 2 and numbers 3 and 4. By symmetry of the statement, it is enough to prove the assertion concerning b^*. We shall rewrite it in a form which will be more convenient. First note that

$$\mathrm{R}\Gamma(T, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T) = (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)_z = (DL)_x \overset{\mathrm{L}}{\otimes}_\Lambda M_y = 0,$$

because $M_y = 0$, and similarly

$$\mathrm{R}\Gamma(B, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)) = (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)_z = 0.$$

Consequently, the relative cohomology triangles give isomorphisms

$$\mathrm{R}\Gamma(T - A, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T - A) \xrightarrow{\sim} \mathrm{R}\Gamma_A(T, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T)[1],$$

$$\mathrm{R}\Gamma(B - z, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|B - z) \xrightarrow{\sim} \mathrm{R}\Gamma_z(B, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M))[1],$$

under which the arrow b^* of 2.5, shifted by 1, is identified with the restriction

$$b^* : \mathrm{R}\Gamma(T - A, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T - A) \longrightarrow \mathrm{R}\Gamma(B - z, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|B - z). \tag{2.6.1}$$

Since $L_x = 0$, $M_y = 0$, one may write

$$L = i_! P, \qquad M = j_! Q, \tag{2.6.2}$$

where $i : X - x \to X$, $j : Y - y \to Y$ are the inclusions, and $P \in \operatorname{ob} D_{\mathrm{ctf}}(X - x)$, $Q \in \operatorname{ob} D_{\mathrm{ctf}}(Y - y)$. By the duality isomorphism (SGA 4 XVIII 3.19.6), together with passage to the limit, one has

$$DL = Ri_* DP = Ri_* P,$$

hence

$$DL \overset{\mathrm{L}}{\otimes}_\Lambda M = Ri_* P \overset{\mathrm{L}}{\otimes}_\Lambda j_! Q.$$

Put $X \times_S Y = Z$, and identify $X \times \{y\}$ and $\{x\} \times Y$ with closed subschemes of Z (resp. T), which we shall simply denote by X and Y. Let

$$f' : Z - Y \hookrightarrow Z, \qquad g' : Z - (X \cup Y) \hookrightarrow Z - Y,$$

$$f : T - Z \hookrightarrow T, \qquad g : T - (X \cup Y) \hookrightarrow T - Y$$

be the inclusions. The Künneth formula (III 1.6.4) gives, by passage to the limit,

$$DL \overset{\mathrm{L}}{\otimes}_\Lambda M = Rf'_* g'_! (P \overset{\mathrm{L}}{\otimes}_\Lambda Q),$$

and hence

$$(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T = Rf_* g_! (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)|T - (X \cup Y). \tag{2.6.3}$$

With this identification, 2.6.1 becomes

$$b^* : \mathrm{R}\Gamma(T - (Y \cup A), g_!(E)|T - (Y \cup A)) \longrightarrow \mathrm{R}\Gamma(B - z, g_!(E)|B - z), \tag{2.6.4}$$

where

$$E = (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)|T - (X \cup Y).$$

In the next two numbers, we shall show that the arrow 2.6.4 is zero: first we shall reduce to the case where the cohomology sheaves of E have tame ramification, more precisely where the inertia action occurs through an ℓ-group, for a prime number $\ell \neq p$, and then we shall treat that case directly by means of a purity theorem.

3. Reduction to the tame case

3.1

Recall that the base ring Λ is annihilated by an integer prime to p, hence is a direct product of noetherian rings Λ_i, each Λ_i being annihilated by a power of a prime number $\ell_i \neq p$. For every scheme S' over S, the derived category $D(S', \Lambda)$ is the product of the categories $D(S', \Lambda_i)$, so that, in order to prove the vanishing of 2.6.4, we may suppose that Λ is annihilated by a power of a prime number $\ell \neq p$. We may also suppose that P (resp. Q) is bounded and has locally free components of finite type.

3.2

Put $X - x = U$, $Y - y = V$. Let U''/U be a finite Galois covering with group G which trivializes the components of P, let H be an ℓ-Sylow subgroup of G, let $U' = U''/H$ be the quotient covering, and let X' be the normalized strictly local trait of X in U'. Then X'/X is a finite connected covering of degree d_1 prime to ℓ, and the representation of $\pi_1(U')$ defined by the components of $P' = P|U'$ factors through a finite ℓ-group. Proceeding similarly with Q, one finds a finite connected covering Y'/Y of degree d_2 prime to ℓ, inducing on V an étale covering V' such that the representation of $\pi_1(V')$ defined by the components of $Q' = Q|V'$ factors through a finite ℓ-group. Let

$$d = d_1 e_1 = d_2 e_2 = \operatorname{lcm}(d_1, d_2).$$

After adjoining an e_1-th root (resp. an e_2-th root) of a uniformizer of X' (resp. Y'), we may suppose that X' (resp. Y') has the same properties as above, and moreover that $d_1 = d_2 = d$.

Let x' (resp. y') be the closed point of X' (resp. Y'), T' the strict localization of $X' \times_S Y'$ at (x', y'), z' the closed point of T', A' (resp. B') the inverse image of A (resp. B) in T', and

$$E' = (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)|T' - (X' \cup Y') = (P' \overset{\mathrm{L}}{\otimes}_\Lambda Q')|T' - (X' \cup Y').$$

We then have a commutative square of restriction arrows

$$\begin{tikzcd}
\mathrm{R}\Gamma(T - (Y \cup A), g_!(E)|T - (Y \cup A)) \arrow[r, "b^*"] \arrow[d] & \mathrm{R}\Gamma(B - z, g_!(E)|B - z) \arrow[d, "(1)"] \\
\mathrm{R}\Gamma(T' - (Y' \cup A'), g'_!(E')|T' - (Y' \cup A')) \arrow[r, "b'^*"] & \mathrm{R}\Gamma(B' - z', g'_!(E')|B' - z'),
\end{tikzcd}$$

where $b' : B' \to T'$, $g' : T' - (X' \cup Y') \to T' - Y'$ are the inclusions. The covering T'/T is finite, of degree d^2, and étale outside $X \cup Y$, hence induces an étale covering of degree d^2 from $B' - z'$ onto $B - z$. The composite of (1) and the trace morphism

$$\mathrm{R}\Gamma(B' - z', g'_!(E')|B' - z') \longrightarrow \mathrm{R}\Gamma(B - z, g_!(E)|B - z)$$

is therefore multiplication by d^2, which is an isomorphism, since ℓ does not divide d. To prove that the arrow 2.6.4 is zero, it is therefore enough to prove that $b'^* = 0$. We are thus reduced to the case where the representation of $\pi_1(U)$ (resp. $\pi_1(V)$) defined by the components of P (resp. Q) factors through a finite ℓ-group, provided one ensures that A', B' satisfy the general-position hypotheses of 1.2. This will follow from the following lemma.

Lemma 3.3. Under the hypotheses of 1.2, a and b being closed immersions, let X'/X, Y'/Y be finite coverings of strictly local S-traits, of the same degree d. Let A' (resp. B') denote the inverse image of A (resp. B) in $X' \times_S Y'$. Then the pairs (A', B'), (A', X'), (Y', B') are each in general position in the sense of 1.2.

The proof of 3.3 will be given after a few preliminaries.

Lemma 3.4. Let $X \xleftarrow{s_1} W \xrightarrow{s_2} Y$ be finite morphisms of S-traits, let $s = (s_1, s_2) : W \to X \times_S Y$, and let u, v, w be uniformizers of X, Y, W respectively, so that

$$s_1^* u = a w^m \pmod{w^{m+1}}, \qquad s_2^* v = b w^n \pmod{w^{n+1}},$$

with $a, b \in k^*$. Then: (i) if $m < n$, $s(W)$ is tangent to X at $z = (x, y)$; (ii) if $m = n$, $s(W)$ is tangent to the subscheme of $X \times_S Y$ with equation

$$ap_2^*(v) - bp_1^*(u) = 0.$$

Let $Z = X \times_S Y$, and let W' be the reduced closed subscheme which is the image of s. In the tangent space to Z at z, the slope, with respect to the tangent to X, of the tangent to W' is the value at the origin of the function induced on W' by $p_2^*(v)/p_1^*(u)$; this is also the value at the origin of the function induced by $p_2^*(v)/p_1^*(u)$ on W, and is therefore 0 (resp. b/a) in case (i) (resp. (ii)).

In the situation of 3.4, we shall define the slope of W with respect to $((X, u), (Y, v))$ as the element of $k \cup \{\infty\}$ equal to 0 if $m < n$, to ∞ if $m > n$, and to b/a if $m = n$. This element is independent of the choice of w.

Lemma 3.5. Let $f : X' \to X$, $g : Y' \to Y$ be finite morphisms of S-traits with the same ramification index d, and let u, v, u', v' be uniformizers of X, Y, X', Y' respectively. Then there exists $C \in k^*$ such that, for every commutative diagram of finite morphisms of S-traits

$$\begin{tikzcd} X' & W' \arrow[l, "s_1'"'] \arrow[r, "s_2'"] \arrow[d, "h"] & Y' \arrow[d, "g"] \\ X & W \arrow[l, "s_1"] \arrow[r, "s_2"'] & Y, \end{tikzcd}$$

one has $\lambda = C\lambda'^d$, where λ (resp. λ') is the slope of W (resp. W') with respect to $((X, u), (Y, v))$ (resp. $((X', u'), (Y', v'))$).

By hypothesis,

$$f^*u = \alpha u'^d \pmod{u'^{d+1}}, \qquad g^*v = \beta v'^d \pmod{v'^{d+1}},$$

with $\alpha, \beta \in k^*$. We show that $C = \beta/\alpha$ has the required properties. Let w (resp. w') be a uniformizer of W (resp. W'). We have

$$s_1^*u = aw^m \pmod{w^{m+1}}, \qquad s_2^*v = bw^n \pmod{w^{n+1}},$$

$$s_1'^*u' = a'w'^{m'} \pmod{w'^{m'+1}}, \qquad s_2'^*v' = b'w'^{n'} \pmod{w'^{n'+1}},$$

$$h^*w = \gamma w'^r \pmod{w'^{r+1}},$$

with $a, b, a', b', \gamma \in k^*$. Commutativity of the squares in the diagram implies

$$m'd = rm, \qquad n'd = rn,$$

hence $m'/n' = m/n$. Suppose first that $m = n$. Then $\lambda = b/a$, $\lambda' = b'/a'$. If $N = m'd = rm = n'd = rn$, the preceding congruences give

$$(fs_1')^*u = \alpha a'^d w'^N \pmod{w'^{N+1}}, \qquad (s_1h)^*u = a\gamma^m w'^N \pmod{w'^{N+1}},$$

whence $\alpha a'^d = a\gamma^m$. Similarly $\beta b'^d = b\gamma^m$, whence $\lambda = C\lambda'^d$, for $\lambda \neq 0, \infty$. But, since $m'/n' = m/n$, $\lambda = 0$ (resp. ∞) is equivalent to $\lambda' = 0$ (resp. ∞), which completes the proof.

We now prove 3.3. Let $Z = X \times_S Y$, $Z' = X' \times_S Y'$, let $\widetilde{A}$ (resp. $\widetilde{B}$) be the normalization of A (resp. B), and let $\widetilde{A}'$ (resp. $\widetilde{B}'$) be the normalization of A' (resp. B'). Since $\widetilde{A}'$ (resp. $\widetilde{B}'$) is also the normalization of $\widetilde{A} \times_Z Z'$ (resp. $\widetilde{B} \times_Z Z'$), we have commutative squares

$$\begin{tikzcd} \widetilde{A}' \arrow[r] \arrow[d] & Z' & \widetilde{B}' \arrow[l] \arrow[d] \\ \widetilde{A} \arrow[r] & Z & \widetilde{B}. \arrow[l] \end{tikzcd}$$

Let $(x_i)_{1 \leq i \leq m}$ (resp. $(y_j)_{1 \leq j \leq n}$) be the points of $\widetilde{A}$ (resp. $\widetilde{B}$) above x (resp. y), and let $(x'_i)_{1 \leq i \leq m'}$ (resp. $(y'_j)_{1 \leq j \leq n'}$) be the points of $\widetilde{A}'$ (resp. $\widetilde{B}'$) above x' (resp. y'). Since A and B are strictly local, we have decompositions

$$\widetilde{A} = \coprod_{1 \leq i \leq m} A_i, \qquad \widetilde{B} = \coprod_{1 \leq j \leq n} B_j, \qquad \widetilde{A}' = \coprod_{1 \leq i \leq m'} A'_i, \qquad \widetilde{B}' = \coprod_{1 \leq j \leq n'} B'_j,$$

where A_i, B_j, A'_i, B'_j are strictly local traits with respective closed points x_i, y_j, x'_i, y'_j. Choose uniformizers u, v, u', v' of X, Y, X', Y' respectively; let λ_i (resp. μ_j) be the slope of A_i (resp. B_j) with respect to $((X, u), (Y, v))$, and let λ'_i (resp. μ'_j) be that of A'_i (resp. B'_j) with respect to $((X', u'), (Y', v'))$. The general-position hypotheses on A and B are expressed by

$$\lambda_i \neq 0 \text{ for all } i, \qquad \mu_j \neq \infty \text{ for all } j, \qquad \lambda_i \neq \mu_j \text{ for all } i, j, \tag{1}$$

and we have to show that

$$\lambda'_i \neq 0 \text{ for all } i, \qquad \mu'_j \neq \infty \text{ for all } j, \qquad \lambda'_i \neq \mu'_j \text{ for all } i, j. \tag{2}$$

Let $i' \in [1, m']$, $j' \in [1, n']$, and let x_i (resp. y_j) be the image of $x'_{i'}$ (resp. $y'_{j'}$). The preceding diagram provides commutative squares of finite morphisms of S-traits

$$\begin{tikzcd} X' & A'_{i'} \arrow[l] \arrow[r] \arrow[d] & Y' \arrow[d] \\ X & A_i \arrow[l] \arrow[r] & Y \end{tikzcd} \qquad \begin{tikzcd} X' & B'_{j'} \arrow[l] \arrow[r] \arrow[d] & Y' \arrow[d] \\ X & B_j \arrow[l] \arrow[r] & Y. \end{tikzcd}$$

By 3.5, there exists $C \in k^*$ such that

$$\lambda_i = C {\lambda'_{i'}}^d, \qquad \mu_j = C {\mu'_{j'}}^d.$$

Thus (1) implies (2), which completes the proof of 3.3.

4. End of the proof of 1.2

4.1

To complete the proof of 2.5, hence of 1.2, it remains to establish the vanishing of 2.6.4 under the following additional hypotheses: Λ is annihilated by a power of a prime number $\ell \neq p$, and

P and Q are bounded, with locally free components of finite type, such that the corresponding representations of the fundamental groups factor through finite ℓ-groups. We have

$$g_!(E)|T-(Y\cup A)=g_{1!}(E_1),$$

where $g_1 : T-(X\cup Y\cup A)\hookrightarrow T-(Y\cup A)$ is the inclusion, and

$$E_1=(P\overset{\mathrm{L}}{\otimes}_\Lambda Q)|T-(X\cup Y\cup A).$$

Now E_1 is a complex with bounded, constructible, locally constant cohomology such that the corresponding representation of $\pi_1(T-(X\cup Y\cup A))$ factors through an ℓ-group. The assertion to be proved will therefore follow from the following more general statement.

Proposition 4.2. Let T be the strict localization at a closed point of a smooth S-scheme of dimension 2, let t be the closed point of T, and let A, B, X be closed subschemes of dimension 1 of T. Let $i : T-(A\cup X)\hookrightarrow T-A$ be the inclusion. Suppose that X is regular, and that A is in general position with respect to B and X; that is, the intersection of the tangent cones at t to A and $B\cup X$ is reduced to t. Let $E\in\operatorname{ob}D^b(T-(A\cup X))$ be such that $H^*(E)$ is constructible and locally constant, and that the corresponding representation of $\pi_1(T-(A\cup X))$ factors through an ℓ-group. Then the restriction arrow

$$\mathrm{R}\Gamma(T-A,i_!(E))\longrightarrow\mathrm{R}\Gamma(B-t,i_!(E)|B-t)$$

is zero.

The proof will rely on the following lemma.

Lemma 4.3. Let Z be a connected regular noetherian scheme of dimension 2, with residual characteristics prime to ℓ, and let X, Y be regular divisors meeting transversely at a point z, giving a diagram of inclusions

$$\begin{tikzcd} Z-(X\cup Y) \arrow[r, "i'"] \arrow[d, "j"'] & Z-Y \arrow[d, "j"] \\ Z-X \arrow[r, "i"'] \arrow[d, "m"'] & Z \arrow[d, "m"] \\ Y-z \arrow[r, "i_1"'] & Y. \end{tikzcd}$$

Assume the purity hypothesis

$$(P)\qquad R^qj_*(\mathbb{Z}/\ell)=0\quad\text{for } q>1.$$

Let $L\in\operatorname{ob}D^b(Z-(X\cup Y),\Lambda)$ have constructible locally constant cohomology, with the corresponding representation of $\pi_1(Z-(X\cup Y))$ factoring through an ℓ-group. Then:

a) There is a canonical isomorphism

$$i_{1!}m'^*Rj_*i'_!L\xrightarrow{\sim}m^*Rj_*(i'_!L),$$

and the complex $m'^*Rj_*i'_!L$ has constructible, locally constant cohomology.

b) Suppose that Y is connected, and let $\bar{s}$ be a geometric point of Y localized at a closed point $s \neq z$. Then the restriction arrow

$$\mathrm{R}\Gamma(Y, m^* Rj_*(i'_! L)) \longrightarrow (Rj_*(i'_! L))_{\bar{s}} \tag{4.3.1}$$

is zero.

c) Under the hypotheses of b), let C be a closed subscheme of dimension 1 of Z such that $Y \cap C = s$, and let $\widetilde{C}$ be the strict localization of C at $\bar{s}$. Then the restriction arrow

$$\mathrm{R}\Gamma(Z - Y, i'_! L) \longrightarrow \mathrm{R}\Gamma(\widetilde{C} - \bar{s}, i'_!(L) | \widetilde{C} - \bar{s}) \tag{4.3.2}$$

is zero.

The trivial isomorphism

$$i^* Rj_* i'_! L \xrightarrow{\sim} Rj_*(i'^* i'_! L) = Rj_* L$$

gives an isomorphism

$$m'^* Rj_* i'_! L \xrightarrow{\sim} i^* m^* Rj_* i'_! L,$$

hence, by adjunction, an arrow

$$i_{1!} m'^* Rj_* i'_! L \longrightarrow m^* Rj_*(i'_! L),$$

which induces an isomorphism outside z. We show that it is an isomorphism, that is, that if $\bar{z}$ is a geometric point above z, then

$$(Rj_*(i'_! L))_{\bar{z}} = 0.$$

For this, one may replace Λ by $\mathbb{Z}/\ell^\nu$, where ℓ^ν annihilates Λ, and by dévissage and passage to the limit reduce to the case where L is a locally constant $\mathbb{Z}/\ell^\nu$-module of finite type corresponding to a representation of π_1 through an ℓ-group. Since the only finite abelian group of ℓ-torsion which is simple under the action of an ℓ-group is $\mathbb{Z}/\ell$ with trivial action, a further dévissage reduces to the case where L is the constant sheaf $\mathbb{Z}/\ell$.

Let $\widetilde{Z}$ be the strict localization of Z at $\bar{z}$, $\widetilde{X} = X \times_Z \widetilde{Z}$, $\widetilde{Y} = Y \times_Z \widetilde{Z}$, and let $i' : \widetilde{Z} - (\widetilde{X} \cup \widetilde{Y}) \hookrightarrow \widetilde{Z} - \widetilde{Y}$ be the inclusion. One has

$$(Rj_*(i'_!(\mathbb{Z}/\ell)))_{\bar{z}} = \mathrm{R}\Gamma(\widetilde{Z} - \widetilde{Y}, i'_!(\mathbb{Z}/\ell)),$$

so it remains to prove that the restriction arrow

$$H^n(\widetilde{Z} - \widetilde{Y}, \mathbb{Z}/\ell) \longrightarrow H^n(\widetilde{X} - z, \mathbb{Z}/\ell)$$

is an isomorphism for all n. For $n \leq 1$ this is a consequence of Abhyankar's lemma, and for $n > 1$ both sides vanish by the purity hypothesis. The first assertion of a) is therefore proved. We prove the second. By dévissage, we may suppose L concentrated in degree 0. The hypothesis (P) implies, by dévissage, that $R^q j_* L = 0$ for $q > 2$. It remains to prove constructibility and local constancy of $m'^* R^q j_* i'_! L$ for $q \leq 1$, which is standard. For constructibility, resolving L on the right by sheaves of the form $p_* p^* L$, where $p : Z' \to Z$ is a finite covering tamely ramified along Y, reduces to the case where L is constant, of value L_0: then

$$m'^* Rj_* i'_! L = (L_0)_{Y-z}, \qquad m'^* R^1 j_* i'_! L = (L_0)_{Y-z}(-1)$$

by Abhyankar. For local constancy, one has to see that the specialization arrows are isomorphisms. One may, for this, replace Λ by $\mathbb{Z}/\ell^\nu$ as above, and then reduce by dévissage to the case where L is the constant sheaf $\mathbb{Z}/\ell$; the conclusion follows from Abhyankar's lemma.

We have therefore proved a). We prove b). Let $\bar\eta$ be a geometric generic point of Y. Choose specialization arrows

$$a : \bar\eta \to Y(\bar z), \qquad b : \bar\eta \to Y(\bar s),$$

where $Y(\bar z)$ (resp. $Y(\bar s)$) is the strict localization of Y at $\bar z$ (resp. $\bar s$). Put

$$M = m^* Rj_*(i'_! L).$$

The commutative square

$$\begin{tikzcd} Y \arrow[r] & Y(\bar s) \\ Y(\bar z) \arrow[u] \arrow[r, "a"'] & \bar\eta \arrow[u, "b"'] \end{tikzcd}$$

induces a commutative square

$$\begin{tikzcd} \mathrm{R}\Gamma(Y, M) \arrow[r, "4.3.1"] \arrow[d] & M_{\bar s} \arrow[d, "b^*"] \\ M_{\bar z} \arrow[r, "a^*"'] & M_{\bar\eta}. \end{tikzcd}$$

Since $M|Y - z$ has locally constant cohomology, b^* is an isomorphism. On the other hand, since $M = i_{1!} i_1^* M$, one has $M_{\bar z} = 0$, hence the arrow 4.3.1 is zero. It remains to prove c). We have $\widetilde{C} = C \times_Z \widetilde{Z}$, where $\widetilde{Z}$ denotes, as above, the strict localization of Z at $\bar z$. Put $\widetilde{Y} = Y \times_Z \widetilde{Z}$. Then there is a commutative diagram of restriction arrows

$$\begin{tikzcd} \mathrm{R}\Gamma(Z - Y, i'_! L) = \mathrm{R}\Gamma(Z, Rj_*(i'_! L)) \arrow[r, "4.3.2"] \arrow[d] & \mathrm{R}\Gamma(\widetilde{C} - \bar s, i'_!(L)|\widetilde{C} - \bar s) \\ \mathrm{R}\Gamma(Y, m^* Rj_*(i'_! L)) \arrow[r, "4.3.1"'] & (Rj_*(i'_! L))_{\bar s} = \mathrm{R}\Gamma(\widetilde{Z} - \widetilde{Y}, (i'_! L)|\widetilde{Z} - \widetilde{Y}) \arrow[u] \end{tikzcd}$$

The vanishing of 4.3.2 therefore follows from b), completing the proof of 4.3.

We prove 4.2. Let $f : T' \to T$ be the blowing-up of the maximal ideal of T, let $D = f^{-1}(z) (\simeq \mathbb{P}^1_S)$ be the exceptional divisor, and let A', B', X' be the respective strict transforms of A, B, X. Then T' is regular, connected, of dimension 2, X' is a regular divisor meeting D transversely at one point z', and the general-position hypotheses on A, B, X mean that A' is disjoint from $B' \cup X'$. Let E' be the inverse image of E on $T' - (A' \cup D \cup X')$, and let

$$i' : T' - (A' \cup D \cup X') \hookrightarrow T' - (A' \cup D)$$

be the inclusion. We must prove that the restriction arrow

$$\mathrm{R}\Gamma(T' - (A' \cup D), i'_!(E')) \longrightarrow \mathrm{R}\Gamma(B' \cup D, i'_!(E')|B' - (B' \cup D)) \tag{*}$$

is zero. But B' is a disjoint union of strictly local schemes B'_j $(1 \le j \le n)$, of dimension 1, such that for every j, $B'_j \cap D$ is reduced to a point b_j. Applying 4.3 c) to $(Z = T' - A', X = X', Y =$

$D-(D\cap A'), L=E', C=B'_j)$, the hypothesis (P) being verified by virtue of the relative purity theorem (SGA 4 XVI 3), since (Z,Y) is a filtered projective limit of smooth pairs of codimension 1 over S, with affine étale transition morphisms, one obtains that the restriction arrow

$$\mathrm{R}\Gamma(T'-(A'\cup D), i'_!(E')) \longrightarrow \mathrm{R}\Gamma(B'_j-b_j, i'_!(E')|B'_j-b_j)$$

is zero. Therefore (*) is zero, which completes the proof of 4.2, hence of 2.5 and 1.2.

5.9. Traces and extension of scalars

Let $A\to B$ be a morphism of flat K-algebras. It induces a morphism of K-algebras $A^e\to B^e$, hence an extension-of-scalars functor

$$-\overset{\mathrm{L}}{\otimes}_{A^e} B^e : D(A^e)\longrightarrow D(B^e).$$

For E,F as in 5.3.5, if $F\overset{\mathrm{L}}{\otimes}_{A^e} B^e\in \operatorname{ob} D^b(B^e)$ and $\mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\in \operatorname{ob} D^b((B^e)^\circ)$, there is a commutative square in $D(K)$

$$\begin{array}{ccc}
\mathrm{RHom}_A(E,F)\overset{\mathrm{L}}{\otimes}_{A^e} E & \xrightarrow{(1)} & F\overset{\mathrm{L}}{\otimes}_{A^e} A\\
\Big\downarrow{\scriptstyle (3)} & & \Big\downarrow{\scriptstyle (4)}\\
\mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_{B^e}(B\overset{\mathrm{L}}{\otimes}_A E) & \xrightarrow[(2)]{} & (F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_{B^e} B,
\end{array}\tag{5.9.1}$$

where the arrows (1) and (2) are defined by 5.3.6, (4) by $A\to B$, and (3) by the canonical arrow $F\to F\overset{\mathrm{L}}{\otimes}_{A^e} B^e$. The verification is immediate. Similarly, for E,F,G as in 5.4.2, if $F\overset{\mathrm{L}}{\otimes}_{A^e} B^e\in \operatorname{ob} D^+(B^e)$ and $(F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_B (B\overset{\mathrm{L}}{\otimes}_A G)\in \operatorname{ob} D^+(B)$, one has a commutative square in $D(K)$

$$\begin{array}{ccc}
\mathrm{RHom}_A(E,F)\overset{\mathrm{L}}{\otimes}_A G & \xrightarrow{(1)} & \mathrm{RHom}_A(E,F\overset{\mathrm{L}}{\otimes}_A G)\\
\Big\downarrow{\scriptstyle (3)} & & \Big\downarrow{\scriptstyle (4)}\\
\mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_B (B\overset{\mathrm{L}}{\otimes}_A G) & \xrightarrow[(2)]{} & \mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, (F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_B (B\overset{\mathrm{L}}{\otimes}_A G)),
\end{array}\tag{5.9.2}$$

where (1) and (2) are defined by 5.4.2, and (3) and (4) by $F\to F\overset{\mathrm{L}}{\otimes}_{A^e} B^e$. From this one deduces, for E,F as in 5.6.1, with $F\overset{\mathrm{L}}{\otimes}_{A^e} B^e\in \operatorname{ob} D^b(B^e)$, a commutative square in $D(K)$, analogous to the preceding ones, in which the vertical arrows are defined by 5.6.1:

$$\begin{array}{ccc}
\mathrm{RHom}_A(E,F\overset{\mathrm{L}}{\otimes}_A E) & \xrightarrow{\mathrm{Tr}_A} & F\overset{\mathrm{L}}{\otimes}_{A^e} A\\
\Big\downarrow & & \Big\downarrow\\
\mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, (F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_B (B\overset{\mathrm{L}}{\otimes}_A E)) & \xrightarrow[\mathrm{Tr}_B]{} & (F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_{B^e} B.
\end{array}\tag{5.9.3}$$

Finally, if one is given $F' \in \operatorname{ob} D^b(B^e)$ such that $\mathrm{RHom}_A(E, F') \in \operatorname{ob} D^b((B^\circ))$, and $f : F \overset{\mathrm{L}}{\otimes}_{A^e} B^e \to F'$, then there is a compatibility slightly more general than 5.9.1, expressed by an analogous commutative square, with F' in place of $F \overset{\mathrm{L}}{\otimes}_{A^e} B^e$ in the lower row and with the vertical arrows defined by f. One has similarly commutative squares analogous to 5.9.2 and 5.9.3 with F' in place of $F \overset{\mathrm{L}}{\otimes}_{A^e} B^e$. Taking in particular $F = A$ and, for $f : A \overset{\mathrm{L}}{\otimes}_{A^e} B^e \to B$, the arrow defined by $A \to B$, one obtains a commutative square

$$\begin{tikzcd} \mathrm{RHom}_A(E,E) \arrow[r, "\mathrm{Tr}_A"] \arrow[d] & A \overset{\mathrm{L}}{\otimes}_{A^e} A \arrow[d] \\ \mathrm{RHom}_B(B \overset{\mathrm{L}}{\otimes}_A E, B \overset{\mathrm{L}}{\otimes}_A E) \arrow[r, "\mathrm{Tr}_B"'] & B \overset{\mathrm{L}}{\otimes}_{B^e} B, \end{tikzcd} \tag{5.9.4}$$

where the left vertical arrow, respectively the right vertical arrow, is given by extension of scalars, respectively is the canonical arrow deduced from $A \to B$.

5.10. Traces and restriction of scalars

Let $A \to B$ be a morphism of flat K-algebras such that B, as a left A-module, is flat and of finite presentation, i.e. locally a direct factor of a free A-module of finite type. This is the case for example if $A \to B$ is the morphism $K[H] \to K[G]$ defined by an injective morphism of discrete groups $H \hookrightarrow G$ of finite index. The scalar-restriction homomorphism

$$\mathrm{Hom}_B(B, B) \longrightarrow \mathrm{Hom}_A(B, B),$$

where B is considered as a left module, identifies, by the canonical isomorphisms

$$B \xrightarrow{\sim} \mathrm{Hom}_B(B, B), \qquad \mathrm{Hom}_A(B, A) \otimes_A B \xrightarrow{\sim} \mathrm{Hom}_A(B, B),$$

with an arrow of B-bimodules

$$B \longrightarrow \mathrm{Hom}_A(B, A) \otimes_A B. \tag{5.10.1}$$

The composite of 5.10.1 with the evaluation arrow $\mathrm{Hom}_A(B, A) \otimes_A B \to A_\natural$ (5.3.3) is the homomorphism $f \mapsto \mathrm{Tr}_A(f)$ ($= \mathrm{Tr}_A(d_f)$, where d_f is right multiplication by f); hence, by the formula $\mathrm{Tr}_A(f_g) = \mathrm{Tr}_A(gf)$ (5.8.5), it factors through $B_\natural$ as an arrow again denoted $\mathrm{Tr}_{B/A} : B_\natural \to A_\natural$. Thus there is a commutative square

$$\begin{tikzcd} B \arrow[r, "\mathrm{Tr}_B"] \arrow[d, "5.10.1"'] & B_\natural \arrow[d, "\mathrm{Tr}_{B/A}"] \\ \mathrm{Hom}_A(B, A) \otimes_A B \arrow[r, "\mathrm{ev}"'] & A_\natural. \end{tikzcd} \tag{5.10.2}$$

In the situation $({}_B E, {}^B P_A)$, one has a homomorphism of right B-modules

$$\mathrm{Hom}_B(E, P) \otimes_A B \longrightarrow \mathrm{Hom}_B(E, P \otimes_A B), \qquad u \otimes b \longmapsto (x \longmapsto \nu(x) \otimes b), \tag{5.10.3}$$

which is an isomorphism, as is checked immediately: it is enough to see that the underlying K-linear homomorphism is an isomorphism, and this follows from the fact that B is locally a direct factor of a free A-module of finite type. This isomorphism derives to an isomorphism of $D(B)$

$$\mathrm{RHom}_B(E,P)\overset{\mathrm{L}}{\otimes}_A B \xrightarrow{\sim} \mathrm{RHom}_B(E,P\overset{\mathrm{L}}{\otimes}_A B), \tag{5.10.4}$$

for $E \in \mathrm{ob}\, D^-(B)$, $P \in \mathrm{ob}\, D^+(B \otimes_K A^\circ)$: resolve E by modules of the form $B \otimes_A L$, where L is flat, and P by injective $(B \otimes_K A^\circ)$-modules, which are necessarily injective as B-modules, since A is flat over K; this allows one to remove the R on the left, and the resolution chosen for E also allows one to remove it on the right.

For $E \in \mathrm{ob}\, D^-(B)$, $M \in \mathrm{ob}\, D^+(K)$, $F \in \mathrm{ob}\, D^-(B)$, define an arrow in $D(K)$

$$\mathrm{RHom}_B(E,B\overset{\mathrm{L}}{\otimes}_K M)\overset{\mathrm{L}}{\otimes}_B F \longrightarrow \mathrm{RHom}_A(E,A\overset{\mathrm{L}}{\otimes}_K M)\overset{\mathrm{L}}{\otimes}_A F \tag{5.10.5}$$

as the composite of the arrows

$$\begin{aligned}
\mathrm{RHom}_B(E,B\overset{\mathrm{L}}{\otimes}_K M)\overset{\mathrm{L}}{\otimes}_B F &\xrightarrow{(1)} \mathrm{RHom}_B(E,\mathrm{Hom}_A(B,A)\overset{\mathrm{L}}{\otimes}_A (B\overset{\mathrm{L}}{\otimes}_K M))\overset{\mathrm{L}}{\otimes}_B F \\
&\xrightarrow{(2)} \mathrm{RHom}_B(E,\mathrm{Hom}_A(B,A)\overset{\mathrm{L}}{\otimes}_K M)\overset{\mathrm{L}}{\otimes}_A B\overset{\mathrm{L}}{\otimes}_B F \\
&\xrightarrow{(3)} \mathrm{RHom}_B(E,\mathrm{Hom}_A(B,A\overset{\mathrm{L}}{\otimes}_K M))\overset{\mathrm{L}}{\otimes}_A F \\
&\xrightarrow{(4)} \mathrm{RHom}_A(E,A\overset{\mathrm{L}}{\otimes}_K M)\overset{\mathrm{L}}{\otimes}_A F,
\end{aligned}$$

where (1) is defined by 5.10.1, (2) by the inverse isomorphism of 5.10.4 with $P = \mathrm{Hom}_A(B,A)\overset{\mathrm{L}}{\otimes}_K M$, (3) by the canonical isomorphism

$$\mathrm{Hom}_A(B,A)\overset{\mathrm{L}}{\otimes}_K M \xrightarrow{\sim} \mathrm{Hom}_A(B,A\overset{\mathrm{L}}{\otimes}_K M)$$

(5.4.2), and (4) by the adjunction isomorphism 5.2.2.

Proposition 5.10.6. Suppose that $M\overset{\mathrm{L}}{\otimes}_K F \in \mathrm{ob}\, D^+(B)$. Then there is a commutative square in $D(K)$

$$\begin{tikzcd}
\mathrm{RHom}_B(E,B\overset{\mathrm{L}}{\otimes}_K M)\overset{\mathrm{L}}{\otimes}_B F \arrow[r, "5.4.2"] \arrow[d, "5.10.5"'] & \mathrm{RHom}_B(E,M\overset{\mathrm{L}}{\otimes}_K F) \arrow[d, "\mathrm{res}"] \\
\mathrm{RHom}_A(E,A\overset{\mathrm{L}}{\otimes}_K M)\overset{\mathrm{L}}{\otimes}_A F \arrow[r, "5.4.2"'] & \mathrm{RHom}_A(E,M\overset{\mathrm{L}}{\otimes}_K F),
\end{tikzcd} \tag{5.10.7}$$

where the right vertical arrow is restriction of scalars.

One has to verify that there is a commutative square

$$\begin{tikzcd}
E \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B F \arrow[r] \arrow[d] & M \overset{\mathrm{L}}{\otimes}_K F \arrow[d, "\mathrm{res}"] \\
E \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A F \arrow[r] & M \overset{\mathrm{L}}{\otimes}_K F,
\end{tikzcd}$$

where the horizontal arrows are given by the evaluations (5.3.6). This square is deduced, by applying $- \overset{\mathrm{L}}{\otimes}_B F$, from the analogous square with $F = B$, so that one may suppose $F = B$. The commutativity of 5.10.7 in this case is equivalent to that of the triangle

$$\begin{tikzcd}
\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \arrow[d, "(1)"'] \arrow[dr, "(2)"] & \\
\mathrm{RHom}_B(E, \mathrm{Hom}_A(B, B \overset{\mathrm{L}}{\otimes}_K M)) \arrow[r, "(3)"'] & \mathrm{RHom}_A(E, B \overset{\mathrm{L}}{\otimes}_K M),
\end{tikzcd}$$

where (1) is defined by 5.10.1, (2) is restriction of scalars, and (3) is the adjunction isomorphism. But the arrow $B \overset{\mathrm{L}}{\otimes}_K M \to \mathrm{Hom}_A(B, B \overset{\mathrm{L}}{\otimes}_K M)$ given by 5.10.1 is nothing other than the adjunction arrow corresponding to the functors restriction of scalars and $\mathrm{Hom}_A(B, -)$; the commutativity follows at once.

Let $E \in \mathrm{ob}\, D^-(B)$, $M \in \mathrm{ob}\, D^b(K)$ be such that $\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \in \mathrm{ob}\, D^b(B^\circ)$. Then, by 5.3.5, there is an evaluation arrow

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_{B^e} E \longrightarrow M \overset{\mathrm{L}}{\otimes}_K (B \overset{\mathrm{L}}{\otimes}_{B^e} B),$$

which, by composition with the canonical arrow $B \overset{\mathrm{L}}{\otimes}_{B^e} B \to B_\natural$, gives an arrow

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E \longrightarrow M \overset{\mathrm{L}}{\otimes}_K B_\natural. \tag{5.10.8}$$

Proposition 5.10.9. Suppose that A is of finite presentation and flat as a K-module, and that $A_\natural$, $B_\natural$ are flat over K. Let $E \in \mathrm{ob}\, D^-(B)$, $M \in \mathrm{ob}\, D^b(K)$ be such that $\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \in \mathrm{ob}\, D^b(B^\circ)$. Then $\mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \in \mathrm{ob}\, D^b(A^\circ)$, and there is a commutative square

$$\begin{tikzcd}
\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E \arrow[r, "5.10.8"] \arrow[d, "5.10.5"'] & M \overset{\mathrm{L}}{\otimes}_K B_\natural \arrow[d, "M \overset{\mathrm{L}}{\otimes} \mathrm{Tr}_{B/A}"] \\
\mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A E \arrow[r, "5.10.8"'] & M \overset{\mathrm{L}}{\otimes}_K A_\natural.
\end{tikzcd} \tag{5.10.10}$$

The additional hypotheses on A and B (verified, for example, if A and B are K-algebras of finite groups) are no doubt useless, and one should have a commutative square analogous to 5.10.10 with $A_\natural$ (resp. $B_\natural$) replaced by $A \overset{\mathrm{L}}{\otimes}_{A^e} A$ (resp. $B \overset{\mathrm{L}}{\otimes}_{B^e} B$).

The first assertion of 5.10.9 is trivial. We prove the second. We may suppose that M has injective components and that E has components of the form $B \otimes_K L$, with L flat. Then

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E = \mathrm{Hom}^\bullet_B(E, B \otimes_K M) \otimes_B E,$$

$$\mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A E = \mathrm{Hom}^\bullet_A(E, A \otimes_K M) \otimes_A E,$$

and the arrows in 5.10.10 are defined at the level of complexes. Thus we are reduced to verifying that, if L, M are K-modules and E a left B-module, there is a commutative square

$$\begin{tikzcd}
\mathrm{Hom}_B(B \otimes_K L, B \otimes_K M) \otimes_B (B \otimes_K L) \arrow[r, "5.3.3"] \arrow[d, "(1)"'] & M \otimes_K B_\natural \arrow[d, "M \otimes_K \mathrm{Tr}_{B/A}"] \\
\mathrm{Hom}_A(B \otimes_K L, A \otimes_K M) \otimes_A (B \otimes_K L) \arrow[r, "5.3.3"'] & M \otimes_K A_\natural,
\end{tikzcd}$$

where the arrow (1) is defined by sending $B \otimes_K M$ into $\mathrm{Hom}_A(B, A) \otimes_A B \otimes_K M$ by 5.10.1 and composing with an isomorphism of the type 5.10.3 and an adjunction isomorphism. Since A is flat and of finite presentation as a K-module, the tensor-product arrows

$$B \otimes_K \mathrm{Hom}_K(L, M) = \mathrm{Hom}_B(B, B) \otimes_K \mathrm{Hom}_K(L, M) \to \mathrm{Hom}_B(B \otimes_K L, B \otimes_K M),$$

$$\mathrm{Hom}_A(B, A) \otimes_K \mathrm{Hom}_K(L, M) \to \mathrm{Hom}_A(B \otimes_K L, A \otimes_K M)$$

are isomorphisms, by which (1) identifies with the tensor product over K of 5.10.1 by $\mathrm{Hom}_K(L, M) \otimes_K L$. On the other hand, the upper horizontal arrow, respectively the lower horizontal arrow, of the square identifies under these isomorphisms with the tensor product over K of $\mathrm{Tr}_B : B \to B_\natural$ (resp. $\mathrm{ev} : \mathrm{Hom}_A(B, A) \otimes_A B \to A_\natural$) by the evaluation arrow $\mathrm{Hom}_K(L, M) \otimes_K L \to M$. The commutativity therefore follows from that of 5.10.2, completing the proof of 5.10.9.

Corollary 5.10.11. Under the hypotheses of 5.10.9, for $E \in \mathrm{ob}\, D(B)_{\mathrm{parf}}$, $M \in \mathrm{ob}\, D^b(K)$, there is a commutative square in $D(K)$

$$\begin{tikzcd}
\mathrm{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E) \arrow[r, "\mathrm{Tr}_B"] \arrow[d] & M \overset{\mathrm{L}}{\otimes}_K B_\natural \arrow[d, "M \overset{\mathrm{L}}{\otimes} \mathrm{Tr}_{B/A}"] \\
\mathrm{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E) \arrow[r, "\mathrm{Tr}_A"'] & M \overset{\mathrm{L}}{\otimes}_K A_\natural,
\end{tikzcd} \tag{5.10.12}$$

where the left vertical arrow is restriction of scalars.

It is enough indeed to conjugate 5.10.6 and 5.10.9.

5.11. Application to finite groups: divisibility properties

Let G be a finite group, with neutral element denoted e, and put $K[G] = B$. The K-module $B_\natural$ is free with basis the set $G_\natural$ of conjugacy classes of G; every section s of $B_\natural$ is written uniquely

$$s = \sum_{x \in G_\natural} s(x) x.$$

Proposition 5.11.1. Cf. SGA $4^{1/2}$, Rapport 4.5. Let $E \in \operatorname{ob} D(B)_{\mathrm{parf}}$, $M \in \operatorname{ob} D^b(K)$. For $u \in \operatorname{Hom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E)$, one has, in $H^0(T, M)$,

$$\operatorname{Tr}_K(u) = |G| \operatorname{Tr}_B(u)(e). \tag{5.11.1}$$

It is enough to apply 5.10.11 with $A = K$, noting that $\operatorname{Tr}_{B/K} : B_\natural \to K$ is given by

$$\operatorname{Tr}_{B/K}(g) = \begin{cases} |G|, & g = e, \\ 0, & g \neq e. \end{cases} \tag{5.11.2}$$

Proposition 5.11.3. Under the hypotheses of 5.11.1, let $g \in G$, let Z_g be the centralizer of g in G, and put $A = K[Z_g]$. Let $u \in \operatorname{Hom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E)$, and let $gu \in \operatorname{Hom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E)$ be the composite of the image of u in $\operatorname{Hom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E)$ with left multiplication by g. Then, in $H^0(T, M)$,

$$\operatorname{Tr}_A(gu)(e) = \operatorname{Tr}_B(u)(g^{-1}). \tag{5.11.4}$$

We shall show more generally that there is a commutative square

$$\begin{tikzcd} \operatorname{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E) \arrow[r, "\operatorname{Tr}_B"] \arrow[d] & M \overset{\mathrm{L}}{\otimes}_K B_\natural \arrow[r, "x \mapsto x(g^{-1})"] & M \\ \operatorname{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E) \arrow[r, "\operatorname{Tr}_A"'] & M \overset{\mathrm{L}}{\otimes}_K A_\natural \arrow[ur, "x \mapsto x(e)"'] & \end{tikzcd}$$

where the left vertical arrow is the composite of restriction of scalars and left multiplication by g. This square is the boundary of the diagram obtained by inserting $\operatorname{Tr}_{B/A}$ and the left multiplication s_g by g; the central square commutes by 5.10.11, and the lower one by the definition of Tr_A. It remains only to prove that, for $h \in G$,

$$\operatorname{Tr}_A(x \mapsto gxh)(e) = \text{the value at } g^{-1} \text{ of the image of } h \text{ in } B_\natural,$$

i.e.

$$\operatorname{Tr}_A(x \mapsto gxh)(e) = \begin{cases} 1, & g^{-1} \text{ is conjugate to } h, \\ 0, & \text{otherwise.} \end{cases} \tag{*}$$

For this one may replace T by the punctual topos and K by $\mathbb{Z}$, and it is then enough to prove (*) after multiplication by $|Z_g|$. In view of 5.11.1, we are therefore reduced to showing that

$$\operatorname{Tr}_{\mathbb{Z}}(x \mapsto gxh) = \begin{cases} |Z_g|, & g^{-1} \text{ is conjugate to } h, \\ 0, & \text{otherwise.} \end{cases}$$

But this follows from the fact that the left hand side is the number of fixed points of the map $x \mapsto gxh$ from G to G, whence the commutativity of 5.11.4 and the conclusion of 5.11.3.

Corollary 5.11.5. Under the hypotheses of 5.11.3, one has

$$\operatorname{Tr}_K(gu) = |Z_g| \operatorname{Tr}_B(u)(g^{-1}) = |Z_g| \operatorname{Tr}_A(gu)(e). \tag{5.11.5}$$

5.12. Traces and external tensor products

Let, for $i = 1, 2$, A_i be a flat K-algebra, and put $A = A_1 \otimes_K A_2$. Let $E_i \in \operatorname{ob} D^-(A_i)$, $F_i \in \operatorname{ob} D^b(A_i^e)$ be such that $\mathrm{RHom}_{A_i}(E_i, F_i) \in \operatorname{ob} D^b(A_i^\circ)$, let $F = F_1 \overset{\mathrm{L}}{\otimes}_K F_2 \in \operatorname{ob} D^b(A^e)$, and suppose $\mathrm{RHom}_A(E, F) \in \operatorname{ob} D^b(A^\circ)$, where $E = E_1 \overset{\mathrm{L}}{\otimes}_K E_2$. Then there is a commutative square

$$\begin{tikzcd}
(\mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_{A_1} E_1) \overset{\mathrm{L}}{\otimes}_K (\mathrm{RHom}_{A_2}(E_2, F_2) \overset{\mathrm{L}}{\otimes}_{A_2} E_2) \arrow[r, "(1)"] \arrow[d, "(2)"'] & (F_1 \overset{\mathrm{L}}{\otimes}_{A_1^e} A_1) \overset{\mathrm{L}}{\otimes}_K (F_2 \overset{\mathrm{L}}{\otimes}_{A_2^e} A_2) \arrow[d, "(3)"] \\
\mathrm{RHom}_A(E, F) \overset{\mathrm{L}}{\otimes}_A E \arrow[r] & F \overset{\mathrm{L}}{\otimes}_{A^e} A,
\end{tikzcd} \tag{5.12.1}$$

where (1) is the tensor product of the arrows 5.3.5, (2) is defined by the tensor-product arrow

$$\mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_{A_2}(E_2, F_2) \longrightarrow \mathrm{RHom}_A(E, F), \tag{5.12.2}$$

and (3) is the evident canonical isomorphism. The verification is essentially trivial. One checks in the same way that, for $E_i \in \operatorname{ob} D^-(A_i)$, $F_i \in \operatorname{ob} D^+(A_i^e)$, $G_i \in \operatorname{ob} D^-(A_i)$, with $F_i \overset{\mathrm{L}}{\otimes}_{A_i} G_i \in \operatorname{ob} D^b(A_i)$, $\mathrm{RHom}_{A_i}(E_i, F_i) \in \operatorname{ob} D^b(A_i^\circ)$, $\mathrm{RHom}_{A_i}(E_i, F_i \overset{\mathrm{L}}{\otimes}_{A_i} G_i) \in \operatorname{ob} D^b(K)$, $F \overset{\mathrm{L}}{\otimes}_A G \in \operatorname{ob} D^b(A)$, where $G = G_1 \overset{\mathrm{L}}{\otimes}_K G_2$, there is a commutative square

$$\begin{tikzcd}
(\mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_{A_1} G_1) \overset{\mathrm{L}}{\otimes}_K (\mathrm{RHom}_{A_2}(E_2, F_2) \overset{\mathrm{L}}{\otimes}_{A_2} G_2) \arrow[r, "(1)"] \arrow[d, "(2)"'] & \mathrm{RHom}_{A_1}(E_1, F_1 \overset{\mathrm{L}}{\otimes}_{A_1} G_1) \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_{A_2}(E_2, F_2 \arrow[d, "(3)"] \\
\mathrm{RHom}_A(E, F) \overset{\mathrm{L}}{\otimes}_A G \arrow[r, "5.4.2"'] & \mathrm{RHom}_A(E, F \overset{\mathrm{L}}{\otimes}_A G),
\end{tikzcd} \tag{5.12.3}$$

where (2) and (3) are tensor-product arrows of the type 5.12.2, and (1) is the tensor product of the arrows 5.4.2. From the commutativity of the squares 5.12.1 and 5.12.3 it follows that, for $E_i \in \operatorname{ob} D(A_i)_{\mathrm{parf}}$, $F_i \in \operatorname{ob} D^b(A_i^e)$ such that $F = F_1 \overset{\mathrm{L}}{\otimes}_K F_2 \in \operatorname{ob} D^b(A^e)$, there is a commutative square

$$\begin{tikzcd}
\mathrm{RHom}_{A_1}(E_1, F_1 \overset{\mathrm{L}}{\otimes}_{A_1} E_1) \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_{A_2}(E_2, F_2 \overset{\mathrm{L}}{\otimes}_{A_2} E_2) \arrow[r, "(1)"] \arrow[d, "(2)"'] & (F_1 \overset{\mathrm{L}}{\otimes}_{A_1^e} A_1) \overset{\mathrm{L}}{\otimes}_K (F_2 \overset{\mathrm{L}}{\otimes}_{A_2^e} A_2) \arrow[d, "(3)"] \\
\mathrm{RHom}_A(E, F \overset{\mathrm{L}}{\otimes}_A E) \arrow[r, "5.6.1"'] & F \overset{\mathrm{L}}{\otimes}_{A^e} A,
\end{tikzcd} \tag{5.12.4}$$

where (1) is the tensor product of the arrows Tr_{A_i} (5.6.1), (2) is defined by 5.12.2, and (3) is the canonical isomorphism appearing in 5.12.1. Consequently, for $u_i \in \mathrm{Hom}_{A_i}(E_i, F_i \overset{\mathrm{L}}{\otimes}_{A_i} E_i)$, one has

$$\mathrm{Tr}_A(u_1 \overset{\mathrm{L}}{\otimes}_K u_2) = \mathrm{Tr}_{A_1}(u_1)\, \mathrm{Tr}_{A_2}(u_2), \tag{5.12.5}$$

the product on the right being defined by the canonical arrow

$$H^0(T, F_1 \overset{\mathrm{L}}{\otimes}_{A_1^e} A_1) \otimes H^0(T, F_2 \overset{\mathrm{L}}{\otimes}_{A_2^e} A_2) \longrightarrow H^0(T, F \overset{\mathrm{L}}{\otimes}_{A^e} A).$$

5.13. Traces of semilinear endomorphisms

Let A be a K-algebra and φ an endomorphism of A. If E, F are left A-modules, recall that an additive homomorphism $u : E \to F$ is called φ-A-linear if $u(ax) = \varphi(a)u(x)$ for all $a \in A$, $x \in E$, i.e. if u is an A-linear homomorphism from E to the A-module ${}_{(\varphi)}F$ deduced from F by restriction of scalars along φ. If F is an A-bimodule and G a left A-module, there is an evident isomorphism

$$ {}_{(\varphi)}F \otimes_A G \xrightarrow{\sim} {}_{(\varphi)}(F \otimes_A G), $$

which derives trivially to

$$ {}_{(\varphi)}F \overset{\mathrm{L}}{\otimes}_A G \xrightarrow{\sim} {}_{(\varphi)}(F \overset{\mathrm{L}}{\otimes}_A G). $$

Assuming A flat over K, one deduces from 5.4.2 the arrow

$$ \mathrm{RHom}_A(E, {}_{(\varphi)}F) \overset{\mathrm{L}}{\otimes}_A G \longrightarrow \mathrm{RHom}_A(E, {}_{(\varphi)}(F \overset{\mathrm{L}}{\otimes}_A G)) \tag{5.13.1} $$

for $E \in \operatorname{ob} D^-(A)$, $F \in \operatorname{ob} D^+(A^e)$, $G \in \operatorname{ob} D^-(A)$, with $F \overset{\mathrm{L}}{\otimes}_A G \in \operatorname{ob} D^+(A)$. For $E \in \operatorname{ob} D(A)_{\mathrm{parf}}$, 5.13.1 is an isomorphism, which allows one to define, by 5.3.5, a trace arrow

$$ \mathrm{Tr}_{A,\varphi} : \mathrm{RHom}_A(E, {}_{(\varphi)}(F \overset{\mathrm{L}}{\otimes}_A E)) \longrightarrow {}_{(\varphi)}F \overset{\mathrm{L}}{\otimes}_{A^e} A. \tag{5.13.2} $$

In particular, for $F = A$, one obtains

$$ \mathrm{Tr}_{A,\varphi} : \mathrm{Hom}_A(E, {}_{(\varphi)}E) \longrightarrow H^0(T, A_{\natural,\varphi}), \tag{5.13.3} $$

where $A_{\natural,\varphi} = {}_{(\varphi)}A \otimes_{A^e} A$ is the quotient K-module of A by the submodule locally generated by local sections of the form $ab - \varphi(b)a$. In the case where $A = K[G]$, G is a discrete group, and φ is defined by an endomorphism of G, $A_{\natural,\varphi}$ is the free K-module with basis $G_{\natural,\varphi}$, where $G_{\natural,\varphi}$ is the quotient of G by the action $(g, x) \mapsto \varphi(g)xg^{-1}$; the orbits are sometimes called φ-conjugacy classes of G. The traces 5.13.2 coincide, for T punctual, with those considered by Grothendieck and discussed in Exposé XII.

In the situation of 5.10, if φ, ψ are respective endomorphisms of A, B, compatible with $A \to B$, there is a unique arrow

$$ \mathrm{Tr}_A : B_{\natural,\psi} \longrightarrow A_{\natural,\varphi} $$

making the square, analogous to 5.10.2,

$$ \begin{tikzcd} {}_{(\psi)}B = \mathrm{Hom}_B(B, {}_{(\psi)}B) \arrow[r, "\mathrm{Tr}_{B,\psi}"] \arrow[d] & B_{\natural,\psi} \arrow[d, "\mathrm{Tr}_A"] \\ \mathrm{Hom}_A(B, {}_{(\varphi)}A) \otimes_A B \arrow[r, "\mathrm{ev}"'] & A_{\natural,\varphi}, \end{tikzcd} \tag{5.13.4} $$

commute, where ev is the evaluation arrow, and the left vertical arrow is restriction of scalars. One defines an arrow analogous to 5.10.5, with $A \otimes_K M$ (resp. $B \otimes_K M$) replaced by ${}_{(\varphi)}A \otimes_K M$ (resp. ${}_{(\psi)}B \otimes_K M$), and there are compatibilities analogous to 5.10.6, 5.10.9, 5.10.11, which we leave to the reader to state.

Let G be a finite group and φ an endomorphism of G. For $g \in G$, let $Z_{g,\varphi}$ be the φ-centralizer of g in G, i.e. the set of $x \in G$ such that $\varphi(x)gx^{-1} = g$. Put $A = K[G]$, and let $E \in \operatorname{ob} D(A)_{\mathrm{parf}}$,

$M \in \operatorname{ob} D^b(K)$. The preceding compatibilities easily imply that, for $u \in \operatorname{Hom}_A(E, {}_{(\varphi)}(M \overset{\mathrm{L}}{\otimes}_K E))$ and $g \in G$,

$$\operatorname{Tr}_K(gu) = |Z_{g^{-1},\varphi}| \operatorname{Tr}_{A,\varphi}(u)(g^{-1}), \tag{5.13.5}$$

the details of the verification being left to the reader.

6. Equivariant Correspondences and Divisibility of Local Terms

6.1. Notation

In this number, S denotes the spectrum of a field of characteristic p, and Λ a noetherian commutative ring annihilated by an integer not divisible by p. The S-schemes considered will be assumed separated and of finite type. The Λ-algebras considered will be associative, unital, and of finite type, but not necessarily commutative. Most often, and in particular for everything concerning "non-commutative Lefschetz–Verdier local terms," they will be assumed flat; in applications, they will be algebras of finite groups.

If X is an S-scheme and A a Λ-algebra, we write $D({}_AX)$ (resp. $D(X_A)$) for the derived category of the category of sheaves of left (resp. right) A-modules on X. For A commutative, we shall also write $D(X, A)$ for $D({}_AX)$, and $D(X) = D(X, \Lambda)$. We denote by $D_c(-)$ (resp. $D_{\mathrm{ctf}}(-)$) the full subcategory formed by complexes with constructible cohomology (resp. finite tor-dimension and constructible cohomology). If A, B are Λ-algebras, we write $D({}_AX_B)$ for the derived category of sheaves of (A, B)-bimodules, left over A and right over B. Finally, we use the notations A^o, A^e, $A_\natural$ of 5.1.

6.2

The formalism of (III 1, 2, 3) extends without difficulty to the "non-commutative" case. We briefly indicate the main points of this generalization.

Let X be an S-scheme and let A, B be Λ-algebras. If $L \in \operatorname{ob} D_{\mathrm{ctf}}(X_A)$ and $M \in \operatorname{ob} D({}_AX_B)$, with M of finite tor-dimension and constructible cohomology as an object of $D(X_B)$, then

$$L \overset{\mathrm{L}}{\otimes}_A M \in \operatorname{ob} D_{\mathrm{ctf}}(X_B)$$

as is immediate. On the other hand, it follows from (SGA $4^{1/2}$ Finiteness 1.6) that, for M as above and $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_AX)$, one has

$$\operatorname{RHom}_A(L, M) \in \operatorname{ob} D_{\mathrm{ctf}}(X_B).$$

It also follows from loc. cit. that, if f is an S-morphism, $D_{\mathrm{ctf}}({}_A-)$ is stable under the operations $(f^*, f_*, f_!, f^!)$, the case of f^* being trivial.

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism,

$$f = f_1 \times_S f_2 : X = X_1 \times_S X_2 \longrightarrow Y = Y_1 \times_S Y_2,$$

and let A be a Λ-algebra. For $L_1 \in \operatorname{ob} D^-(X_{1A})$, $L_2 \in \operatorname{ob} D^-(X^o_{2A})$, there is a Künneth arrow in $D^-(Y)$

$$f_{1!}L_1 \overset{\mathrm{L}}{\otimes}_{S,A} f_{2!}L_2 \longrightarrow f_*(L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2), \tag{6.2.1}$$

where

$$L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \overset{\text{dfn}}{=} p_1^* L_1 \overset{\mathrm{L}}{\otimes}_A p_2^* L_2.$$

As in (III 1.6), one deduces that it is an isomorphism. If B, C are Λ-algebras, with B or C assumed flat over Λ in order to define

$$\overset{\mathrm{L}}{\otimes}_A : D^-(X_{BA}) \times D^-(X_{AC}) \longrightarrow D^-(X_{BC}),$$

there is an isomorphism analogous to 6.2.1 in $D^-(Y_{BC})$, for $L_1 \in \mathrm{ob}\, D^-(X_{B1A})$, $L_2 \in \mathrm{ob}\, D^-(X_{A2B})$. One checks similarly that, for $M_1 \in \mathrm{ob}\, D^-(Y_{1A})$, $M_2 \in \mathrm{ob}\, D^-(Y^o_{2A})$, there is a Künneth isomorphism in $D^-(X)$

$$f^! M_1 \overset{\mathrm{L}}{\otimes}_{S,A} f^! M_2 \longrightarrow f^!(M_1 \overset{\mathrm{L}}{\otimes}_{S,A} M_2), \tag{6.2.2}$$

and an analogous isomorphism in $D^-({}_B X_C)$ for $M_1 \in \mathrm{ob}\, D^-(Y_{B1A})$, $M_2 \in \mathrm{ob}\, D^-(Y_{A2C})$.

Finally, let, for $i = 1, 2$, A_i be a Λ-algebra, A_2 being flat, $E_1 \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{A_1}X_1)$, $F_2 \in \mathrm{ob}\, D^+(X_{A_2})$, $F_1 \in \mathrm{ob}\, D({}_{A_1}X_{1A_2})$, with F_1 of finite tor-dimension and constructible cohomology as an object of $D(X_{1A_2})$. Then

$$\mathrm{RHom}_{A_1}(E_1, F_1) \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_{1A_2}),$$

as seen above, and there is a canonical arrow in $D^+(X)$

$$\mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_{S,A_2} F_2 \longrightarrow \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1 \overset{\mathrm{L}}{\otimes}_{S,A_2} F_2), \tag{6.2.3}$$

defined by composing the "inverse-image" arrow (III 2.1.1)

$$p_1^* \mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2 \longrightarrow \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2$$

with the arrow analogous to 5.4.2

$$\mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2 \longrightarrow \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1 \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2),$$

adjoint, by 5.2.3, to the arrow

$$p_1^* E_1 \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2 \longrightarrow p_1^* F_1 \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2$$

defined by the evaluation arrow 5.3.6. One deduces from 6.2.1, as in (III 2.3), that 6.2.3 is an isomorphism. In particular, for $A_2 = \Lambda$, $F_2 = \Lambda_{X_2}$, 6.2.3 gives an isomorphism

$$p_1^* \mathrm{RHom}_{A_1}(E_1, F_1) \xrightarrow{\sim} \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1). \tag{6.2.4}$$

6.3

If $t : T \to S$ is an S-scheme, and A is a Λ-algebra, we put

$$t^!_A = KA_T \qquad (\in \mathrm{ob}\, D^+({}_A T_A)). \tag{6.3.1}$$

For $A = \Lambda$, we write K_T instead of $K\Lambda_T$. It follows from 6.2.2 that there is a canonical isomorphism in $D^+({}_A T_A)$

$$K_T \overset{\mathrm{L}}{\otimes}_\Lambda A \xrightarrow{\sim} KA_T. \tag{6.3.2}$$

We denote by D_A the functor

$$\mathrm{RHom}_A(-, KA_T).$$

It sends $D_{\mathrm{ctf}}({}_AT)$ into $D_{\mathrm{ctf}}(T_A)$ and conversely, and the proof of (SGA $4^{1/2}$ Finiteness Theorem 4) shows that, for $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_AT)$, the canonical arrow

$$L \longrightarrow D_A D_A L$$

is an isomorphism. For $A = \Lambda$, we write D instead of D_A.

6.4

Let A be a flat Λ-algebra, and let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, $p_i : X \to X_i$ the projections, and $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_AX_i)$. By 6.2.3 applied to $A_1 = A_2 = A$, $E_1 = L_1$, $F_2 = L_2$, $E_2 = KA_{X_1}$, there is an isomorphism

$$D_A L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \xrightarrow{\sim} \mathrm{RHom}_A(p_1^* L_1, KA_{X_1} \overset{\mathrm{L}}{\otimes}_{S,A} L_2),$$

whence, by composing with the Künneth isomorphism 6.2.2

$$KA_{X_1} \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \xrightarrow{\sim} p_2^! L_2,$$

an isomorphism

$$D_A L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \xrightarrow{\sim} \mathrm{RHom}_A(p_1^* L_1, p_2^! L_2), \tag{6.4.1}$$

which generalizes (III 3.1.1).

Let $c : C \to X$ be an S-morphism, and put $c_i = p_i c$. By the induction formula (SGA 4 XVIII 3.1.12.2), there is a canonical isomorphism, analogous to (III 3.2.1),

$$c^! \mathrm{RHom}_A(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \mathrm{RHom}_A(c_1^* L_1, c_2^! L_2), \tag{6.4.2}$$

and hence, by composing with 6.4.1, an isomorphism

$$H^0(X, c_* c^! (D_A L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2)) \xrightarrow{\sim} \mathrm{Hom}_A(c_1^* L_1, c_2^! L_2). \tag{6.4.3}$$

The elements of $\mathrm{Hom}_A(c_1^* L_1, c_2^! L_2)$ will be called cohomological correspondences from L_1 to L_2 with support in c. The terminology differs slightly from that of loc. cit.; the two terminologies coincide when c is proper, which is the case in practice.

Given S-morphisms $f_i : X_i \to Y_i$ $(i = 1, 2)$ and a commutative square (III 3.7.1) with c proper, one defines, as in loc. cit., direct-image arrows analogous to (III 3.7.5), (III 3.7.6), for $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_AX_i)$.

6.5

Let A be a flat Λ-algebra, X an S-scheme, $X^2 = X \times_S X$, $p_1, p_2 : X^2 \to X$ the canonical projections, $\delta : X \to X^2$ the diagonal, and

$$c : C \longrightarrow X^2$$

an S-morphism. Put $c_i = p_i c$, and let $X^c = C \times_{X^2} X$ be the scheme of fixed points of c, giving a cartesian square

$$\begin{tikzcd} X^c \arrow[r, "i^c"] \arrow[d, "\delta^c"'] & X \arrow[d, "\delta"] \\ C \arrow[r, "c"'] & X^2. \end{tikzcd} \tag{6.5.1}$$

Put

$$KA_{X,L_\natural} = K_X \overset{\mathrm{L}}{\otimes}_\Lambda (A \overset{\mathrm{L}}{\otimes}_{A^e} A), \qquad KA_{X,\natural} = K_X \overset{\mathrm{L}}{\otimes}_\Lambda A_\natural. \tag{6.5.2}$$

Let $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X)$. We define a "local trace" homomorphism

$$\mathrm{Tr}_A : \delta^* c_* \mathrm{RHom}_A(p_1^* L, p_2^! L) \longrightarrow KA_{X,L_\natural}, \tag{6.5.3}$$

by composing the isomorphism

$$\delta^* \mathrm{RHom}_A(p_1^* L, p_2^! L) \xrightarrow{\sim} \delta^*(D_A L \overset{\mathrm{L}}{\otimes}_{S,A} L) \tag{6.5.4}$$

deduced from the inverse of 6.4.1 by applying δ^*, the trivial isomorphism

$$\delta^*(D_A L \overset{\mathrm{L}}{\otimes}_{S,A} L) = D_A L \overset{\mathrm{L}}{\otimes}_A L, \tag{6.5.5}$$

and the evaluation arrow 5.3.5

$$D_A L \overset{\mathrm{L}}{\otimes}_A L \longrightarrow KA_{X,L_\natural}. \tag{6.5.6}$$

It follows at once from the definition that, for $X = S$, 6.5.3 coincides with the arrow

$$\mathrm{Tr}_A : \mathrm{RHom}_A(L, L) \longrightarrow A \overset{\mathrm{L}}{\otimes}_{A^e} A$$

of 5.6.1. Composing the arrow deduced from 6.5.3 by applying $i^c c^!$ with the arrow

$$\delta^* c^c_* \mathrm{RHom}_A(c_1^* L, c_2^! L) \longrightarrow i^c_* i^{c!} \delta^* \mathrm{RHom}_A(p_1^* L, p_2^! L) \tag{6.5.7}$$

deduced from 6.4.2 and the base-change arrow

$$\delta^* c_* c^! \longrightarrow i^c_* i^{c!} \delta^*,$$

one obtains an arrow

$$\mathrm{Tr}_A : \delta^* c_* \mathrm{RHom}_A(c_1^* L, c_2^! L) \longrightarrow i^c_* i^{c!} KA_{X,L_\natural} = i^c_* KA_{X^c,L_\natural}, \tag{6.5.8}$$

and hence the arrow

$$\mathrm{Tr}_A : \mathrm{Hom}_A(c_1^* L, c_2^! L) \longrightarrow H^0(X^c, KA_{X^c,L_\natural}). \tag{6.5.9}$$

We again write Tr_A for the composites of 6.5.3, 6.5.8, 6.5.9 with the arrows deduced from the canonical arrow

$$KA_{X,L_\natural} \longrightarrow KA_{X,\natural}.$$

One checks easily that, for $A = \Lambda$, one has, for $u \in \mathrm{Hom}(c_1^* L, c_2^! L)$,

$$\mathrm{Tr}(u) = \langle u, 1 \rangle, \tag{6.5.10}$$

with the notation of (III 4.2.5), where 1 denotes the diagonal correspondence. More precisely, the arrow

$$c_* \mathrm{RHom}(c_1^* L, c_2^! L) \longrightarrow \delta_* i^c_* K_{X^c}$$

adjoint to 6.5.8 is the one defined by (III 4.2.4') with $X_1 = X_2$, $L_1 = L_2$, $d = \delta$, and the diagonal correspondence $1 \in \mathrm{Hom}(L, L)$.

6.6

Let A be a flat Λ-algebra, and let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X_i)$. We define a pairing

$$\mathrm{RHom}_A(p_1^* L_1, p_2^! L_2) \overset{\mathrm{L}}{\otimes}_\Lambda \mathrm{RHom}_A(p_2^* L_2, p_1^! L_1) \longrightarrow KA_{X, L_\natural} \tag{6.6.1}$$

as follows. One identifies the left hand side with

$$\mathrm{RHom}_A(p_1^* L_1, p_1^* KA_{X_1}) \overset{\mathrm{L}}{\otimes} p_2^* L_2 \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_A(p_2^* L_2, p_2^* KA_{X_2}) \overset{\mathrm{L}}{\otimes} p_1^* L_1$$

by the isomorphisms 6.4.1, then with

$$A \overset{\mathrm{L}}{\otimes}_{A^e} \mathrm{RHom}_A(\ ) \overset{\mathrm{L}}{\otimes}_\Lambda p_2^* L_2 \overset{\mathrm{L}}{\otimes}_\Lambda \mathrm{RHom}_A(\ ) \overset{\mathrm{L}}{\otimes}_\Lambda p_1^* L_1 \overset{\mathrm{L}}{\otimes}_{A^e} A$$

by 5.3.1. This object is sent to

$$A \overset{\mathrm{L}}{\otimes}_{A^e} (KA_{X_1} \overset{\mathrm{L}}{\otimes}_\Lambda KA_{X_2}) \overset{\mathrm{L}}{\otimes}_{A^e} A = R$$

by the tensor product of the evaluation arrows 5.3.6. Finally R is identified with

$$A \overset{\mathrm{L}}{\otimes}_{A^e} (KA_{X_1} \overset{\mathrm{L}}{\otimes}_\Lambda KA_{X_2})$$

by 5.8.2, and then with $KA_{X, L_\natural}$ by 6.5.2 and (III 1.7.6).

If $c : C \to X$, $d : D \to X$ are correspondences, and

$$e = c \times_X d : E \to X,$$

one deduces from 6.6.1, as in (III 4.2), a pairing

$$c_* \mathrm{RHom}_A(c_1^* L_1, c_2^! L_2) \overset{\mathrm{L}}{\otimes}_\Lambda d_* \mathrm{RHom}_A(d_2^* L_2, d_1^! L_1) \longrightarrow e_* KA_{E, L_\natural}, \tag{6.6.2}$$

and hence a cup-product

$$\langle\ ,\ \rangle_A : \mathrm{Hom}_A(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}_A(d_2^* L_2, d_1^! L_1) \longrightarrow H^0(E, KA_{E, L_\natural}). \tag{6.6.3}$$

It is not difficult to extend to the non-commutative case the notion of composition of correspondences studied in (III 5.2), and one checks that, for

$$u \in \mathrm{Hom}_A(c_1^* L_1, c_2^! L_2), \qquad v \in \mathrm{Hom}_A(d_2^* L_2, d_1^! L_1),$$

one has the formula analogous to (III 5.2.9)

$$\langle u, v \rangle_A = \mathrm{Tr}_A(\bar{v} u) = \mathrm{Tr}_A(u \bar{v}). \tag{6.6.4}$$

The Lefschetz formula (III 4.4) generalizes without difficulty in this context. Given proper S-morphisms $f_i : X_i \to Y_i$ $(i = 1, 2)$ and a commutative cube (III 4.4.0), for $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X_i)$ one has a commutative square analogous to (III 4.4.1), with K_C (resp. K_D) replaced by $KA_{C, L_\natural}$ (resp. $KA_{D, L_\natural}$), and a formula analogous to (III 4.5.1), in $H^0(D, KA_{D, L_\natural})$:

$$\langle f_* u, f_* v \rangle_A = g_* \langle u, v \rangle_A. \tag{6.6.5}$$

The arrow

$$f_* c_* KA_{C, L_\natural} \longrightarrow d_* KA_{D, L_\natural}$$

is deduced from arrow (4) of (III 4.4.1) by tensoring with

$$A \overset{\mathrm{L}}{\otimes}_{A^e} A.$$

6.7

Let X and c be as in 6.5, and let $A \to B$ be a morphism of flat Λ-algebras. With respect to extension and restriction of scalars, the arrows Tr_A (6.5.3, 6.5.8, 6.5.9) behave like the arrows Tr_A of 5.6.1, 5.6.2. The point is that, if f is an S-morphism, the functors $(f^*, f_*, f_!, f^!)$ commute with extension and restriction of scalars; consequently, the study of the behavior of Tr_A (6.5.3) with respect to extension and restriction of scalars reduces to the behavior of the "tensor-product" arrow

$$\mathrm{RHom}_A(p_1^*L, p_1^*KA_X) \overset{\mathrm{L}}{\otimes}_A p_2^*L \longrightarrow \mathrm{RHom}_A(p_1^*L, p_1^*KA_X \overset{\mathrm{L}}{\otimes}_A p_2^*L)$$

and of the "evaluation" arrow

$$\mathrm{RHom}_A(L, KA_X) \overset{\mathrm{L}}{\otimes}_A L \longrightarrow KA_{X,L_\natural}.$$

Thus the commutativity of the squares 5.9.1, 5.9.2 implies the commutativity of the square

$$\begin{tikzcd}
\delta^* c_* \mathrm{RHom}_A(c_1^*L, c_2^!L) \arrow[r, "\mathrm{Tr}_A"] \arrow[d] & i_*^c KA_{X^c, L_\natural} \arrow[d] \\
\delta^* c_* \mathrm{RHom}_B(c_1^*(B \overset{\mathrm{L}}{\otimes}_A L), c_2^!(B \overset{\mathrm{L}}{\otimes}_A L)) \arrow[r, "\mathrm{Tr}_B"'] & i_*^c KB_{X^c, L_\natural},
\end{tikzcd} \tag{6.7.1}$$

where the left vertical arrow is extension of scalars and the right vertical arrow is defined by the canonical arrow

$$A \overset{\mathrm{L}}{\otimes}_{A^e} A \longrightarrow B \overset{\mathrm{L}}{\otimes}_{B^e} B.$$

It follows that, for $u \in \mathrm{Hom}_A(c_1^*L, c_2^!L)$,

$$\mathrm{Tr}_B(B \overset{\mathrm{L}}{\otimes}_A u) = \varphi\, \mathrm{Tr}_A(u), \tag{6.7.2}$$

where

$$\varphi : H^0(X^c, KA_{X^c, L_\natural}) \longrightarrow H^0(X^c, KB_{X^c, L_\natural})$$

is the canonical map.

Assume A is finite type and flat as a Λ-module, B finite type and flat as a left A-module, and $A_\natural$, $B_\natural$ flat over Λ. There is a map $\mathrm{Tr}_{B/A} : B_\natural \to A_\natural$; we shall again denote by

$$\mathrm{Tr}_{B/A} : KB_{X,\natural} \longrightarrow KA_{X,\natural} \tag{6.7.3}$$

the arrow deduced from it by applying $K_X \overset{\mathrm{L}}{\otimes} -$ (6.5.2). It follows from 5.10.6 and 5.10.9 that, for $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_BX)$, there is a commutative square

$$\begin{tikzcd}
\delta^* c_* \mathrm{RHom}_B(c_1^*L, c_2^!L) \arrow[r, "\mathrm{Tr}_B"] \arrow[d] & i_*^c KB_{X^c, \natural} \arrow[d, "\mathrm{Tr}_{B/A}"] \\
\delta^* c_* \mathrm{RHom}_A(c_1^*L, c_2^!L) \arrow[r, "\mathrm{Tr}_A"'] & i_*^c KA_{X^c, \natural},
\end{tikzcd} \tag{6.7.4}$$

where the left vertical arrow is restriction of scalars. Consequently, for $u \in \mathrm{Hom}_B(c_1^*L, c_2^!L)$, one has, in $H^0(X^c, KA_{X^c,\natural})$,

$$\mathrm{Tr}_A(u) = \mathrm{Tr}_{B/A}(\mathrm{Tr}_B(u)). \tag{6.7.5}$$

6.8

Let A, B be flat Λ-algebras, X, Y S-schemes, $Z = X \times_S Y$, and let $c : C \to X^2$, $d : D \to Y^2$ be correspondences. Let

$$e = c \times_S d : E = C \times_S D \longrightarrow Z^2$$

be their product. Then

$$Z^e = X^c \times_S Y^d.$$

Put $R = A \otimes_\Lambda B$. As in (III 5.3), one defines the external tensor product

$$\mathrm{Hom}_A(c_1^* L, c_2^! L) \otimes \mathrm{Hom}_B(d_1^* M, d_2^! M) \xrightarrow{\overset{\mathrm{L}}{\otimes}_S} \mathrm{Hom}_R(e_1^*(L \overset{\mathrm{L}}{\otimes}_S M), e_2^!(L \overset{\mathrm{L}}{\otimes}_S M)), \tag{6.8.1}$$

and one deduces easily from the commutativity of the squares 5.12.1, 5.12.3 that, for

$$u \in \mathrm{Hom}_A(c_1^* L, c_2^! L), \qquad v \in \mathrm{Hom}_B(d_1^* M, d_2^! M),$$

one has, in $H^0(Z^e, KR_{Z^e, L_\natural})$,

$$\mathrm{Tr}_R(u \overset{\mathrm{L}}{\otimes}_S v) = \mathrm{Tr}_A(u)\,\mathrm{Tr}_B(v), \tag{6.8.2}$$

the product on the right being defined by the canonical arrow

$$H^0(X^c, KA_{X^c, L_\natural}) \otimes H^0(Y^d, KB_{Y^d, L_\natural}) \longrightarrow H^0(Z^e, KR_{Z^e, L_\natural}), \tag{6.8.3}$$

coming from the isomorphism

$$KA_{X^c, L_\natural} \overset{\mathrm{L}}{\otimes}_S KB_{Y^d, L_\natural} \longrightarrow K(A \otimes B)_{Z^e, L_\natural}$$

given by (III 1.7.6) and arrow (4) of 5.9.1.

6.9

Let G be a finite group. If X is a G-S-scheme, that is, an S-scheme endowed with an action of G by S-automorphisms, and A is a Λ-algebra, we denote by $D(G, {}_AX)$ the derived category of the category of sheaves of left G-A-modules on X. For the notions of G-sheaf of sets, rings, modules, and so on, on X for the étale topology, we refer the reader to (X 1). We denote by the subscript c (resp. ctf) the full subcategory formed by complexes of G-A-modules which, as complexes of A-modules, have constructible cohomology (resp. finite tor-dimension and constructible cohomology).

Similarly, if B is a Λ-algebra, we denote by $D(G, {}_AX_B)$ the derived category of sheaves of G-(A, B)-bimodules on X. If G acts trivially on X, a left G-A-module on X is just a left $A[G]$-module on X without operators, hence

$$D(G, {}_AX) = D({}_{A[G]}X).$$

Since G is finite, one also has

$$D_c({}_AX) = D_c({}_{A[G]}X),$$

but only an inclusion

$$D_{\mathrm{ctf}}({}_{A[G]}X) \subset D_{\mathrm{ctf}}(G, {}_AX).$$

The duality formalism of (SGA 4 XVII, XVIII) extends without difficulty to G-S-schemes. Since the functor forgetting the action of G is faithful, it is enough to extend naturally the definition of the "six operations" ($\overset{\mathrm{L}}{\otimes}, \mathrm{RHom}, f^*, f_*, f_!, f^!$) and the canonical arrows considered in loc. cit. (base change, Künneth, duality, etc.). This extension presents no difficulty: for the definition of the functors $f_!$, $f^!$, it is enough to use equivariant compactifications. We leave to the reader the task of paraphrasing the definitions and results 6.2 to 6.8 in the setting of G-S-schemes.

If, for $i = 1, 2$, X_i is a G-S-scheme, $c : C \to X = X_1 \times_S X_2$ is a G-S-morphism, and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(G, {}_A X_i)$, the elements of

$$\mathrm{Hom}_A(c_1^* L_1, c_2^! L_2),$$

where Hom_A is taken in the category $D(G, {}_A-)$, will be called equivariant cohomological correspondences from L_1 to L_2 with support in c. Note that, when G acts trivially on X_i ($i = 1, 2$) and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_{A[G]} X_i)$, the isomorphism 6.4.1 in $D(G, X)$ refines to an isomorphism

$$D_{A[G]} L_1 \overset{\mathrm{L}}{\otimes}_{S, A[G]} L_2 \xrightarrow{\sim} \mathrm{RHom}_{A[G]}(p_1^* L_1, p_2^! L_2).$$

It follows that, in a G-equivariant situation of type 6.5, with trivial action of G on the spaces, and with $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_{A[G]} X) \subset D_{\mathrm{ctf}}(G, {}_A X)$, one may, for

$$u \in \mathrm{Hom}_{A[G]}(c_1^* L, c_2^! L),$$

define a local term

$$\mathrm{Tr}_{A[G]}(u) \in H^0(X^c, KA[G]_{X^c, L_\natural}),$$

finer, in the evident sense, than the local term

$$\mathrm{Tr}_A(u) \in H^0(G, X^c, KA_{X^c, L_\natural})$$

obtained by using only finite tor-dimension of L with respect to A.

6.10

Let X, c be as in 6.5, and let $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]} X)$. For

$$u \in \mathrm{Hom}_{\Lambda[G]}(c_1^* L, c_2^! L),$$

6.5.9 gives a local trace

$$\mathrm{Tr}_{\Lambda[G]}(u) \in H^0(X^c, K\Lambda[G]_{X^c, \natural}) = H^0(X^c, K_{X^c} \otimes \Lambda[G]_\natural), \tag{6.10.1}$$

where $G_\natural$ is the set of conjugacy classes of G. For every $x \in G_\natural$, one may consider the "value of $\mathrm{Tr}_{\Lambda[G]}(u)$ at x,"

$$\mathrm{Tr}_{\Lambda[G]}(u)(x) \in H^0(X^c, K_{X^c}),$$

the image of $\mathrm{Tr}_{\Lambda[G]}(u)$ under the projection

$$H^0(X^c, K_{X^c} \otimes \Lambda[G]_\natural) \longrightarrow H^0(X^c, K_{X^c})$$

defined by x. It follows from 6.7.5 and 5.11.2 that

$$\mathrm{Tr}_\Lambda(u) = |G| \ \mathrm{Tr}_{\Lambda[G]}(u)(e), \tag{6.10.2}$$

where e denotes the identity element of G.

More generally, let $g \in G$, let Z_g be the centralizer of g, and let

$$gu \in \mathrm{Hom}_{\Lambda[Z_g]}(c_1^* L, c_2^! L)$$

be the correspondence obtained by composing u with left multiplication by $g : c_1^* L \to c_1^* L$. We have $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda[Z_g]}X)$, and hence 6.10.2 implies

$$\mathrm{Tr}_\Lambda(gu) = |Z_g|\ \mathrm{Tr}_{\Lambda[Z_g]}(gu)(e). \tag{6.10.3}$$

By the same argument as in 5.11.3, one deduces from 6.7.4 the formula

$$\mathrm{Tr}_{\Lambda[Z_g]}(gu)(e) = \mathrm{Tr}_{\Lambda[G]}(u)(g^{-1}), \tag{6.10.4}$$

which allows 6.10.3 to be rewritten, if desired, in the form

$$\mathrm{Tr}_\Lambda(gu) = |Z_g|\ \mathrm{Tr}_{\Lambda[G]}(u)(g^{-1}). \tag{6.10.5}$$

6.11

Let $f : X \to Y$ be a proper G-S-morphism, and let

$$\begin{tikzcd} C \arrow[r, "c"] \arrow[d] & X \times_S X \arrow[d, "f \times_S f"] \\ D \arrow[r, "d"'] & Y \times_S Y \end{tikzcd} \tag{6.11.1}$$

be a commutative equivariant square of G-S-schemes, with c and d proper. Assume that G acts trivially on Y and D. We denote by

$$f^{c/d} \quad (\text{or } f^c) : X^c \longrightarrow Y^d \tag{6.11.2}$$

the morphism induced by 6.11.1 on the schemes of fixed points of c and d, with the notation of 6.5.1. Let $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(G, {}_A X)$, and let

$$u \in \mathrm{Hom}(c_1^* L, c_2^! L)$$

be a G-equivariant correspondence with support in c. By direct image, it gives a G-equivariant correspondence

$$f_* u \in \mathrm{Hom}_{\Lambda[G]}(d_1^* f_* L, d_2^! f_* L).$$

For $g \in G$, let Z_g denote the centralizer of g, and let

$$gu \in \mathrm{Hom}((gc)_1^* L, c_2^! L) \tag{6.11.3}$$

be the Z_g-equivariant correspondence with support in

$$gc \overset{\mathrm{dfn}}{=} (gc_1 = c_1 g, c_2) : C \to X \times_S X, \tag{6.11.4}$$

defined as the composite of the arrows

$$g^*(c_1^* L) \xrightarrow{\alpha_g} c_1^* L \xrightarrow{u} c_2^! L,$$

where α_g is the arrow giving the action of G on c_1^*L. It is clear that

$$f_*(gu) = g(f_*u). \tag{6.11.5}$$

Proposition 6.12. Under the hypotheses of 6.11, assume that

$$f_*L \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}Y).$$

Then, for every $g \in G$,

$$f_*^{gc} \operatorname{Tr}_\Lambda(gu) = |Z_g| \ \operatorname{Tr}_{\Lambda[G]}(f_*u)(g^{-1}), \tag{6.12.1}$$

where

$$f_*^{gc} : H^0(X^{gc}, K_{X^{gc}}) \longrightarrow H^0(Y^d, K_{Y^d})$$

is the Gysin morphism defined by the adjunction arrow

$$f_*^{gc} K_{X^{gc}} = f_*^{gc} f^{gc!} K_{Y^d} \longrightarrow K_{Y^d}.$$

Indeed, by 6.10.5 applied to f_*u, and by 6.11.5, one has

$$\operatorname{Tr}_\Lambda(f_*(gu)) = |Z_g| \ \operatorname{Tr}_{\Lambda[G]}(f_*u)(g^{-1}),$$

and 6.12.1 therefore follows from the Lefschetz formula (III 4.5.1).

6.13

Let ℓ be a prime number distinct from p. For every integer $n \geq 1$, put

$$\Lambda_n = \mathbb{Z}/\ell^n.$$

Under the hypotheses of 6.11, let $L = (L_n)$ be a projective system of objects of

$$D_{\mathrm{ctf}}(G, X, \Lambda_n)$$

such that

$$L_m \overset{\mathrm{L}}{\otimes}_{\Lambda_m} \Lambda_n \xrightarrow{\sim} L_n \quad (m \geq n),$$

and let

$$u \in \operatorname{Hom}(c_1^*L, c_2^!L)$$

be a projective system of equivariant correspondences $u_n \in \operatorname{Hom}(c_1^*L_n, c_2^!L_n)$ such that, for $m \geq n$,

$$u_m \overset{\mathrm{L}}{\otimes}_{\Lambda_m} \Lambda_n = u_n$$

modulo the isomorphism $L_m \overset{\mathrm{L}}{\otimes}_{\Lambda_m} \Lambda_n = L_n$. By compatibility of local traces with extension of scalars, the local terms, for $g \in G$,

$$\operatorname{Tr}_{\Lambda_n}(gu_n) \in H^0(X^{gc}, K\Lambda_{n,X^{gc}})$$

define an element

$$\operatorname{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(X^{gc}, K_{\ell,X^{gc}}) \overset{\mathrm{dfn}}{=} \varprojlim_n H^0(X^{gc}, K\Lambda_{n,X^{gc}}),$$

and the terms

$$f_*^{gc}\operatorname{Tr}_{\Lambda_n}(gu_n)=\operatorname{Tr}_{\Lambda_n}(gf_*u_n)$$

an element

$$f_*\operatorname{Tr}_{\mathbb{Z}_\ell}(gu)\in H^0(Y^d,K_{\ell,Y^d})\overset{\text{dfn}}{=}\varprojlim_n H^0(Y^d,K\Lambda_{n,Y^d}).$$

Assume moreover that

$$f_*M_n\in\operatorname{ob}D_{\mathrm{ctf}}({}_{\Lambda_n[G]}Y)$$

for all n. Then the local traces

$$\operatorname{Tr}_{\Lambda_n[G]}(f_*u_n)$$

similarly define an element

$$\operatorname{Tr}_{\mathbb{Z}_\ell[G]}(f_*u)\in H^0(Y^d,K\mathbb{Z}_\ell[G]_{Y^d,\natural})\overset{\text{dfn}}{=}\varprojlim_n H^0(Y^d,K\Lambda_n[G]_{Y^d,\natural}).$$

It follows from 6.12 that

$$f_*^{gc}\operatorname{Tr}_{\mathbb{Z}_\ell}(gu)=|Z_g|\ \operatorname{Tr}_{\mathbb{Z}_\ell[G]}(f_*u)(g^{-1}).\tag{6.13.1}$$

6.14

Let X be a G-S-scheme, let $j:U\to X$ be an equivariant open immersion, and let

$$c:C\to X\times_S X$$

be a morphism of G-S-schemes such that $c_2=(p_2c):C\to X$ is quasi-finite and of finite tor-dimension. Suppose that

$$c_1^{-1}(U)\subset c_2^{-1}(U).$$

Then there are commutative squares

$$\begin{array}{ccccc}
U & \longleftarrow & c_1^{-1}(U) & \longleftarrow & c_2^{-1}(U)\longrightarrow U\\
\downarrow & & \downarrow & & \downarrow\\
X & \xleftarrow{c_1} & C & \xrightarrow{c_2} & X\\
\uparrow & & \uparrow & & \uparrow\\
X-U & \longleftarrow & c_2^{-1}(X-U) & \longrightarrow & X-U
\end{array}\tag{6.14.1}$$

The trace morphism

$$\operatorname{Tr}_{c_2}:c_{2!}\Lambda_C\longrightarrow\Lambda_X$$

then defines a filtered G-equivariant correspondence

$$\underline{c}\in\operatorname{Hom}(c_1^*\Lambda_X,c_2^!\Lambda_X),\tag{6.14.2}$$

where Λ_X is endowed with the filtration defined by the submodule $i_!\Lambda_U$, that is,

$$F^n\Lambda_X=\Lambda_X\quad(n\le 0),\qquad F^1\Lambda_X=i_!\Lambda_U,\qquad F^n\Lambda_X=0\quad(n\ge 1).$$

The associated graded consists of two G-equivariant correspondences

$$\operatorname{gr}^0\underline{c}=\underline{c}_{X-U,X}\in\operatorname{Hom}(c_1^*\Lambda_{X-U,X},c_2^!\Lambda_{X-U,X}),$$

$$\operatorname{gr}^1 \underline{c} = \underline{c}_{U,X} \in \operatorname{Hom}(c_1^*\Lambda_{U,X}, c_2^!\Lambda_{X-U,X}), \tag{6.14.3}$$

where we have put

$$\Lambda_{X-U,X} = j_*\Lambda_{X-U}, \qquad \Lambda_{U,X} = i_!\Lambda_U.$$

We have

$$\underline{c}_{X-U,U} = j_*\underline{c}_{X-U}, \tag{6.14.4}$$

where

$$\underline{c}_{X-U} \in \operatorname{Hom}(c_1^*\Lambda_{X-U}, c_2^!\Lambda_{X-U})$$

is the correspondence defined by $\operatorname{Tr}_{c_2,X-U}$. We write $\underline{c}$, $\underline{c}_{U,X}$, and so on, when we want to specify the base ring.

Forgetting the action of G, there are local terms

$$\operatorname{Tr}_\Lambda(\underline{c}), \quad \operatorname{Tr}_\Lambda(\underline{c}_{U,X}), \quad \operatorname{Tr}_\Lambda(\underline{c}_{X-U,X}) \in H^0(X^c, K_{X^c}), \tag{6.14.5}$$

which, by additivity of traces (III 4.13.1), satisfy the relation

$$\operatorname{Tr}_\Lambda(\underline{c}) = \operatorname{Tr}_\Lambda(\underline{c}_{U,X}) + \operatorname{Tr}_\Lambda(\underline{c}_{X-U,X}). \tag{6.14.6}$$

Moreover, by 6.14.4,

$$\operatorname{Tr}_\Lambda(\underline{c}_{X-U,X}) = j_*^c \operatorname{Tr}_\Lambda(\underline{c}_{X-U}), \tag{6.14.7}$$

where

$$j_*^c : H^0((X-U)^c, K_{(X-U)^c}) \longrightarrow H^0(X^c, K_{X^c})$$

is the direct-image morphism defined by $j^c : (X-U)^c \to X^c$. Recall (III 4.3) that, if X is smooth, c a closed immersion, and X^c finite, $\operatorname{Tr}_\Lambda(\underline{c})$ is the image in Λ of the intersection multiplicity (C, X) of C with the diagonal. If, moreover, $X-U$ is smooth, $\operatorname{Tr}_\Lambda(\underline{c}_{X-U})$ is the image in Λ of $(c_2^{-1}(X-U), X-U)$. Note also that, if $X-U$ is proper over S and $c|X-U$ is the diagonal, then the Lefschetz formula (III 4.7) implies that

$$\int_{X-U} \operatorname{Tr}(\underline{c}_{X-U}) \in \Lambda$$

is the Euler–Poincaré characteristic of $X-U$ with values in Λ.

6.15

Let there be a G-equivariant diagram 6.11.1, with f, c, d proper, and an equivariant open immersion $U \to X$. Suppose that c_2 is quasi-finite and of finite tor-dimension, that

$$c_1^{-1}(U) \subset c_2^{-1}(U),$$

that G acts trivially on C and D, and that

$$f|U : U \longrightarrow f(U) = V$$

is an étale Galois covering with group G. This last hypothesis implies that

$$Rf_*(\Lambda_{U,X}) \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}Y) :$$

indeed $Rf_*(\Lambda_{U,X})$ is concentrated in degree 0 and is extension by zero of a sheaf on V locally isomorphic to $\Lambda[G]$. Hence one may apply 6.12 to $L = \Lambda_{U,X}$, $u = \underline{c}_{U,X}$, and, in view of 6.14.6 and 6.14.7, one obtains:

Proposition 6.16. Under the hypotheses of 6.15, for every $g \in G$ one has

$$f_*^{gc} \operatorname{Tr}_\Lambda(g\underline{c}) - (fj)_*^{gc} \operatorname{Tr}_\Lambda(g\underline{c}_{X-U}) = |Z_g| \operatorname{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1}), \tag{6.16.1}$$

where $j : X - U \to X$ is the inclusion, and Z_g denotes the centralizer of g.

Applying 6.16 to the projective system of correspondences $\underline{c}_\ell = (\underline{c}_{\Lambda_n})$, where $\Lambda_n = \mathbb{Z}/\ell^n$ with ℓ prime distinct from p, and taking into account that 6.14.6 and 6.14.7 give, after passage to the limit and application of f_*^{gc},

$$f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_\ell) = f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,U,X}) + (fj)_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,X-U}),$$

one deduces, by 6.13:

Corollary 6.17. Under the hypotheses of 6.15, for every $g \in G$ one has

$$f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_\ell) - (fj)_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,X-U}) = |Z_g| \operatorname{Tr}_{\mathbb{Z}_\ell[G]}(f_*\underline{c}_{\ell,U,X})(g^{-1}), \tag{6.17.1}$$

with the notation of 6.16.

In view of the compatibility of local traces with localization and of the remarks made in 6.14, one sees that 6.17 gives an affirmative solution, in a more general formulation, to Grothendieck's divisibility conjecture (XII 4.5), in the case where the endomorphism φ of G considered in loc. cit. is the identity.

In fact, by following 5.13, it is not difficult to give divisibility statements analogous to 6.12, 6.13, 6.16, 6.17, in the case where c is not G-equivariant, but where c_2 is and c_1 satisfies

$$c_1(xg) = c_1(x)\varphi(g)$$

for every point x of G, φ being a given endomorphism of G. The divisibility formulas are then written as above, with Z_g replaced by the φ-centralizer of g^{-1} (5.13.5).

6.18

Put

$$N_G = \sum_{g \in G} g \in \Lambda[G]. \tag{6.18.1}$$

Let X be an S-scheme. For $L \in \operatorname{ob} D({}_{\Lambda[G]}X)$, left multiplication by N_G induces an arrow in $D(X)$

$$N_G : \Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} L \longrightarrow \operatorname{RHom}_{\Lambda[G]}(\Lambda, L), \tag{6.18.2}$$

where Λ is regarded as a $\Lambda[G]$-algebra by the natural augmentation sending every $g \in G$ to 1.

Lemma 6.18.3. If $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}X)$, the arrow 6.18.2 is an isomorphism.

By dévissage, one reduces to the case where L is perfect, then to $L = \Lambda[G]$, and the assertion follows from the fact that N_G (6.18.1) is a basis of the module of G-invariants of the left $\Lambda[G]$-module $\Lambda[G]$.

Proposition 6.19. In the situation of 6.11, let $M \in \operatorname{ob} D_{\mathrm{ctf}}(Y)$,

$$v \in \operatorname{Hom}(d_1^* M, d_2^! M),$$

and let $a : M \to f_* L$ be an arrow in $D({}_{\Lambda[G]}Y)$, where M is regarded as an object of $D({}_{\Lambda[G]}Y)$ by the augmentation $\Lambda[G] \to \Lambda$, such that the square

$$\begin{tikzcd} d_1^* M \arrow[r, "v"] \arrow[d] & d_2^! M \arrow[d] \\ d_1^* f_* L \arrow[r, "f_* u"'] & d_2^! f_* L \end{tikzcd}$$

is commutative. Suppose that $f_* M \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}Y)$ and that a induces an isomorphism

$$M \xrightarrow{\sim} \operatorname{RHom}_{\Lambda[G]}(\Lambda, f_* L).$$

Then

$$\operatorname{Tr}_\Lambda(v) = \sum_{g \in G_\natural} \operatorname{Tr}_{\Lambda[G]}(f_* u)(g^{-1}). \tag{6.19.1}$$

By 6.18.3, a, composed with N_G^{-1}, gives an isomorphism

$$M \xrightarrow{\sim} \Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} f_* L,$$

under which v is identified with

$$\Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} f_* u.$$

By 6.7.2, $\operatorname{Tr}_\Lambda(v)$ is therefore the image of $\operatorname{Tr}_{\Lambda[G]}(f_* u)$ under the arrow induced by the augmentation

$$\Lambda[G] \longrightarrow \Lambda.$$

This augmentation, by definition, sends every conjugacy class to 1, whence the conclusion.

6.20

Let, for $i = 1, 2$, X_i be a G-S-scheme, and let

$$X = X_1 \times_S X_2, \qquad c : C \to X$$

be a G-S-morphism such that c_2 is quasi-finite and of finite tor-dimension. Let $L_i \in \operatorname{ob} D^+(G, {}_A X_i)$. The trace morphism

$$\operatorname{Tr}_{c_2} : c_{2!} c_2^! L_2 \longrightarrow L_2$$

defines by adjunction an arrow in $D(G, X)$, again denoted

$$\operatorname{Tr}_{c_2} : c_2^! L_2 \longrightarrow c_2'^! L_2, \tag{6.20.1}$$

which associates to every G-equivariant morphism

$$u \in \mathrm{Hom}(c_1^*L_1, c_2^*L_2)$$

a G-equivariant correspondence

$$u^! = \mathrm{Tr}_{c_2} \circ u \in \mathrm{Hom}(c_1^*L_1, c_2^!L_2). \tag{6.20.2}$$

Proposition 6.21. Let $f : X \to Y$ be a proper G-morphism, and let 6.11.1 be an equivariant commutative square, with trivial action of G on Y and D. Suppose c, d are proper, c_2, d_2 quasi-finite and of finite tor-dimension, and the square

$$\begin{tikzcd} C \arrow[r, "c_2"] \arrow[d, "h"'] & X \arrow[d, "f"] \\ D \arrow[r, "d_2"'] & Y \end{tikzcd}$$

is cartesian and tor-independent (SGA 6 III 1.5). Let $M \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$,

$$v \in \mathrm{Hom}(d_1^*M, d_2^*M),$$

hence a G-equivariant homomorphism

$$h^*v \in \mathrm{Hom}(c_1^*f^*M, c_2^*f^*M),$$

and G-equivariant correspondences

$$(h^*v)^! \in \mathrm{Hom}(c_1^*f^*M, c_2^!f^*M), \qquad f_*((h^*v)^!) \in \mathrm{Hom}(d_1^*(f_*f^*M), d_2^!f_*f^*M).$$

Then there is a commutative square of arrows in $D({}_{\Lambda[G]}Y)$, where M is regarded as an object of $D({}_{\Lambda[G]}Y)$ by the trivial action,

$$\begin{tikzcd} d_1^*M \arrow[r, "v^!"] \arrow[d] & d_2^!M \arrow[d] \\ d_1^*f_*f^*M \arrow[r, "f_*((h^*v)^!)"'] & d_2^!f_*f^*M, \end{tikzcd}$$

where the vertical arrows are given by the adjunction arrow $M \to f_*f^*M$.

Put $h^*v = u$, $f^*M = L$, and consider the diagram

$$\begin{tikzcd} d_1^*M \arrow[r, "v"] \arrow[d] & d_2^*M \arrow[r, "\mathrm{Tr}_{d_2}"] \arrow[d] & d_2^!M \arrow[d] & \\ d_1^*f_*L \arrow[r] & h_*c_1^*L \arrow[r, "h_*\,\mathrm{Tr}_{c_2}"] & h_*c_2^!L \arrow[r] & d_2^!f_*L. \end{tikzcd}$$

The arrows (1) and (3) are given by the adjunction arrow $M \to f_*L$, (2) by the adjunction arrow $d_2^*M \to h_*h^*(d_2^*M) = h_*c_2^*L$, (4) by the base-change arrow, and (5) is the canonical isomorphism

(SGA 4 XVIII 3.1.12.3). By definition, $v^!$ is the composite of the upper row. On the other hand, it follows from the interpretation of the direct image of a correspondence (III 3.5 b), 3.7) that $f_*u^!$ is the composite of the lower row. It remains only to prove commutativity of the squares (A) and (B). The commutativity of (A) follows immediately from the definition of the base-change arrow. As for (B), it follows from the fact that, in view of the tor-independence hypothesis, Tr_{d_2} is compatible with base change by f (SGA $4^{1/2}$ Cycle 2.3.3).

Applying 6.19, one obtains:

Corollary 6.22. Under the hypotheses of 6.21, assume that

$$f_*f^*M \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}Y),$$

and that the adjunction arrow

$$M \to f_*f^*M$$

induces an isomorphism

$$M \xrightarrow{\sim} \mathrm{RHom}_{\Lambda[G]}(\Lambda, f_*f^*M).$$

Then

$$\mathrm{Tr}_\Lambda(v^!) = \sum_{g\in G_\natural} \mathrm{Tr}_{\Lambda[G]}(f_*((f^*v)^!))(g^{-1}). \tag{6.22.1}$$

Proposition 6.23. Under the hypotheses of 6.21, suppose moreover that there is an open immersion

$$i : V \to Y$$

such that

$$d_1^{-1}(V) = d_2^{-1}(V),$$

and that f induces a principal Galois covering with group G,

$$f_V : U = f^{-1}(V) \to V.$$

Suppose also that M is of the form $M = i_!E$, where $E \in \operatorname{ob} D(V)$, and that there is an isomorphism

$$\alpha : f^*E \xrightarrow{\sim} \Lambda_X \overset{\mathrm{L}}{\otimes}_\Lambda E_0$$

in $D(G, X)$, with $E_0 \in \operatorname{ob} D(\Lambda[G])$, perfect over Λ. Let

$$v \in \mathrm{Hom}(d_1^*M, d_2^*M)$$

be as in 6.21, and let

$$v_0 \in \mathrm{Hom}_{\Lambda[G]}(E_0, E_0)$$

be the homomorphism defined, thanks to α, by $f^*v|c_1^{-1}(U)$. Then, if $\underline{c}_{U,X}$ denotes the correspondence defined in 6.14.3, with the notation 6.20.2 one has

$$\mathrm{Tr}_\Lambda(v^!) = \sum_{g\in G_\natural} \mathrm{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1})\,\mathrm{Tr}_\Lambda(gv_0). \tag{6.23.1}$$

Note that, by the central character of Tr_Λ, the right hand side does not depend on the choice of α.

By the projection formula

$$f_*\Lambda_Y \overset{\mathrm{L}}{\otimes} M \xrightarrow{\sim} f_*f^*M$$

and 6.15, M satisfies the hypotheses of 6.22. On the other hand, α defines an isomorphism

$$f^*M \xrightarrow{\sim} \Lambda_{U,X} \otimes E_0$$

under which $(f^*v)^!$ is identified with the external tensor product $\underline{c}_{U,X} \otimes v_0$. By the projection formula, f_*f^*M is therefore identified with $f_*(\Lambda_{U,X}) \otimes E_0$, and $f_*((f^*v)^!)$ with the external tensor product $f_*\underline{c}_{U,X} \otimes v_0$. For $g \in G$, the Z_g-equivariant correspondence $gf_*((f^*v)^!)$ is therefore identified with

$$gf_*\underline{c}_{U,X} \otimes gv_0.$$

Since, by 6.10.4,

$$\mathrm{Tr}_{\Lambda[G]}(f_*((f^*v)^!))(g^{-1}) = \mathrm{Tr}_{\Lambda[Z_g]}(gf_*((f^*v)^!))(e),$$

it is enough, in view of 6.22.1, to prove the formula

$$\mathrm{Tr}_{\Lambda[Z_g]}(gf_*\underline{c}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1})\,\mathrm{Tr}_{\Lambda}(gv_0),$$

or, equivalently by 6.10.4,

$$\mathrm{Tr}_{\Lambda[Z_g]}(gf_*\underline{c}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{\Lambda[Z_g]}(gf_*\underline{c}_{U,X})(e)\,\mathrm{Tr}_{\Lambda}(gv_0). \tag{6.23.2}$$

Replacing G by Z_g if necessary, we may suppose that $g = e$. Returning to the definition of the traces, one sees that it is enough to establish the following lemma.

Lemma 6.23.3. (cf. SGA $4^{1/2}$ Report 4.7). Let P be a G_V-module locally a direct factor of a free module of finite type, let $P' = i_!P$, and let Q be a complex of $\Lambda[G]$-modules, perfect over Λ. Then $P \overset{\mathrm{L}}{\otimes}_\Lambda Q$, considered as an object of $D({}_{\Lambda[G]}V)$ by the diagonal action of G, is perfect over $\Lambda[G]$, and there is a commutative square

$$\begin{tikzcd}
\delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P', p_2^!P') \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_{\Lambda[G]}(Q,Q) \arrow[r] \arrow[d] & K_Y \arrow[d, equal] \\
\delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P' \overset{\mathrm{L}}{\otimes} Q, p_2^!(P' \overset{\mathrm{L}}{\otimes} Q)) \arrow[r] & K_Y,
\end{tikzcd}$$

where $\delta : Y \to Y \times_S Y$ is the diagonal, the upper horizontal arrow is the tensor product of the arrows

$$\delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P', p_2^!P') \xrightarrow{6.5.3} K\Lambda[G]_{Y,\natural} \xrightarrow{x\mapsto x(e)} K_Y$$

and

$$\mathrm{RHom}_{\Lambda[G]}(Q,Q) \xrightarrow{\mathrm{Tr}_\Lambda} \Lambda,$$

the vertical arrow is deduced from the external tensor product by restriction along the diagonal

$$\Lambda[G] \to \Lambda[G] \otimes \Lambda[G],$$

and the lower horizontal arrow is the composite of 6.5.3 and of $x \mapsto x(e)$.

The first assertion is evident, for one reduces to $P = \Lambda[G]_V$, and $P \overset{\mathrm{L}}{\otimes}_\Lambda Q$ is then isomorphic to the tensor product of P by Q considered with the trivial action of G. For the second, first note that, by 6.5.4 and 6.5.5, one has

$$\delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^* P', p_2^! P') \xrightarrow{\sim} D_{\Lambda[G]}(i_! P) \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} i_! P \xrightarrow{\sim} i_!(D_{\Lambda[G]} P \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} P),$$

so that we are reduced to checking that the square induced by the preceding square on V commutes. We may therefore suppose that $V = Y$. The square then reads

$$\begin{array}{ccc}
\mathrm{RHom}_{\Lambda[G]}(P, K_Y \overset{\mathrm{L}}{\otimes} P) \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_{\Lambda[G]}(Q, Q) & \longrightarrow & K_Y \\
\downarrow & & \| \\
\mathrm{RHom}_{\Lambda[G]}(P \overset{\mathrm{L}}{\otimes} Q, K_Y \overset{\mathrm{L}}{\otimes} P \overset{\mathrm{L}}{\otimes} Q) & \longrightarrow & K_Y,
\end{array}$$

where the vertical arrow is the external tensor product, the upper horizontal arrow is given by

$$\mathrm{Tr}_{\Lambda[G]}(-)(e) \otimes \mathrm{Tr}_\Lambda,$$

and the lower horizontal arrow by $\mathrm{Tr}_{\Lambda[G]}(-)(e)$. By descent, one reduces to $P = \Lambda[G]_Y$, and, by the initial remark, one may then suppose that G acts trivially on Q. The assertion is then an immediate consequence of 5.12.5.

6.24

If Λ is annihilated by a power of a prime number $\ell \neq p$, the right hand side of 6.23.1 may be rewritten in the form

$$\sum_{g \in G_\natural} \mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_* \underline{c}_{\ell,U,X})(g^{-1}) \, \mathrm{Tr}_\Lambda(g v_0),$$

with the notation of 6.17. Formula 6.17.1 shows that the local terms defined by Grothendieck in (XII 6.2) coincide with Verdier's local terms, at least when the endomorphism φ of G considered in loc. cit. is the identity; but, as already observed, this restriction is not serious. If Y is proper over S, then by conjugating 6.23.1 with the Lefschetz formula for $Y \to S$ (III 4.7), one obtains a formula generalizing (XII 6.3), and a fortiori the Lefschetz formula for Frobenius proved in (SGA $4^{1/2}$ Report) and Part I of the present exposé.

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In SGA $4^{1/2}$:

Finiteness Theorem: finiteness theorems in ℓ-adic cohomology;

Cycle: the cohomology class associated with a cycle;

Report: report on the trace formula.